Subset Sums in a Fourfold Sumset

Proof Supplement for the Upper Bound 86

5 October 2026

This supplement proves that every finite real set A whose subset sums lie in its fourfold sumset has at most 86 elements. It supplies the complete descent used as an input to the earlier report establishing $71 \le N \le 85$; the later exclusion of size 86 and the lower construction are outside its scope.

The complete finite calculations have been replayed using exact integer or rational arithmetic. Their inputs, coverage obligations and verification record are specified in Section 39.

Write $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$, with the empty sum included, and $4A = \{a_1 + a_2 + a_3 + a_4 : a_i \in A\}$, with repetition allowed. The notation N denotes the supremum of $|A|$ over feasible finite real sets. The preliminary bound proved here shows that this supremum is finite. We write $T_r = r(r+1)/2$ for triangular numbers and use AP to mean an arithmetic progression.

The argument separates the proof that actual sets satisfy each finite system from the exact nonnegative row combinations that contradict those systems. The electronic certificate data are an integral part of this computer-assisted proof.

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How the proof is organized

The proof begins with a universal finite cap and then excludes every remaining cardinality in descending order. It never assumes that deleting arbitrary elements preserves feasibility.

1. Part I proves the real-to-integer reduction, covering and peeling estimates, and the preliminary bound $N \le 115$.
2. Part II proves all exclusions from size 115 through size 97, including the common-lattice, defect and hole lemmas used later.
3. Part III proves the successive exclusions of sizes $96, 95, \dots, 87$, with each actual-set reduction and every selected finite alternative specified.
4. The appendices give the fixed-slot certificate and the complete smaller coefficient pools, together with exact specifications of the larger electronic pools and branch trees.

The only externally cited mathematical inputs are the published statements explicitly reproduced with their hypotheses in the proof. References to prior versions of the research are not used as mathematical premises.

1 The logical role of the finite certificates

The finite calculations in this proof are exact arithmetic arguments, not numerical evidence for an inequality. We state explicitly how they are used. None of the following principles asserts that a feasible point of a necessary model is realized by a set.

Lemma 1.1 (Exact nonnegative row combination)**.** *Let $M \in \mathbb{Q}^{p\times q}$ and $b \in \mathbb{Q}^p$. If $w \in \mathbb{Q}^p_{\ge 0}$ satisfies*

$$w^{\mathsf{T}}M \ge 0 \quad \textit{coordinatewise}, \qquad w^{\mathsf{T}}b < 0,$$

then there is no $x \in \mathbb{R}^q_{\ge 0}$ satisfying $Mx \le b$.

Proof. Such an x would give $0 \le (w^{\mathsf{T}}M)x \le w^{\mathsf{T}}b < 0$. Multiplying a rational w by a common positive denominator permits all weights to be integers. Zero weights may be omitted. □

Lemma 1.2 (Exhaustive necessary models)**.** *Fix a cardinality n. Suppose every feasible integer n-element set containing zero determines a member of a specified finite set of roots. For each root, suppose the written construction provides a finite tree of necessary systems with the following properties.*

1. *Every coordinate assigned to an actual set has the asserted sign and satisfies every row at the root.*
2. *At every internal node, every actual set satisfying its hypotheses satisfies the hypotheses of at least one child. All these children are retained. Additional rows are proved under that child's hypotheses.*
3. *Every leaf has a row combination as in Lemma 1.1 for its actual necessary system.*

Then no such integer set exists.

Proof. An alleged set selects a root and then, by the second property, a child at each internal node. Finiteness yields a leaf. Its actual coordinates contradict Lemma 1.1. □

Overlapping alternatives are harmless: disjointness of the mathematical children is unnecessary. Their union must cover the parent. If an integer-valued expression f is split at an integer t, the

alternatives $f \leq t$ and $f \geq t + 1$ are exhaustive. This split is invalid for a merely real auxiliary variable; for a real expression one must instead use alternatives whose union covers all real values. The integrality of each split expression is therefore part of the applicability proof.

The programs regenerate the labeled coefficient rows from the current root and branch, read the supplied multipliers, and perform the indicated integer or rational operations. A successful multiplier check does not prove that its rows apply to an actual set; that is the first and second property of Lemma 1.2, established in the written chapters. Conversely, a written classification without a contradiction at every leaf does not exclude its roots.

For the compact endpoints, assignment words select an ordinary multiplier or mark a root for a supplemental tree. Completeness requires equality between the actual marked root set and the union of the supplemental parent sets, with no missing parent. The checker also verifies the complete required child sets and the contradiction at every leaf. Counts alone, an archived success field, a digest, or a previous run's terminal message cannot substitute for these mathematical checks.

Part I
Global reductions and a preliminary bound

2 Definitions and the cardinality reduction

For a finite real set A, write

$$\mathrm{FS}(A) = \left\{ \sum_{x \in X} x : X \subseteq A \right\}, \qquad 4A = \{a_1 + a_2 + a_3 + a_4 : a_i \in A\}.$$

The empty subset contributes zero, and repetitions are allowed in $4A$. Call A feasible if $\mathrm{FS}(A) \subseteq 4A$. We will first prove the finite bound $|A| \leq 115$. The subsequent endpoint arguments exclude every cardinality from 115 down through 87. The reduction below explains why this proves the bound for all finite real sets.

2.1 Adjoining zero

Lemma 2.1 (Adjoining zero)**.** *If* $\mathrm{FS}(A) \subseteq 4A$ *and* $0 \notin A$*, then* $A \cup \{0\}$ *also satisfies the inclusion.*

Proof. Adjoining zero creates no new subset-sum value, while the fourfold sumset can only grow:

$$\mathrm{FS}(A \cup \{0\}) = \mathrm{FS}(A) \subseteq 4A \subseteq 4(A \cup \{0\}).$$

□

It is therefore enough, for a uniform upper bound, to consider sets containing zero.

2.2 Integralization

Theorem 2.2 (Integralization)**.** *For every* $n \geq 1$*, a real* n*-element solution exists if and only if an integral* n*-element solution exists. Zero can be preserved.*

Proof. Only the forward direction needs proof. Write $A = \{a_1 < \cdots < a_n\}$. For every $X \subseteq A$, choose one representation

$$\sum_{a_i \in X} a_i = a_{j_1} + a_{j_2} + a_{j_3} + a_{j_4},$$

with repetition of indices allowed. These finitely many identities form a homogeneous linear system $Mx = 0$ with integer coefficients, and $a = (a_1, \ldots, a_n)$ is a solution. The kernel is a rational linear subspace, so its rational points are dense. The relatively open region $x_1 < \cdots < x_n$ containing a therefore contains a rational solution. Clearing denominators gives distinct integers and preserves all selected representations. If $a_i = 0$ is to be preserved, append $x_i = 0$ before taking the rational point. □

Rationalization is an existence argument. It preserves distinctness and a prescribed zero; it does not imply that the original real set lies on a one-dimensional rational lattice. Clearing denominators means multiplying by a positive common denominator, so all strict order inequalities survive.

Lemma 2.3 (Descending exact-cardinality reduction)**.** *Let* $m \geq 1$*. Suppose every feasible real set has at most* m *elements and there is no feasible integer set of exactly* m *elements containing zero. Then every feasible real set has at most* $m - 1$ *elements.*

Proof. A counterexample would have exactly m elements. If it omitted zero, Lemma 2.1 would produce a feasible set of size $m+1$, contrary to the assumed bound. It therefore contains zero, and Theorem 2.2 produces an integer counterexample with the same size and with zero preserved. This contradicts the exact-size exclusion. □

Equivalently, a global finite cap M together with exclusions of every zero-containing integer size in $\{87, \dots, M\}$ proves $|A| \le 86$: adjoin zero first, if necessary, and rationalize the resulting set. Its cardinality is between 87 and M for any putative counterexample. Neither formulation deletes arbitrary elements from a feasible set. In particular, later subsets D, D_1, E need not themselves be feasible; all uses of feasibility are applied to subset sums of the original A.

3 The four-extreme-element covering count

For a finite set F, put

$$R(F) = \{x + y : x, y \in F,\ x \neq y\}, \qquad \Delta(F) = |2F \setminus R(F)|.$$

Every value in $2F \setminus R(F)$ is a diagonal value $2x$, and $x \mapsto 2x$ is injective. Hence

$$\Delta(F) \le |F|. \tag{1}$$

From a feasible set A containing zero, take the four largest positive elements and the four smallest negative elements, taking every element on a side that has fewer than four. Call their union B, and put

$$D = A \setminus B, \qquad d = |D|.$$

Then $0 \in D$ and $|B| \le 8$.

Lemma 3.1 (Four-extreme-element covering)**.** *If* $\mathrm{FS}(A) \subseteq 4A$*, then*

$$\boxed{|\mathrm{FS}(D)| \le 2|2D| + 18d + 85.} \tag{2}$$

Proof. We first work on the positive side. Let $p_1 > p_2 > p_3 > p_4 > 0$ be the four largest positive elements, and put $T_p = p_1 + p_2 + p_3 + p_4$. If $x \in \mathrm{FS}(D)$ and $x > 0$, then $T_p + x \in 4A$. Write a representation in decreasing order:

$$T_p + x = z_1 + z_2 + z_3 + z_4, \qquad z_1 \ge z_2 \ge z_3 \ge z_4.$$

Classifying by the first failure of $z_i \le p_i$ gives only

2	$2p_1 + u + v$,
3	$p_1 + 2p_2 + u$ or $3p_2 + u$,
4	$p_1 + p_2 + 2p_3,\ p_1 + 3p_3,\ 2p_2 + 2p_3,\ p_2 + 3p_3,\ 4p_3$.

Subtracting T_p and substituting $A = D \sqcup B$ gives the table

source	translates of $2D$	translates of D	fixed slots
$(2p_1 - T_p) + A + A$	1	8	36
$(p_1 + 2p_2 - T_p) + (A \setminus \{p_1\})$	0	1	7
$(3p_2 - T_p) + (A \setminus \{p_1\})$	0	1	7
five fixed shapes	0	0	5
total	1	10	55

The negative side gives the same table. Without deletions, the two sides and zero yield

$$|\mathrm{FS}(D)| \le 2|2D| + 20d + 111. \tag{3}$$

If one side has fewer than four elements, all elements of that sign lie in B, and D has no subset sum of that sign. The single remaining table is already stronger than (2). We may therefore assume that both sides have four extreme elements. Write the absolute values on the negative side as $q_1 > q_2 > q_3 > q_4 > 0$.

Among the twenty translates of D, Appendix A gives two distinct pairs of centers $u \le v$. For any such pair,

$$|(u + D) \cap (0, \infty)| + |(v + D) \cap (-\infty, 0)| \le d, \tag{4}$$

because for one $z \in D$ the conditions $z > -u$ and $z < -v$ cannot both hold. The two pairs reduce the linear contribution from $20d$ to $18d$.

There are 110 fixed slots in the two tables. Appendix A places them in one common universe and gives 26 mutually disjoint groups. The total score in every group is coordinatewise nonpositive in eight strictly positive gap variables, so at least one slot in each group lies on the wrong open half-line. At most $110 - 26 = 84$ fixed slots survive. Adding zero gives the constant 85, proving (2). □

4 Two four-element peeling steps

Lemma 4.1 (Four-element peeling)**.** *Let F be a finite real set containing zero and at least four nonzero elements. Four nonzero elements can be deleted to obtain F' such that*

$$|\mathrm{FS}(F)| \ge |\mathrm{FS}(F')| + |R(F)| - 1.$$

Proof. Choose the two largest positive and the two smallest negative elements, filling with as-yet unselected nonzero elements if one side has fewer than two. Call the selected set H, put $F' = F \setminus H$, and define

$$\alpha = \sum_{h \in H,\, h<0} h, \qquad \beta = \sum_{h \in H,\, h>0} h, \qquad t = \beta - \alpha > 0.$$

The extreme choice gives $R(F) \subseteq [\alpha, \beta]$, while

$$\alpha + \mathrm{FS}(F') \subseteq \mathrm{FS}(F), \qquad \beta + \mathrm{FS}(F') \subseteq \mathrm{FS}(F).$$

In every residue class modulo t occupied by $\mathrm{FS}(F')$, the union of these two translates contains at least one more point than $\mathrm{FS}(F')$. In the length-t interval $[\alpha, \beta]$, each residue class occurs at most once, except that the endpoints have the same residue. Thus $R(F)$ occupies at least $|R(F)| - 1$ residue classes. Add one point of $R(F)$ from every class not already supplied by the two translates. The stated inequality follows. □

If $d \le 8$, then $|A| \le d + 8 \le 16$, so the desired upper bound is already trivial. Hence assume $d \ge 9$. The set D then has at least eight nonzero elements, and after the first deletion the remaining set still has at least four. We may therefore apply the lemma twice:

$$D \supset D_1 \supset E, \qquad |D_1| = d - 4, \qquad |E| = d - 8, \qquad 0 \in E.$$

Deleting four elements affects at most

$$4(d - 4) + \binom{4}{2} = 4d - 10$$

unordered pairs of distinct elements. Hence

$$|R(D_1)| \geq |R(D)| - (4d - 10).$$

If $s = |\mathrm{FS}(E)|$, the two peeling inequalities give

$$|\mathrm{FS}(D)| \geq s + 2|R(D)| - 4d + 8. \tag{5}$$

Since $|2D| = |R(D)| + \Delta(D)$, comparison with (2) eliminates $R(D)$ and gives

$$\boxed{s \leq 22d + 2\Delta(D) + 77.} \tag{6}$$

5 The rigorous upper bound $N \leq 115$

Write

$$E = P \cup \{0\} \cup (-Q),$$

where $P, Q \subset \mathbb{R}_{>0}$, and put

$$a = |P|, \qquad b = |Q|, \qquad a + b = d - 9.$$

For an r-element positive real set X, define its subset-sum excess and its doubled-value defect by

$$\mu(X) = |\mathrm{FS}(X)| - \left(1 + \frac{r(r+1)}{2}\right), \qquad \delta_0(X) = |\{x \in X : 2x \notin R(X \cup \{0\})\}|.$$

The elementary increasing-chain argument gives $\mu(X) \geq 0$.

We use the following published stability theorem in precisely its stated range: if $r \geq 4$, $0 \leq M \leq r - 4$, and

$$|\mathrm{FS}(X)| \leq 1 + \frac{r(r+1)}{2} + M,$$

then $X = cB$ for some $c > 0$ and an r-element set $B \subset \mathbb{Z}_{>0}$ satisfying

$$\sum_{b \in B} b \leq \frac{r(r+1)}{2} + M.$$

This is Theorem 1.1 of Carpenter–Defant–Kravitz [1].

We also use Lemma 3.6 and Theorem 1.5 of Balandraud–Girard–Griffiths–Hamidoune [2]. For a finite subset of a finite abelian group, these give

$$U \cap (-U) = \varnothing \quad \Longrightarrow \quad |\mathrm{FS}(U)| \geq 2|U|, \tag{7}$$

and, if $S = -S$, $|S| = 2p \geq 4$, and $\mathrm{FS}(S)$ is aperiodic, then

$$|\mathrm{FS}(S)| \geq p(p-1) + 5. \tag{8}$$

Here a finite set is *aperiodic* if only translation by zero fixes it.

Lemma 5.1 (Aperiodic subset sums)**.** *Let G be an abelian group and let $R \subseteq G \setminus \{0\}$ have $m \geq 2$ elements. If $\mathrm{FS}(R)$ is aperiodic, then*

$$\boxed{|\mathrm{FS}(R)| \geq 2m - 2.} \tag{9}$$

Proof. First suppose that G is finite. No member of R has order two: otherwise FS of that singleton, and hence $\mathrm{FS}(R)$, has a nonzero period. Decompose

$$R = S \sqcup U, \qquad S = -S, \quad |S| = 2p, \quad U \cap (-U) = \varnothing, \quad |U| = u.$$

The subset-sum set of every subset of R is aperiodic, since any of its periods would also be a period after addition of the complementary subset-sum set. Kneser's addition theorem, in the form $|C + D| \geq |C + H| + |D + H| - |H|$ with H the stabilizer of $C + D$, therefore has trivial stabilizer here and gives

$$|\mathrm{FS}(R)| \geq |\mathrm{FS}(S)| + |\mathrm{FS}(U)| - 1.$$

If $p = 0$, (7) gives the stronger bound $2m$. If $p = 1$, then $|\mathrm{FS}(S)| = 3$, and the right side is at least $2u + 2 = 2m - 2$ (with the empty-U case even larger). If $p \geq 2$ and $u \geq 1$, equations (7) and (8) give

$$p(p-1) + 5 + 2u - 1 = 2m - 2 + (p-2)(p-3) \geq 2m - 2.$$

If $p \geq 2$ and $u = 0$, the difference between the bound $p(p-1) + 5$ and $2m - 2 = 4p - 2$ is $p^2 - 5p + 7 > 0$.

For a general G, work in the finitely generated subgroup generated by R and put $K = \mathrm{FS}(R)$. A finitely generated abelian group is a product of a finite group and a free abelian group; reduction of the free coordinates modulo a sufficiently large integer gives a finite quotient whose map is injective on the finite set $K + K - K$. It also preserves aperiodicity: a nonzero period of the image of K can be written as the image of $k_1 - k_0$; injectivity on $K + K - K$ then lifts every translated point back to K, producing a nonzero period of K. The finite-group result in this quotient proves (9). □

Order a positive set as $X_i = \{x_1 < \cdots < x_i\}$ and put

$$d_i = |(x_i + \mathrm{FS}(X_{i-1})) \setminus \mathrm{FS}(X_{i-1})|, \qquad y_i = d_i - i.$$

Indeed, modulo x_i the numbers $0, x_1, \ldots, x_{i-1}$ lie in distinct residue classes. Each occupied class of $\mathrm{FS}(X_{i-1})$ has a largest point, whose translate by x_i is new. Thus $d_i \geq i$, so $y_i \geq 0$, and telescoping gives

$$\mu(X_i) = \sum_{j=1}^{i} y_j. \tag{10}$$

We shall also use the exact equality case in Lemma 2.1 of Carpenter–Defant–Kravitz [1]: for $i \geq 4$,

$$y_i = 0 \quad \Longleftrightarrow \quad X_i = t\{1, 2, \ldots, i\} \quad \text{for some } t > 0. \tag{11}$$

Lemma 5.2 (Unit-increment inverse lemma)**.** *If $i \geq 6$ and $y_i = 1$, then there are $t > 0$ and $q \in \{1, \ldots, i\}$ such that*

$$X_i = t(\{1, 2, \ldots, i+1\} \setminus \{q\}). \tag{12}$$

Proof. Put $x = x_i$, reduce $x_1, \ldots, x_{i-1}$ modulo x in the circle group $\mathbb{R}/x\mathbb{Z}$, and call the resulting $(i-1)$-element set R. Let $\rho = |\mathrm{FS}(R)|$. Distinct occupied residues give distinct maximal x-chains in $\mathrm{FS}(X_{i-1})$, while $0, x_1, \ldots, x_{i-1}$ occupy distinct residues. Hence

$$i \leq \rho \leq d_i = i + 1.$$

If $\mathrm{FS}(R)$ were aperiodic, Lemma 5.1 would give $\rho \geq 2i - 4 > i + 1$, a contradiction.

Let H be the nontrivial stabilizer of $\mathrm{FS}(R)$, write $h = |H|$ and $\rho = hq_0$, and put $v = |R \cap H|$. The stabilizer H is finite, since it acts faithfully by translations on the finite nonempty set $\mathrm{FS}(R)$. Iterated Kneser applied to the $(i-1)$ sets $\{0, r\}$ gives

$$\rho \geq h(i - v), \qquad v \leq h - 1, \qquad h + q_0 \geq i + 1.$$

If $q_0 \geq 2$, then

$$\rho = hq_0 \geq 2(h+q_0) - 4 \geq 2i - 2 > i + 1,$$

again impossible. Thus $q_0 = 1$, so $\mathrm{FS}(R) = H$ and $R \subseteq H$. The circle group has a unique subgroup of order h; with $t = x/h$ and representatives in $(0, x)$,

$$X_i = t(\{1, \ldots, h\} \setminus D), \qquad |D| = h - i, \qquad i \leq h \leq i + 1,$$

where $D \subseteq \{1, \ldots, h-1\}$. The case $h = i$ is the homogeneous progression and has $y_i = 0$. Therefore $h = i + 1$ and $D = \{q\}$, proving (12). □

Lemma 5.3 (Uniform defect versus excess)**.** *For every finite positive real set X,*

$$\boxed{2\delta_0(X) - \mu(X) \leq 10.} \tag{13}$$

Proof. Use the prefixes above and write $\delta_i = \delta_0(X_i)$ and $\Phi_i = 2\delta_i - \mu(X_i)$. Adding x_i can remove old defects and can create only the one new defect x_i, so

$$\Phi_i \leq \Phi_{i-1} + 2 - y_i. \tag{14}$$

For $i \leq 5$, the trivial bounds $\delta_i \leq i$ and $\mu(X_i) \geq 0$ give $\Phi_i \leq 10$.

Proceed by induction for $i \geq 6$. If $y_i \geq 2$, equation (14) preserves the bound. If $y_i = 0$, the published increment-equality case (11) makes X_i a homogeneous progression; then $\delta_i = 1$, $\mu(X_i) = 0$, and $\Phi_i = 2$.

It remains to consider $y_i = 1$. By Lemma 5.2, after scaling we have $X_i = [1, i+1]_{\mathbb{Z}} \setminus \{q\}$. Every present $k \in [2, i-1]_{\mathbb{Z}}$ has two disjoint representations

$$2k = (k-1) + (k+1) = (k-2) + (k+2),$$

where 0 is allowed; deleting one q cannot destroy both. The largest element $i + 1$ is defective. The only possible additional defect is 1 when $q = 2$, or i when $q = i - 1$. Thus $\delta_i \leq 2$, and $\delta_i = 2$ is possible only for these two values of q.

For $q = 2$ and $q = i - 1$, the integer totals are respectively

$$T_i + i - 1 \quad \text{and} \quad T_i + 2, \qquad T_i = \frac{i(i+1)}{2}.$$

Both exceed $T_i + 1$. If $\mu(X_i) \leq 1$, the published stability theorem with $M = \mu(X_i) \leq 1 \leq i - 4$ would give an integer normalization of total at most $T_i + 1$. Since both displayed sets contain 1, every integer normalization has total at least the displayed total, a contradiction. Hence $\mu(X_i) \geq 2$ whenever $\delta_i = 2$. In all cases $\Phi_i \leq 2$, completing the induction and the proof. □

Put $\mu_P = \mu(P)$, $\mu_Q = \mu(Q)$, and $\mu = \mu_P + \mu_Q$. For finite real sets U, V, the standard increasing chain gives $|U + V| \geq |U| + |V| - 1$. Hence

$$s = |\mathrm{FS}(P) - \mathrm{FS}(Q)| \geq \Lambda + \mu, \qquad \Lambda := 1 + \frac{a(a+1)}{2} + \frac{b(b+1)}{2}. \tag{15}$$

The set $D \setminus E$ has eight elements. If $x \in P$ contributes to $\Delta(D)$, then $2x \notin R(P \cup \{0\})$; the analogous statement holds on the negative side. Counting the eight deleted elements and zero separately therefore gives

$$\Delta(D) \leq \delta_0(P) + \delta_0(Q) + 9. \tag{16}$$

By Lemma 5.3,

$$\begin{aligned} 2\Delta(D) - \mu &\leq (2\delta_0(P) - \mu_P) + (2\delta_0(Q) - \mu_Q) + 18 \\ &\leq 10 + 10 + 18 = 38. \end{aligned} \tag{17}$$

Combining (6), (15), and (17) yields

$$\Lambda \le 22d + 115. \tag{18}$$

For fixed $a + b = d - 9$, Λ is minimized when a and b are as balanced as possible, so

$$\Lambda \ge \frac{(d-8)^2 + 3}{4}.$$

Consequently

$$\frac{(d-8)^2 + 3}{4} \le 22d + 115, \qquad \text{or equivalently} \qquad d^2 - 104d - 393 \le 0.$$

The quadratic is $39 > 0$ at $d = 108$ and is strictly increasing for $d \ge 108$. Hence $d \le 107$. Since $|B| \le 8$,

$$|A| = d + |B| \le 107 + 8 = 115.$$

We have proved this bound for zero-containing real sets. Applying Lemma 2.1 to a set that omits zero yields the same bound (and in fact the bound 114 for such a set). Thus no cardinality greater than 115 can occur.

Theorem 5.4 (Preliminary global bound)**.** *Every finite real feasible set has at most* 115 *elements.*

The following sections sharpen the structural estimates before excluding $108, \dots, 115$ in one argument.

Lemma 5.5 (Small-increment inverse lemma)**.** *Let $i \ge 6$ and $0 \le y_i \le i - 5$. There are an integer h and a set $D_0 \subseteq [1, h-1]_{\mathbb{Z}}$ such that, for some $t > 0$,*

$$i \le h \le i + y_i, \qquad |D_0| = h - i, \qquad X_i = t\big([1,h]_{\mathbb{Z}} \setminus D_0\big).$$

Proof. Repeat the circle-group argument in Lemma 5.2, now with $d_i = i + y_i$. If the projected subset-sum set were aperiodic, its cardinality ρ would satisfy

$$\rho \ge 2i - 4 > i + y_i.$$

For a nontrivial stabilizer, retain the notation h, q_0, v from that proof. If $q_0 \ge 2$, then

$$\rho = hq_0 \ge 2(h + q_0) - 4 \ge 2i - 2 > i + y_i.$$

Thus $q_0 = 1$, the projected subset sums form the subgroup of order $h = \rho$, and $i \le h \le i + y_i$. Taking the representatives in $(0, x_i)$ gives the asserted interval with $h - i$ deletions. □

Part II
The descent from 115 to 96

6 Scope and the descending implication

The preceding chapter proves integralization, the covering and peeling lemmas, the positive-prefix increment estimates, and $N \le 115$. We now exclude every integer cardinality from 115 down through 97. At an endpoint n, adjoining zero is used only after $n+1$ has been excluded. All inverse theorems are applied after integralization.

The finite verifications below have two logically separate parts. First, the arguments assign every actual feasible set to an explicitly listed linear system and prove that each branch covers its parent. Second, nonnegative integer combinations contradict every resulting system. The complete smaller multiplier pools are printed, and the larger pools and trees are identified exactly in Appendix F. A numerical optimization status is never a premise of this proof.

7 Two stronger defect estimates

For a positive set X, write

$$\Phi(X) = 2\delta_0(X) - \mu(X), \qquad \varrho(X) = 2 - \Phi(X) = \mu(X) + 2 - 2\delta_0(X).$$

The uniform estimate already proved is $\Phi \le 10$.

Lemma 7.1 (Deleted intervals)**.** *Let $X = [1, r+q]_{\mathbb{Z}} \setminus D$, where $|D| = q$, $r+q \notin D$, and $r \ge \max(5, 4q+1)$. Then $\Phi(X) \le 2$. For $q = 1, 2$ and $r \ge 7$, one also has $\delta_0(X) \le 2q$.*

Proof. The case $q = 0$ is immediate. Put $h = r+q$ and $C = \{h - d : d \in D\}$. The integer total excess is

$$e = \sum X - T_r = \sum_{c \in C} c - T_{q-1}.$$

There are $\min(k, h-k)$ disjoint representations of $2k$ by distinct members of $[0,h]_{\mathbb{Z}}$ when $k \ne h/2$; for $k = h/2$, use $0 + h$. Therefore a defective present k is either in $[1,q]$ or in $[r,h]$. There are at most $q+1$ high defects. For $q \ge 2$ there are at most $q-1$ low defects: having all of $1, \dots, q$ present and defective would make 2 both absent (from defectivity of 1) and present. For $q = 1$, the direct one-deletion argument above gives at most two defects. These observations prove $\delta_0(X) \le 2q$ and the last assertion.

Since $r > q$, no integer greater than one can divide all r elements of this interval with q deletions; hence X has gcd one. If $\mu(X) \le r - 4$, the stability theorem in Section 5 implies $\mu(X) = e$: any integer normalization of a primitive integer set is an integer dilation, and the support lies in $[0, \sum X]$. Suppose first that there are no low defects. Apart from h, let the m high defects be $h - t$. Their presence and defectivity imply $t \notin C$ and $2t \in C$. Replacing these distinct depths $2t$ by their absent halves t leaves q distinct positive integers. Hence

$$e \ge q + \sum t \ge q + m \ge 2m = 2\delta_0(X) - 2.$$

If there are $\ell \ge 1$ low defects, their forced holes $2k$ have distinct depths at least $r - q$, spaced by at least two. Thus

$$e \ge \ell(r-q) + \ell(\ell-1) + T_{q-\ell} - T_{q-1}.$$

Subtracting $2q + 2\ell$ gives

$$\ell(r - 2q - 2) + \tfrac{3}{2}\ell(\ell-1) - q \ge 0$$

under $r \geq 4q+1$. Since $\delta_0(X) \leq q+1+\ell$, this again implies $e \geq 2\delta_0(X)-2$. Finally, if $\mu(X) > r-4$, then

$$\Phi(X) \leq 4q-(r-3) \leq 2.$$

All cases give the claim. □

Theorem 7.2 (Threshold 14)**.** *For every positive real set X of size $r \geq 14$,*

$$\boxed{\Phi(X) \leq 2, \qquad \varrho(X) \geq 0.}$$

Proof. For $9 \leq i \leq 14$, put $q_i = \lfloor (i-1)/4 \rfloor$. If $y_i \leq q_i$, the small-increment inverse lemma expresses X_i, up to scaling, as an initial integer interval with at most q_i deletions. Lemma 7.1 gives $\Phi(X_i) \leq 2$. If $y_i \geq q_i+1$, the defect recurrence gives $\Phi(X_i) \leq \Phi(X_{i-1})+1-q_i$. Starting from the uniform bound 10 at size 8 gives the following successive upper bounds:

i	8	9	10	11	12	13	14
$\Phi(X_i)$ upper bound	10	9	8	7	6	4	2.

After size 14, increments $y_i \geq 2$ preserve the bound. For $y_i = 0$ or 1, the equality and unit-increment arguments in Section 5 already give $\Phi(X_i) \leq 2$. □

Theorem 7.3 (Penalty for a non-progression support)**.** *If X is a positive real set of size $r \geq 6$ and $\mathrm{FS}(X)$ is not an arithmetic progression, then*

$$\boxed{\varrho(X) \geq r-14.}$$

Proof. At $r=6$ the uniform bound gives $\varrho \geq -8 = r-14$. Induct on $r \geq 7$. If $y_r \leq 2$, the small-increment inverse lemma gives a primitive deleted interval with at most two deletions. It has at most four defects. Its non-progression support implies $\mu \geq r-3$ by the stability theorem, since $\mu \leq r-4$ would force a full interval in the primitive normalization. Consequently $\varrho \geq r-9 \geq r-14$.

Now let $y_r \geq 3$. If the preceding support is not an arithmetic progression, the induction hypothesis and $\varrho(X_r) \geq \varrho(X_{r-1}) + y_r - 2$ give the result. If the preceding support is $t[0,L]_{\mathbb{Z}}$, its translate by x_r must be disjoint from it; an overlap would produce an arithmetic progression again. Hence

$$y_r = |\,\mathrm{FS}(X_{r-1})| - r \geq \frac{(r-1)(r-2)}{2}.$$

Using $\varrho(X_{r-1}) \geq -8$ gives $\varrho(X_r) \geq (r-1)(r-2)/2 - 10 \geq r-14$. □

8 A sharper effective hole threshold

For a finite integer set S, its hull-hole count is $\max S - \min S + 1 - |S|$. For a set on a specified lattice $u\mathbb{Z}$, divide by u before counting holes.

We use Lev's central-interval theorem [4]: if $B \subseteq [1,L]_{\mathbb{Z}}$, $|B| = s$, and $L \leq 3s/2-2$, then

$$[2L-2s+1,\ \textstyle\sum B - (2L-2s+1)]_{\mathbb{Z}} \subseteq \mathrm{FS}(B).$$

Write $\mathcal{S}(L,H) = [0,L]_{\mathbb{Z}} \setminus (H \cup (L-H))$. The endpoint bridge proved in Lemma 10.1 states that an integer prefix of size m with support $\mathcal{S}(L,H)$ preserves the same endpoint holes upon adjoining its next element with excess at most E, if $2h < L$ and

$$m \geq M(h,q,E) := \max\left\{h, 1+\left\lfloor \frac{1+\sqrt{9+8(E+2q+2h)}}{2} \right\rfloor\right\}, \quad h = \max H, \quad q = |H|.$$

Here $h = q = 0$ is allowed. We need this explicit bridge, not an unspecified "sufficiently large" threshold from a classification theorem.

Lemma 8.1 (Small increments with counted holes)**.** *If $i \geq 2E+5$ and $0 \leq y_i = e \leq E$, then*

$$X_i = t([1, i+e]_{\mathbb{Z}} \setminus D), \qquad |D| = e,$$

and, after division by t,

$$\mathrm{FS}(t^{-1}X_i) = \mathcal{S}(L, H), \qquad H \subseteq [1, 2e]_{\mathbb{Z}}, \qquad |H| \leq e.$$

Proof. For $e = 0$, the increment equality case gives the assertion directly. Assume $e > 0$. The small-increment inverse lemma first gives the same form with δ deletions, $0 \leq \delta \leq e$, and largest coefficient $i + \delta$. Remove that coefficient. The remaining $i - 1$ integers lie in $[1, i + \delta - 1]$ and satisfy Lev's density condition, because $i \geq 2\delta + 5$. Their support therefore has endpoint state $H \subseteq [1, 2\delta]$. Its total L_0 satisfies

$$L_0 \geq T_{i-1} > i + \delta + 4\delta.$$

Adding $i + \delta$ overlaps the two central intervals and leaves exactly the same endpoint holes. Its support increment is $i + \delta$; comparison with $i + e$ gives $\delta = e$. Finally every lower hole in $[1, 2e]$ is absent from the coefficient set itself. Since $2e < i + e$ and there are exactly e deleted coefficients, $H \subseteq D$ and $|H| \leq e$. This last count halves the former bound $|H| \leq 2e$. □

Theorem 8.2 (Effective hole bound)**.** *Let X be a positive real set of size $r \geq 35$ with $\mathrm{FS}(X) \subseteq u\mathbb{Z}$. If its hull-hole count h satisfies*

$$14 < h < |\mathrm{FS}(X)| - 1,$$

then

$$\boxed{\mu(X) \geq r + 103.}$$

Proof. Suppose $\mu(X) \leq r + 102$. Divide by u to obtain an integer set B. It is primitive: a gcd of at least two would leave at least one missing lattice point between each two successive support values, contradicting $h < |\mathrm{FS}(X)| - 1$. The support is not a full integer interval.

We first obtain a tail of positive increments with a uniform budget. If no prefix of size at least three has an arithmetic-progression support, then $y_j \geq 1$ for $j \geq 4$, and $\sum_{j=4}^{r}(y_j - 1) \leq 105$. Otherwise let $k \geq 3$ be the last size with such a support. The next translate is disjoint, so

$$y_{k+1} \geq T_k - k, \qquad \mu(B) \geq r - 1 + \frac{k(k-3)}{2}.$$

Thus $k(k-3)/2 \leq 103$, whence $k \leq 15$. All later increments are positive, and $\sum_{j=k+1}^{r}(y_j - 1) \leq 102 + k \leq 117$. In either case there is $r_0 \leq 16$ such that

$$y_j \geq 1 \ (j \geq r_0), \qquad \sum_{j=r_0}^{r}(y_j - 1) \leq 117. \tag{19}$$

The window $19 \leq j \leq r$ has at least 17 indices. If every increment there were at least eight, it would contribute at least $7 \cdot 17 = 119$ to (19). Therefore some $i_0 \geq 19$ has $1 \leq y_{i_0} = e \leq 7$. Lemma 8.1 gives

$$B_{i_0} = t([1, i_0 + e]_{\mathbb{Z}} \setminus D), \quad H \subseteq [1, 14], \quad |H| \leq 7.$$

The coefficient set has gcd one since $i_0 > e$, so t is a positive integer. If $t > 1$, primitivity of B gives a first later coefficient outside $t\mathbb{Z}$. At that step the two supports are disjoint, and

$$y_j - 1 \geq \frac{i_0(i_0 - 1)}{2} - 1 \geq 170 > 117,$$

contradicting (19). Hence $t = 1$.

Every subsequent increment is at most 118, and

$$M(14, 7, 118) = 19, \qquad T_{19} = 190 > 28.$$

The endpoint bridge now preserves H through the whole remaining tail. The final number of holes is $2|H| \leq 14$, contradicting $h > 14$. □

9 A common argument for all eight endpoints

Lemma 9.1 (Unpaired absolute values)**.** *Let $0 \in F$, let $e = |F|$, and write the distinct nonzero absolute values as $0 < b_1 < \cdots < b_r$. Let $m_i \in \{1, 2\}$ count how many of $\{b_i, -b_i\}$ lie in F. If*

$$p = |\{i : m_i = 2\}|, \qquad q = |\{i : m_i = 1\}|,$$

and

$$J = |\{(i, j) : i < j,\ m_i = 1,\ m_j = 2\}|,$$

then

$$|\mathrm{FS}(F)| \geq \frac{e^2 + 3 + q^2}{4} + J. \tag{20}$$

Proof. Translate $\mathrm{FS}(F)$ by the sum of the absolute values of the negative members of F. The resulting set is

$$\left\{ \sum_{i=1}^{r} k_i b_i : 0 \leq k_i \leq m_i \right\}.$$

Before b_i is introduced, the residues represented modulo b_i by $0, b_1, \ldots, b_{i-1}$ are distinct. Extending each corresponding maximal b_i-chain by the m_i new coefficient levels creates at least im_i new sums. Therefore

$$|\mathrm{FS}(F)| \geq 1 + \sum_{i=1}^{r} im_i.$$

If the paired positions are $s_1 < \cdots < s_p$, then

$$\sum_{i=1}^{r} im_i = \frac{r(r+1)}{2} + \sum_{j=1}^{p} s_j = \frac{r(r+1)}{2} + \frac{p(p+1)}{2} + J.$$

Using $r = p + q$ and $e = 2p + q + 1$ gives (20). □

Lemma 9.2 (Four-extreme restricted-sum defect)**.** *Let*

$$F = \{x_1 < \cdots < x_m\} \subseteq \mathbb{Z}, \qquad m \geq 2,$$

and let

$$a_1 < a_2 < x_1 < \cdots < x_m < b_2 < b_1, \qquad D = F \cup \{a_1, a_2, b_2, b_1\}.$$

Let η_F, η_D be the hole counts of $R(F), R(D)$ in their endpoint integer hulls, and put

$$g_H = 4m + 6 - \big(|R(D)| - |R(F)|\big).$$

Then

$$\boxed{\eta_D + \eta_F + g_H \geq 2m - 8.} \tag{21}$$

Proof. Put $L = x_1 + x_2$, $U = x_{m-1} + x_m$, and let

$$h_- = L - (a_1 + a_2), \qquad h_+ = (b_1 + b_2) - U, \qquad h = h_- + h_+.$$

Let W_+ be the union of the portions of $b_1 + F$ and $b_2 + F$ strictly above U. Write

$$y = b_2, \quad z = b_1, \quad u = y - x_m, \quad v = z - y, \quad e = x_m - x_{m-1}.$$

All three last quantities are positive and $h_+ = 2u + v + e$. If p, q count the terms above U in $y + F, z + F$, respectively, then adjacency of x_{m-1}, x_m gives $p \leq u + 1$, while $q \leq m$. Also $W_+ \subseteq [U + 1, z + x_m]_{\mathbb{Z}}$. Combining the cardinality and interval bounds gives

$$2|W_+| \leq h_+ + m + 1.$$

The symmetric lower-end set W_- satisfies $2|W_-| \le h_- + m + 1$.

Let N_{out} count the new restricted sums outside $[L, U]$. Besides W_+ and W_- only the six pairs among the four new elements can occur, so

$$2N_{\text{out}} \le h + 2m + 14.$$

If $N = |R(D)| - |R(F)|$, at most η_F further new values can lie inside the old hull; hence $N \le N_{\text{out}} + \eta_F$. Since

$$\eta_D = h + \eta_F - N, \qquad g_H = 4m + 6 - N,$$

we obtain

$$\eta_D + \eta_F + g_H = h + 2\eta_F + 4m + 6 - 2N \ge h + 4m + 6 - 2N_{\text{out}} \ge 2m - 8.$$

□

Lemma 9.3 (Sharper four-extreme defect)**.** *In the notation of Lemma 9.2,*

$$\boxed{\eta_D + \eta_F + g_H \ge 2m - 4.}$$

Consequently, whenever the common-lattice and overlapping-translate reductions have been established, the lower bounds for T improve to (the slack quantities are defined in Proposition 9.5)

$$T \ge \kappa + G - 3z \ge G - 2z, \qquad T \ge \lceil G/3 \rceil, \qquad G = 2d - 12. \tag{22}$$

Proof. Keep $L = x_1 + x_2$, $U = x_{m-1} + x_m$, h_-, h_+ and W_-, W_+ from that lemma. Define the integer intervals

$$I_+ = (U, b_1 + x_m] \cap \mathbb{Z}, \qquad I_- = [a_1 + x_1, L) \cap \mathbb{Z},$$

and their unused capacities $H_\pm = |I_\pm| - |W_\pm| \ge 0$. For the upper end, put $u = b_2 - x_m$, $v = b_1 - b_2$, and $e = x_m - x_{m-1}$. The two bounds used previously are

$$|W_+| \le u + 1 + m, \qquad |W_+| = u + v + e - H_+.$$

Adding gives $2|W_+| \le h_+ + m + 1 - H_+$; reflection gives the lower-end version. This retains the capacity discarded in the earlier proof.

Every mixed new-new sum $a_i + b_j$ outside $[L, U]$ lies in $I_- \cup I_+$. Let q count the distinct such values not already in $W_- \cup W_+$. Then

$$q \le 4, \qquad q \le H_- + H_+.$$

The other two new-new sums, $a_1 + a_2$ and $b_1 + b_2$, lie beyond these intervals. Thus the number of new restricted sums outside the old hull is $N_{\text{out}} = |W_-| + |W_+| + 2 + q$, and

$$2N_{\text{out}} \le h_- + h_+ + 2m + 6 - (H_- + H_+) + 2q \le h_- + h_+ + 2m + 10.$$

As before, $N = |R(D)| - |R(F)| \le N_{\text{out}} + \eta_F$. Substitution into

$$\eta_D + \eta_F + g_H = h_- + h_+ + 2\eta_F + 4m + 6 - 2N$$

proves $2m - 4$. Apply this with $F = D_1$, $m = d - 4$, and use $\eta_2 \le z + \ell_2$, $\eta_1 \le 2z + \ell_1$ to obtain (22). Here $z \le \kappa \le T$, so $3T \ge G$. □

Lemma 9.4 (Universal core-span estimate)**.** *If $X \subset \mathbb{Z}_{>0}$ has $r \ge 15$ elements, then*

$$\max X \le r + \varrho(X) + h(X) + 2.$$

Neither primitivity nor full subset-sum support is required.

Proof. Let X' be the first $r-1$ elements, write $\max X = r+q$, and let v be the number of old diagonal defects destroyed by adjoining this maximum. The new maximum is defective, so $\delta_0(X) = \delta_0(X') + 1 - v$. Also

$$\mu(X) - \mu(X') = q - h(X) + h(X').$$

It follows exactly that

$$\varrho(X) = \varrho(X') + q - h(X) + h(X') - 2 + 2v.$$

Here $\varrho(X') \geq 0$ because $r - 1 \geq 14$, and $h(X'), v \geq 0$. Rearranging proves the claim. □

Proposition 9.5 (One common-lattice reduction for all endpoints)**.** *Let $97 \leq n \leq 115$, and suppose a feasible n-element set contains zero. After integralization and an existence-preserving dilation there are sets*

$$A = B \sqcup D, \qquad D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$$

with $|B| = 8$, $d = |D| = n - 8$, $|D_1| = d - 4$, $e = |E| = n - 16$, and $a = |P|, b = |Q| \geq 17$. The set D is integral on one common lattice. Put $s = |\operatorname{FS}(E)|$, $\Lambda = 1 + T_a + T_b$, and define

$$\begin{aligned}
\kappa &= s - \Lambda - \mu(P) - \mu(Q), & \rho_P &= \varrho(P), \quad \rho_Q = \varrho(Q),\\
\vartheta &= \delta_0(P) + \delta_0(Q) + 8 - \Delta(D), & c &= 2|2D| + 18d + 85 - |\operatorname{FS}(D)|,\\
\ell_1 &= |\operatorname{FS}(D)| - |\operatorname{FS}(D_1)| - |R(D)| + 1, & \ell_2 &= |\operatorname{FS}(D_1)| - s - |R(D_1)| + 1,\\
g &= 4d - 10 - (|R(D)| - |R(D_1)|), & T &= \kappa + \ell_1 + \ell_2 + g.
\end{aligned}$$

All these quantities are nonnegative, and the exact budget is

$$\rho_P + \rho_Q + 2\vartheta + c + T = 22d + 97 - \Lambda =: \Omega. \tag{23}$$

Let h_P, h_Q, z, w be the hull-hole counts of $\operatorname{FS}(P), \operatorname{FS}(Q), \operatorname{FS}(E), \operatorname{FS}(D)$, respectively, on this lattice. Then

$$h_P, h_Q \leq \kappa, \qquad z = h_P + h_Q - \kappa, \qquad 0 \leq z \leq \min(h_P, h_Q) \leq \kappa, \tag{24}$$

$$0 \leq w \leq 4z, \qquad T \geq \kappa + G - 3z \geq G - 2z, \qquad T \geq \lceil G/3 \rceil, \qquad T \geq z, \quad G = 2d - 12. \tag{25}$$

If $z > 0$, both one-sided supports are non-progressions and

$$\rho_P + \rho_Q \geq a + b - 28 = n - 45, \qquad \mu(P) + \mu(Q) \geq a + b - 6 = n - 23.$$

Proof. The notation B, D, D_1, E is obtained by the covering and two peeling steps. If $|B| \leq 7$, one sign is absent from D; the nonzero part of E is then one-sided and has at least $n - 16 \geq 81$ elements. Hence

$$s \geq 1 + T_{81} = 3322 > 24 \cdot 115 + 77 \geq 24d + 77,$$

contradicting the preliminary core bound. Thus $|B| = 8$. The unpaired-value lemma and $s \leq 24e + 269$ give

$$q^2 \leq 96e + 1073 - e^2 \leq 2288 \qquad (81 \leq e \leq 99).$$

The number of opposite pairs is $(e - 1 - q)/2 \geq 17$; the endpoint $e = 81$ uses the fact that q is even. Thus $a, b \geq 17$ and E contains an opposite pair. Zero is consequently not a diagonal exception in D, so $\Delta(D) \leq \delta_0(P) + \delta_0(Q) + 8$. The covering, peeling and restricted-pair estimates prove nonnegativity of the displayed slacks; $\rho_P, \rho_Q \geq 0$ follows from Theorem 7.2. Substituting their definitions and cancelling $|2D| = |R(D)| + \Delta(D)$ gives (23) exactly.

The inverse theorem on the core. We use the following sufficient form of Stanchescu's two-set theorem [3, Corollary 2]. If finite integer sets U, V, each of size at least two, satisfy

$$|U+V| < |U| + |V| + \min(|U|, |V|) - 3,$$

then their hull cardinalities on a common difference lattice are at most $|U+V| - |V| + 1$ and $|U+V| - |U| + 1$, respectively. Apply it to $U = \mathrm{FS}(P)$ and $V = -\mathrm{FS}(Q)$ after integralization. Write $m = a + b = n - 17$, $(a, b) = (\lfloor m/2 \rfloor - k, \lceil m/2 \rceil + k)$ up to reflection, and

$$I = k^2 + (m \bmod 2)k, \qquad \Lambda_0 = 1 + T_{\lfloor m/2 \rfloor} + T_{\lceil m/2 \rceil}, \qquad \Omega_0 = 22d + 97 - \Lambda_0.$$

Then $\Lambda = \Lambda_0 + I$, $\Omega = \Omega_0 - I$. Nonnegativity gives $I \le \Omega_0$. The table below includes every possible integer k, from zero to its stated maximum. Its inverse margin is the minimum of $T_{\lfloor m/2 \rfloor - k} - 2 - (\Omega_0 - I)$ over precisely that range. It is strictly positive, so $\kappa \le \Omega < T_{\min(a,b)} - 2 \le \min(|U|, |V|) - 3$. This is a slightly stronger sufficient condition than required by the inverse theorem. Both U, V contain zero; after division by the common step they, and hence P, Q, are integral. The two hull bounds give $h_P, h_Q \le \kappa$. Comparing the hull of $U + V$ with the two individual hulls gives $z = h_P + h_Q - \kappa$, proving (24).

Forcing both peeled layers onto that lattice. Only P, Q, E are asserted integral at this stage. Set $\rho = \rho_P + \rho_Q$, $\mu = \mu(P) + \mu(Q)$ and $R_p = \max P, R_q = \max Q$. Lemma 9.4, applied to the integer core sides, gives

$$R_p + R_q \le m + \rho + h_P + h_Q + 4 = m + \rho + \kappa + z + 4.$$

Since $R(E)$ is integral, $|R(E)| \le 2(R_p + R_q) + 1$. Regardless of whether the four elements of $D_1 \setminus E$ are on the lattice, their restricted pairs add at most $4e + 6$ values. Thus, using $s = \Lambda + \mu + \kappa$ and $C = 2m + 4e + 14 - \Lambda$, we obtain

$$\begin{aligned} |\,\mathrm{FS}(D_1)| - 2s &\le C + 2\rho - \mu + \kappa + 2z + \ell_2 \\ &\le C + 3\Omega - \rho - \mu. \end{aligned}$$

The last inequality uses $z \le \kappa$ and $\rho + \kappa + \ell_2 \le \Omega$ once, keeping the same quantities. If $z = 0$, the first line is at most $C + 2\Omega$, because $\rho \le \Omega$ and $\mu \ge 0$. If $z > 0$, each one-sided support has positive hull-hole count smaller than its cardinality minus one. An integer arithmetic progression of step at least two has at least that many holes, so neither support is an arithmetic progression. The non-progression theorem gives $\rho \ge m - 28$. The stability theorem in the foundation chapter gives $\mu(X) \ge r - 3$ for a non-progression support of an r-element set: otherwise $\mu(X) \le r - 4$ places its support in an interval of exactly $1 + T_r + \mu(X)$ integers, forcing it to fill that interval. Hence $\mu \ge m - 6$.

The two relevant balanced upper bounds are $C_0 + 2\Omega_0$ and $C_0 + 3\Omega_0 - (2m - 34)$, where $C_0 = 2m + 4e + 14 - \Lambda_0$. Imbalance decreases them by $3I$ and $4I$. Both are strictly negative in every row below. Therefore $|\,\mathrm{FS}(D_1)| < 2s$. Any element of $D_1 \setminus E$ off the core lattice would put two disjoint s-point translates of $\mathrm{FS}(E)$ inside $\mathrm{FS}(D_1)$, a contradiction. This proves integrality of D_1.

For the other layer,

$$|\,\mathrm{FS}(D)| - 2|\,\mathrm{FS}(D_1)| \le 4d - 10 + \ell_1 - \ell_2 - s \le 4d - 10 + \Omega - s < 0.$$

The last strict inequality follows from the last column of the table, which gives $\Lambda_0 - (4d - 10 + \Omega_0) > 0$. The same disjoint-translate argument puts all of D on the lattice.

n	Λ_0	Ω_0	$k_{\max}$	Inv.	$z=0$	$z>0$	Layer 1
115	2451	0	0	1223	−1845	−2007	2033
114	2402	27	4	981	−1748	−1881	1961
113	2353	54	7	854	−1651	−1755	1889
112	2305	80	8	770	−1557	−1633	1819
111	2257	106	10	695	−1463	−1511	1749
110	2210	131	10	643	−1372	−1393	1681
109	2163	156	12	581	−1281	−1275	1613
108	2117	180	12	535	−1193	−1161	1547
107	2071	204	14	486	−1105	−1047	1481
106	2026	227	14	446	−1020	−937	1417
105	1981	250	15	408	−935	−827	1353
104	1937	272	16	371	−853	−721	1291
103	1893	294	17	335	−771	−615	1229
102	1850	315	17	299	−692	−513	1169
101	1807	336	18	264	−613	−411	1109
100	1765	356	18	230	−537	−313	1051
99	1723	376	19	196	−461	−215	993
98	1682	395	19	163	−388	−121	937
97	1641	414	20	131	−315	−27	881

All entries are evaluations of the displayed quadratic formulas; no set search or optimization enters this table.

Hole propagation and the common lower budget. At the second peeling step write α_2, β_2 for the sums of the two removed negative and positive elements. Then $R(D_1)$ has endpoints α_2, β_2. The translates $\alpha_2 + \mathrm{FS}(E)$ and $\beta_2 + \mathrm{FS}(E)$ are inside $\mathrm{FS}(D_1)$. The strict $2s$ inequality proves they overlap. Their hulls cover the new subset-sum hull and leave at most $2z$ holes. If $u = \beta_2 - \alpha_2$, their union has at least $s + u - z$ points. Comparing with $|\mathrm{FS}(D_1)| = s + |R(D_1)| - 1 + \ell_2$ gives $\eta_2 \le z + \ell_2$, where η_2 is the hull-hole count of $R(D_1)$. The first peeling step similarly gives at most $4z$ subset-sum holes and $\eta_1 \le 2z + \ell_1$ restricted-sum holes. Thus

$$\eta_2 \le z + \ell_2, \qquad \eta_1 \le 2z + \ell_1. \tag{26}$$

Lemma 9.3, applied with $F = D_1$, $|F| = d - 4$, gives $\eta_1 + \eta_2 + g \ge 2d - 12 = G$. Hence $T \ge \kappa + G - 3z \ge G - 2z$. Also $z \le \kappa \le T$, so $3T \ge G$ and $T \ge z$. This proves all the claimed conclusions. □

Theorem 9.6 (Excluding sizes 108 through 115)**.** *No feasible set containing zero has cardinality* $108, 109, \ldots, 115$*. Consequently every feasible set has at most 107 elements.*

Proof. Use Proposition 9.5 and deliberately weaken its sharp constant to $G = 2d - 16$. This preserves every inequality $T + \rho_P + \rho_Q \le \Omega$, $T \ge G - 2z$ and $T \ge \lceil G/3 \rceil$ used below. Put $\rho = \rho_P + \rho_Q$. *The endpoint arithmetic.* The minimum Λ_0 occurs when a, b are as balanced as possible. Set $\Omega_0 = 22d + 97 - \Lambda_0$, so $\Omega \le \Omega_0$.

n	d	e	Λ_0	Ω_0	G	$\lceil G/3 \rceil$
115	107	99	2451	0	198	66
114	106	98	2402	27	196	66
113	105	97	2353	54	194	65
112	104	96	2305	80	192	64
111	103	95	2257	106	190	64
110	102	94	2210	131	188	63
109	101	93	2163	156	186	62
108	100	92	2117	180	184	62

The first three rows are already impossible since $T \le \Omega_0$. In all remaining rows $G > \Omega_0$, so (25) forces $z > 0$. Both h_P, h_Q are then positive. They are at most 180 and are smaller than $|U| - 1, |V| - 1$. Thus neither U nor V can itself be an arithmetic progression: a progression of step at least two on this lattice has at least $|U| - 1$, respectively $|V| - 1$, holes. Theorem 7.3 therefore gives

$$\rho_P + \rho_Q \ge a + b - 28 = n - 45. \tag{27}$$

For $n = 112$, this leaves $T \le 80 - 67 = 13 < 64$; for $n = 111$, it leaves $T \le 106 - 66 = 40 < 64$.

Only $108, 109, 110$ remain. Combining (25) with the exact imbalance in Λ gives the following bounds:

n	$\min(a, b)$	T upper bound	z lower bound	$\rho_P + \rho_Q$ lower bound	$T + \rho_P + \rho_Q$
110	≥ 39	66	61	117	$\ge 180 > 131$
109	≥ 37	92	47	118	$\ge 180 > 156$
108	≥ 35	117	34	119	$\ge 181 > 180$

For completeness, write $(a, b) = (46-k, 47+k)$ at 110, $(46-k, 46+k)$ at 109, and $(45-k, 46+k)$ at 108, allowing reflection. The respective excesses $\Lambda - \Lambda_0$ are $k(k+1), k^2, k(k+1)$. The requirement $\Omega \ge \lceil G/3 \rceil$ bounds k by $7, 9, 10$, which proves the size column. The next two columns follow from $T \le \Omega_0 - (n - 45)$ and $2z \ge G - T$.

In every row $h_P, h_Q \ge z > 14$. Theorem 8.2 applies to each side and gives $\mu(P) \ge a + 103$ and $\mu(Q) \ge b + 103$. Using $\delta_0(P) \le a$ and $\delta_0(Q) \le b$ gives

$$\rho_P + \rho_Q \ge (105 - a) + (105 - b) = 227 - n.$$

This proves the strong lower-bound column. The last column contradicts (23) in each case, including a strict one-unit contradiction at 108. All eight endpoints are excluded.

The preliminary bound $N \le 115$ and these eight exclusions give the claim for sets containing zero. A feasible set omitting zero extends by zero to a larger feasible set, so it too has at most 107 elements. □

10 An elementary endpoint bridge and a parameterized hole test

Lemma 10.1 (Elementary endpoint bridge)**.** *The endpoint bridge stated in Section 8 follows from its explicit threshold $M(h, q, E)$.*

Proof. Let $S = \mathcal{S}(L, H)$ be the support of m positive integers. Then $L \ge T_m$, $|S| = L + 1 - 2q$, and $x > \max B \ge m \ge h$. If $x \ge L - 2h$, the part of $x + S$ above L has at least $\min(x, L+1) - 2q \ge L - 2h - 2q$ points. But the definition of M gives

$$T_m - 2h - 2q > m + 1 + E,$$

contradicting the permitted increment. Thus $h < x < L - 2h$. The integer intervals $[h+1, L-h-1]$ and $[x + h + 1, x + L - h - 1]$ overlap or abut, because the integral $x < L - 2h$. They cover the whole central interval $[h + 1, L + x - h - 1]$. Since $x > h$, the lower holes stay H and the upper holes are their reflections about $(L + x)/2$. Consequently $S \cup (x + S) = \mathcal{S}(L + x, H)$, its cardinality increases by x, and $x = m + 1 + e_x$ follows. This also supplies a short self-contained proof of the bridge used in the earlier hole estimate. □

Lemma 10.2 (Parameterized effective hole test)**.** *Let $C \ge 0$ and $b \ge 1$ be integers. Define*

$$K = \max\{k \ge 3 : k(k-3)/2 \le C + 1\}, \quad D = C + \max(3, K), \quad E = D + 1.$$

Suppose integers J, r satisfy

$$J \ge \max\{K+1, 2b+5, M(2b,b,E)\},$$
$$(r-J+1)b > D, \qquad T_{J-1}-1 > D.$$

If a positive r-element set has, on a specified lattice, a hull-hole count h satisfying $2b < h < |\,\mathrm{FS}\,| - 1$, then $\mu \ge r + C + 1$.

Proof. Assume instead $\mu \le r + C$, and divide by the lattice step. The strict upper bound on h makes the integer set primitive, and its support is not an arithmetic progression. The tail argument of Theorem 8.2, with 102 replaced by C, gives a starting index at most $K+1$ (or 4 if no prefix of size at least three has progression support) after which

$$y_j \ge 1, \qquad \sum (y_j - 1) \le D.$$

Here a last progression prefix of size k costs at least $r - 1 + k(k-3)/2$, so $k \le K$; without such a prefix the budget is $C + 3$. The window $J, \dots, r$ contains some increment $1 \le y_i \le b$. Lemma 8.1 gives a deleted interval, with endpoint holes in $[1, 2b]$ and at most b lower holes. Its scale is an integer. A first later coefficient off this scale would cost $y_j - 1 \ge T_{i-1} - 1 \ge T_{J-1} - 1 > D$; primitivity therefore makes the scale one. Every later increment is at most E. The endpoint bridge preserves the holes, since $i \ge J$ and $L \ge T_J > 4b$. The final hole count is at most $2b$, a contradiction. □

The following are exact applications, with no asymptotic size convention:

$r \ge$	C	b	K	D	J	$M(2b,b,D+1)$	Conclusion when $2b < h < \lvert\,\mathrm{FS}\,\rvert - 1$
35	102	7	15	117	19	19	$\mu \ge r + 103$
36	111	8	16	127	21	20	$\mu \ge r + 112$
37	118	8	17	135	21	20	$\mu \ge r + 119$

For example, the last two window costs are $16 \cdot 8 = 128 > 127$ and $17 \cdot 8 = 136 > 135$, while $T_{20} - 1 = 209$ exceeds both budgets.

11 The complete-core reduction at size 107

The following reduction supplies the complete core used to exclude size 107.

Theorem 11.1 (The complete-core reduction at size 107)**.** *A hypothetical feasible 107-element set has an existence-preserving normalization in which D, D_1, E are integral, their subset-sum supports are full intervals, and*

$$(a,b) = (45-k, 45+k) \quad \textit{up to reflection}, \qquad 0 \le k \le 4, \qquad \rho_P + \rho_Q + \kappa + c \le 22 - k^2.$$

Proof. Size 108 is already excluded, so a feasible 107-element set contains zero. Use Proposition 9.5, retaining the weaker $G = 182$ for compatibility with the following exact arithmetic. Then $\Lambda = 2071 + k^2$, $\Omega = 204 - k^2$, and $T \ge 61$.

If $z > 0$, non-progression gives $\rho \ge 62$, so $k \le 9$, $a, b \ge 36$, $T \le 142$ and $z \ge 20$. The size-36 hole row gives $\rho \ge 138$; then $k^2 \le 5$ and $a, b \ge 43$. The size-37 row gives $\rho \ge 152$, contradicting $61 + 152 > 204$. Thus $z = 0$.

The stronger inequality $T \ge \kappa + 182$ now gives $\rho + \kappa + c \le 22 - k^2$, hence $k \le 4$ and $a, b \ge 41$. If h_P or h_Q were positive, the corresponding support would be a non-progression and would have defect at least $41 - 14 = 27 > 22$. Thus $h_P = h_Q = \kappa = 0$. The overlap and hole propagation in Proposition 9.5 give full interval supports for E, D_1, D. □

12 A further general improvement: $N \le 104$

This section closes the three cases $107, 106, 105$. The reduction at size 107 in Theorem 11.1 is used as an intermediate implication; that size is no longer an unresolved candidate. The new argument keeps the *sum* of the losses in the covering translates, and uses the total length of the subset-sum hull. It does not require an exhaustive search of integer sets or any new external inverse theorem.

12.1 Lattice and hole bookkeeping at sizes 105 and 106

Lemma 12.1. *Suppose a feasible integer set of size $n \in \{105, 106\}$ contains zero. The notation and exact budget identity of Theorem 9.6 continue to apply, with $d = n - 8$, $e = n - 16$, and $a + b = n - 17$. After a common lattice normalization, D is integral. If w is the hull-hole count of $\mathrm{FS}(D)$, then*

$$0 \le w \le 4z, \qquad T \ge \kappa + G - 3z \ge G - 2z, \qquad T \ge \lceil G/3 \rceil, \quad G = 2d - 16, \tag{28}$$

$$\rho_P + \rho_Q + c + T \le \Omega = 22d + 97 - \Lambda. \tag{29}$$

If $z > 0$, then $\rho_P + \rho_Q \ge n - 45$.

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

Three additional applications of Lemma 10.2 will be used below. They are displayed with all threshold data; the earlier rows are included to make the successive deductions checkable.

$r \ge$	C	b	K	D	J	$M(2b, b, D+1)$	$(r - J + 1)b$
33	89	7	15	104	19	18	105
35	102	7	15	117	19	19	119
36	111	8	16	127	21	20	128
37	118	8	17	135	21	20	136
38	126	8	17	143	21	21	144
40	143	9	18	161	23	22	162

Each last entry exceeds D, and $T_{J-1} - 1 > D$. In each row the conclusion is $\mu(X) \ge r + C + 1$ if $2b < h < |\mathrm{FS}(X)| - 1$. In particular, applying one row to both sides gives

$$\rho_P + \rho_Q \ge 2C + 6 - (a + b), \tag{30}$$

using only $\delta_0(X) \le |X|$.

12.2 A certificate that uses all translate losses

The following lemma is the main new estimate. Its two inequalities are explicit nonnegative combinations of covering constraints; numerical linear programming is not a premise of the proof.

Lemma 12.2 (Outer translate certificate)**.** *In the covering setup for a feasible A, use the four outer elements on each side, $p_1 > p_2 > p_3 > p_4 > 0$ and $-q_1 < -q_2 < -q_3 < -q_4 < 0$. Suppose*

D, E are integral, $E = P \cup \{0\} \cup (-Q)$, and $D \setminus E$ has four positive and four negative elements, all beyond the corresponding extreme of E. Write

$$R_p = \max P, \quad R_q = \max Q, \quad M_p = \max D, \quad M_q = -\min D,$$
$$S_p = \sum_{x\in D,\ x>0} x, \qquad S_q = \sum_{x\in D,\ x<0} |x|, \qquad q_p = R_p - a, \quad q_q = R_q - b,$$
$$F = S_p + S_q + c - 2(M_p + M_q) - 2(R_p + R_q).$$

Assume $c < d$. Then there are exactly eighteen contributing D-translate slots, counted with their original multiplicities, after the two pairings. Their individual losses sum to at most c. All eight outer elements are integers. Reflect the whole set so that the first paired slot that contributes is the positive one.

If all eighteen centers lie in their respective intervals $[0, S_p]$ or $[0, S_q]$, one of the following bounds holds, with $\xi = 0$:

$$F \geq F_A(a,b) - \xi, \qquad F_A(a,b) = \tfrac{1}{2}T_a + T_b + 6a + 8b + 85, \tag{31}$$
$$F \geq F_B(a,b) - \xi, \qquad F_B(a,b) = \tfrac{2}{3}T_a + T_b + 6a + \tfrac{22}{3}b + \tfrac{236}{3}. \tag{32}$$

The same alternatives hold with $\xi = \max(q_p, q_q)$ if at most one center is outside its respective interval and $c < d - 4$. In all cases, if w is the hull-hole count of $\mathrm{FS}(D)$, then

$$F \leq 18d + 126 + w. \tag{33}$$

Proof. For a positive slot with center v, count points in $(v + D) \cap [1, S_p]$; on the negative side use $(v - D) \cap [1, S_q]$. These hull counts dominate the actual contributions to $\mathrm{FS}(D)$.

The two pairings and integrality. The two centers in the first pair, expressed on positive half-lines, are $c_p - q_1$ and $c_q - p_1$. Their sum is $-(p_2 + p_3 + p_4 + q_2 + q_3 + q_4) < -(M_p + M_q)$, so at most one of the two slots even meets its open half-line. The sixteen unpaired slots must all contribute, since an empty one would cost $d > c$.

If $\beta_p + \beta_q \leq 0$, the second pair is β_p, β_q. The unpaired centers $c_p - q_2$ and $c_q - p_2$ contribute, so their sum is greater than $-(M_p + M_q)$. Consequently

$$\beta_p + \beta_q < M_p + M_q - 2(p_3 + p_4 + q_3 + q_4) < -(M_p + M_q).$$

If $\beta_p + \beta_q > 0$, the second paired centers instead have sum $-(\beta_p + \beta_q) - 2(p_3 + p_4 + q_3 + q_4) < -(M_p + M_q)$. Thus at most one member of each pair contributes. Neither pair can be empty, again because that would cost d. The bound is now the sum of eighteen actual slot counts, $2|2D|$, 84 fixed scores, and zero. If λ_v denotes d minus the appropriate hull count, this proves

$$\sum_v \lambda_v \leq c. \tag{34}$$

Every active center is integral, since it has an intersection with the integral set $\mathrm{FS}(D)$. In particular $b_i = c_p + p_i$ is integral for all four i. The identities $c_p = -b_1 + b_2 + b_3 + b_4$ and $p_i = b_i - c_p$ make all p_i integral; the reflected identities do the same for q_i.

Quantile inequalities. Split $\lambda_v = l_v + u_v$ into losses below 1 and above the appropriate S. For a positive-side center $0 \leq v \leq S_p$, distinctness and integrality give

$$v \geq b + 5 - l_v, \qquad v \leq S_p - a - 4 + u_v. \tag{35}$$

For example, at most $v - 1$ negative magnitudes can be smaller than the positive integer v; the case $v = 0$ includes zero among the lower losses. The negative-side versions interchange a, b and p, q.

If $v < 0$ and $c < d - 4$, the slot has more than four points. Therefore it uses a core point, and $-v < R_p$. Among the integers $1, \dots, -v$, at least $-v - q_p$ belong to P. Counting them as lower losses gives

$$v \geq b + 5 - l_v - q_p.$$

If $v > S_p$, the same argument on Q gives $v \leq S_p - a - 4 + u_v + q_q$. These are the only corrections needed. On the negative side the roles of q_p, q_q reverse. If there is at most one exceptional slot, any combination with loss coefficients in $[0, 1]$ costs at most one additional $\xi = \max(q_p, q_q)$.

The two possible configurations. Reflect so that $A_0 = c_p - q_1$ is the active first paired center. In the branch $\beta_p + \beta_q \leq 0$, put $B_0 = c_q - p_2$; both centers meet their half-lines, and

$$\beta_p = -(A_0 + B_0) - 2(p_3 + p_4) - q_2 - q_3 - q_4 < -M_p.$$

Thus the active second center is β_q. In the other branch, β_p is unpaired and contributes, and $p_2 - q_1 = A_0 + \beta_p + 2(p_3 + p_4) > 2M_p$. It follows that $c_q - p_2 < -M_q$, so the active second paired center is $c_p - q_2$. This classification did not assume that centers are positive.

In the first configuration use the lower inequalities for the five slots

$$c_p - q_1, \quad c_q - p_2, \quad c_q - p_3, \quad \alpha_p, \quad \beta_q,$$

each with coefficient one, and the upper inequality for $b_1 = c_p + p_1$ with coefficient $1/2$. Subtracting the lower centers from $b_1/2$ gives

$$\tfrac{1}{2}p_2 + \tfrac{5}{2}p_3 + \tfrac{3}{2}p_4 + 3q_3 + 3q_4.$$

Equations (34)–(35), with the possible correction ξ, therefore give

$$\tfrac{1}{2}S_p + c \geq \tfrac{9}{2}M_p + 6M_q + \tfrac{7}{2}a + 2b + 44 - \xi. \tag{36}$$

Here $p_2 \geq M_p + 3$, $p_3 \geq M_p + 2$, $p_4 \geq M_p + 1$, and their negative-side analogues were used.

In the second configuration use lower centers $c_p - q_1, c_q - p_3, c_q - p_4, \beta_q$ with coefficient one, and β_p with coefficient $1/3$, together with b_1's upper inequality with coefficient $1/3$. The corresponding difference of centers is $2p_3 + 2p_4 + 3q_3 + 3q_4$. Hence

$$\tfrac{1}{3}S_p + c \geq 4M_p + 6M_q + \tfrac{10}{3}a + \tfrac{4}{3}b + 38 - \xi. \tag{37}$$

All used losses have coefficient at most one and refer to distinct slots.

Finally,

$$S_p \geq T_a + 3R_p + 6 + M_p, \quad S_q \geq T_b + 3R_q + 6 + M_q,$$
$$M_p \geq R_p + 4, \quad M_q \geq R_q + 4, \quad R_p \geq a, \quad R_q \geq b.$$

Substitution in (36) or (37) gives (31) or (32), respectively.

For the upper bound, every sum involving E, and every mixed-sign pair from $D \setminus E$, lies in $[-M_q - R_q, M_p + R_p]$. Only the ten positive-positive and ten negative-negative outlier pairs can lie outside. Thus

$$|2D| \leq M_p + M_q + R_p + R_q + 21.$$

Since $|\mathrm{FS}(D)| = S_p + S_q + 1 - w$, the definition of c gives $S_p + S_q + c = 2|2D| + 18d + 84 + w$. This proves (33). □

Lemma 12.3 (A small combined excess budget)**.** *For a positive set of size $r \geq 39$ with $\varrho(X) \leq 23$,*

$$\mu(X) \leq \varrho(X) + 14 + 2\lfloor \varrho(X)/3 \rfloor.$$

Consequently, if $z = 0$, $a, b \geq 39$, and $\rho_P + \rho_Q + \kappa \leq 23$, then $q_p + q_q \leq 65$.

Proof. Let j be the last index in $[18, r]$ with $y_j \le 4$. If it did not exist, nonnegativity of $\varrho(X_{17})$ and the increment recurrence would give $\varrho(X_r) \ge 3(r-17) > 23$. The later increments give $r - j \le \lfloor \varrho(X_r)/3 \rfloor \le 7$, so $j \ge 32$. The small-increment inverse lemma and the deleted-interval defect bound give $\delta_0(X_j) \le 8$. Each later coefficient adds at most one defect, proving the first formula. On the common lattice, $q_p \le \mu(P) + h_P$ and $q_q \le \mu(Q) + h_Q$. Since $h_P + h_Q = \kappa$ when $z = 0$, summing gives $q_p + q_q \le 23 + 28 + 2\lfloor 23/3 \rfloor = 65$. □

12.3 Excluding 107, 106, and 105

Theorem 12.4 (Improved general upper bound)**.** *Every feasible real set has at most 104 elements. In particular,*

$$\boxed{N \le 104.}$$

Proof. Theorem 9.6 gives $N \le 107$. We exclude the next three sizes in descending order. At each size a set omitting zero would extend to the already excluded next size, so it suffices to treat sets containing zero. Integralization is applied before any lattice inverse theorem; the reductions remain existence-preserving.

Size 107. Theorem 11.1 gives $z = w = 0$, $(a, b) = (45 - k, 45 + k)$ up to reflection, $0 \le k \le 4$, and $c \le 22 - k^2$. Every active center is in its hull interval: a center below zero or above its S loses at least $\min(a + 5, b + 5) > c$ points. Lemma 12.2 applies with $\xi = 0$. The minimum of F_A over both orientations is $4361/2$, and that of F_B is 2351; both exceed $18 \cdot 99 + 126 = 1908$. Thus 107 is impossible.

Size 106, many core holes. Here

$$(a, b) = (44 - k, 45 + k), \quad \Omega = 227 - k(k+1), \quad G = 180, \quad T \ge 60,$$

up to reflection. If $z \ge 15$, the non-progression penalty gives $\rho_P + \rho_Q \ge 61$, hence $k \le 9$ and $a, b \ge 35$. The size-35 hole row gives $\rho_P + \rho_Q \ge 121$. Now $k(k+1) \le 46$, so $a, b \ge 38$, and $T \le 106$ forces $z \ge 37$. The size-38 row gives $\rho_P + \rho_Q \ge 169$. This contradicts $60 + 169 > 227$.

Size 106, at most fourteen core holes. If $z = 0$, then $c \le 47 - k(k+1)$ and $k \le 6$. If $1 \le z \le 14$, then

$$c \le 47 - k(k+1) + 2z - 61 \le 14 - k(k+1),$$

so $k \le 3$. In either case $38 \le a, b \le 51$, $w \le 56$, and $c < \min(a + 5, b + 5)$. All centers are in their hull intervals. The minimum certificate bounds are $F_A \ge 2099$ and $F_B \ge 6865/3$, while $F \le 18 \cdot 98 + 126 + 56 = 1946$. This excludes 106.

Size 105, many core holes. Now

$$(a, b) = (44 - k, 44 + k), \quad \Omega = 250 - k^2, \quad G = 178, \quad T \ge 60.$$

Suppose $z \ge 15$. The initial penalty is $\rho_P + \rho_Q \ge 60$, giving $k \le 11$ and $a, b \ge 33$. Applying the hole rows successively gives the following exact deductions. The indicated minimum size is available *before* its row is applied.

Minimum side size	C	New $\rho_P + \rho_Q \ge$	New $k \le$	Further information
33	89	96	9	Next side size at least 35
35	102	122	8	$z \ge 25$, next size at least 36
36	111	140	7	$z \ge 34$, next size at least 37
37	118	154	6	$z \ge 41$, next size at least 38
38	126	170	4	$z \ge 49$, next size at least 40
40	143	204	–	$60 + 204 > 250$

The first two rows need $z > 14$, the next three $z > 16$, and the last $z > 18$; the displayed deductions meet these strict conditions. All one-sided hole counts are below their support cardinalities minus one. Thus this case is impossible.

Size 105, one to fourteen core holes. The budget now gives

$$c \le 72 - k^2 + 2z - 60 \le 40 - k^2.$$

Hence $k \le 6$, $38 \le a, b \le 50$, and every active center lies in its hull interval. The certificate gives $F_A \ge 4135/2$ and $F_B \ge 6745/3$, both larger than $18 \cdot 97 + 126 + 56 = 1928$.

Size 105, no core holes. Here $w = 0$, $k \le 8$, and the stronger form $T \ge \kappa + 178$ yields

$$\rho_P + \rho_Q + \kappa + c \le 72 - k^2. \tag{38}$$

We have $c \le 72 < d - 4 = 93$. A center outside its hull interval loses at least $49 - k$ points. There is at most one such center, since $2(49 - k) > 72 - k^2$ for every integer k. If there is none, $36 \le a, b \le 52$ and the minimum bounds $F_A \ge 2040$, $F_B \ge 6718/3$ already exceed 1872.

If there is one exceptional center, then $c \ge 49 - k$ and

$$\rho_P + \rho_Q + \kappa \le 23 + k - k^2 \le 23.$$

Nonnegativity forces $k \le 5$, so $a, b \ge 39$. Lemma 12.3 gives $\xi = \max(q_p, q_q) \le 65$. The corrected certificate yields

$$F \ge \min\{F_A - 65, F_B - 65\} \ge \min\{4037/2, 6571/3\} > 1872,$$

again contradicting (33). This excludes 105 and completes the proof. □

For audit purposes, all the last numerical comparisons are collected here. The ranges include both orientations after the reflection in the certificate. Each minimum follows by substituting $b = n - 17 - a$ into an explicit quadratic; these quadratics decrease throughout the listed ranges of a.

Case	Range of a	ξ	$\min(F_A - \xi)$	$\min(F_B - \xi)$	Upper bound for F
107	41–49	0	4361/2	2351	1908
106, $0 \le z \le 14$	38–51	0	2099	6865/3	1946
105, $1 \le z \le 14$	38–50	0	4135/2	6745/3	1928
105, $z = 0$, no exception	36–52	0	2040	6718/3	1872
105, $z = 0$, one exception	39–49	65	4037/2	6571/3	1872

12.4 An additional reduction for the next case, 104

The lattice step itself can also be extended to the remaining endpoint. This proposition alone does not exclude that endpoint. It supplies the lattice step used in the exclusion in Section 13.

Proposition 12.5. *If a feasible real set has 104 elements, integralization followed by a common-core-lattice normalization gives the lattice and hole conclusions of Lemma 12.1, with*

$$d = 96, \quad e = 88, \quad (a, b) = (43 - k, 44 + k), \quad \Omega = 272 - k(k + 1), \quad G = 176, \quad T \ge 59,$$

up to reflection. In particular D is integral in this normalization and $w \le 4z$. Integrality of the outer eight elements in this normalization is not asserted here.

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

With the older constant, the balanced zero-hole loss budget is $\Omega - G = 96 = d$. Section 13 improves the four-extreme constant, reduces this budget to 92, and handles all out-of-hull centers. This closes the obstruction left by the preceding version.

13 Two more exclusions: a general upper bound of 102

We exclude sizes 104 and 103. The first exclusion is analytic. For the second, the only additional ingredient is a finite family of *explicit integer linear certificates*, printed in Appendix B. Their verification uses exact arithmetic and does not depend on a numerical optimization solver. All earlier theorems are retained as intermediate results, with their original weaker constants.

13.1 Improving the four-extreme estimate by four

The sharp four-extreme estimate was established in Lemma 9.3.

13.2 Absorbing every out-of-hull center

The earlier outer certificate required all centers, or all but one, to lie in their target hulls. That restriction is unnecessary if the full span of D is retained until the end.

Lemma 13.1 (Unrestricted active-center certificate)**.** *In the setup and notation of Lemma 12.2, suppose all sixteen unpaired covering units and both paired units are nonempty. Then all eight outer elements are integral on the common lattice. After the reflection specified there, either*

$$F \ge F_A(a,b) \quad \text{or} \quad F \ge F_B(a,b), \qquad F \le 18d + 126 + w.$$

There is no restriction on the number of centers outside their hulls. In particular, the conclusion applies whenever $c < d$.

Proof. The two-pair classification and integrality proof in Lemma 12.2 use only nonemptiness, not $c < d$ itself. They leave eighteen individual active slots with total loss at most c. Set

$$U = M_p - (a+4), \qquad V = M_q - (b+4), \qquad q_p = R_p - a, \quad q_q = R_q - b.$$

All are nonnegative, and $q_p \le U$, $q_q \le V$. For an active positive-side center v, let l_v, u_v count the elements of $v + D$ below 1 and above S_p, respectively. Then

$$\begin{aligned} v &\ge b + 5 - l_v - U\,\mathbf{1}_{v<0}, \\ v &\le S_p - a - 4 + u_v + V\,\mathbf{1}_{v>S_p}. \end{aligned} \tag{39}$$

For example, if $v < 0$, nonemptiness gives $-v < M_p$. Among the integers $1, \dots, -v$, at least $-v - U$ occur in the positive part of D; together with all negative elements and zero this gives the first inequality. If $v > S_p$, use the negative magnitudes smaller than $v - S_p$ for the second. In-hull centers satisfy the old inequalities. Reflection exchanges U, V and a, b.

In branch A, the five lower constraints used in the old certificate have at most two U corrections and three V corrections; the upper constraint has coefficient $1/2$ and costs at most $V/2$. Thus their total correction is at most $2U + 7V/2$. Retaining the span terms when substituting the mass bounds into (36) gives

$$F \ge F_A + 3U + 5V - \tfrac{1}{2}q_p + q_q - (2U + \tfrac{7}{2}V) \ge F_A + \tfrac{1}{2}U + \tfrac{3}{2}V \ge F_A.$$

In branch B, the total correction is at most $4U/3 + 10V/3$. The analogous substitution in (37) gives

$$F \ge F_B + \tfrac{8}{3}U + 5V + q_q - (\tfrac{4}{3}U + \tfrac{10}{3}V) \ge F_B + \tfrac{4}{3}U + \tfrac{5}{3}V \ge F_B.$$

These estimates allow every selected center to be exceptional at once. The upper bound for F is unchanged. □

13.3 Extending the common lattice to size 103

Proposition 13.2 (The 103-element lattice reduction)**.** *Suppose sizes at least 104 have been excluded. A feasible 103-element set can be replaced, preserving existence, by an integer set containing zero. After common-core normalization, D is integral and*

$$d = 95, \quad e = 87, \quad (a, b) = (43 - k, 43 + k) \text{ up to reflection},$$
$$\Lambda = 1893 + k^2, \quad \Omega = 294 - k^2, \quad G = 178, \quad T \geq 60,$$
$$w \leq 4z, \quad T \geq \kappa + 178 - 3z, \quad \rho_P + \rho_Q + c + T \leq 294 - k^2.$$

If $z > 0$, then $\rho_P + \rho_Q \geq 58$.

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

13.4 Many holes: asymmetric use of the effective test

It is useful to apply different parameter rows to the two sides. A row (r_0, C, b_0) in the following table means that a side of size $r \geq r_0$ with $2b_0 < h < |\,\mathrm{FS}\,| - 1$ has $\mu \geq r + C + 1$ and therefore $\varrho \geq C + 3 - r$. The full threshold data for Lemma 10.2 are included.

r_0	C	b_0	K	D	J	M	$(r_0 - J + 1)b_0$
30	70	7	13	83	19	17	84
32	83	7	14	97	19	18	98
33	89	7	15	104	19	18	105
34	96	8	15	111	21	19	112
36	111	8	16	127	21	20	128
38	126	8	17	143	21	21	144
39	134	9	18	152	23	21	153
40	143	9	18	161	23	22	162
41	151	9	19	170	23	22	171
42	160	9	19	179	23	23	180
43	142	7	18	160	21	21	161
43	156	8	19	175	22	22	176
43	168	9	19	187	23	23	189
44	147	7	18	165	21	21	168
44	176	9	20	196	23	23	198

Every window entry exceeds D, and every row also has $T_{J-1} - 1 > D$. Applying rows with parameters C_-, C_+ to the smaller and larger side gives

$$\rho_P + \rho_Q \geq C_- + C_+ + 6 - (a + b). \tag{40}$$

Lemma 13.3. *For sizes 104 and 103, once their lattice reductions are available, $z \geq 15$ is impossible.*

Proof. At size 104, write $(a, b) = (43 - k, 44 + k)$ up to reflection. The new constants are $G = 180$, $T \geq 60$, and $\Omega = 272 - k(k + 1)$. The initial penalty $\rho \geq 59$ gives $k \leq 11$, so the smaller side is at least 32 and the larger at least 44. The following successive rows use (40):

Available side sizes	C_-	C_+	New $\rho \geq$	New $k \leq$	Further consequence
32,44	83	147	149	7	$z \geq 29$, smaller side ≥ 36
36,44	111	176	206	2	$z \geq 57$, both sides ≥ 41
41,41	151	151	221	–	$60 + 221 > 272$

The successive hole thresholds are $z > 14$, $z > 18$, $z > 18$ and are all met. For example, after the first row $T \leq 123$ and $T \geq 180 - 2z$ give $z \geq 29$. This closes the 104-element many-hole case.

At size 103, $\Omega = 294 - k^2$, $G = 178$, $T \geq 60$. The initial penalty $\rho \geq 58$ gives $k \leq 13$, a smaller side at least 30, and a larger side at least 43. The analogous deductions are

Available side sizes	C_-	C_+	New $\rho \geq$	New $k \leq$	New $z \geq$
30,43	70	142	132	10	15
33,43	89	142	151	9	18
34,43	96	156	172	7	28
36,43	111	168	199	5	42
38,43	126	168	214	4	49
39,43	134	168	222	3	53
40,43	143	168	231	1	58
42,43	160	168	248	–	Contradiction

Here the first two rows require $z > 14$, the third $z > 16$, and the remaining rows $z > 18$. All conditions hold before their use. The last contradiction is $60 + 248 > 294$. □

13.5 The one-empty-unit reduction

Lemma 13.4. *In the lattice setup of Proposition 13.2, there is no feasible 103-element set for which $z = 0$ and at least one of the eighteen covering units is empty.*

Proof. The exact budget and (22) give

$$\rho_P + \rho_Q + \kappa + c \leq 116 - k^2. \tag{41}$$

An empty unit costs $d = 95$, so $k \leq 4$. Moreover $c \leq 116 < 190 = 2d$; there is exactly one empty unit. Put $\delta = c - 95$. Then

$$a + b = 86, \qquad 39 \leq a, b \leq 47, \qquad 0 \leq \delta \leq 21 - k^2 \leq 21. \tag{42}$$

Each other unit has at least $95 - \delta \geq 74$ actual contributions.

Outer integrality. Every nonempty individual slot has an integral center, because both D and $\mathrm{FS}(D)$ are integral. If no one of the eight unpaired centers $c_p + p_i, c_q + q_i$ is missing, the usual identities give all outer elements integral. If, for example, $c_p + p_j$ is missing, all four $c_q + q_i$ remain, so every q_i is integral. The unpaired centers $c_p - q_3, c_p - q_4$ both remain, giving $c_p \in \mathbb{Z}$. The other three p_i and then the identity $c_p = p_1 - p_2 - p_3 - p_4$ give the last one. The other case is its reflection. Thus all strict order gaps are at least one.

Pair choices and possible splitting. The first pair always has at most one contributing member. When this unit is present, reflect to make its positive member $c_p - q_1$ active. In branch B ($\beta_p + \beta_q > 0$), the second pair also always has at most one contributing member. In branch A ($\beta_p + \beta_q \leq 0$), the same is true whenever both unpaired cross-centers $c_p - q_2, c_q - p_2$ remain. If either of these is the empty unit, we impose no individual constraint on the second pair; allowing it to split only enlarges the possible set. If the first pair is empty, both orientations of the second are retained. Thus no orientation has been silently discarded.

The relaxed inequalities. For every remaining unpaired slot, and every paired slot known to have one contributing member, the hull loss is at most $\delta \leq 21$. A center outside $[0, S_p]$ or $[0, S_q]$

would lose at least $\min(a+5, b+5) \geq 44$ points. All these centers are therefore in their hulls and obey the uncorrected quantile inequalities. The sum of all their lower and upper losses is at most δ; dropping a possibly split pair preserves this inequality.

Let $a = 39 + t$, $b = 47 - t$, where we relax to the continuous range $0 \leq t \leq 8$. Use $\delta \leq 21$ and the lower tangent

$$T_r \geq \tfrac{87}{2} r - \tfrac{1849}{2}, \qquad T_r - \left(\tfrac{87}{2} r - \tfrac{1849}{2}\right) = \tfrac{1}{2}(r-43)^2.$$

Together with the order, span, mass and quantile constraints, these give the integer coefficient systems printed in Appendix B. There are 33 systems in branch A and 35 in branch B. The appendix supplies 21 explicit nonnegative integer row combinations covering all 68 systems. In 30 systems they imply a negative upper bound for a nonnegative linear form. In the remaining 38 they imply

$$F = S_p + S_q + 95 + \delta - 2(M_p + M_q) - 2(R_p + R_q) \geq 1940.$$

But $z = 0$ gives $w = 0$ and $F \leq 18 \cdot 95 + 126 = 1836$. This contradiction proves the lemma. The certificate identities, including their nonnegative residual coefficients, can be checked directly from the printed tables or by the supplied standard-library verifier. □

13.6 The new upper bound

Theorem 13.5 (General upper bound 102)**.** *Every finite real set A satisfying* $\mathrm{FS}(A) \subseteq 4A$ *has at most 102 elements. Hence*

$$\boxed{N \leq 102.}$$

Proof. Theorem 12.4 gives $N \leq 104$. We exclude 104 first, then 103; at each stage adjoining zero and integralization are legitimate.

For 104, Proposition 12.5 provides the common lattice, and Lemma 9.3 improves its constants to $G = 180$, $T \geq 60$. Lemma 13.3 excludes $z \geq 15$. If $z = 0$, then $c \leq 92 - k(k+1) < 96 = d$ and $k \leq 9$. If $1 \leq z \leq 14$, then

$$c \leq 92 - k(k+1) + 2z - 59 \leq 61 - k(k+1) < d,$$

so $k \leq 7$. Lemma 13.1 applies in both cases.

For 103, Proposition 13.2 now applies and the many-hole lemma again excludes $z \geq 15$. If $1 \leq z \leq 14$, then

$$c \leq 116 - k^2 + 2z - 58 \leq 86 - k^2 < 95 = d,$$

so $k \leq 9$ and the active-center certificate applies. If $z = 0$, then $k \leq 10$. When all eighteen units are nonempty, the same certificate applies regardless of whether $c < d$. Any empty-unit case is excluded by Lemma 13.4.

For completeness, all active-center comparisons are displayed below. The ranges include both orientations after reflection.

Size	Core holes	Range of a	$\min F_A$	$\min F_B$	Upper bound for F
104	$z = 0$	34–53	3971/2	2195	1854
104	$1 \leq z \leq 14$	36–51	2008	6596/3	1910
103	$z = 0$, all units active	33–53	3887/2	6461/3	1836
103	$1 \leq z \leq 14$	34–52	1953	6461/3	1892

Each row contradicts the upper bound. Both remaining sizes are excluded. □

14 Retaining core mass: a general upper bound of 100

The next exclusions are 102 and 101. The principal change is to retain the exact size and the exact mass of the core, rather than replacing both by their balanced lower bounds. This simplifies the lattice step and makes the many-hole case substantially stronger. The remaining finite certificates allow one empty covering unit and, when necessary, one out-of-hull center.

14.1 A shorter, stronger common-lattice argument

Write $\mu = \mu(P) + \mu(Q)$ and $\rho = \rho_P + \rho_Q$. The exact identities and bounds already established include

$$s = \Lambda + \mu + \kappa, \quad R_p + R_q \le a + b + \mu + \kappa + z, \quad \mu \le \rho + 2(a+b) - 4. \tag{43}$$

The last inequality follows directly from $\delta_0(P) \le a$ and $\delta_0(Q) \le b$.

Proposition 14.1. *For $n \in \{102, 101, 100\}$, any feasible integer set containing zero admits the common-core normalization in which D is integral. In this normalization all the exact budget and hole conclusions apply, including*

$$w \le 4z, \quad T \ge \kappa + G - 3z \ge G - 2z, \quad T \ge \lceil G/3 \rceil, \qquad G = 2d - 12.$$

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

14.2 A mass bonus in the active-center certificate

In the common normalization put

$$\varepsilon_p = \mu(P) + h_P = \sum P - T_a, \qquad \varepsilon_q = \mu(Q) + h_Q = \sum Q - T_b.$$

These are exact equalities, not stability approximations.

Lemma 14.2. *When all eighteen covering units are nonempty, the alternatives in Lemma 13.1 strengthen to*

$$F \ge F_A(a,b) + \tfrac{1}{2}\varepsilon_p + \varepsilon_q, \quad \text{or}$$
$$F \ge F_B(a,b) + \tfrac{2}{3}\varepsilon_p + \varepsilon_q.$$

Both remain valid with arbitrarily many out-of-hull centers. If $z > 0$ in the reductions of Proposition 14.1, then

$$\mu(P) \ge a - 3, \quad \mu(Q) \ge b - 3, \qquad \varepsilon_p \ge a - 3 + z, \quad \varepsilon_q \ge b - 3 + z.$$

Proof. The mass estimates in the original certificate can be replaced by

$$S_p \ge T_a + \varepsilon_p + 3R_p + 6 + M_p, \quad S_q \ge T_b + \varepsilon_q + 3R_q + 6 + M_q.$$

Their coefficients in the two substitutions are $(1/2, 1)$ and $(2/3, 1)$. All the span terms that absorb exceptional-center errors are unchanged. This proves the two strengthened alternatives.

For a positive r-element set, the stability theorem stated in Section 5 shows that $\mu \le r - 4$ forces progression support: take $M = \mu$. In its integer normalization the support has $1 + T_r + \mu$ points, while the sum of all elements is at most $T_r + \mu$; therefore the support fills that entire integer interval. Consequently a non-progression support has $\mu \ge r - 3$. When $z > 0$, the inverse-theorem bounds ensure $0 < h_P, h_Q < |\mathrm{FS}(P)| - 1, |\mathrm{FS}(Q)| - 1$ respectively, so both supports are non-progressions. Finally $h_P, h_Q \ge z$. □

14.3 Excluding the many-hole branches at 102 and 101

Only nine applications of the effective-hole lemma are needed, all with $b_0 = 7$. Thus every row is available whenever $z \geq 15$.

r_0	C	K	D	J	M	$7(r_0 - J + 1)$
29	63	12	75	19	16	77
32	83	14	97	19	18	98
34	96	15	111	19	19	112
35	102	15	117	19	19	119
36	109	16	125	19	19	126
38	116	16	132	20	20	133
39	122	17	139	20	20	140
40	128	17	145	20	20	147
42	135	18	153	21	21	154

Every last entry exceeds D, and $T_{J-1} - 1 > D$ in each row. Let $C(r)$ be the largest available C with $r_0 \leq r$, or -4 if none is available. The default -4 denotes the non-progression bound $\mu \geq r - 3$. Define

$$u(r) = r + C(r) + 1, \qquad v(r) = \max\{r - 14,\ u(r) + 2 - 2r\}.$$

Thus $\mu(P) \geq u(a)$, $\mu(Q) \geq u(b)$ and $\rho \geq v(a) + v(b)$ in the many-hole branch.

Lemma 14.3. *A feasible set of size 102 or 101 cannot have $z \geq 15$ in its common normalization.*

Proof. Put $H = \Omega - v(a) - v(b)$ and $t_0 = \lceil G/3 \rceil$. The budget gives $T + c \leq H$, so $z \leq T \leq H$ and $c \leq H - t_0$. If $H < t_0$, this is already impossible. Otherwise the table below verifies $H - t_0 < d$, so all eighteen units are active.

By Lemma 14.2, the two differences between the lower bound for F and its upper bound $18d + 126 + 4z$ are bounded below by

$$F_A(a,b) + \tfrac{1}{2}u(a) + u(b) - (18d + 126) - \tfrac{5}{2}H,$$
$$F_B(a,b) + \tfrac{2}{3}u(a) + u(b) - (18d + 126) - \tfrac{7}{3}H.$$

Indeed both expressions decrease as z increases, and $z \leq H$. Let $\mathcal{M}$ be the minimum of these two expressions and their two reflected versions. The following exact table checks every remaining imbalance. For 102 the initial non-progression budget gives $k \leq 13$; $k = 12, 13$ yield $H = 40, 14 < t_0 = 59$. For 101 it gives $k \leq 14$; $k = 13, 14$ yield $H = 47, 44 < t_0 = 58$.

$n = 102$, $d = 94$				$n = 101$, $d = 93$			
k	H	$c \leq$	$\mathcal{M}$	k	H	$c \leq$	$\mathcal{M}$
0	124	65	389/2	0	144	86	269/2
1	129	70	305/2	1	150	92	179/2
2	125	66	283/2	2	147	89	151/2
3	125	66	116	3	148	90	47
4	123	64	97	4	147	89	25
5	120	61	81	5	145	87	6
6	108	49	96	6	134	76	18
7	101	42	93	7	128	70	12
8	91	32	100	8	119	61	16
9	86	27	89	9	115	57	2
10	66	7	130	10	96	38	40
11	64	5	215/2	11	95	37	29/2
				12	72	14	131/2

Every margin is strictly positive, a contradiction. No search over sets A is involved in this table. □

14.4 Low holes and one possible exceptional center

After the preceding lemma, $0 \le z \le 14$. The budgets are

Size	$z = 0$	$1 \le z \le 14$
102	$\rho + \kappa + c \le 139 - k(k+1)$	$c \le 110 - k(k+1)$
101	$\rho + \kappa + c \le 162 - k^2$	$c \le 134 - k^2$

The positive-hole bounds use $\rho \ge n - 45$ and $T \ge G - 2z$. In particular $c < 2d$: there is at most one empty covering unit.

When no unit is empty, the strengthened active-center certificate gives the following complete comparisons. For positive z we use only $\varepsilon_p \ge a - 2$, $\varepsilon_q \ge b - 2$ and $w \le 56$.

Size	Core holes	Range of a	First minimum	Second minimum	Upper F
102	$z = 0$	31–54	3791/2	6337/3	1818
102	$1 \le z \le 14$	32–53	1958	6530/3	1874
101	$z = 0$	30–54	3713/2	6215/3	1800
101	$1 \le z \le 14$	31–53	1917	6406/3	1856

Thus only exactly one empty unit needs further treatment.

Lemma 14.4. *The one-empty-unit cases at sizes 102 and 101 are impossible.*

Proof. Put $\delta = c - d$. All other units are nonempty and the sum of their losses is at most δ. The outer-integrality argument in Lemma 13.4 requires only that at most one unit is empty, so it applies here unchanged. The same is true of the pair choices: reflect to select the positive first pair when it is present, retain both possibilities for a nonempty second pair, and drop individual second-pair constraints in branch A if one of its unpaired cross-centers is missing.

Only the centers retained by those rules are used below. Their hull losses sum to at most δ. An out-of-hull retained center loses at least $\min(a, b) + 5$ points.

Size 102. If $z = 0$, then

$$k \le 6, \quad a, b \ge 36, \quad \delta \le 45 - k(k+1) < 47 - k = \min(a, b) + 5.$$

If $z > 0$, then $k \le 3$ and $\delta \le 16 - k(k+1)$, an even smaller bound. Thus every retained center is in its hull. Both cases are covered by the single relaxed regime

$$a + b = 85, \quad 36 \le a, b \le 49, \quad 0 \le \delta \le 45, \quad F \le 1874.$$

For this regime the finite certificates in Appendix C give a contradiction in every mode, or $F \ge 9378/5 > 1874$.

Size 101 with $z > 0$. Here

$$k \le 6, \quad 36 \le a, b \le 48, \qquad \delta \le 41 - k^2 < 47 - k.$$

All retained centers again lie in their hulls. The exact mass bonus gives $\varepsilon_p \ge a - 2$, $\varepsilon_q \ge b - 2$. In this regime the certificates give a contradiction or $F \ge 1910 > 1856$.

Size 101 with $z = 0$. Initially

$$k \le 8, \quad 34 \le a, b \le 50, \quad \delta \le 69 - k^2.$$

If all retained centers are in their hulls, the certificate lower bound is $F \ge 9103/5 > 1800$, or the constraints are inconsistent. Otherwise, since

$$2(47 - k) - (69 - k^2) = (k - 1)^2 + 24 > 0,$$

there is at most one exceptional retained center. Its loss is at least $47-k$, so $k^2-k\le 22$, whence $k\le 5$ and $a,b\ge 37$.

We retain the improved range $37\le a,b\le 47$ in this exceptional regime. If the exceptional center is on the positive side, the corrected quantile bounds of (39) become

$$\begin{aligned} v<0:&\quad -v-l_v-M_p\le -d, \qquad l_v\ge b+5,\\ v>S_p:&\quad v-u_v-S_p-M_q\le -(d-1), \qquad u_v\ge a+5. \end{aligned}$$

The reflected formulas cover the negative side. The unmodified opposite quantile bound is still valid. Appendix C considers every retained center and both endpoints; it does not guess the exceptional location. Every such system is inconsistent or has $F\ge 10957/6>1800$.

All four regimes are exact relaxations of the corresponding configurations. Their 2512 systems are certified by 42 printed nonnegative integer row combinations. The exhaustive mode-generation rule and a solver-free verification procedure are included in the appendix. □

Theorem 14.5 (General upper bound 100)**.** *Every feasible finite real set has at most 100 elements. Thus*

$$\boxed{N\le 100}.$$

Proof. Theorem 13.5 gives $N\le 102$. Exclude size 102 first, then 101. At each size a set omitting zero could be extended to the already excluded next size; integralization preserves existence. Proposition 14.1 supplies the common lattice. Lemma 14.3 excludes $z\ge 15$; the displayed active-center comparisons handle $0\le z\le 14$ when no unit is empty, and Lemma 14.4 handles the only remaining possibility. □

15 Excluding size 100: a general upper bound of 99

The new exclusion retains the integer imbalance in both the loss budget and the exact triangular mass bounds. It also treats two phenomena that were absent from the preceding proof: two exceptional retained centers, and two empty covering units. The latter case requires no assertion that all eight outer elements are integral.

At size 100, adjoining zero is legitimate because size 101 is already excluded. Integralization and Proposition 14.1 give

$$\begin{gathered} d=92,\quad e=84,\quad a+b=83,\quad \{a,b\}=\{41-k,42+k\},\quad I=k(k+1),\\ \Lambda=1765+I,\quad \Omega=356-I,\quad G=172,\\ \rho+2\vartheta+c+T=\Omega,\quad T\ge\kappa+172-3z\ge 172-2z,\quad T\ge 58,\quad T\ge z,\quad w\le 4z. \end{gathered}\tag{44}$$

Here $\rho=\rho_P+\rho_Q$, and all slack variables are nonnegative. If $z>0$, the non-progression penalty is $\rho\ge 55$.

15.1 The many-hole branch

For each $r\in[26,57]_{\mathbb{Z}}$ use the two effective-hole rows $(r,C_7(r),7)$ and $(r,C_9(r),9)$ printed in Appendix D. Each satisfies the strict hypotheses of Lemma 10.2. For $z\ge 15$ put

$$u(r,z)=r+1+\max\big(\{C_7(r)\}\cup\{C_9(r):z\ge 19\}\big), \qquad v(r,z)=\max\{r-14,u(r,z)+2-2r\}.$$

These give $\mu(P)\ge u(a,z)$, $\mu(Q)\ge u(b,z)$ and $\rho\ge v(a,z)+v(b,z)$. The initial budget $55+58\le 356-I$ implies $k\le 15$, so every required side size is in the stated range. Define

$$C_k(z)=356-I-v(a,z)-v(b,z)-\max(58,z,172-2z).$$

Then $c \le C_k(z)$. For all states with $C_k(z) \ge 0$ the following table gives $C_k(z) < 92$. Thus all eighteen units are active.

For each orientation $(x, y) = (a, b)$ or (b, a), the two lower bounds for F are

$$F_A(x, y) + \tfrac{1}{2}\big(u(x, z) + z\big) + u(y, z) + z, \quad F_B(x, y) + \tfrac{2}{3}\big(u(x, z) + z\big) + u(y, z) + z.$$

Their minimum, minus $1782 + 4z$, is denoted by $M_k(z)$, with both orientations included. Exact evaluation of the integer range $15 \le z \le 356 - I$ gives:

k	Smallest possible z	Largest possible z	Maximum $C_k(z)$	Minimum $M_k(z)$
0	15	122	64	185
1	15	120	62	164
2	15	117	59	146
3	15	118	60	117
4	15	110	52	115
5	36	100	42	120
6	41	91	33	123
7	47	78	20	139
8	56	61	3	167
9–15	No state has $C_k(z) \ge 0$			

The listed endpoints need not assert that every intervening z survives; all integer values in the original range are checked. Every surviving margin is positive. Consequently $z \ge 15$ is impossible.

15.2 Low holes and the possible empty units

Assume henceforth $0 \le z \le 14$. At $z = 0$, the stronger first inequality for T in (44) gives

$$\rho + \kappa + 2\vartheta + c \le 184 - I. \tag{45}$$

For $z > 0$, the weaker but useful bound is

$$c \le 129 - I + 2z \le 157 - I. \tag{46}$$

Hence there are at most two empty units, and two are possible only when $z = 0$. If no unit is empty, the mass-bonus lemma applies. The minima of its first alternative are

$$\min_{28 \le a \le 55} F_A(a, 83 - a) = 1815 > 1782$$

and

$$\min_{29 \le a \le 54} \big(F_A(a, 83 - a) + (a - 2)/2 + (81 - a)\big) = 3743/2 > 1838.$$

The second alternative is larger: indeed $F_B - F_A = T_a/6 - 2b/3 - 19/3 > 0$ throughout these ranges, and its mass coefficient is also larger. The upper bounds are $F \le 1782$ at $z = 0$ and $F \le 1838$ for $1 \le z \le 14$. This excludes the no-empty case.

Suppose exactly one unit is empty and write $\delta = c - 92$. The integrality and pair-choice arguments in Lemma 13.4 apply unchanged. All retained centers are integral, and the sum of their hull losses is at most δ. A retained center outside its hull loses at least

$$L = \min(a, b) + 5 = 46 - k$$

points. The following exhaustive distinctions will be used:

Regime	Residual budget	Range of a	Exceptional retained centers
Z	$\delta \le 92 - I$	32–51	None
P	$\delta \le 65 - I$	34–49	None
ZE	$\delta \le 92 - I$	35–48	Exactly one
PE	$\delta \le 65 - I$	37–46	Exactly one
ZZ	$\delta \le 92 - I$	40–43	Exactly two

The intervals are derived rather than assumed. Nonnegative residual budget gives $k \le 9$ in Z and $k \le 7$ in P. One exceptional center requires $L \le \delta$, giving $k^2 \le 46$ in ZE and $k^2 \le 19$ in PE. Further,

$$2L - (65 - I) = (k - 1/2)^2 + 107/4 > 0, \qquad 3L - (92 - I) = (k - 1)^2 + 45 > 0.$$

Thus there are at most one and two exceptions, respectively.

15.3 Rigidity when two retained centers are exceptional

In regime ZZ, $2L \le \delta \le 92 - I$ implies $k(k-1) \le 0$. Therefore $k = 0$ or 1, $\delta = 2L$, and equality in (45) forces

$$\rho_P = \rho_Q = \kappa = 0. \tag{47}$$

We need the following elementary equality consequence.

Lemma 15.1 (A zero-defect core)**.** *Let $X \subset \mathbb{Z}_{>0}$ have $r \ge 15$ elements, suppose $\mathrm{FS}(X) = [0, \sum X]_{\mathbb{Z}}$, and suppose $\varrho(X) = 0$. Then*

$$X = [1, r]_{\mathbb{Z}} \quad \text{or} \quad X = [1, r+1]_{\mathbb{Z}} \setminus \{r-1\}.$$

In particular $\max X \le r + 1$.

Proof. The increment recurrence and $\varrho(X_{r-1}) \ge 0$ give $y_r \le 2$. The small-increment inverse lemma therefore writes X, up to scaling, as $[1, r+q] \setminus D_0$ with $q \le 2$ deletions and the last element present. Since $1 \in \mathrm{FS}(X)$, the smallest element is 1; the scale is 1 in the integral normalization. More explicitly, the deleted interval is primitive, its scale is an integer because X is integral, and then the smallest element condition forces that integer to be 1.

Put $h = r + q$ and $C = \{h - d : d \in D_0\}$. The full-support hypothesis gives

$$\mu = \sum X - T_r = \sum C - T_{q-1} = 2\delta_0(X) - 2.$$

For $q = 0$ the assertion is immediate. For $q = 1, 2$, the deleted-interval lemma gives $\delta_0(X) \le 2q$, hence $\mu \le 6$. A low defect $x \le q$ would force $2x \in D_0$ and a depth at least $h - 2q \ge r - 2$ in C, implying $\mu \ge r - 3 > 6$. So there are no low defects. Apart from the maximum h, let the m high defects be $h - t$. Then $t \notin C$, while $2t \in C$; these m forced depths are distinct. Replacing each $2t$ by t leaves q distinct positive depths. Hence

$$2m = \mu \ge q + \sum t \ge q + T_m, \qquad 0 \le m \le q \le 2.$$

The only possibilities are $(q, m) = (0, 0)$ or $(1, 1)$. In the latter, $\mu = 2$ forces $C = \{2\}$, giving the second displayed set. □

In (47), $h_P, h_Q \le \kappa = 0$, so both core supports are full intervals. The lemma gives $R_p - a, R_q - b \le 1$. Moreover, each retained center has loss at most $L \le 46$, because the other exceptional center already costs L; its hull count is therefore at least $92 - 46 = 46 > 4$. The core-point version of the quantile argument in Lemma 12.2 is valid for every retained center. Allowing a correction of 1 at both ends, even for nonexceptional centers, gives the simultaneous relaxed inequalities

$$-v - l_v \le -b - 4, \qquad v - u_v - S_p \le -a - 3 \tag{48}$$

on the positive side, and the reflected versions on the negative side. This handles all locations of the two exceptions without enumerating them.

15.4 Two empty units: use zero loss, not outer integrality

If two units are empty, their combined cost is 184. Equation (45) forces

$$k = 0, \quad \{a, b\} = \{41, 42\}, \quad c = 184, \quad \rho_P = \rho_Q = \kappa = \vartheta = 0.$$

Every other unit contributes its full capacity of 92. For every retained individual center the entire translate of D therefore lies in its target interval. In positive-side notation this gives the stronger inequalities

$$v \geq M_q + 1, \qquad v \leq S_p - M_p, \tag{49}$$

and, on the negative side, $v \geq M_p + 1$, $v \leq S_q - M_q$.

Both empty units are enumerated among the same eighteen units as before. The first pair has at most one contributing member and can be oriented positively when present. In branch B the second pair also has at most one. In branch A this remains true if both cross-centers are present; if either cross-center is missing, all individual constraints for the second pair are dropped. There are 258 modes in branch A and 289 in branch B, hence 547 modes in all, and both values $a = 41, 42$ are retained.

The outer elements need not be proved integral here. The models relax their strict order to weak order, $p_i \geq p_{i+1}$, $q_i \geq q_{i+1}$, $p_4 \geq M_p$, $q_4 \geq M_q$. Likewise branch B uses $\beta_p + \beta_q \geq 0$. The inner set D is integral, so $M_p \geq R_p + 4$ and its reflection remain valid. Every retained center satisfies (49), independently of outer integrality. This is the regime denoted by EE below.

15.5 Exact finite certificates and the conclusion

Appendix D specifies every inequality and every mode for the six regimes. The masses use the exact T_a, T_b; the budget uses the same integer $I = k(k+1)$, with neither expression replaced by a global minimum. In P and PE, the additional core masses are at least $a - 2, b - 2$. In ZE and PE the exceptional center and both possible directions are enumerated, with the corrected quantile row and its necessary loss bound.

The 52 printed nonnegative integer row combinations give the following complete certificate results. A bound is listed only for systems not certified inconsistent; F has constant term 92 in the first five rows and 184 in the last.

Regime	Systems	Bound certificates	Inconsistency certificates	Least certified F	Upper F
Z	1360	1280	80	5357/3	1782
P	1088	1024	64	1840	1838
ZE	32312	29541	2771	1783	1782
PE	23080	21002	2078	1840	1838
ZZ	272	256	16	1795	1782
EE	1094	1002	92	5374/3	1782

Every finite certificate is verified coefficient by coefficient using arbitrary-precision integers. The 59206 models exhaust the low-hole cases with an empty unit. The numerical optimizer was used only to discover candidate multipliers; it is absent from the delivered verification path.

Theorem 15.2 (General upper bound 99)**.** *Every finite real set A with $\mathrm{FS}(A) \subseteq 4A$ has at most 99 elements. Consequently*

$$\boxed{N \leq 99}.$$

Proof. The previous theorem gives $N \leq 100$. A hypothetical set of size 100 can be assumed to contain zero, since otherwise adjoining zero gives the already excluded size 101. The lattice reduction and (44) apply. The many-hole table excludes $z \geq 15$. For $0 \leq z \leq 14$, the active-center bounds exclude no empty unit, and the six certificate regimes exclude one or two empty units. There are no other possibilities. □

15.6 An additional reduction at the next size, 99

Proposition 15.3. *At size 99 the common-lattice and hole conclusions still hold, with*

$$d = 91, \quad e = 83, \quad (a, b) = (41 - k, 41 + k), \quad \Lambda = 1723 + k^2, \quad \Omega = 376 - k^2, \quad G = 170.$$

This proposition does not exclude size 99.

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

16 Retaining residual loss: a general upper bound of 98

We exclude size 99. The distinction from the preceding endpoint is essential: two empty units now leave a residual budget as large as 24, so the full-translate inequalities used at size 100 cannot be reused. Instead we prove a span estimate for small-defect cores, bound $2D$ using several nested interval envelopes, and retain nonemptiness of a potentially split paired unit in the only mode where dropping it is insufficient.

By Theorem 15.2, a hypothetical 99-element set can be assumed to contain zero. Proposition 15.3 gives

$$\begin{gathered} d = 91, \quad e = 83, \quad a + b = 82, \quad \{a, b\} = \{41 - k, 41 + k\}, \\ \Lambda = 1723 + k^2, \quad \Omega = 376 - k^2, \quad G = 170, \\ \rho + 2\vartheta + c + T = \Omega, \qquad T \geq \kappa + 170 - 3z \geq 170 - 2z, \\ T \geq 57, \quad T \geq z, \quad w \leq 4z, \qquad \rho = \rho_P + \rho_Q. \end{gathered} \tag{50}$$

All slack quantities are nonnegative. For $z > 0$, both side supports are non-progressions and $\rho \geq 54$.

16.1 A stronger small-defect estimate

Lemma 16.1. *Let X be a positive real set of size $r \geq 15$, and let $\rho = \varrho(X)$. If $0 \leq \rho < 5(r - 14)$, then*

$$\mu(X) \leq \rho + 22 + 2\lfloor \rho/5 \rfloor \leq \tfrac{7}{5}\rho + 22.$$

Consequently, a lower bound $\mu(X) \geq u$ implies

$$\rho \geq \min\left(5(r - 14), \left\lceil \frac{5(u - 22)}{7} \right\rceil\right).$$

Proof. Use the ordered prefixes X_j and the increments y_j from the defect recurrence. The counting part of Lemma 7.1, independently of its stronger hypothesis for the estimate on Φ, gives $\delta_0([1, j + q] \setminus H) \leq 2q$ for $q \geq 2$ and $j > q$. Indeed every defect belongs to $[1, q]$ or $[j, j + q]$, and there are at most $q - 1$ defects in the former band: if all of $1, \ldots, q$ were present and defective, defectivity of 1 would require 2 to be absent. There are at most $q + 1$ in the latter band. For $q = 1$ the one-deletion argument gives at most two defects (at the prefix sizes used here); for $q = 0$ there is one.

If every y_j for $15 \leq j \leq r$ were at least 7, nonnegativity of $\varrho(X_{14})$ and the recurrence would give $\rho \geq 5(r - 14)$. Choose the largest $j \geq 15$ with $y_j \leq 6$. The small-increment inverse lemma applies, since $6 \leq j - 5$, and makes X_j, up to scaling, an interval with at most six deletions. The

preceding count gives $\delta_0(X_j) \le 12$. Each subsequent step adds at most one defect, while all its increments are at least 7; hence $r - j \le \lfloor \rho/5 \rfloor$ and $\delta_0(X) \le 12 + \lfloor \rho/5 \rfloor$. Now use $\mu = \rho - 2 + 2\delta_0$. The final assertion follows by separating $\rho \ge 5(r-14)$ from its complement. □

For $25 \le r \le 57$, take the effective-hole rows with parameters $\beta = 7, 9$ in Appendix E. Put

$$u(r,z) = r + 1 + \max(\{C_7(r)\} \cup \{C_9(r) : z \ge 19\}),$$
$$v(r,z) = \max\left(r - 14,\ u(r,z) + 2 - 2r,\ \min\left(5(r-14), \left\lceil \frac{5(u(r,z) - 22)}{7} \right\rceil\right)\right).$$

For $z \ge 15$ these give $\mu(P) \ge u(a,z)$, $\mu(Q) \ge u(b,z)$ and $\rho \ge v(a,z) + v(b,z)$. The initial bound $54 + 57 \le 376 - k^2$ gives $k \le 16$, so the side-size range is valid. Define

$$C_k(z) = 376 - k^2 - v(a,z) - v(b,z) - \max(57, z, 170 - 2z).$$

The minimum of the two mass-bonus alternatives, in both orientations, minus $1764 + 4z$ is denoted by $M_k(z)$. Exact evaluation of all $15 \le z \le 376 - k^2$ gives:

k	Smallest surviving z	Largest surviving z	Maximum $C_k(z)$	Minimum $M_k(z)$
0	15	130	73	151
1	15	129	72	255/2
2	15	127	70	213/2
3	15	121	64	98
4	15	121	64	143/2
5	30	111	54	153/2
6	36	99	42	177/2
7	41	88	31	197/2
8	48	74	17	117
9–16	No state has $C_k(z) \ge 0$			

The verifier checks all 4658 states, not just the displayed endpoints. Since $c \le 73 < 91$, all eighteen units are active. Each displayed margin is positive, so Lemma 14.2 excludes every many-hole case.

16.2 The low-hole classification

For $z = 0$, equation (50) yields

$$\rho + \kappa + 2\vartheta + c \le 206 - k^2. \tag{51}$$

For $1 \le z \le 14$ it gives $c \le 152 - k^2 + 2z \le 180 - k^2$. Thus there are at most two empty units, and two are possible only at $z = 0$. If none is empty, the mass-bonus lemma gives

$$\min_{27 \le a \le 55} F_A(a, 82 - a) = 1779 > 1764,$$

and, for $z > 0$,

$$\min_{28 \le a \le 54} (F_A(a, 82 - a) + (a - 2)/2 + (80 - a)) = 3667/2 > 1820.$$

The F_B alternative is larger in these ranges. These comparisons allow arbitrarily many exceptional centers and exclude the no-empty case.

For one empty unit put $\delta = c - 91$; for two put $\delta = c - 182$. A retained center outside its target hull loses at least $L = \min(a,b) + 5 = 46 - k$ points. The exhaustive remaining regimes are:

Regime	Budget on δ	Range of a	Exceptional retained centers
Z	$115-k^2$	31–51	None
P	$89-k^2$	32–50	None
ZE	$115-k^2$	33–49	Exactly one
PE	$89-k^2$	34–48	Exactly one
ZZ	$115-k^2$	36–46	Exactly two
EE	$24-k^2$	37–45	None

The ranges follow from $\delta \ge 0$, $\delta \ge L$, or $\delta \ge 2L$ as appropriate. In particular

$$3L-(115-k^2)=(k-3/2)^2+83/4>0, \quad 2L-(89-k^2)=(k-1)^2+2>0.$$

In EE the entire residual budget is smaller than L. With one empty unit, the outer-integrality and pair-choice proof of Lemma 13.4 applies. With two, only weak outer order is used.

16.3 Small defect controls the core span

Lemma 16.2. *Suppose $X \subset \mathbb{Z}_{>0}$ has $r \ge 16$ elements, $\mathrm{FS}(X)=[0,\sum X]_{\mathbb{Z}}$, and $0 \le \varrho(X) \le r-5$. Then*

$$\max X \le r+\varrho(X)+1.$$

Proof. Every prefix has full interval support: otherwise its first missing integer below the next element can never be supplied by later elements. Equivalently, fullness forces $x_1=1$ and $x_i \le 1+x_1+\cdots+x_{i-1}$ at every step. Put $q=x_r-r\ge 0$. The last excess increment is $y_r=q$. If v counts old diagonal defects destroyed by adjoining x_r, then

$$\varrho(X)=\varrho(X_{r-1})+q-2+2v.$$

Nonnegativity gives $q \le \varrho(X)+2$. If equality holds, then $\varrho(X_{r-1})=v=0$ and $2 \le q \le r-3$. Lemma 15.1 says that the prefix is either $[1,r-1]$ or $[1,r]\setminus\{r-2\}$. In the first case its defective maximum $r-1$ acquires the representation $2(r-1)=x_r+(r-q-2)$, with the second summand in the prefix. In the second case r and $r-1$ are defective. If $q\ne 2$, the identity $2r=x_r+(r-q)$ destroys the former defect; if $q=2$, the identity $2(r-1)=x_r+(r-4)$ destroys the latter. All these are sums of distinct present elements. Thus $v>0$, a contradiction. The equality case is impossible, proving the stated integer bound. □

In ZZ, two exceptions and (51) imply

$$\rho+\kappa \le 23+2k-k^2, \qquad 0 \le k \le 5.$$

In EE they imply $\rho+\kappa+\delta \le 24-k^2$, $0 \le k \le 4$. Each bound on ρ is strictly below $\min(a,b)-14$, so both side supports are arithmetic progressions by Theorem 7.3. An integer progression of step at least two and cardinality s has at least $s-1$ hull holes. Since $h_P, h_Q \le \kappa \le 24$ and $s-1 \ge T_{\min(a,b)} > 24$, both steps are one. Thus $h_P=h_Q=\kappa=0$. The bounds on ρ also meet the hypotheses of Lemma 16.2. Adding its two conclusions to the residual budget gives

$$R_p+R_q+\delta \le \begin{cases} 199-k^2, & \text{ZZ}, \\ 106-k^2, & \text{EE}. \end{cases} \tag{52}$$

In ZZ every retained center loses at most $\delta-L \le 69+k-k^2 \le 69$, so its hull count exceeds four. The core-point quantile correction is therefore valid at every center. For positive centers its relaxed form is

$$-v-l_v-R_p \le -87, \qquad v-u_v-S_p-R_q \le -86,$$

with reflected inequalities for negative centers. These bounds use the actual core spans, not an unjustified claim that they are at most $a+1, b+1$.

16.4 Two additional mass and sumset estimates

In all regimes, distinct positive integers in the core give $\sum P \geq T_{a-1} + R_p$. The four positive elements of $D \setminus E$ are at least $R_p + 1, R_p + 2, R_p + 3, M_p$. Consequently

$$S_p \geq T_{a-1} + 4R_p + 6 + M_p, \qquad S_q \geq T_{b-1} + 4R_q + 6 + M_q. \tag{53}$$

These inequalities are imposed alongside, not added to, the earlier mass inequalities.

A useful discrete improvement holds if $M_q - R_q = 4$. Then the four negative magnitudes outside the core are exactly $R_q + 1, \ldots, R_q + 4$, and their ten unordered pairs with repetition have only seven possible sums. All other sums of two elements of D lie in $[-M_q - R_q, M_p + R_p]$, except possibly the ten positive-positive outlier pairs. Hence

$$|2D| \leq M_p + M_q + R_p + R_q + 18, \qquad F \leq 1758 \quad (z = 0). \tag{54}$$

This improves the usual upper bound 1764 by six. At the one ZE mode specified in the appendix we split the integer gap into $M_q - R_q = 4$ and $M_q - R_q \geq 5$. The first uses (54); the second adds its gap inequality to the model. This avoids turning the non-strict bound $F \geq 1764$ into a false contradiction.

For EE expose the four inner extremes on each side:

$$x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p, \quad y_0 = R_q < y_1 < y_2 < y_3 < y_4 = M_q.$$

All are integers and successive gaps are at least one. Also

$$S_p \geq T_{a-1} + R_p + x_1 + x_2 + x_3 + x_4, \quad S_q \geq T_{b-1} + R_q + y_1 + y_2 + y_3 + y_4.$$

Lemma 16.3 (Nested interval envelopes)**.** *For $j = 0, 1, 2, 3$, put $W_j = x_j + y_j$ and $t_j = 2(4 - j)$. Then*

$$|2D| \leq 2W_j + 1 + \binom{t_j + 1}{2} + \sum_{i=j+1}^{4} (\min(W_j + 1, x_i - x_{i-1}) + \min(W_j + 1, y_i - y_{i-1})). \tag{55}$$

Proof. Let D_j consist of E and the first j added elements on each side. Replace D_j by its integer hull $[-y_j, x_j]$. Its double sum occupies $2W_j + 1$ integers. Adjoining the translates $x_i + [-y_j, x_j]$ in increasing order enlarges their union with this initial interval by at most $\min(W_j + 1, x_i - x_{i-1})$ at step i. For the first step the preceding upper endpoint is $2x_j$; for later steps it is $x_{i-1} + x_j$. The negative translates give the reflected bounds. Adding the two one-sided enlargement bounds is still an upper bound on their combined union, even if some intervals overlap. The only remaining sums use two of the t_j external elements, with repetition allowed. There are at most $\binom{t_j+1}{2}$ such sums. This covers every element of $2D$ and proves the inequality. □

In the model put $K = |2D|$. The minima in (55) can be replaced by nonnegative variables g bounded above by both of their arguments: an actual set realizes the model by taking g equal to the minimum. Since $w = 0$ in EE, the exact identity defining c gives

$$S_p + S_q + \delta - 2K = 18 \cdot 91 + 84 - 182 = 1540. \tag{56}$$

We impose only the corresponding upper inequality, a valid relaxation.

16.5 The split pair and complete finite certification

Use the mode labels of Appendix D. Among the EE modes, one needs an extra necessary condition: branch A, missing units p_0, q_5. The second pair β_p, β_q is present, though its individual constraints were dropped to allow splitting. Its combined contribution is at least $91 - \delta > 0$.

Thus at least one of its two individual slots contributes. If the positive one contributes, its center is integral and its largest translated point is $\beta_p + M_p \geq 1$. If the negative one contributes, likewise $\beta_q + M_q \geq 1$. It is therefore enough to check the two relaxations

$$\beta_p + M_p \geq 1 \qquad \text{or} \qquad \beta_q + M_q \geq 1. \tag{57}$$

No assertion about which member contributes, or whether both do, is made. In particular this step does not require integrality of all eight outer elements.

Appendix E gives all rows and all 77 nonnegative integer row combinations. They certify the following complete case enumeration:

Regime	Systems	Strict lower-bound certificates	Inconsistency certificates
Z	1428	1344	84
P	1292	1216	76
ZE	39237	36361	2876
PE	34620	32056	2564
ZZ	748	704	44
EE	4932	4129	803
Total	82257	75810	6447

For every system the checker regenerates its rows, verifies a nonnegative integer combination coefficient by coefficient, and compares the resulting bound with that system's upper bound using exact fractions. The gap-four ZE system uses 1758; all other zero-hole systems use 1764, and the positive regimes use 1820. Thus the displayed system count includes one additional gap branch and nine additional split-pair branches, and no endpoint or orientation is inferred by interpolation.

Theorem 16.4. *Every finite real set satisfying* $\mathrm{FS}(A) \subseteq 4A$ *has at most 98 elements. Consequently*

$$\boxed{N \leq 98}.$$

Proof. The preceding theorem already excludes all sizes greater than 99. At size 99, adjoining zero is legitimate because size 100 has been excluded. The common-lattice reduction, the many-hole table, and the no-empty-unit comparisons leave exactly the six regimes above. The small-span, mass, quantile, envelope, and split-pair arguments prove that any remaining actual configuration realizes at least one enumerated linear system. Every such system has a strict exact certificate of impossibility. Hence size 99 is impossible. □

16.6 A further common-lattice reduction at size 98

Proposition 16.5. *At size 98 the common-lattice and hole conclusions hold, with*

$$d = 90, \quad e = 82, \quad (a, b) = (40 - k, 41 + k), \quad I = k(k + 1),$$
$$\Lambda = 1682 + I, \quad \Omega = 395 - I, \quad G = 168.$$

This is a structural reduction, not an exclusion of size 98.

Proof. Apply Proposition 9.5 at the stated size(s). The displayed parameters are obtained from $d = n - 8$, $e = n - 16$, $a + b = n - 17$, $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. Where the statement uses $G = 2d - 16$ rather than $2d - 12$, it is a weaker consequence of the same proved inequalities. If needed, zero is present because the next larger size has already been excluded; the proposition itself assumes zero and does not use that endpoint exclusion. □

17 Two further exclusions: the general upper bound 96

We exclude sizes 98 and 97, in that order. The new finite proof uses integer inconsistency certificates and, for 19 roots at size 97, exhaustive binary disjunctions. The disjunctions are proved for actual sets; no claim that they partition every point of the relaxed polyhedron is needed or made. The certificate files accompanying this report are part of the finite proof.

17.1 A span estimate that permits holes

For a positive integer set X, write $h(X) = \sum X + 1 - |\operatorname{FS}(X)|$. This is its number of missing integer points in the full subset-sum hull, on the specified unit lattice.

The universal span estimate of Lemma 9.4 applies here.

Put $m = a + b = n - 17$, $d = n - 8$, $\rho = \rho_P + \rho_Q$, and $\mu = \mu(P) + \mu(Q)$. On the common core lattice, the lemma gives

$$R_p + R_q \le m + \rho + \kappa + z + 4. \tag{58}$$

After D has been placed on that lattice, write j for the number of empty covering units and $\delta = c - jd$. Combining the exact budget with $T \ge \kappa + G - 3z$, where $G = 2d - 12$, gives the useful joint bound

$$R_p + R_q + \delta \le m + 4 + \Omega - G - jd + 4z. \tag{59}$$

The same z occurs in the loss, mass, cardinality, and span constraints. Replacing it by unrelated worst-case values can lose the contradiction.

Lemma 17.1 (Flexible defect bound with an integer inverse)**.** *Let X be a positive real set of size $r \ge 15$, and let $2 \le \beta \le 10$ be an integer. If $0 \le \varrho(X) < (\beta - 1)(r - 14)$, then*

$$\mu(X) \le \varrho(X) + 4\beta - 2 + 2\left\lfloor \frac{\varrho(X)}{\beta - 1} \right\rfloor.$$

For an integer lower bound $\mu(X) \ge u$, put $v = \max(0, u - 4\beta + 2)$ and write $v = (\beta + 1)q + s$, $0 \le s \le \beta$. Then

$$\varrho(X) \ge \min\big((\beta - 1)(r - 14), (\beta - 1)q + \min(s, \beta - 1)\big). \tag{60}$$

Proof. If all increments y_i for $15 \le i \le r$ exceed β, the recurrence $\rho_i \ge \rho_{i-1} + y_i - 2$ contradicts the hypothesis. Choose the last index $i \ge 15$ with $y_i \le \beta$. Since $\beta \le i - 5$, Lemma 5.5 gives an interval with at most β deletions. The defect-counting argument in Lemma 16.1 uses only $i > \beta$ and gives at most 2β defects; the zero-deletion case has one defect. Every subsequent step adds at most one defect and raises ρ by at least $\beta - 1$. Nonnegativity at index i therefore gives $r - i \le \lfloor \varrho(X)/(\beta - 1) \rfloor$. Use $\mu = \varrho - 2 + 2\delta_0$. Finally, the least nonnegative integer t satisfying $t + 2\lfloor t/(\beta - 1) \rfloor \ge v$ is $(\beta - 1)q + \min(s, \beta - 1)$: each complete block of $\beta - 1$ in t adds $\beta + 1$ to the left side. This proves the stated inverse, including the alternative where the small-defect hypothesis fails. □

17.2 The lattice reduction at sizes 97 and 96

Proposition 17.2. *For $n \in \{97, 96\}$, a feasible integer set of size n containing zero admits the common-core normalization in which D is integral and*

$$w \le 4z, \quad T \ge \kappa + G - 3z \ge G - 2z, \quad T \ge \lceil G/3 \rceil, \quad T \ge z.$$

The parameters are

n	d	e	m	Λ_0	Ω_0	I	G
97	*89*	*81*	*80*	*1641*	*414*	k^2	*166*
96	*88*	*80*	*79*	*1601*	*432*	$k(k+1)$	*164*

Here $(a,b) = (\lfloor m/2 \rfloor - k, \lceil m/2 \rceil + k)$*, up to reflection,* $\Lambda = \Lambda_0 + I$*, and* $\Omega = \Omega_0 - I$*.*

Proof. The deficient-sign argument forces $|B| = 8$: otherwise a one-sided core of at least 80 nonzero elements would give $s \geq 1 + T_{80} = 3241 > 24 \cdot 97 + 77$. The unpaired-value bound is $q^2 \leq 2288$ at $n = 97$ and $q^2 \leq 2353$ at $n = 96$. Parity gives $q \leq 46, 47$, respectively, so both sides have at least 17,16 elements. The nonnegative budget applies and implies $k \leq 20$. For all these integer k, $\Omega < T_{\min(a,b)} - 2$. Thus the simultaneous inverse theorem supplies the common core lattice and $h_P, h_Q \leq \kappa$, $z = h_P + h_Q - \kappa \leq \kappa$.

It remains to put the peeled elements on the same lattice. Apply (58) in the second-layer comparison, before asserting the integrality of D. With $C = 2m + 4e + 14 - \Lambda$ this gives

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\leq C + 2\rho - \mu + \kappa + 2z + \ell_2 \\ &\leq C + 3\Omega - \rho - \mu, \qquad (61) \\ |\operatorname{FS}(D_1)| - 2s &\leq C + \Omega + \rho - \mu + 2z. \qquad (62) \end{aligned}$$

These follow from $\rho + \kappa + \ell_2 \leq \Omega$ and $z \leq \kappa$. The maximum of each positive side is always defective; hence $\rho \leq \mu$.

At $n = 97$, when $z = 0$, the first line above also gives the coarse bound $C + 2\Omega = -315 - 3I < 0$ (use $\rho \leq \Omega$ and $\mu \geq 0$). When $z > 0$, both side supports are non-progressions: a progression with a larger integer step would have at least $|\operatorname{FS}| - 1$ holes, contrary to $h \leq \kappa < |\operatorname{FS}| - 1$. Consequently $\rho \geq m - 28 = 52$, $\mu \geq m - 6 = 74$, and (61) is at most $-27 - 4I < 0$.

At $n = 96$, equation (62) is at most $C + \Omega + 28 = -649 - 2I < 0$ for $z \leq 14$. For $z \geq 15$, non-progression gives $\rho + \mu \geq 2m - 34 = 124$. As $C + 3\Omega = 187 - 4I$, a remaining case must have $k \leq 3$ and $a, b \geq 36$. The effective-hole test with $(r_0, C, \beta) = (36, 109, 7)$ gives $\mu(P) \geq a + 110$, $\mu(Q) \geq b + 110$. Its strict hypotheses are verified by the row at $r = 36$ in Appendix F; they continue to hold for larger r. Thus $\rho + \mu \geq 51 + 299 = 350 > 187$, again strictly. The first-layer comparisons are $1641 > 4 \cdot 89 - 10 + 414 = 760$ and $1601 > 4 \cdot 88 - 10 + 432 = 774$. The earlier overlap and hole-propagation arguments now apply, followed by the four-extreme estimate. This proves all stated conclusions. □

The proposition assumes zero is already present. In its use below, zero is adjoined only after the larger endpoint has been excluded. The result at size 96 is a structural reduction, not an exclusion of that size.

17.3 One model format for both exclusions

The outer centers and eighteen covering units are those of Appendix D. Their exact labels and the new row formulas are recorded in Appendix F. We recall the validity issues that are important for completeness.

There are two pairing branches, according as $\beta_p + \beta_q \leq 0$ or > 0. Reflect to put the contributing first-pair member on the positive side when that pair is nonempty. If it is empty, retain both possibilities for the second pair. With zero or one empty unit, the outer elements are integral by the contributing-center argument in Lemma 13.4. With two empty units, only weak outer order is imposed. Every retained contributing center is still integral, since both D and $\operatorname{FS}(D)$ are integral.

In pairing branch A, if an unpaired center p_5 or q_5 is missing, the second pair may have two contributors. Its individual centers are dropped from the usual loss rows. If this paired

unit is present, choose one *actually contributing* member, denoted by $s = p$ or q, and retain the necessary inequality

$$\beta_s + M_s \geq 1. \tag{63}$$

This row is not asserted to be sufficient for contribution. The choice is made from the actual set, not from the solutions of the relaxation. This distinction will justify the sole-contributor disjunction below. The resulting numbers of modes with $j = 0, 1, 2$ empty units are $4, 70, 578$, including these member choices.

For every retained center let $\lambda_v = l_v + u_v$ be its number of hull losses. The covering inequality gives $\sum \lambda_v \leq \delta = c - jd$, also when a split pair has been dropped. For a positive center the following relaxed quantile rows are available:

Type	Lower row	Upper row
Ordinary	$-v - l_v \leq -b - 5$	$v - u_v - S_p \leq -a - 4$
Global	$-v - l_v - M_p \leq -d$	$v - u_v - S_p - M_q \leq -(d-1)$
Core	$-v - l_v - R_p \leq -(m+5)$	$v - u_v - S_p - R_q \leq -(m+4)$

Reflection gives the negative rows. Ordinary rows apply when the relevant endpoint is inside the hull. Global rows apply to any nonempty center. Core rows apply when its hull count exceeds four. Indeed, an out-of-hull translate with more than four points must use a core point, since there are only four removed positive and four removed negative elements. The core-gap counting in the earlier quantile proof then gives these rows; in-hull ordinary rows imply them using $R_p \geq a$, $R_q \geq b$.

Let $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$ be the positive core endpoint and the four positive elements of $D \setminus E$; define y_i by reflection. All eight gaps are at least one. The mass and four nested-envelope inequalities of Section 16 are imposed. In particular, with $K = |2D|$ and $\varepsilon_p = \mu(P) + h_P$,

$$S_p \geq T_a + \varepsilon_p + 3R_p + 6 + M_p, \qquad S_p \geq T_{a-1} + 4R_p + 6 + M_p, \tag{64}$$

$$S_p \geq T_a + \varepsilon_p + \sum_{i=1}^{4} x_i, \qquad S_p \geq T_{a-1} + R_p + \sum_{i=1}^{4} x_i, \tag{65}$$

$$S_p + S_q + \delta - 2K \leq 18d + 84 - jd + 4z. \tag{66}$$

The last inequality follows from the exact identity with w in place of $4z$. It does not rely on an estimate of the hull length of $2D$.

17.4 Excluding size 98

By the preceding upper bound, a hypothetical 98-element set contains zero after adjoining it if necessary. Integralization and the corrected Proposition 16.5 give

$$d = 90, \quad e = 82, \quad m = 81, \quad (a, b) = (40-k, 41+k), \quad I = k(k+1), \quad \Omega = 395 - I, \quad G = 168.$$

For $z = 0$, $c \leq 227 - I$. For $1 \leq z \leq 14$, both supports are non-progressions, so $\rho \geq 53$ and $c \leq 174 - I + 2z \leq 202 - I$. There are at most two empty units in this range. Set $B_0 = 227 - j \cdot 90 - I$. The zero-hole models use

$$\delta \leq B_0, \quad R_p + R_q + \delta \leq 85 + B_0, \quad \varepsilon_p, \varepsilon_q \geq 0.$$

The positive low-hole models retain a single variable $1 \leq z \leq 14$ and use

$$\delta \leq B_0 - 53 + 2z, \qquad R_p + R_q + \delta \leq 85 + B_0 + 4z,$$
$$\varepsilon_p \geq a - 3 + z, \qquad \varepsilon_q \geq b - 3 + z. \tag{67}$$

The mass and cardinality rows use that same z.

For one empty unit, $\delta \le 137$ or 112, both less than $2(d-4) = 172$. Therefore at most one retained center can have $\lambda_v \ge 86$. If none does, all core quantiles are valid. If one does, designate it and use both global endpoint rows for it, with $\lambda_v \ge 86$; all other centers use core rows. For two empty units, $\delta \le 47$ or 22, so every retained center has hull count greater than four. In the positive-hole case the budget is even below $\min(a,b)+5$, giving ordinary quantiles at all centers.

The finite low-hole root list is:

Regime	j	Range of a	Quantile rule	Roots
Z0	0	26–55	Global	120
P0	0	27–54	Global	112
Z1	1	29–52	Core	1680
P1	1	30–51	Core	1540
ZL	1	34–47	One designated large loss	16604
PL	1	36–45	One designated large loss	11860
Z2	2	34–47	Core	8092
P2	2	36–45	Ordinary	5780

Z means $z = 0$, P means $1 \le z \le 14$, and L designates the large-loss center. These are relaxations: a listed model need not impose the strict small-loss bounds used to select it from an actual set.

For $z \ge 15$, the initial penalty gives $k \le 16$ and $24 \le a, b \le 57$. Use the hole rows in Appendix F to define

$$u(r,z) = \max\big(\{r-3\} \cup \{r + C_\beta(r) + 1 : \beta \in \{7,9,10\},\ z > 2\beta,\ C_\beta(r) \text{ listed}\}\big),$$
$$v(r,u) = \max\{r-14,\ u+2-2r,\ L_8(r,u),\ L_9(r,u)\},$$

where L_β is the right side of (60). Then $\mu(P) \ge u(a,z)$ and $\rho_P \ge v(a,u(a,z))$, with reflected inequalities on Q. Consequently

$$c \le C_k(z) := 395 - I - v(a,u(a,z)) - v(b,u(b,z)) - \max(56, z, 168-2z).$$

Evaluate every integer $15 \le z \le 395 - I$, $0 \le k \le 16$: there are 4845 such states. Whenever $C_k(z) \ge 0$, the four possible all-active mass-bonus margins, using both orientations, have minimum at least $63/2 > 0$. Thus no all-active state survives. A state with an empty unit requires $C_k(z) \ge 90$, and the only possibilities are

$$(k, z, C_k(z), u(a,z), u(b,z)) = (0, 56, 91, 184, 193), \quad (0, 57, 90, 184, 193).$$

Both are covered by a single family H1, in both orientations, using

$$\delta \le 1, \quad R_p + R_q + \delta \le 450, \quad (\varepsilon_p, \varepsilon_q) \ge (240, 249), \quad w \le 228,$$

or its reflection, and ordinary quantiles. Its cardinality row has right side 1842. There are $2 \cdot 70 = 140$ roots in H1.

Proposition 17.3. *No feasible real set has 98 elements. Hence $N \le 97$.*

Proof. The preceding classification assigns every actual set to one of 45928 roots. The model rows are necessary by the span, mass, envelope, loss, and pairing arguments. The 179 integer certificates in `upper97_exact.json`, containing 5888 nonzero weights, give inconsistency of every root. Explicitly, for its rows $Ax \le b$, $x \ge 0$, the selected nonnegative integer vector w satisfies $w^{\mathsf{T}}A \ge 0$ coordinatewise and $w^{\mathsf{T}}b < 0$. Thus $0 \le w^{\mathsf{T}}Ax \le w^{\mathsf{T}}b < 0$, a contradiction. The independent checker regenerates the complete ordered root list and checks each combination using arbitrary-precision integers. □

17.5 Classifying all size-97 roots

Zero can now be adjoined in a hypothetical 97-element set. Proposition 17.2 gives $d = 89$, $e = 81$, $m = 80$, $(a, b) = (40 - k, 40 + k)$, $I = k^2$, $\Omega = 414 - I$, and $G = 166$. For $z = 0$, $c \le 248 - I$; for $1 \le z \le 14$, $c \le 196 - I + 2z \le 224 - I$. Again there are at most two empty units. Put $B_0 = 248 - j \cdot 89 - I$. Zero-hole rows are

$$\delta \le B_0, \quad R_p + R_q + \delta \le 84 + B_0, \quad \varepsilon_p, \varepsilon_q \ge 0.$$

The positive-hole rows are

$$\begin{aligned} \delta \le B_0 - 52 + 2z, \quad R_p + R_q + \delta \le 84 + B_0 + 4z, \quad 1 \le z \le 14, \\ \varepsilon_p \ge a - 3 + z, \qquad \varepsilon_q \ge b - 3 + z. \end{aligned} \tag{68}$$

The cardinality row is $S_p + S_q + \delta - 2K \le 1686 - j \cdot 89 + 4z$. With one empty unit, $\delta \le 159$ or 135, both strictly less than $2(d - 4) = 170$. At most one retained center has loss at least 85. With two empty units, $\delta \le 70$ or 46, both below 85, so every retained center uses core quantiles. Unlike at size 98, ordinary quantiles cannot be assumed throughout the positive two-empty family.

Regime	j	Range of a	Quantile rule	Roots
Z0	0	25–55	Global	124
P0	0	26–54	Global	116
Z1	1	28–52	Core	1750
P1	1	29–51	Core	1610
ZL	1	32–48	One designated large loss	20162
PL	1	33–47	One designated large loss	17790
Z2	2	32–48	Core	9826
P2	2	34–46	Core	7514

There are 58892 low-hole roots. Each large-loss family loops over every retained center; the 70 one-empty modes together have 1186 such centers.

For $z \ge 15$, the initial inequality $52 + 56 \le 414 - k^2$ gives $k \le 17$, so $23 \le a, b \le 57$. Use the same functions u, v as above and put

$$C_k(z) = 414 - k^2 - v(a, u(a, z)) - v(b, u(b, z)) - \max(56, z, 166 - 2z).$$

The mass-test margin $M_k(z)$ is the minimum of the four expressions

$$\begin{aligned} F_A(a, b) + \tfrac{1}{2}(u(a, z) + z) + u(b, z) + z, \\ F_B(a, b) + \tfrac{2}{3}(u(a, z) + z) + u(b, z) + z, \end{aligned}$$

and their reflections, minus $1728 + 4z$. The exact enumeration retains every state with $C_k(z) \ge 0$ and finds $C_k(z) \le 118 < 178$. Thus at most one unit is empty. For no empty unit, states with $M_k(z) > 0$ are excluded by Lemma 14.2. There remain 128 (k, z) states, or 251 oriented (a, z) states. For one empty unit, $C_k(z) \ge 89$ leaves 175 (k, z) states, or 305 oriented states. In the latter cases $\delta \le C_k(z) - 89 \le 29 < \min(a, b) + 5$, so all retained centers obey ordinary quantiles.

For each remaining high-hole parameter state, fix z to that integer; use $\delta \le C_k(z) - j \cdot 89$, $\varepsilon_p \ge u(a, z) + z$, $\varepsilon_q \ge u(b, z) + z$, and retain (59) and (66) with the same fixed z. Use global quantiles for H0, ordinary ones for H1. These give $251 \cdot 4 = 1004$ H0 roots and $305 \cdot 70 = 21350$ H1 roots. Together with the low-hole families there are 81246 roots.

17.6 Three valid disjunctions and the 19 remaining roots

Most size-97 roots are already inconsistent. The remaining 19 are covered by finite binary trees. Each split has one of the following three proofs.

An ordinary endpoint or an outside endpoint. For a positive contributing center v, the lower endpoint satisfies

$$-v - l_v \le -b - 5 \quad \text{or} \quad (l_v \ge b + 5 \text{ and } v \le -1). \tag{69}$$

If $v \ge 0$, the ordinary counting proof applies. Otherwise the integral center has $v \le -1$, and all $b + 4$ negative points and zero are lost below 1. Similarly, the upper endpoint satisfies

$$v - u_v - S_p \le -a - 4 \quad \text{or} \quad (u_v \ge a + 5 \text{ and } v \ge S_p + 1). \tag{70}$$

Reflection gives both negative-side versions. Only integrality of the contributing center and D is used, not integrality of every outer element.

Inner extreme or endpoint loss. For each $i = 1, 2, 3, 4$, the positive center satisfies

$$l_v \ge i \text{ or } v \ge y_{5-i} + 1, \qquad u_v \ge i \text{ or } v \le S_p - x_{5-i}. \tag{71}$$

For example, if the integer count $l_v < i$, at least one of the i largest negative magnitudes remains above the lower endpoint; in particular the smallest of those i gives $v - y_{5-i} \ge 1$. The upper version is the reflected counting assertion. Again include the versions with p, q exchanged. These disjunctions apply to actual integer loss counts. They are not asserted for arbitrary fractional loss variables.

Both paired members contribute, or the selected one is sole. Consider a two-empty mode with a split second pair and a member s chosen from actual contributors as in (63). Either the other member t also contributes, in which case $\beta_t + M_t \ge 1$, or s is the sole contributor. In the latter case the paired unit's capacity loss is at least its individual hull loss. Introduce $l_*, u_* \ge 0$ for this hull loss and strengthen the total-loss row to

$$\sum_{v \text{ already retained}} (l_v + u_v) + l_* + u_* \le \delta.$$

Its hull count is at least $d - \delta > 4$, because the two-empty budgets are at most 70 and 46. It therefore satisfies its own core quantiles:

$$-\beta_s - l_* - R_s \le -85, \qquad \beta_s - u_* - S_s - R_t \le -84. \tag{72}$$

Only two unused nonnegative coordinates are needed for l_*, u_*. All six trees using this split finish after its two children. In particular, the argument does not replace actual contribution by the merely necessary numerical row (63).

The trees and all assignments are explicit in `upper96_exact.json`. A summary of every branched root is printed in Appendix F. There are 19 such roots, 130 internal nodes and 149 leaves. The other 81227 roots need no split. Thus there are exactly 81376 certified leaf systems, counting the ordinary roots. At every internal node the checker visits *both* children and adds the corresponding proved rows. No child is pruned on numerical evidence.

Theorem 17.4. *Every feasible finite real set has at most 96 elements. Consequently,*

$$\boxed{N \le 96}.$$

Proof. Proposition 17.3 supplies $N \le 97$. A hypothetical 97-element set can therefore be assumed to contain zero, then integralized. Its common-lattice normalization satisfies one of the 81246 root models just classified. Follow any satisfied child at each node of its finite tree. The disjunction proofs guarantee such a child for the actual configuration, with the auxiliary hull losses chosen from that configuration.

The selected leaf is inconsistent: the 303 certificates in `upper96_exact.json` have 10514 positive integer weights in total, and certify $w^{\mathsf{T}}A \geq 0$, $w^{\mathsf{T}}b < 0$ at every leaf. The standard-library verifier independently regenerates every root and every allowed split and checks all these inequalities exactly. Hence there is no feasible set of size 97. The general upper bound follows. □

The finite proof at sizes 98 and 97 consists of 127174 roots in the two exclusions and 127304 leaf contradictions. Numerical linear programming was used only to discover multiplier vectors and useful disjunctions; none is called by the delivered verification program. The universal reductions, the assignment of actual sets to roots, and the exhaustiveness of each split are separate mathematical arguments given above.

Part III
The descent from 96 to 86

18 The descending chain from 96 to 86

Theorem 17.4 establishes the global bound $N \le 96$. We now exclude sizes $96, 95, \dots, 87$ in that order. The only cardinality premise at stage n is the already established bound $N \le n$. A feasible n-set omitting zero would produce a feasible $(n+1)$-set by adjoining zero; therefore a hypothetical size-n set contains zero. Integralization preserves its cardinality and this zero. No feasibility assertion for an arbitrary subset is used.

Finite certificates. This part supplies the actual-set reductions, every added inequality family, the exhaustive case and split rules, and the exact finite lemma specifications. The accompanying certificate data and exact checkers are part of the proof. Their complete verification is recorded in Section 39; its mathematical role is Lemma 1.2.

Inputs already proved in the earlier chapters. We use the universal four-extreme covering and its 110 fixed labels with 26 disjoint nonpositive deletion groups; both peeling inequalities; the increasing-chain, threshold-14, small-increment and flexible-defect inverse lemmas; the parameterized effective-hole theorem with the corrected endpoint bridge; the eighteen-unit pairing and contributor grammar; the all-active mass certificate; and the size-96 core-lattice theorem. These are mathematical lemmas of the preceding $N \le 96$ proof, not a second assumed upper bound. They are stated and proved in Parts I and II. The sharp four-extreme theorem and all stronger ingredients first needed below are proved here.

18.1 Core notation and the exact budget

Let A be feasible, integral, and contain zero. Remove its four largest positive and four smallest negative elements, taking all elements of a sign if fewer than four exist; call this removed set B, and put $D = A \setminus B$, $d = |D|$. Peel the two largest positive and two smallest negative elements twice more, obtaining

$$D \supset D_1 \supset E = P \cup \{0\} \cup (-Q), \quad |D_1| = d - 4, \quad e = |E| = d - 8,$$

where P, Q are positive sets. The endpoint arguments below ensure both signs have enough elements for these choices. Put

$$a = |P|, \quad b = |Q|, \quad m = a + b = e - 1, \quad \Lambda = 1 + T_a + T_b, \quad s = |\operatorname{FS}(E)|.$$

For any positive r-element set X, define

$$\begin{aligned} \mu(X) &= |\operatorname{FS}(X)| - 1 - T_r, \\ \delta_0(X) &= |\{x \in X : 2x \notin R(X \cup \{0\})\}|, \\ \rho(X) &= \mu(X) + 2 - 2\delta_0(X). \end{aligned}$$

Write $\mu = \mu(P) + \mu(Q)$ and $\rho = \rho(P) + \rho(Q)$. The increasing-chain bound gives $\mu(X) \ge 0$. Theorems 7.2 and 7.3 give

$$\rho(X) \ge 0 \ (r \ge 14), \qquad \rho(X) \ge r - 14 \quad \text{if } r \ge 6 \text{ and } \operatorname{FS}(X) \text{ is not an AP.} \tag{73}$$

The stability theorem of Carpenter–Defant–Kravitz [1] additionally gives $\mu(X) \ge r - 3$ for a non-AP support in the range used here. Since the maximum of X is defective, $\rho(X) \le \mu(X)$.

The covering and peeling inequalities are

$$|\operatorname{FS}(D)| \le 2|2D| + 18d + 85, \tag{74}$$
$$|\operatorname{FS}(F)| \ge |\operatorname{FS}(F')| + |R(F)| - 1, \tag{75}$$
$$s \le 22d + 2\Delta(D) + 77 \le 24d + 77. \tag{76}$$

Here F' is a four-extreme peel of F. The proof of (74) includes a fixed, explicitly checked 110-slot covering and 26 disjoint deletion groups; it is not an asymptotic estimate. The last inequality uses $\Delta(D) \le d$.

If $a, b \ge 14$ and E has an opposite pair, define

$$\begin{aligned} \kappa &= s - \Lambda - \mu, & \vartheta &= \delta_0(P) + \delta_0(Q) + 8 - \Delta(D), \\ c &= 2|2D| + 18d + 85 - |\operatorname{FS}(D)|, \\ \ell_1 &= |\operatorname{FS}(D)| - |\operatorname{FS}(D_1)| - |R(D)| + 1, \\ \ell_2 &= |\operatorname{FS}(D_1)| - s - |R(D_1)| + 1, \\ g &= 4d - 10 - (|R(D)| - |R(D_1)|), & T &= \kappa + \ell_1 + \ell_2 + g. \end{aligned}$$

All these quantities are nonnegative. For κ, use the elementary sumset bound $|U + V| \ge |U| + |V| - 1$. For ϑ, zero is not a diagonal exception, the eight members of $D \setminus E$ account for at most eight exceptions, and any remaining exception is counted in $\delta_0(P)$ or $\delta_0(Q)$. For g, count the $4(d-4)+6 = 4d-10$ distinct-element pairs that meet the removed layer. The other signs follow from (74)–(75). Direct cancellation gives the exact identity

$$\boxed{\kappa + \rho + 2\vartheta + c + \ell_1 + \ell_2 + g = 22d + 97 - \Lambda =: \Omega.} \tag{77}$$

For clarity, $c + \ell_1 + \ell_2 + g = 22d + 77 + 2\Delta(D) - s$ and $\rho + 2\vartheta = \mu + 20 - 2\Delta(D)$, which proves the cancellation without an inequality or a duplicated expenditure of slack.

19 A sharp four-extreme defect inequality

Theorem 19.1. *Let $F = \{x_1 < \cdots < x_m\} \subset \mathbb{Z}$, $m \ge 7$, and let*

$$a_1 < a_2 < x_1 < \cdots < x_m < b_2 < b_1, \qquad D = F \cup \{a_1, a_2, b_2, b_1\}.$$

Let η_F, η_D count the holes in the endpoint hulls of $R(F), R(D)$, and put $g_H = 4m+6-(|R(D)|-|R(F)|)$. Then

$$\boxed{\eta_D + \eta_F + g_H \ge 2m - 2.} \tag{78}$$

The constant is sharp for every $m \ge 7$.

Proof. Write $L = x_1 + x_2$ and $U = x_{m-1} + x_m$. Let W_+ be $((b_1 + F) \cup (b_2 + F)) \cap (U, \infty)$, and let W_- be the corresponding union of the two lower translates below L. Put

$$I_+ = (U, b_1 + x_m] \cap \mathbb{Z}, \quad I_- = [a_1 + x_1, L) \cap \mathbb{Z}, \quad H_\pm = |I_\pm| - |W_\pm|.$$

Let q count the distinct mixed sums $a_i + b_j$ that lie outside $[L, U]$ and are not already in $W_- \cup W_+$. These sums lie in $I_- \cup I_+$, so $0 \le q \le 4$ and $q \le H_- + H_+$.

Translate x_1 to zero and write

$$x_m = M, \quad x_2 = s_0, \quad x_{m-1} = M - t_0,$$
$$a_2 = -w_0, \quad a_1 = -w_0 - v_-, \quad b_2 = M + u, \quad b_1 = M + u + v_+.$$

All six gap parameters are positive integers. At most $u+1$ members of b_2+F lie above U: at most u eligible points are at or below x_{m-1}, followed by x_m. Thus the integers

$$S_+ = u+1+m-|W_+|, \qquad S_- = w_0+1+m-|W_-|$$

are nonnegative. Set $H = H_+ + H_-$ and $S = S_+ + S_-$. We claim

$$H + S \geq 2q-2. \tag{79}$$

For $q \leq 2$ this follows from $H \geq q$. The same bound proves the case $q=3, S \geq 1$ and the case $q=4, S \geq 2$.

Suppose first that $S=0$. Since $|W_\pm| \leq 2m$, we have $u, w_0 \leq m-1$. The mixed value $B_0 = a_2+b_2 = M+u-w_0$ satisfies

$$M-m+2 \leq B_0 \leq M+m-2.$$

Now $s_0 \leq M-m+2$ and $x_{m-1} \geq m-2$, so $B_0 \in [s_0, 2M-t_0] = [L,U]$. In particular $q=4, S=0$ is impossible.

If $q=3, S=0$, the other mixed values

$$A_0 = B_0 - v_-, \qquad C_0 = B_0 + v_+ - v_-, \qquad D_0 = B_0 + v_+$$

are all outside $[L,U]$. Thus $A_0 < L$ and $D_0 > U$. Reflection allows us to assume $C_0 < L$. The definitions give

$$H_+ = v_+ + t_0 - m - 1, \qquad H_- = v_- + s_0 - m - 1.$$

Using integrality in $D_0 > U$ and $C_0 < L$ yields, respectively,

$$u - w_0 \geq M - H_+ - m, \qquad u - w_0 \leq t_0 - M + H_- - H_+ - 1.$$

Consequently

$$H_- \geq 2M - m - t_0 + 1 = M + x_{m-1} - m + 1 \geq M - 1 \geq m - 2 \geq 5.$$

This proves (79) in that case.

Finally, suppose $q=4, S=1$. Reflect if necessary so that $S_+ = 1, S_- = 0$. The cardinality bounds now give $u \leq m$ and $w_0 \leq m-1$, so $B_0 \geq M-m+2 \geq s_0$. Since B_0 lies outside the old hull, $B_0 > U$, whence

$$u - w_0 > x_{m-1} \geq m-2, \qquad u - w_0 \leq m-1.$$

Equality forces $u=m$, $w_0=1$, and $x_{m-1} = m-2$. The first $m-1$ members of F are therefore $0, 1, \ldots, m-2$, so $s_0 = 1$, and $B_0 = M+m-1 = U+1$. Because $v_- \geq 1$, the value $A_0 = B_0 - v_-$ is at most U; it must consequently be less than 1. Thus $v_- \geq M+m-1$ and

$$H_- = v_- + s_0 - m - 1 = v_- - m \geq M - 1 \geq m - 2 \geq 5.$$

Together with $S=1$, this proves (79) in the final case.

Put $h_- = L - (a_1+a_2)$, $h_+ = (b_1+b_2) - U$, and $h = h_- + h_+$. The definitions give the exact identity

$$2(|W_-| + |W_+|) = h + 2m + 2 - H - S.$$

Every new restricted sum outside $[L,U]$ lies in $W_- \cup W_+$, is one of the q new mixed values, or is one of a_1+a_2, b_1+b_2. The latter values are distinct and outside $I_- \cup I_+$. Hence the number N_{out} of these sums satisfies

$$2N_{\text{out}} = h + 2m + 6 - H - S + 2q \leq h + 2m + 8.$$

Write $N_0 = |R(D)| - |R(F)|$. At most η_F new sums are inside the old hull, so $N_0 \le N_{\text{out}} + \eta_F$. Also $\eta_D = h + \eta_F - N_0$. Therefore

$$\begin{aligned}\eta_D + \eta_F + g_H &= h + 2\eta_F + 4m + 6 - 2N_0 \\ &\ge h + 4m + 6 - 2N_{\text{out}} \ge 2m - 2.\end{aligned}$$

For sharpness, take $F = [0, m-1]_{\mathbb{Z}}$ and $\{a_1, a_2, b_2, b_1\} = \{-m-2, -1, m, 2m+1\}$. Here $R(F) = [1, 2m-3]_{\mathbb{Z}}$ and $R(D) = [-m-3, 3m+1]_{\mathbb{Z}}$. For the latter assertion the four translates of F fill the ranges $[-m-2, -3]$, $[-1, m-2]$, $[m, 2m-1]$, $[2m+1, 3m]$; $R(F)$ fills the central gap, and the sums of outer elements supply $-m-3, -2, 2m, 3m+1$. Thus both hole counts vanish and $g_H = 2m - 2$. □

20 Lattice reductions at sizes 95 and 94

This section extends the zero-containing common-core reduction from sizes 97 and 96 in Proposition 17.2. It does not exclude either new size.

20.1 Explicit effective hole inputs

For increasing prefixes X_i of a positive set, write $y_i = |\operatorname{FS}(X_i)| - |\operatorname{FS}(X_{i-1})| - i$; these increments are nonnegative and sum to $\mu(X)$. For reproducibility, the exact finite parameter test used below is as follows. For integers $C \ge 0$ and $\beta \ge 1$, define

$$K_0 = \max\{k \ge 3 : k(k-3)/2 \le C+1\}, \qquad D_0 = C + \max(3, K_0),$$
$$E_0 = D_0 + 1, \qquad M_0 = \max\left\{2\beta, 1 + \left\lfloor \frac{1+\sqrt{9+8(E_0+6\beta)}}{2} \right\rfloor\right\},$$
$$J = \max(K_0 + 1, 2\beta + 5, M_0).$$

The parameterized hole test, Lemma 10.2, states that if

$$(r - J + 1)\beta > D_0, \qquad T_{J-1} - 1 > D_0, \tag{80}$$

then a positive r-element set on a specified lattice with $2\beta < h < |\operatorname{FS}(X)| - 1$ satisfies $\mu(X) \ge r + C + 1$. No implicit "sufficiently large" threshold is used.

Here is the mechanism of that proved above test. If instead $\mu \le r + C$, the last AP prefix and the nonnegative support increments give a tail with $y_j \ge 1$ and $\sum(y_j - 1) \le D_0$, starting no later than $K_0 + 1$. If there is no AP prefix of size at least three, it starts at 4 with budget $C + 3$. In the other case a last AP prefix of size k followed by a non-AP support costs at least $r - 1 + k(k-3)/2$, so $k \le K_0$. The first condition in (80) forces an index $i \ge J$ with $1 \le y_i \le \beta$. The small-increment inverse and Lev's central-interval theorem [4] identify a deleted integer interval whose lower holes lie in $[1, 2\beta]$ and number at most β. The condition $h < |\operatorname{FS}| - 1$ makes the final integer set primitive. A first later element outside the prefix scale would make the two supports disjoint and cost at least $T_{J-1} - 1 > D_0$; hence the scale is one. Every later increment is at most E_0. The elementary endpoint bridge, with threshold M_0, preserves these holes to the final support, leaving at most 2β holes, a contradiction. The earlier chapters prove these inputs with the stated strict thresholds.

Only $\beta = 7$ and the following $C = C_7(r)$ rows are needed here:

r	31	32	33	34	35	36	37	38
$C_7(r)$	76	83	89	96	102	109	110	116
r	39	40	41	42	43	44	45	46
$C_7(r)$	122	128	129	135	142	147	149	155

Every row satisfies both strict inequalities in (80), using exact integer square roots. A row remains valid at every larger r with its same C, β, J. The finite arithmetic checker must check all sixteen rows, rather than using decimal square-root approximations.

20.2 The lattice reduction

We use the universal span estimate in Lemma 9.4, with $h(X) = \sum X + 1 - |\operatorname{FS}(X)|$. The notation $\rho(X)$ here denotes the same defect $\varrho(X)$ used in that lemma.

Theorem 20.1. *Let A be a feasible integer set containing zero with $n \in \{95, 94\}$. With the peeling notation of Section 18.1, $|B| = 8$ and there is a common normalization of the core lattice in which D is integral. Let h_P, h_Q, z be the hull-hole counts of $\operatorname{FS}(P), \operatorname{FS}(Q), \operatorname{FS}(E)$ on this lattice, and let w be the hull-hole count of $\operatorname{FS}(D)$. Then*

$$w \le 4z, \qquad T \ge \kappa + G - 3z \ge G - 2z, \qquad T \ge \lceil G/3 \rceil, \qquad T \ge z, \quad G = 2d - 10. \tag{81}$$

The exact balanced parameters are

n	d	e	m	(a_0, b_0)	Λ_0	Ω_0	I	G
95	*87*	*79*	*78*	$(39, 39)$	*1561*	*450*	k^2	*164*
94	*86*	*78*	*77*	$(38, 39)$	*1522*	*467*	$k(k+1)$	*162*

Here, after reflection, $(a, b) = (a_0 - k, b_0 + k)$, $\Lambda = \Lambda_0 + I$, and $\Omega = \Omega_0 - I$.

Proof. Obtaining the core lattice. If $|B| \le 7$, one sign is absent from D, and E has at least 78 nonzero elements. The one-sided chain bound gives $s \ge 1 + T_{78} = 3082$, contradicting $s \le 24d + 77 \le 24 \cdot 95 + 77 = 2357$. Thus $|B| = 8$.

If q counts the nonzero absolute values occurring only once in E, the unpaired-value inequality is $s \ge (e^2 + 3 + q^2)/4$; the sharper version has an additional nonnegative inversion count. Together with (76) and $d = e + 8$, this gives $q^2 \le 96e + 1073 - e^2$, namely 2416 at size 95 and 2477 at size 94. Parity gives $q \le 48$ and $q \le 49$, respectively. Thus E has at least 15 and 14 opposite pairs, respectively. In particular $a, b \ge 14$, zero is not a diagonal exception, and the nonnegative exact budget (77) applies.

Since $\Omega \ge 0$, $k \le 21$ in either case, improving the minimum side size to 18 and 17, respectively. For every such integer k, the Stanchescu sufficient margin $T_a - 2 - \Omega$ is

$$\frac{3k^2 - 79k + 656}{2} \quad (n = 95), \qquad \frac{3k^2 - 75k + 544}{2} \quad (n = 94).$$

Their minima on $0 \le k \le 21$ are 68 and 38. Therefore the simultaneous inverse theorem [3, Corollary 2] applies to $U = \operatorname{FS}(P)$ and $V = -\operatorname{FS}(Q)$: indeed $|U + V| = |U| + |V| - 1 + \kappa$ and $\kappa \le \Omega < T_a - 2 \le \min(|U|, |V|) - 3$. It places U, V in progressions on one common lattice, with hull cardinalities at most $|U| + \kappa, |V| + \kappa$. Since both contain zero, normalize the common lattice to $\mathbb{Z}$. Then $E \subset \mathbb{Z}$ and

$$h_P, h_Q \le \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \le z \le \min(h_P, h_Q) \le \kappa. \tag{82}$$

No lattice membership of D_1 or D has yet been assumed.

Forcing the second peeled layer onto this lattice. Put $R_p = \max P$, $R_q = \max Q$. Lemma 9.4 and (82) give

$$R_p + R_q \le m + \rho + \kappa + z + 4. \tag{83}$$

The integer set $R(E)$ has at most $2(R_p + R_q) + 1$ values; adding four new elements creates at most $4e + 6$ further restricted sums, regardless of their lattice membership. Thus, with $C = 2m + 4e + 14 - \Lambda$,

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\le C + 2\rho - \mu + \kappa + 2z + \ell_2 \\ &\le C + 3\Omega - \rho - \mu, \qquad (84) \\ |\operatorname{FS}(D_1)| - 2s &\le C + \Omega + \rho - \mu + 2z. \qquad (85) \end{aligned}$$

The last two lines follow from $\rho + \kappa + \ell_2 \le \Omega$ and $z \le \kappa$. In particular neither uses G.

At size 95, $C = -1075 - I$ and $C + 3\Omega = 275 - 4I$; at size 94 they are $-1042 - I$ and $359 - 4I$. If $z \le 14$, use $\rho \le \mu$ in (85): its upper bound is $-597 - 2I$ or $-547 - 2I$, respectively, and is strictly negative.

Suppose $z \ge 15$. Both side supports have positive holes, and $h \le \kappa < |\operatorname{FS}| - 1$. They cannot be progressions with step one; a progression with larger integer step would have at least $|\operatorname{FS}| - 1$ holes. The non-progression inputs therefore give

$$\rho \ge m - 28, \qquad \mu \ge m - 6, \qquad \rho + \mu \ge 2m - 34.$$

This already makes (84) strict except possibly $k = 0, \dots, 6$ at size 95 and $k = 0, \dots, 7$ at size 94. At size 95 these remaining side sizes are at least 33. The effective-hole row $(r, C, \beta) = (33, 89, 7)$ yields $\mu \ge 78 + 180 = 258$ and $\rho \ge 50$. Hence $\rho + \mu \ge 308 > 275 - 4I$. At size 94 the exact comparisons are

k	a	b	$C_7(a)$	$C_7(b)$	$\rho + \mu \ge$	$C + 3\Omega$	Margin
0	38	39	116	122	366	359	7
1	37	40	110	128	366	351	15
2	36	41	109	129	366	335	31
3	35	42	102	135	365	311	54
4	34	43	96	142	366	279	87
5	33	44	89	147	364	239	125
6	32	45	83	149	360	191	169
7	31	46	76	155	359	135	224

Here the lower bound is $128 + C_7(a) + C_7(b)$. Every margin is positive. We have proved $|\operatorname{FS}(D_1)| < 2s$ in all cases. An element $x \in D_1 \setminus E$ outside $\mathbb{Z}$ would supply the two disjoint s-point sets $\operatorname{FS}(E)$ and $x + \operatorname{FS}(E)$ inside $\operatorname{FS}(D_1)$, a contradiction. Thus $D_1 \subset \mathbb{Z}$.

Forcing the first peeled layer. Writing $F_1 = |\operatorname{FS}(D_1)| = s + |R(D_1)| - 1 + \ell_2$ gives

$$|\operatorname{FS}(D)| - F_1 \le |R(D_1)| + 4d - 11 + \Omega.$$

The right side is less than F_1 because

$$1561 > 4 \cdot 87 - 10 + 450 = 788, \qquad 1522 > 4 \cdot 86 - 10 + 467 = 801.$$

Thus $|\operatorname{FS}(D)| < 2|\operatorname{FS}(D_1)|$. The same disjoint-translate argument proves $D \subset \mathbb{Z}$ in this normalization.

Propagating holes, only after integrality. Let α_2, β_2 be the sums of the two negative and two positive members of $D_1 \setminus E$. The extreme translates $\alpha_2 + \operatorname{FS}(E)$ and $\beta_2 + \operatorname{FS}(E)$ must overlap, by the strict $2s$ bound. Their hulls cover the whole subset-sum hull of D_1, leaving at most $2z$ holes. If $u_2 = \beta_2 - \alpha_2$, their union has at least $s + u_2 - z$ points. Since $R(D_1)$ has endpoints α_2, β_2, its hole count η_2 satisfies

$$|\operatorname{FS}(D_1)| = s + u_2 - \eta_2 + \ell_2, \qquad \eta_2 \le z + \ell_2.$$

Repeat at the first peeled layer, using at most $2z$ holes in $\mathrm{FS}(D_1)$, to obtain $w \le 4z$ and $\eta_1 \le 2z + \ell_1$ for $R(D)$. Theorem 19.1, now legitimately applied to $F = D_1$, gives

$$\eta_1 + \eta_2 + g \ge 2(d-4) - 2 = 2d - 10 = G.$$

Thus $T \ge \kappa + G - 3z \ge G - 2z$. Also $T \ge \kappa \ge z$, so $3T \ge G$ and all of (81) follows. □

Corollary 20.2. *At size 96, the proved above common-lattice reduction similarly allows $G = 166$ in place of 164. At sizes 95 and 94 the new values are 164 and 162. The corresponding lower bounds $T \ge \lceil G/3 \rceil$ are 56, 55 and 54. More generally, wherever the existing reduction has established the same lattice and hole propagation, use $G = 2d - 10$.*

If j of the retained covering units are empty and $\delta = c - jd$ is the residual covering slack, combining (83), (77) and (81) yields

$$R_p + R_q + \delta \le m + 4 + \Omega - G - jd + 4z. \tag{86}$$

Indeed, $\rho + c \le \Omega - T$; substitute $T \ge \kappa + G - 3z$ into (83). Thus the new theorem lowers this upper bound by exactly two. The same variable z must be retained in all simultaneous constraints.

Remark 20.3. Theorem 20.1 puts D on the core lattice. It does not by itself place the eight original outer elements B on that lattice. For real feasible sets, integralization transfers the existence problem before the theorem is applied. To cover all real sets of an exact size by a zero-containing endpoint argument, the larger sizes must first be excluded. The cases 95 and 94 are structural reductions, not such exclusions.

21 Further necessary inequalities

Throughout this section D is already integral on the common lattice. Write

$$E = P \cup \{0\} \cup (-Q), \quad |E| = e, \quad |D| = d = e + 8, \quad K = |2D|,$$

$$x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p, \qquad y_0 = R_q < y_1 < y_2 < y_3 < y_4 = M_q,$$

where the eight non-core elements are $x_1, \dots, x_4$ and $-y_1, \dots, -y_4$. These statements allow holes in both core supports.

Proposition 21.1 (Independent same-sign cutoffs)**.** *For $0 \le i, h \le 4$,*

$$K \le M_p + M_q + x_i + y_h + 1 + \binom{5-i}{2} + \binom{5-h}{2}.$$

Proof. Every mixed-sign sum, and every same-sign sum with one summand at or below the corresponding exposed endpoint, lies in $[-M_q - y_h, M_p + x_i]_{\mathbb{Z}}$. Only pairs among the remaining positive outliers, or among the remaining negative outliers, can be outside. There are at most the displayed numbers of unordered pairs with repetition. Counting these pairs and the integer interval proves the claim, even if some pairs have equal sums. □

Proposition 21.2 (Cardinality-aware envelopes)**.** *For $0 \le j \le h \le 4$ and $0 \le k \le l \le 4$, put $t = 8 - j - k$, $r = d - t$, and $u = 8 - h - l$. Then*

$$K \le x_j + x_h + y_k + y_l + 1 + ur + \frac{t(t+1)}{2}. \tag{87}$$

These choices give 225 valid linear inequalities.

Proof. Let $C = E \cup \{x_1, \dots, x_j, -y_1, \dots, -y_k\}$. It has r members and lies in $[-y_k, x_j]$. Put $O = D \setminus C$, of size t. The sumset $2C$ and the translates $C + x_i$ for $j < i \le h$ and $C - y_i$ for $k < i \le l$ all lie in $[-y_k - y_l, x_j + x_h]_{\mathbb{Z}}$. Each of the u other translates of C has exactly r points. Finally $2O$ has at most $t(t+1)/2$ points. These sets cover $2D$, proving (87). There are 15 choices of (j, h) and 15 of (k, l). □

21.1 An elementary odd-pivot defect bound

Lemma 21.3. *Let X be a set of r positive integers, let $R = \max X$, and assume that X contains an odd element. Then*

$$\delta_0(X) \le R - r + 2.$$

If R is odd, the stronger bound $\delta_0(X) \le R - r + 1$ holds.

Proof. Let t be the largest odd element, put $s = |X \cap (t, R]|$, and $r_t = r - s$. The s elements above t are all even, so $2s \le R - t + 1$. For each defective $x < t$, define

$$f(x) = \begin{cases} 2x, & x < t/2, \\ 2x - t, & x > t/2. \end{cases}$$

Equality $x = t/2$ is impossible. Each image lies in $[1, t-1]$ and is absent from X: otherwise $2x = 0 + f(x)$ or $2x = t + f(x)$ would be a representation by two distinct members of $X \cup \{0\}$. Both branches are injective; their images have opposite parity and are therefore disjoint. There are only $t - r_t$ missing integers in $[1, t]$, so at most that many defective elements lie below t. There is at most one defect at t and at most s above it. Therefore

$$\delta_0(X) \le t - r_t + 1 + s = t - r + 2s + 1 \le R - r + 2.$$

If R is odd, $t = R$ and $s = 0$, giving the stronger conclusion. □

In the strict common-core reductions, $h(X) < |\operatorname{FS}(X)| - 1$ forces each core to be primitive: a gcd of at least two would leave at least one hole between every two successive support values. Thus each core has an odd element, and the lemma gives the necessary linear bound

$$\rho(X) \ge \mu(X) + 2r - 2R - 2. \tag{88}$$

With a certified constant $\mu(X) \ge u$, replace $\mu(X)$ by u. On each side independently, choose this bound or any bound proved above constant lower bound for $\rho(X)$. The four choices in $\rho(P) + \rho(Q) + c + T \le \Omega$ give four valid linear budget rows. This lemma is proved without finite sampling or a solver. Its validity by itself does not exclude the next cardinality.

21.2 Leading twofold translate losses

Let $S_p = \sum(D \cap \mathbb{R}_{>0})$, $S_q = -\sum(D \cap \mathbb{R}_{<0})$. The leading positive covering slot is $c_p + 2D$, where $c_p = p_1 - p_2 - p_3 - p_4$ for the four original positive outer elements $p_1 > p_2 > p_3 > p_4$. The reflected negative slot uses $c_q = q_1 - q_2 - q_3 - q_4$. Define the actual losses

$$L_p = K - |(c_p + 2D) \cap \operatorname{FS}(D) \cap (0, \infty)|,$$

and define L_q by reflection. For a retained individual translate, let λ_v be the number of its points outside the appropriate open half of the subset-sum hull. The covering keeps these slots separate from the paired units. Thus, after j empty units,

$$L_p + L_q + \sum_v \lambda_v \le \delta = c - jd. \tag{89}$$

This follows by subtracting the actual slot contributions from their nominal capacities in the same union bound that proves (74); discarded paired-unit deficits are nonnegative.

Proposition 21.4. *If $\delta < K$, then c_p, c_q are integers and*

$$c_p + 2M_p + L_p \geq K, \qquad S_p + 2M_q - c_p + L_p \geq K - 1, \qquad S_p + L_p \geq K,$$

with the reflected inequalities on the negative side. If also $\delta < 2e + 7$, then for $i = 0, \ldots, 4$,

$$c_p + 2x_i + L_p \geq 2e + 2i + 7,$$
$$S_p + 2y_i - c_p + L_p \geq 2e + 2i + 6,$$

and likewise by reflection.

Proof. Since $L_p, L_q \leq \delta < K$, each leading translate contributes at least one point. All values in $2D$ and $\mathrm{FS}(D)$ are integers, so each of c_p, c_q is integral. The $K - L_p$ contributing positive values lie in $[1, c_p + 2M_p]$, in $[c_p - 2M_q, S_p]$, and in $[1, S_p]$. Counting integer points in these three intervals gives the first three inequalities.

For the first nested inequality use $F = E \cup \{-y_1, \ldots, -y_4, x_1, \ldots, x_i\}$. Then $|F| = e + 4 + i$ and $|2F| \geq 2|F| - 1 = 2e + 2i + 7$ by the elementary ordered-chain sumset bound. At most L_p of these values fail to contribute through the leading slot; all contributors lie in $[1, c_p + 2x_i]$. The hypothesis $L_p \leq \delta < 2e + 7$ ensures that this interval has a contributor, so its integer cardinality gives the assertion. For the second use all four positive outliers and only the first i negative outliers. The contributing interval is $[c_p - 2y_i, S_p]$, which has at most $S_p + 2y_i - c_p + 1$ integer points. Reflection gives the remaining claims. □

At size 96, $K \geq 2d - 1 = 175$ and $2e + 7 = 167$. Thus bounds $\delta < 175$ and $\delta < 167$ suffice for the respective hypotheses. They must be established in each modeled regime; they are not automatic from the cardinality alone.

21.3 Overlap of retained translates

The following generic form makes the overlap argument independent of particular center labels or a numerical solver.

Proposition 21.5. *Let $v_1, \ldots, v_k$ be the integral centers of k distinct retained same-sign translate slots $v_i + D$, with minimum $v_{\min}$ and maximum $v_{\max}$. Let H be the target open half of the integer subset-sum hull, and put $\lambda_i = |(v_i + D) \setminus H|$. Define*

$$H_G = ke - (R_p + R_q + v_{\max} - v_{\min} + 1).$$

Their joint capacity deficit satisfies

$$kd - \left| H \cap \bigcup_{i=1}^{k} (v_i + D) \right| \geq H_G + \max_i \lambda_i.$$

Disjoint groups of counted slots can be added in the total covering budget. Their centers need not have distinct numerical values.

Proof. The union $U = \bigcup_i (v_i + D)$ is contained in the union of the core hull $[v_{\min} - R_q, v_{\max} + R_p]_{\mathbb{Z}}$ and at most $k(d-e)$ non-core points. Hence $kd - |U| \geq H_G$. Also $U \setminus H$ contains each $(v_i + D) \setminus H$, so $|U \setminus H| \geq \max_i \lambda_i$. Adding the two inequalities proves the claim. For disjoint groups of slot labels, sum the corresponding union bounds along with all ungrouped capacities and losses. No assumption that the core fills its hull is made. □

Every proposition in this section is a necessary condition on an actual configuration. The next section combines them with the original rows and complete exact certificates to obtain an endpoint exclusion.

22 An exact exclusion of size 96

Theorem 22.1. *No feasible real set has 96 elements. Consequently*

$$\boxed{N \le 95.}$$

Proof. The preceding global bound $N \le 96$ implies that a hypothetical size-96 example already contains zero: otherwise adjoining zero produces a forbidden size-97 example. Integralize it, then apply Proposition 17.2 and Corollary 20.2. Thus

$$d = 88, \quad e = 80, \quad m = 79, \quad a + b = 79, \quad I = T_a + T_b - T_{39} - T_{40}, \quad \Omega = 432 - I, \quad G = 166.$$

Every actual configuration falls in the exhaustive root families below. Every root is contradicted by an exact positive-integer certificate, either directly or at every leaf of a proved exhaustive binary tree. The remaining subsections give the full finite reduction and verification rule; the companion `exact95` directory supplies the complete integer rows, root assignments, multipliers and trees. These finite data are part of the proof. □

22.1 The conservative low-hole root list

Retain $19 \le a \le 60$, the side-size range given by the size-96 structural reduction. For $z = 0$, even the older value $G = 164$ gives $c \le 268 - I$. For $1 \le z \le 14$, both supports are non-progressions and $\rho \ge 51$, so $c \le 217 - I + 2z \le 245 - I$. If j units are empty, put $B_0 = 268 - I - 88j$. We enumerate

$$\begin{aligned} Z_j &: \quad z = 0, && 0 \le j \le 3, \quad B_0 \ge 0, \\ P_j &: \quad 1 \le z \le 14, && 0 \le j \le 2, \quad B_0 - 23 \ge 0. \end{aligned}$$

For every such (a, j) use every pairing mode and every necessary contributor choice from the eighteen-unit construction of Appendix F, with the center and pairing definitions in Appendix D. The mode counts for $j = 0, 1, 2, 3$ are exactly $4, 70, 578, 2992$. Both pairing branches and all choices of empty units are retained; split-pair contributor choices are selected from actual contributors, as in that construction. The complete counts are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2
Roots	128	1820	11560	11968	128	1820	9248

Their sum is 36672. This is a conservative root list: using the stronger G later in a root's inequalities does not remove cases from the list.

The basic and extra model stages retain the valid older bounds

$$\delta \le B_0 \ (Z), \qquad \delta \le B_0 - 51 + 2z \ (P), \qquad R_p + R_q + \delta \le 83 + B_0 + 4z.$$

The overlap, group and leading stages reduce both relevant right sides by two, using the proved $G = 166$. The common cardinality row is

$$S_p + S_q + \delta - 2K \le 1668 - 88j + 4z.$$

Mass, inner-gap and quantile rows retain the same single z coordinate. For a retained positive center v with lower and upper hull losses l_v, u_v, the available quantile rows are

Type	Lower row	Upper row
Ordinary	$v + l_v \ge b + 5$	$v - u_v \le S_p - a - 4$
Global	$v + l_v + M_p \ge 88$	$v - u_v - M_q \le S_p - 87$
Core	$v + l_v + R_p \ge 84$	$v - u_v - R_q \le S_p - 83$

Reflection supplies the negative rows. The global rows need only a contributing center. Core rows require more than four points in the hull contribution, ensured by a total loss cap below 84. Ordinary rows hold at in-hull endpoints, and a cap below $\min(a,b)+5$ excludes every out-of-hull exception. The exact generator selects only rows justified by these bounds. With three empty units it keeps weak outer order; with at most two it uses strict order only when the integrality argument below applies.

22.2 The high-hole root list

For $z \ge 15$, write $(a,b) = (39-k, 40+k)$ up to reflection. For each side size r and integer β, choose a nonnegative $C_\beta(r)$ satisfying (80), if one is available, and set

$$u(r,z) = \max\left(\{r-3\} \cup \{r + C_\beta(r) + 1 : 1 \le \beta \le \lfloor (r-5)/2 \rfloor,\ 2\beta < z\}\right).$$

Only chosen constants satisfying the exact parameterized theorem are used. The checker must verify all 646 nonnegative parameter rows that may occur; it does not assume that a numerical optimization found the best constant, or restrict to the 103-row table in Appendix F. For $2 \le \beta \le 10$, write

$$v_\beta = \max(0, u - 4\beta + 2) = (\beta+1)q_\beta + s_\beta, \qquad 0 \le s_\beta \le \beta,$$

and define

$$L_\beta(r,u) = \min\big((\beta-1)(r-14), (\beta-1)q_\beta + \min(s_\beta, \beta-1)\big).$$

Lemma 17.1 gives

$$\rho_*(r,u) = \max\{r-14, u+2-2r, L_2(r,u), \dots, L_{10}(r,u)\} \le \rho(X).$$

Consequently, with $u_a = u(a,z)$ and $u_b = u(b,z)$,

$$c \le C(a,z) := 432 - I - \rho_*(a,u_a) - \rho_*(b,u_b) - \max(56, z, 166-2z). \tag{90}$$

The already known bound $\rho \ge 51$ and $T \ge 56$ excludes $k \ge 19$. The exact checker conservatively enumerates all 5662 pairs

$$0 \le k \le 18, \qquad 15 \le z \le 432 - k(k+1).$$

After discarding $C(a,z) < 0$, its largest $C(a,z)$ is 144. In particular $j \le 1$, and both leading-translate thresholds 175 and 167 are satisfied. When $j = 1$, $\delta \le 144 - 88 = 56 < 84$, so the core quantiles are valid.

For $j = 0$, Lemmas 13.1 and 14.2 give a further explicit elimination. Put

$$F_A(a,b) = \tfrac{1}{2}T_a + T_b + 6a + 8b + 85, \qquad F_B(a,b) = \tfrac{2}{3}T_a + T_b + 6a + \tfrac{22b+236}{3}.$$

The quantity $F = S_p + S_q + c - 2(M_p + M_q + R_p + R_q)$ is at least one of

$$F_A(a,b) + \tfrac{1}{2}(u_a + z) + u_b + z, \qquad F_B(a,b) + \tfrac{2}{3}(u_a + z) + u_b + z,$$

or one of their reflected versions. The basic envelope $K \le M_p + M_q + R_p + R_q + 21$ and the cardinality row give $F \le 1710 + 4z$. Thus a positive value of the minimum of those four lower expressions minus $1710 + 4z$ excludes the state.

There remain exactly 795 unoriented (a, j, z) states: 413 with $j = 0$ and 382 with $j = 1$. Group consecutive values of z only when (a, j, u_a, u_b) is constant and the same affine part realizes $\max(56, z, 166 - 2z)$. The resulting 30 intervals, nine for $j = 0$ and 21 for $j = 1$, are listed in `high_intervals.json`. The verifier must expand every interval and checks equality with the complete set of 795 states. After both orientations and every pairing mode, they give 72 H_0 roots and 2940 H_1 roots, totaling 3012. Replacing integer z by a real variable in each interval only enlarges its model, so an exact contradiction for that model excludes all covered integer states.

22.3 Additional row validity and exhaustive splits

The model stages use the inequalities of Section 21 and the following elementary refinements, recorded to specify the complete new mathematical content of the generators.

Tail sum cardinalities. Beyond cutoff x_j, the residual positive sums lie in $[2x_{j+1}, 2M_p]$, so their number is at most $2(M_p - x_{j+1}) + 1$, as well as the pair count in Proposition 21.1. For four increasing outliers $t_1 < t_2 < t_3 < t_4$, their twofold sumset has cardinality at most each of

$$2(t_3 - t_1) + 5, \quad 2(t_4 - t_2) + 5, \quad (t_2 - t_1) + (t_4 - t_3) + 7.$$

The first two bounds put the three-element twofold sumset in its integer hull and count four pairs involving the remaining singleton. The third counts at most three within-pair sums for each pair and bounds the cross-pair sums by their hull, of length $(t_2 - t_1) + (t_4 - t_3) + 1$. Combine any positive and negative such bounds with the central interval of Proposition 21.1.

Outer integrality with at most two empty units. Assume $c_p, c_q \in \mathbb{Z}$ and at most two units are empty. Every own center $c_p + p_i$ and $c_q + q_i$ is an unpaired slot, so a nonintegral outer element makes its own slot empty. There cannot be exactly one nonintegral element on a side, because the integral identity $c_p = p_1 - p_2 - p_3 - p_4$ (or its reflection) would force it to be integral. Thus any failure consists of exactly two elements on one side, and exhausts all empty units. On that side, say positive, the always-unpaired cross centers $c_q - p_3, c_q - p_4$ contribute, forcing $p_3, p_4 \in \mathbb{Z}$. In pairing branch A, $c_q - p_2$ is unpaired and contributes. In branch B, $\alpha_p = p_2 - p_3 - p_4$ is unpaired and contributes. Either forces $p_2 \in \mathbb{Z}$, then c_p forces $p_1 \in \mathbb{Z}$, a contradiction. The case of negative failures is its reflection. Here branch A means $\beta_p + \beta_q \le 0$, branch B means > 0, where $\beta_p = 2p_2 - p_1 - p_3 - p_4$ and β_q is its reflection. Leading-center integrality follows from the old contributing-center argument with at most one empty unit; in the other applicable models it follows from $\delta < K$ and Proposition 21.4. Therefore strict outer integer gaps and the branch-B bound $\beta_p + \beta_q \ge 1$ are imposed only in justified regimes.

Groups containing a leading slot. Group the leading slot $c_p + 2D$ with k retained own-center slots $c_p + p_i + D$, and let $p_{\max}$ be their largest p_i. The core pieces $c_p + 2E$ and $c_p + p_i + E$ lie in the common integer hull

$$[c_p - 2R_q, c_p + p_{\max} + R_p]_{\mathbb{Z}}.$$

Since $|2E| \ge 2e-1$, the remaining group points number at most $(K - |2E|) + k(d-e)$. Subtracting this upper bound for the raw union from the nominal capacity $K + kd$ proves a global deficit of at least

$$H_* = (k+2)e - 2 - p_{\max} - R_p - 2R_q.$$

The deficit in the actual positive subset-sum target is at least $H_* + \max(L_p, \lambda_1, \dots, \lambda_k)$: the part of the raw union outside that target contains the loss set of every member, and hull losses are a subset of actual-target losses. Disjoint groups and ungrouped losses may be added in (89). This proves the leading-group rows, including their constant $e(k+2) - 2$. For the grouped-translate recurrence, choose each auxiliary prefix variable to be the actual deficit of that prefix union. The deficit of a union of two disjoint groups of slot labels is at least the sum of their separate deficits. Proposition 21.5 therefore proves every recurrence row, and the final prefix deficits fit in the total loss budget. The use of auxiliary variables is thus existentially valid for every actual configuration.

Normal or outside endpoint. Only the following proved above disjunctions are used in the trees. For a retained integral positive center v, either

$$v + l_v \ge b + 5, \quad \text{or} \quad (l_v \ge b + 5 \text{ and } v \le -1);$$

and either

$$v - u_v \le S_p - a - 4, \quad \text{or} \quad (u_v \ge a + 5 \text{ and } v \ge S_p + 1).$$

For the first, if $v \ge 0$, at most v of the $b+5$ nonpositive members of D can translate to a positive value, proving the ordinary row; if $v < 0$, all $b + 5$ are lost. For the second, if $v \le S_p$, at most $S_p - v$ of the $a + 4$ positive members fit below the upper endpoint; if $v > S_p$, all these members and zero are lost. Integer centers give the displayed strict outside bounds. Reflect for negative centers. Thus each binary split covers every actual set, without requiring LP loss variables themselves to be integral.

22.4 Exact finite lemma for size 96

The required finite assertion has 39684 canonical roots and 39809 leaf contradictions. The data are in `proofs/exact95`. There are 36672 low and 3012 high roots. Of these, 39624 have direct contradictions. The remaining 60 have complete binary trees with 125 internal splits and 185 leaves. The ordinary pool has 863 vectors and the branch pool 81. Every root, current matrix, normal/outside split, both children, and effective-hole witness is reconstructed.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact95.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

23 A further exact exclusion of size 95

Theorem 23.1. *No feasible real set has 95 elements. Consequently*

$$\boxed{N \le 94.}$$

Proof. Apply Theorem 22.1 first. A hypothetical size-95 example contains zero by the adjoining-zero argument; integralization and Theorem 20.1 give the common lattice. Its parameters are

$$d = 87, \quad e = 79, \quad m = 78, \quad a + b = 78, \quad I = (a - 39)^2, \quad \Omega = 450 - I, \quad G = 164.$$

Use the same actual-set rows and finite verification rule as in Section 22, with the exact parameter changes detailed below. The complete root list has 74106 members. Of these, 73951 have direct exact integer contradictions and 155 have complete binary trees with 488 internal splits and 643 contradictory leaves. Thus all 74594 leaf systems are exactly infeasible, excluding size 95. The general upper bound follows. □

23.1 All roots and their parameter changes

For low holes, retain every $18 \le a \le 60$, all $j = 0, 1, 2, 3$, and every pairing mode and contributor choice, subject to

$$B_0 = 286 - I - 87j \ge 0 \quad (z = 0), \qquad B_0 - 22 \ge 0 \quad (1 \le z \le 14).$$

These follow from $c \le 286 - I$ for zero holes and $c \le 236 - I + 2z \le 264 - I$ for positive low holes, since $\rho \ge m - 28 = 50$. The exact root counts are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2	P_3
Roots	132	2030	12138	32912	132	1890	10982	8976

They sum to 69192. The loss and span rows are

$$\delta \le B_0 \ (Z), \qquad \delta \le B_0 - 50 + 2z \ (P), \qquad R_p + R_q + \delta \le 82 + B_0 + 4z.$$

The cardinality right side becomes $1650 - 87j + 4z$. For positive three-empty modes, the model explicitly retains the same $1 \le z \le 14$ variable in its loss, span, mass and cardinality rows; it does not import a zero-hole equality from the preceding template.

For high holes use the same $u(r, z)$ and $\rho_*(r, u)$ defined above, now with the exact cap

$$C_{95}(a, z) = 450 - I - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(55, z, 164 - 2z).$$

The non-progression and T lower bounds exclude $k \ge 19$ for $(a, b) = (39 - k, 39 + k)$. The complete conservative enumeration

$$0 \le k \le 18, \qquad 15 \le z \le 450 - k^2$$

contains 6175 states. The 620 effective-hole rows used in it are rechecked against the two strict integer tests. Its maximum nonnegative covering cap is 171, so at most one unit is empty. For $j = 0$, apply the same four all-active mass lower bounds, now against $1692 + 4z$. Exactly 1552 unoriented (a, j, z) states survive: 804 with $j = 0$ and 748 with $j = 1$. Consecutive states with the same side excess bounds and affine T piece form 48 intervals. They expand exactly to those 1552 states. Both orientations are kept when $a \ne b$; the balanced case is kept once. Every pairing mode is included, giving 84 H_0 roots and 4830 H_1 roots, totaling 4914.

The common model construction in this endpoint uses the following parameter substitutions in the already proved row families:

Row or hypothesis	Size-95 value
Basic twofold lower bound	$K \ge 2d - 1 = 173$
Leading-translate threshold	$\delta < 173$
Nested leading-translate threshold	$\delta < 2e + 7 = 165$
Core quantile threshold	Hull loss $< d - 4 = 83$
Global quantile right sides	87 and 86
Core quantile right sides	83 and 82
Two-slot core overlap constant	$2e - 1 = 157$
k-slot core overlap constant	$ke - 1 = 79k - 1$
Leading-core group constant	$(k + 2)e - 2 = 79(k + 2) - 2$

The 225 cardinality-aware envelopes use their actual $d = 87$ value. Each high interval has constant u_a, u_b and an affine T piece, so its maximum residual cap is at an endpoint; the checker also tests it at every integer z. Since 171 is below 173, all high leading slots contribute. The stronger nested rows are included only for intervals whose residual cap is below 165. At residual cap at least 83 use global quantiles; below 83 use core quantiles; ordinary rows are used only when their stronger loss-cap test applies. These separate threshold checks prevent carrying an invalid size-96 row into size 95.

23.2 One additional exhaustive disjunction

In addition to the normal or outside splits already proved, the size-95 trees permit this split for a retained center:

$$\text{both core quantile rows hold} \qquad \text{or} \qquad l_v + u_v \ge 83.$$

Indeed, if $l_v + u_v < 83$, integrality of the actual loss count and $d = 87$ give at least five points of $v + D$ in the target hull. Since only four members on either side lie outside the core, the core-quantile argument gives both rows. If the strict inequality fails, the second child holds. The two children need not be disjoint; they cover every actual configuration. This is a proved disjunction, not an inference from an LP optimum.

23.3 Exact finite lemma for size 95

The required finite assertion has 74106 canonical roots and 74594 leaf contradictions. The data are in `proofs/exact94`. There are 69192 low and 4914 high roots. The 155 tree roots consist of 147 low and eight high roots; the trees have 488 internal splits and 643 leaves. All other roots are direct contradictions. All 48 high-hole intervals are expanded. The ordinary and branch pools have 1008 and 257 vectors, respectively.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact94.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

24 The odd-pivot bound and an exact exclusion of size 94

Theorem 24.1. *No feasible real set has 94 elements. Consequently*

$$\boxed{N \le 93.}$$

Proof. The preceding exclusion of 95 justifies zero at the new endpoint. Integralization and Theorem 20.1 give

$$d = 86, \quad e = 78, \quad m = 77, \quad a + b = 77, \quad I = T_a + T_b - T_{38} - T_{39}, \quad \Omega = 467 - I, \quad G = 162.$$

The same root-model construction is valid with these parameters, retaining each threshold at its new value. Add the elementary odd-pivot bound of Lemma 21.3 in the one high-hole tree where it is needed, and use the additional inner-extreme disjunction proved below. The exhaustive exact verification supplies 118081 leaf contradictions covering all 116832 root systems. Thus size 94 is impossible; the general upper bound follows. □

24.1 Complete low and high classifications

The low-hole list contains every $17 \le a \le 60$, every $j = 0, 1, 2, 3$, and every pairing mode and contributor choice, subject to

$$B_0 = 305 - I - 86j \ge 0 \quad (z = 0), \qquad B_0 - 21 \ge 0 \quad (1 \le z \le 14).$$

Here $\rho \ge 49$ when $z > 0$, and the loss, span and cardinality rows are

$$\delta \le B_0 \ (Z), \qquad \delta \le B_0 - 49 + 2z \ (P),$$
$$R_p + R_q + \delta \le 81 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1632 - 86j + 4z.$$

The full low-hole counts are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2	P_3
Roots	136	2100	13872	41888	136	1960	12716	29920

Their total is 102728.

For high holes write $(a, b) = (38 - k, 39 + k)$ up to reflection. Use the same certified side bounds u, ρ_* with

$$C_{94}(a, z) = 467 - I - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(54, z, 162 - 2z).$$

The inequalities $\rho \ge 49$ and $T \ge 54$ exclude $k \ge 19$. Enumerate all 6327 pairs $0 \le k \le 18$, $15 \le z \le 467 - k(k + 1)$. The checker revalidates all 627 nonnegative effective-hole constants

used. The maximum nonnegative cap is 196, so $j \le 2$. The all-active mass comparison for $j = 0$ now uses the upper bound $1674 + 4z$. There remain 2242 unoriented states: 1157 with $j = 0$, 980 with $j = 1$, and 105 with $j = 2$. Their canonical grouping gives 54 intervals, respectively 12, 34 and eight of these three types. Both orientations and every mode give 96 H_0 roots, 4760 H_1 roots and 9248 H_2 roots, totaling 14104.

The thresholds are now $K \ge 171$, nested threshold $2e + 7 = 163$, and core-quantile loss threshold $d - 4 = 82$. The high cap 196 is *not* below the leading threshold. Therefore leading and nested capacity rows are inserted only when the actual interval's residual cap is below 171 or 163, respectively. High intervals use global quantiles at cap at least 82, core quantiles below 82, and ordinary quantiles only when their stronger test applies. The overlap constants become $2e - 1 = 155$, $ke - 1 = 78k - 1$, and $(k + 2)e - 2 = 78(k + 2) - 2$. All 225 envelope rows use $d = 86$.

24.2 The parity row and inner-extreme splits

The primitive-core hypothesis of Lemma 21.3 holds here: Stanchescu gave $h_P < |\mathrm{FS}(P)| - 1$ and $h_Q < |\mathrm{FS}(Q)| - 1$. Thus the high-hole side bounds can also use

$$\rho(P) \ge u_a + 2a - 2R_p - 2, \qquad \rho(Q) \ge u_b + 2b - 2R_q - 2.$$

Take either this expression or ρ_* independently on the two sides and substitute in $\rho + c + T \le \Omega$. Besides the original constant row this supplies three additional valid affine rows. The finite proof uses these rows in its parity-enhanced high-hole tree. Their proof is the parity injection, not a numerical classification of dense cores.

The trees may also use, for $i = 1, 2, 3, 4$ and a retained positive center v, the disjunctions

$$l_v \ge i \quad \text{or} \quad v \ge y_{5-i} + 1, \qquad u_v \ge i \quad \text{or} \quad v \le S_p - x_{5-i}.$$

If $l_v < i$, at least one of the i largest negative magnitudes gives a translated value above zero, forcing $v > y_{5-i}$, hence the first integer inequality. If $u_v < i$, at least one of the i largest positive elements fits at or below S_p, giving the second. Reflection supplies the negative versions. These cover actual integer losses; they do not assert integrality of a solution to the relaxed LP. The earlier normal or outside splits remain valid, as does the core or large-loss split with 82 replacing 83.

24.3 Exact finite lemma for size 94

The required finite assertion has 116832 canonical roots and 118081 leaf contradictions. The data are in `proofs/exact93`. There are 102728 low and 14104 high roots. Of these, 116566 are direct contradictions, while 228 low and 38 high roots have trees with 1249 internal splits and 1515 leaves. The ordinary and branch pools contain 1363 and 671 vectors. Every selected parity and inner-extreme row must be reconstructed on its stated actual-set domain.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact93.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

25 Further common-core reductions at sizes 93 and 92

Theorem 25.1. *The zero-containing common-core conclusion of Theorem 20.1 also holds at sizes 93 and 92, with the following parameters:*

n	d	e	m	(a_0, b_0)	Λ_0	Ω_0	I	G
93	*85*	*77*	*76*	$(38, 38)$	*1483*	*484*	k^2	*160*
92	*84*	*76*	*75*	$(37, 38)$	*1445*	*500*	$k(k+1)$	*158*

Again $(a, b) = (a_0 - k, b_0 + k)$ *up to reflection,* $\Lambda = \Lambda_0 + I$*, and* $\Omega = \Omega_0 - I$*. All the inequalities in* (81) *hold. This theorem alone excludes neither cardinality and does not put the original outer set* B *on the core lattice.*

Proof. Bootstrap. If $|B| < 8$, the one-sided core has at least 76 nonzero members, giving $s \geq 1 + T_{76} = 2927 > 24 \cdot 93 + 77 = 2309 \geq 24d + 77$, a contradiction. Hence $|B| = 8$. Before assuming any minimum side size, use the original uniform defect bound, Lemma 5.3, which gives $\Lambda \leq 22d + 115$, or $I \leq \Omega_0 + 18$. At size 93 this implies $k \leq 22$ and $a \geq 16$; at size 92 it gives $k \leq 22$ and $a \geq 15$. Thus threshold 14 is available on both sides. The unpaired bound gives $q^2 \leq 2536, 2593$, respectively. Parity gives $q \leq 50, 49$ and at least 13 opposite pairs, so zero is not a diagonal exception. The nonnegative exact budget (77) now applies. It refines the size-92 range to $k \leq 21$, $a \geq 16$. The bootstrap is necessary: 13 opposite pairs alone would not justify threshold 14.

The size-93 core lattice. The sufficient Stanchescu margin is

$$T_a - 2 - \Omega = \frac{3k^2 - 77k + 510}{2} \geq 8 \qquad (0 \leq k \leq 22).$$

Thus the same inverse theorem gives (82) and the strict bounds $h_P < |\operatorname{FS}(P)| - 1$, $h_Q < |\operatorname{FS}(Q)| - 1$.

The size-92 core lattice when one support is not an AP. Here the crude margin $f(k) = (3k^2 - 73k + 402)/2$ can be -21. Retain the actual excess and defect terms instead of incorrectly using that margin alone. If $U = \operatorname{FS}(P)$ is not an AP, then $\mu(P) \geq a - 3$ and $\rho(P) \geq a - 14$; using $\kappa \leq \Omega - \rho$ gives

$$|U| - 3 - \kappa \geq f(k) + 2a - 17 = \frac{3k^2 - 77k + 516}{2} \geq 11.$$

If U is an AP but $V = -\operatorname{FS}(Q)$ is not, use $\rho(Q) \geq b - 14$ to get

$$|U| - 3 - \kappa \geq f(k) + b - 14 = \frac{3k^2 - 71k + 450}{2} \geq 15.$$

The larger-side margin is positive since $T_b - 2 \geq T_{38} - 2 = 739 > 500 \geq \Omega$. Thus Stanchescu gives (82) and both strict support-cardinality hole bounds.

The size-92 core lattice when both supports are APs. We use a short elementary fact. If two integer APs U, V, containing zero, have sizes $u, v \geq 2$ and $|U + V| = u + v - 1 + \kappa < uv$, their hull-hole counts on their common gcd lattice are at most κ. To prove this, reflect or translate to write $U = p[0, u-1]_{\mathbb{Z}}$ and $V = q[0, v-1]_{\mathbb{Z}}$ with $\gcd(p, q) = 1$. A collision of two distinct pairs exists, so coprimality forces $u - 1 \geq q$ and $v - 1 \geq p$. Counting the V index modulo p, the intervals in each residue class overlap or touch and give

$$|U + V| = pu + qv - pq.$$

In detail, residue $r \in [0, p-1]$ contributes $u + q\lfloor (v - 1 - r)/p \rfloor$ points; their sum is the displayed value. Thus

$$\begin{aligned} h_U &= (p-1)(u-1), & h_V &= (q-1)(v-1), \\ \kappa - h_U &= (q-1)(v-p) \geq 0, & \kappa - h_V &= (p-1)(u-q) \geq 0, \\ z &= h_U + h_V - \kappa = (p-1)(q-1). \end{aligned}$$

In our application, $\kappa \le 500 < T_{16}T_{38} \le (u-1)(v-1)$, so the fact applies. Also $(q-1)(v-1) = h_V \le 500$ while $v - 1 \ge T_{38} = 741$, forcing $q = 1$ and $z = 0$. All lattice and hole identities hold, although this special case need not give $h_P < |U| - 1$. It never enters a positive-hole case. At this point only E, not yet D_1 or D, is asserted to be integral on the normalized lattice.

Forcing D_1 onto the core lattice. Lemma 9.4 applies because $a, b \ge 16$. The already derived comparisons (84)–(85) remain valid, with

n	C_0	$C_0 + \Omega_0$	$C_0 + 3\Omega_0$
93	-1009	-525	443
92	-977	-477	523

The bound $\rho \le \mu$ makes (85) strictly negative for $z \le 262$ at size 93 and $z \le 238$ at size 92. This includes the exceptional both-AP case, where $z = 0$.

For larger holes, both side supports satisfy the strict hypotheses of the effective-hole theorem and are non-progressions. Thus $\rho + \mu \ge 2m - 34$, reducing possible failures of (84) to $k = 0, \dots, 9$ at either size. The following exact parameter rows in (80), followed by the flexible inverse L_γ from Section 22, give $\mu(X) \ge u_r = r + C_h + 1$ and $\rho(X) \ge v_r$:

r	C_h	β	γ	u_r	v_r	r	C_h	β	γ	u_r	v_r
28	59	6	5	88	48	38	126	9	7	165	105
29	64	6	5	94	52	39	134	9	7	174	112
30	70	7	6	101	57	40	143	9	7	184	120
31	76	7	6	108	62	41	151	9	9	193	128
32	83	7	6	116	68	42	160	10	9	203	136
33	89	7	6	123	73	43	169	10	9	213	144
34	96	8	6	131	79	44	179	10	10	224	153
35	103	8	6	139	85	45	188	10	10	234	162
36	111	8	7	148	92	46	198	11	10	245	171
37	118	8	7	156	98	47	208	11	10	256	180

The chosen β are at most 11, while the holes are at least 239, so $h > 2\beta$ holds throughout. Each row satisfies the two strict integer parameter tests. Substituting both sides gives these complete margins:

	$n = 93$			$n = 92$		
k	$C + 3\Omega$	$u_a + v_a + u_b + v_b$	Margin	$C + 3\Omega$	$u_a + v_a + u_b + v_b$	Margin
0	443	540	97	523	524	1
1	439	540	101	515	526	11
2	427	544	117	499	528	29
3	407	545	138	475	531	56
4	379	549	170	443	535	92
5	343	553	210	403	541	138
6	299	561	262	355	547	192
7	247	566	319	299	554	255
8	187	574	387	235	562	327
9	119	582	463	163	572	409

Every margin is strict, including $524 > 523$ in the balanced size-92 case. Hence $|\mathrm{FS}(D_1)| < 2s$, and the disjoint-translate argument forces D_1 onto the same lattice.

Forcing D and propagating holes. The first-layer comparisons are

$$1483 > 4 \cdot 85 - 10 + 484 = 814, \qquad 1445 > 4 \cdot 84 - 10 + 500 = 826.$$

Exactly as in Theorem 20.1, they give $|\mathrm{FS}(D)| < 2|\mathrm{FS}(D_1)|$, putting D on the lattice. The two strict comparisons now give $w \le 4z$, $\eta_2 \le z + \ell_2$ and $\eta_1 \le 2z + \ell_1$. Only at this stage apply

Theorem 19.1 to obtain $G = 2d - 10$, and hence all of (81). No post-lattice estimate was used to force integrality.

Primitivity in the size-92 both-AP branch. Here $z = 0$ and, only after D is integral, the post-lattice budget gives $\kappa \le \Omega - 158$. The already verified lower bound $T_a - 2 - \Omega \ge -21$ implies $T_a - \kappa \ge 139 > 0$. Since $h_P, h_Q \le \kappa$ and $T_b \ge T_a$, both core hole counts are strictly below their support cardinalities minus one. A coefficient gcd at least two would leave at least one missing position between consecutive support values, contradicting these strict bounds. Both cores are therefore primitive on this lattice. This does not place the original outer eight points on that lattice. □

All displayed finite margins, parameter tests and flexible inverses are checked by `check_lattice93_92.py` in the companion material. The elementary two-AP statement is proved above; tests on examples are not substituted for its proof.

26 Finite progression bounds and an exact exclusion of size 93

Theorem 26.1. *No feasible real set has 93 elements. Consequently*

$$\boxed{N \le 92.}$$

Proof. Theorem 24.1 first justifies zero in a hypothetical size-93 example. Integralization and Theorem 25.1 then give the common lattice, with

$$d = 85, \quad e = 77, \quad m = 76, \quad a + b = 76, \quad I = (a - 38)^2, \quad \Omega = 484 - I, \quad G = 160.$$

The exhaustive root reduction below has 157766 systems. Its complete exact certificates give 160550 leaf contradictions. The new ingredients needed by five strengthened high-hole roots are proved in this section: finite three-term-progression bounds, affine partition bounds, exact-mass defect inequalities and integer-width alternatives. Every other row is covered by the preceding proofs. The actual-set-to-model reduction and complete exact contradiction therefore exclude size 93. The preceding exclusions give the stated general upper bound. □

26.1 Complete endpoint classification

For low holes, take every $16 \le a \le 60$, every $j = 0, 1, 2, 3$, and all pairing modes and contributor choices, retaining precisely

$$B_0 = 324 - I - 85j \ge 0 \quad (z = 0), \qquad B_0 - 20 \ge 0 \quad (1 \le z \le 14).$$

These follow from $G = 160$ and, for positive holes, $\rho \ge m - 28 = 48$. The simultaneous loss, span and cardinality bounds are

$$\delta \le B_0 \ (Z), \qquad \delta \le B_0 - 48 + 2z \ (P),$$
$$R_p + R_q + \delta \le 80 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1614 - 85j + 4z.$$

The exact low-root counts are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2	P_3
Roots	148	2170	14450	50864	140	2030	13294	44880

They total 127976. Three-empty models retain weak outer order.

For high holes, write $(a, b) = (38 - k, 38 + k)$ up to reflection and use

$$C_{93}(a, z) = 484 - I - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(54, z, 160 - 2z).$$

Since $\rho \geq 48$ and $T \geq 54$, $c \leq 382 - k^2$, excluding $k \geq 20$. The complete conservative range is $0 \leq k \leq 19$, $15 \leq z \leq 484 - k^2$, containing 6930 states. All 634 effective-hole constants are directly rechecked. The maximum nonnegative cap is 220, so $j \leq 2$. The all-active mass upper bound is now $1656 + 4z$. Exactly 3372 unoriented states remain: 1646 with $j = 0$, 1367 with $j = 1$, and 359 with $j = 2$. Their canonical grouping produces 88 intervals, respectively 14, 53 and 21. Use both orientations except in the balanced case $a = b = 38$, which is used once. All modes then give 108 H_0, 7140 H_1 and 22542 H_2 roots, totaling 29790.

At this endpoint $K \geq 169$, nested leading rows require $\delta < 161$, and core quantiles require hull loss below 81. Per-interval residual caps control these choices, exactly as in the previous endpoints. In particular the high cap 220 does not imply that every leading translate contributes. All added rows respect their separate thresholds and the common z variable.

26.2 Finite three-term-progression bounds

Let $r_3(L)$ be the largest size of a subset of $[1, L]_{\mathbb{Z}}$ with no nonconstant three-term arithmetic progression; set $r_3(0) = 0$. The finite inputs used by the certificates are

L	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$r_3(L)$	0	1	2	2	3	4	4	4	4	5	5	6	6	7	8
L	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
$r_3(L)$	8	8	8	8	8	9	9	9	9	10	10	11	11	11	11

Together with $r_3(30) = 12$, the additional bounds are

$$r_3(35) \leq 13, \qquad r_3(39) \leq 14, \qquad r_3(50) = 16, \qquad r_3(70) \leq 20. \tag{91}$$

These are finite, exactly verified statements, with the following exhaustive algorithms and validity proofs; they are not empirical fits.

The table through 30. Start from the single empty set at length zero. At step n, retain every old set and also adjoin n whenever no pair $a < b$ in it satisfies $a + n = 2b$. Any new progression must have largest element n, so these are exactly the forbidden pairs. Induction proves that the procedure lists every AP-free set once, with or without its new largest point. The integer-mask finite lemma reaches 880288 sets at length 30 and verifies every maximum in the table. Its full finite loops are included as `verify_apfree_blocks.py`.

The upper bounds at 35, 39 and 70. The bounded depth-first search records chosen points and forbidden future points. It branches on exclusion or inclusion of the least available future point x. Upon inclusion it forbids $2x - y$ for each previously chosen $y < x$. Inductively these are exactly the future completions of progressions whose first two points are chosen. It therefore never omits an admissible set. A node is discarded only when the number of chosen points plus either the number of available points or a proved interval-partition bound cannot exceed the proposed bound. The partition bound uses only the already verified table through 30: a suffix of length L can be partitioned into blocks of lengths at most 30, with capacities equal to their table entries. This is an upper bound on every possible completion, regardless of cross-block constraints. Thus pruning is safe. The specified exhaustive calculation has respectively 1925, 6081 and 59770661 nodes and proves the three upper bounds in (91). Its complete implementation is `verify_bounded_ap.py`.

The bound at 50. Any AP-free subset of $[1, 50]$ of size at least 17 restricts to AP-free sets in its two 25-point halves. Since $r_3(25) = 10$, each half has size between seven and ten. The exact length-25 enumeration has 140840 sets; its size-seven, eight, nine and ten classes have respectively 39332, 16061, 1940 and 33 members. For every pair of halves of combined size at least 17, the checker examines cross-block progressions. Two chosen points $a < b$ in the left half forbid $2b - a$ in the right; two chosen points $a < b$ in the right half, in local coordinates, forbid $25 + 2a - b$ in the left. Every progression crossing the division is of one of these forms. All 69865347 candidate pairs fail at least one cross-block test. Hence $r_3(50) \le 16$. The explicitly checked set

$$\{1, 2, 4, 5, 10, 11, 13, 14, 28, 29, 31, 32, 37, 38, 40, 41\}$$

is AP-free, giving equality. This entire calculation is reproduced by `verify_apfree50.py` using integer bit masks only.

26.3 Affine and width-dependent partition bounds

Restricting an AP-free set to consecutive blocks gives $r_3(L_1 + \cdots + L_t) \le \sum_i r_3(L_i)$. Thus any explicit partition with proved block capacities gives a valid bound; the partition need not be optimal. The companion files contain exact partitions proving

$$29r_3(L) \le 11L + 61 \qquad (L \ge 16), \tag{92}$$
$$29r_3(L) \le 11L + 50 \qquad (L \ge 64), \tag{93}$$
$$50r_3(L) \le 16L + 176 \qquad (L \ge 1). \tag{94}$$

Here is the finite-to-universal proof. For the first inequality, there is one checked partition for each $L = 16, \dots, 44$, using blocks of length at most 30. Appending any number of length-29 blocks, each of capacity 11, preserves the inequality and covers every $L \ge 16$. For the second, the same argument starts with $L = 64, \dots, 92$. For the third, checked partitions for $L = 1, \dots, 50$ use capacities derived from the table and the proved bounds at 35, 39 and 50. Append length-50 blocks, each of capacity 16. The 58 witnesses for the first two affine bounds and 50 witnesses for the third are included in `apfree_affine_certificate.json` and `apfree_affine50_16_certificate.json`. The checker must verify that every part is positive, the parts sum to the claimed length, and their proved capacity sum satisfies the displayed integer inequality. These checks establish (92)–(94) in all their stated ranges.

For the width splits also define a finite partition function. Let $\mathcal{C}$ contain the proved capacities for lengths $1, \dots, 30$ and $(35, 13), (39, 14), (50, 16), (70, 20)$. For $1 \le t \le 70$, put

$$U(t) = \min_{(L,c)\in\mathcal{C}} (c + \max(0, t - L)), \qquad g(0) = 0, \quad g(n) = \min_{1\le t\le \min(70,n)} \{g(n - t) + U(t)\}.$$

Every $U(t)$ bounds $r_3(t)$, by restriction when $t \le L$, and by adding at most $t - L$ further points when $t > L$. Induction using consecutive block partitions proves $r_3(n) \le g(n)$. The width certificates use these exactly computed values only through $n = 201$. No monotonicity of g is assumed.

26.4 Defect inequalities and exact core masses

The diagonal-defective elements of an integer set are AP-free: if $a < b < c$ were three such elements with $a + c = 2b$, then b would not be defective. For a positive core X, its defective set together with zero is also AP-free, since $(0, x, 2x)$ would contradict the defectivity of x relative to $X \cup \{0\}$. With $r = |X|$ and $R = \max X$, this gives

$$\delta_0(X) + 1 \le r_3(R + 1).$$

For $r \geq 15$, (92) applies because $R + 1 \geq 16$. Using $\rho(X) = \mu(X) + 2 - 2\delta_0(X)$ gives the core bounds

$$29\rho(X) + 22R \geq 29\mu(X) - 28, \tag{95}$$
$$50\rho(X) + 32R \geq 50\mu(X) - 184. \tag{96}$$

The latter follows from (94) without a minimum-size condition. Neither AP bound needs primitivity. The existing parity bound gives $\rho(X) \geq \mu(X) + 2r - 2R - 2$ when an odd element is present. Each side may independently use a fixed ρ_* bound, parity, (95), or (96); substituting the chosen pair in the exact budget and clearing positive denominators proves all mixed core-defect rows.

For the full set D, write $W = M_p + M_q$ and retain the exact identity

$$\rho + 2\vartheta = \mu + 20 - 2\Delta(D), \qquad c + T + \rho + 2\vartheta = \Omega. \tag{97}$$

Its defect set lies in an integer interval of length $W + 1$ and is AP-free. At this endpoint $W + 1 \geq d = 85 \geq 64$, so (93) and (94) imply

$$29(c + T + \mu) - 22W \leq 29\Omega - 458, \tag{98}$$
$$50(c + T + \mu) - 32W \leq 50\Omega - 616. \tag{99}$$

There is also a full-set parity bound. Translate D by M_q and remove zero, obtaining a positive $(d-1)$-element set with maximum W. It contains an odd element: otherwise the translated set, including M_q because $0 \in D$, would be all even, forcing D all even, contrary to its primitive core. Translation preserves full restricted-sum defects; zero adds precisely one defect. Therefore Lemma 21.3 yields $\Delta(D) \leq W - d + 4$, and

$$c + T + \mu - 2W \leq \Omega - 2d - 12. \tag{100}$$

Equations (98)–(100) are joint bounds on $\rho + 2\vartheta$. One must not subtract fixed lower bounds for ρ again when using them.

The strengthened models also retain exact core mass. Put

$$C_p = \sum P = S_p - \sum_{i=1}^{4} x_i, \qquad C_q = \sum Q = S_q - \sum_{i=1}^{4} y_i,$$

$$\mathcal{B}_p = \Omega + T_a - jd, \quad \mathcal{B}_q = \Omega + T_b - jd, \quad \mathcal{B} = \Omega + T_a + T_b - jd.$$

Because $C_p - T_a = \mu(P) + h_P$, $h_P \leq \kappa \leq T$, and $h_P + h_Q = \kappa + z$, we have

$$T + \mu(P) \geq C_p - T_a, \qquad T + \mu \geq C_p + C_q - T_a - T_b - z. \tag{101}$$

If a side inequality is written $\rho(P) \geq \mu(P) - \lambda_p R_p + \nu_p$ and the other side has fixed lower bound v_q, the original budget and (101) give

$$\delta + C_p - \lambda_p R_p \leq \mathcal{B}_p - v_q - \nu_p. \tag{102}$$

Reflection gives its companion. For two side inequalities, they give

$$\delta + C_p + C_q - z - \lambda_p R_p - \lambda_q R_q \leq \mathcal{B} - \nu_p - \nu_q. \tag{103}$$

For a joint full-set inequality $\rho + 2\vartheta \geq \mu - \lambda W + \nu$, they give

$$\delta + C_p + C_q - z - \lambda W \leq \mathcal{B} - \nu. \tag{104}$$

The parameter pairs (λ, ν) used are

Bound	Parity	29-period AP bound	50-period AP bound
Individual core of size r	$(2, 2r-2)$	$(22/29, -28/29)$	$(32/50, -184/50)$
Whole D	$(2, 2d+12)$	$(22/29, 458/29)$	$(32/50, 616/50)$

These substitutions, after clearing denominators, are exactly the additional mass-defect row families. The derivation uses the original budget once and does not spend its defect or hole terms twice.

26.5 Width alternatives and paired-unit alternatives

In a strengthened high-hole root, let u_p, u_q be its certified core excess bounds and v_p, v_q its fixed defect bounds. If $W \le t$, then

$$\Delta(D) \le r_3(W+1) \le r_3(t+1) \le g(t+1).$$

Thus (97) and (101) justify the left-child rows

$$\begin{aligned} \delta + T &\le \Omega - jd - u_p - u_q - 20 + 2g(t+1), \\ \delta + C_p + C_q - z &\le \mathcal{B} - 20 + 2g(t+1). \end{aligned}$$

The implemented rows replace T by its proved affine lower bound on the current hole interval. The right child is simply $W \ge t+1$. The integer width W ensures that the two children are exhaustive.

If instead $R_p \le t$, the anchored defect set gives $\delta_0(P) + 1 \le g(t+1)$ and hence $\rho(P) \ge \mu(P) + 4 - 2g(t+1)$. The corresponding left rows are

$$\begin{aligned} \delta + T &\le \Omega - jd - u_p - v_q - 4 + 2g(t+1), \\ \delta + C_p &\le \mathcal{B}_p - v_q - 4 + 2g(t+1). \end{aligned}$$

The right child has $R_p \ge t+1$, and reflection gives the other side. The checked thresholds are $140 \le t \le 200$ for W and $40 \le t \le 120$ for a core width. Monotonicity is used only for the true r_3 before applying its upper bound at $t+1$; the possibly nonmonotone partition function g is never assumed monotone.

The trees also extend the earlier paired-member disjunction to three-empty modes. In a split second-pair case of pairing branch A, choose a member s from the *actual contributors*. Either the other member contributes too, implying $\beta_t + M_t \ge 1$, or s is the sole contributor. In the latter case its lower and upper hull losses may be included in the total-loss budget: the paired unit's capacity deficit is at least this sole member's hull loss. Introduce two fresh nonnegative coordinates for those losses. Use global quantiles, or core quantiles only when the residual cap is below 81. These alternatives are valid with two or three empty units and do not assert integrality of every outer element in the three-empty case. The exact checker verifies that the two new loss coordinates were previously unused and adds the correct row to each child.

26.6 Exact finite lemma for size 93

The required finite assertion has 157766 canonical roots and 160550 leaf contradictions. The data are in `proofs/exact92`. There are 127976 low and 29790 high roots. Exactly 157231 are direct contradictions; 444 low and 91 high roots have trees. Their 2784 splits comprise 2328 normal/outside, 231 inner-extreme, 146 core/large-loss, 28 paired-member, and 51 AP-width alternatives. The trees have 3319 leaves. Of the 535 tree roots, 524 use base rows, six add parity rows, and five use the strengthened AP, exact-mass, and width families. No ordinary root needs an AP row. The finite AP theorems and partition witnesses in this chapter are nevertheless required by these five trees.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact92.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

27 Full-set group deficits and an exact exclusion of size 92

Theorem 27.1. *No feasible real set has 92 elements. Consequently*

$$\boxed{N \le 91.}$$

Proof. Theorem 26.1 first forces zero into a hypothetical size-92 example. Integralization and Theorem 25.1 give the common core lattice. The new group-deficit and retained-center integrality lemmas below justify every additional necessary model row. The exhaustive classification has 248060 original roots, and its complete finite certificates give 248068 exact leaf contradictions. Therefore size 92 is impossible. The earlier exclusions give the stated global upper bound. □

27.1 Using full translates in a common hull

The previous leading-group estimate used the smaller core E to measure overlap. The following estimate uses the whole set D, including its eight inner extreme elements. It introduces no new finite progression-free bound.

Proposition 27.2. *Let $D \subset \mathbb{Z}$, $|D| = d$, $\min D = -M_q$, $\max D = M_p$, and $K = |2D|$. On the positive covering side write*

$$F_p = \mathrm{FS}(D) \cap (0, \infty), \qquad U_0 = c_p + 2D, \quad U_i = c_p + p_i + D, \quad V_j = c_p - q_j + D,$$

where $p_i > M_p$, $q_j > M_q$. Choose distinct components of the fixed covering decomposition, with nominal capacities K for U_0 and d for each selected U_i, V_j. Their centers must be integral on the normalized D lattice, and each selected d-capacity slot must be retained and genuinely contribute. Let I, J be the selected own and cross index sets, $k = |I|$, $l = |J|$, and let p_*, q_* be the largest selected outer elements when defined. The raw capacity deficit is at least

$$\begin{aligned} H_{\mathrm{own}} &= K + kd - p_* - M_p - 2M_q - 1 && (k > 0,\ l = 0), \\ H_{\mathrm{cross}} &= K + ld - q_* - 2M_p - M_q - 1 && (k = 0,\ l > 0), \\ H_{\mathrm{mixed}} &= K + (k+l)d - p_* - q_* - M_p - M_q - 1 && (k, l > 0). \end{aligned}$$

If H denotes the applicable bound and λ_i is any valid lower bound for the actual loss of member i outside F_p, then its actual target capacity deficit $G_{I,J}$ satisfies

$$G_{I,J} \ge H + \lambda_i \quad \text{for every selected member } i, \qquad G_{I,J} \ge \sum_i \lambda_i.$$

The leading component is included among these members. All statements reflect to the negative target, interchanging p and q.

Proof. The own-only raw union is contained in $[c_p - 2M_q, c_p + p_* + M_p]_{\mathbb{Z}}$: the leading component ends at $c_p + 2M_p$, which is smaller than its stated upper endpoint since $p_* > M_p$, and each own component starts above $c_p - 2M_q$. Its hull has $p_* + M_p + 2M_q + 1$ points. The cross-only and mixed hulls are respectively

$$[c_p - q_* - M_q, c_p + 2M_p]_{\mathbb{Z}}, \qquad [c_p - q_* - M_q, c_p + p_* + M_p]_{\mathbb{Z}}.$$

Subtract their cardinalities from the total nominal capacities to obtain the three bounds. They remain valid if their right-hand sides are negative.

Write W for the raw union and C for the total capacity. Its deficit against the actual target is

$$C - |W \cap F_p| = (C - |W|) + |W \setminus F_p|.$$

The outside union contains every member's outside set, proving $G_{I,J} \geq H + \lambda_i$. Also $|W \cap F_p| \leq \sum_i |W_i \cap F_p|$, proving $G_{I,J} \geq \sum_i \lambda_i$. Both parts use the same target F_p; losses outside its hull are used only as lower bounds for actual losses. □

27.2 One deficit variable for all subgroups and clipped hulls

For one sign, let $\mathcal{C}_p$ contain the leading component, every retained own slot, and the retained cross slots indexed 3 and 4 in mathematical outer-element notation. Those two cross slots are always unpaired in the fixed decomposition. Introduce an auxiliary Γ_p whose realization in an actual configuration is

$$\Gamma_p = \sum_{W \in \mathcal{C}_p} |W| - \left| \bigcup_{W \in \mathcal{C}_p} W \cap F_p \right|.$$

For any selected subgroup $\mathcal{S} \subseteq \mathcal{C}_p$ containing the leading component, let $G_{\mathcal{S}}$ be its actual deficit. Then

$$\Gamma_p \geq G_{\mathcal{S}} + \sum_{W \in \mathcal{C}_p \setminus \mathcal{S}} \lambda_W. \tag{105}$$

Indeed, the whole target union is contained in the subgroup target union plus the separate target contributions of the outside components. Thus every subgroup inequality simultaneously lower-bounds the same auxiliary. Also $\Gamma_p \geq \sum_{W \in \mathcal{C}_p} \lambda_W$. The corresponding negative family uses distinct covering resources, so $\Gamma_p + \Gamma_q$ and all unmatched resource losses fit within the residual covering slack δ. Geometric overlap between groups causes no problem; their component labels, hence their capacities, must be disjoint.

Proposition 27.3. *Suppose a selected subgroup of total capacity C lies in an integral hull $[L, U]_{\mathbb{Z}}$ and has a nonempty contribution to F_p. Since $F_p \subseteq [1, S_p]_{\mathbb{Z}}$, its actual deficit also satisfies*

$$G_{\mathcal{S}} \geq C - U, \qquad G_{\mathcal{S}} \geq C - S_p + L - 1, \qquad G_{\mathcal{S}} \geq C - S_p.$$

Proof. Nonemptiness implies $U \geq 1$ and $L \leq S_p$. The target union is contained in $[L, U]_{\mathbb{Z}} \cap [1, S_p]_{\mathbb{Z}}$, whose cardinality is at most each of $U - L + 1$, U, $S_p - L + 1$ and S_p. Subtract from C. □

These clipped bounds already measure actual target loss. Unlike a raw hull bound in Proposition 27.2, they must not receive an additional selected member's loss. Only the outside-component losses from (105) are added. In the size-92 implementation every clipped subgroup contains at least one retained own slot, establishing the required nonemptiness. The applied hulls are the own-only or mixed hulls above. Taking suffixes of the retained own and unpaired cross slots supplies all the inequalities used by the certificate.

27.3 Outer integrality from retained centers

Lemma 27.4. *If D lies on the normalized integer lattice and at most two covering units are empty, then all eight outer elements and both leading centers c_p, c_q are integral. No contribution from a leading $2D$ component is required.*

Proof. Order $p_1 > p_2 > p_3 > p_4$ and $q_1 > q_2 > q_3 > q_4$, and recall

$$c_p = p_1 - p_2 - p_3 - p_4, \qquad \alpha_p = p_2 - p_3 - p_4, \qquad \beta_p = 2p_2 - p_1 - p_3 - p_4,$$

with reflected definitions. Every retained unpaired D slot has an integral center, since it meets an integral subset-sum target. The own slots $c_p + p_i, c_q + q_i$, the tail cross slots $c_p - q_3, c_p - q_4, c_q - p_3, c_q - p_4$, and both α slots are always unpaired. In branch A, $c_q - p_2$ is also unpaired; in branch B, β_p is unpaired. Reflection gives the opposite-side statements.

Writing $w_i = c_p + p_i$ gives $c_p = -w_1 + w_2 + w_3 + w_4$. Thus a complete own side makes its four outer elements and leading center integral. Suppose first, after reflection, that the q own side is complete. If exactly one p own slot is missing, at least one of $c_p - q_3, c_p - q_4$ contributes because only one other unit can be empty. It makes c_p integral, then the three retained own slots and the defining identity give every p_i. If two p own slots are missing, all cross and α slots that are unpaired contribute. The centers $c_p - q_3, c_q - p_3, c_q - p_4$ give $c_p, p_3, p_4 \in \mathbb{Z}$. A retained own slot among p_1, p_2 then finishes the argument. If both are missing, branch A gives p_2 from $c_q - p_2$; branch B gives it from α_p. The identity gives p_1.

It remains to have one missing own slot on each side. They consume both empty units, so all unpaired cross, α and unpaired β slots contribute. At least one positive own slot with index $i \in \{3, 4\}$ is retained. Adding its center $c_p + p_i$ to $c_q - p_i$ gives $c_p + c_q \in \mathbb{Z}$. Let k be the missing positive own index. For $i \neq k$, one has $p_i \equiv -c_p \pmod{\mathbb{Z}}$. If $k = 1$, the integral center $\alpha_p \equiv c_p \pmod{\mathbb{Z}}$ gives $c_p \in \mathbb{Z}$. If $k = 3$ or 4, substituting the three known congruences in the c_p identity gives $p_k \in \mathbb{Z}$; then $c_q - p_k$ gives $c_q \in \mathbb{Z}$ and hence $c_p \in \mathbb{Z}$. If $k = 2$, the same substitution gives $p_2 \in \mathbb{Z}$. Branch A uses $c_q - p_2$. In branch B, $\alpha_p \equiv 2c_p$ and $\beta_p \equiv 3c_p \pmod{\mathbb{Z}}$; their integral difference gives $c_p \in \mathbb{Z}$. In every case both leading centers are integral, all retained own coordinates follow, and the missing coordinate on each side follows from its defining identity. □

This strengthens the earlier conditional integrality argument. In particular the zero-hole two-empty residual cap at size 92 can be 174, which exceeds the general bound $K \geq 167$; the preceding lemma justifies all two-empty strict gaps even there. A redundant exact check of all 578 two-empty center patterns found an integer left inverse in every case, but the proof above supplies the universal statement.

Individual certificates for the geometric roots. The new group rows are used only after an explicit integer-combination certificate. Let $\mathbf{p} = (p_1, p_2, p_3, p_4, q_1, q_2, q_3, q_4)^T$. Each retained contributing center is an integer linear form in $\mathbf{p}$: its translate of D meets the integral target, and D is integral. Collect these coefficient rows in an integer matrix C. Include a selected paired contributor's center where the covering mode requires it. The stored certificate is an integer matrix W with

$$WC = I_8.$$

It follows that $\mathbf{p} = W(C\mathbf{p}) \in \mathbb{Z}^8$. The checker reconstructs C from the actual retained-center rows and checks this identity exactly; it does not rely on a numerical rank computation or a discovered Smith normal form. Integral outer coordinates give integral leading centers even if a leading component happens to be empty.

Consequently every strict outer order relation can be strengthened to an integer gap of at least 1. In pairing branch B the strict sign $\beta_p + \beta_q > 0$ can likewise be strengthened to $\beta_p + \beta_q \geq 1$. The six final low-hole trees use these strengthened rows only after their individual $WC = I_8$ identities have been checked. Thus four-empty roots and other cases lacking the earlier general outer-integrality argument are not treated as integral without proof.

27.4 The complete size 92 root classification

The existing upper bound first forces a hypothetical size-92 example to contain zero: otherwise adjoining zero gives the already excluded size 93. The real-to-integer reduction and Theorem 25.1 give

$$d = 84, \quad e = 76, \quad m = 75, \quad a+b = 75, \qquad I = T_a + T_b - T_{37} - T_{38}, \quad \Omega = 500 - I, \quad G = 158.$$

For low holes enumerate $16 \le a \le 59$. The structurally possible side sizes $a = 14, 15$ and their reflections $a = 60, 61$ already have negative low-hole caps and therefore contribute no roots. For $z = 0$ allow $j = 0, 1, 2, 3, 4$; for $1 \le z \le 14$ allow $j = 0, 1, 2, 3$. Retain exactly

$$B_0 = 342 - I - 84j \ge 0 \quad (Z), \qquad B_0 - 19 \ge 0 \quad (P).$$

The positive-hole condition uses $\rho \ge m - 28 = 47$. The corresponding necessary simultaneous bounds are

$$\delta \le B_0 \quad (Z), \qquad \delta \le B_0 - 47 + 2z \quad (P),$$
$$R_p + R_q + \delta \le 79 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1596 - 84j + 4z.$$

All pairing modes and selected contributors give these exact counts:

Empty units j	0	1	2	3	4
$z = 0$	152	2240	15028	59840	65280
$1 \le z \le 14$	144	2100	13872	47872	—

Their sum is 206528. Weak outer order remains in the base model whenever the earlier argument does not prove outer integrality.

For high holes write $(a, b) = (37 - k, 38 + k)$ up to reflection, so $I = k(k + 1)$. Use

$$C_{92}(a, z) = 500 - I - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(53, z, 158 - 2z).$$

The bounds $\rho \ge 47$ and $T \ge 53$ give $c \le 400 - k(k + 1)$, excluding $k \ge 20$. Hence enumerate $0 \le k \le 19$, $15 \le z \le 500 - k(k + 1)$, totaling 7060 states. The effective-hole lemma's hypotheses are directly checked for all 640 constants used. The largest nonnegative cap is 244, so $j \le 2$. The all-active mass upper bound is $1638 + 4z$. Exactly 4198 unoriented states survive: 2013 with $j = 0$, 1586 with $j = 1$, and 599 with $j = 2$. Their canonical consecutive-z grouping gives 106 intervals, respectively 21, 56 and 29. Both orientations are distinct because $a + b = 75$ is odd. With every mode and contributor choice these yield 168 H_0, 7840 H_1, and 33524 H_2 roots, totaling 41532. Thus the complete endpoint has $206528 + 41532 = 248060$ original roots.

Here $K \ge 167$, nested leading rows require $\delta < 159$, and the core-quantile threshold is hull loss below 80. Every quantile or leading-cardinality row is gated by its own residual cap; strict outer gaps use Lemma 27.4 or an individual integer-center certificate. In particular the maximum cap 244 alone does not establish that a leading component contributes. The integer-center certificates and the nonempty own slots justify the full-set rows independently of that stronger assertion.

27.5 Exact finite lemma for size 92

The required finite assertion has 248060 canonical roots and 248068 leaf contradictions. The data are in `proofs/exact91`. There are 246487 ordinary direct roots, 1345 low and 222 high direct geometric roots, and six low geometric tree roots. The latter have eight splits and 14 leaves: four normal/outside and four inner-extreme/loss alternatives. The only added row families are full-translate group bounds, clipped hull bounds, and their simultaneous actual-deficit realization.

Every one of the 1573 geometric roots has a reconstructed integer-center witness. No AP-width split is needed here; the earlier size-93 exclusion still requires its own AP finite lemmas.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact91.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

28 A common core lattice at size 91

The next exclusion needs a new structural theorem. Its proof must handle progression subset-sum supports directly: the previous sufficient simultaneous-inverse inequality can fail at this size.

28.1 Progression summands and a uniform repair

We first record two elementary facts. If $W = \{w_1 < \cdots < w_v\} \subset \mathbb{Z}$ and $J = [0, u-1]_{\mathbb{Z}}$, then

$$|W + J| = v + u - 1 + \sum_{i=1}^{v-1} \min(w_{i+1} - w_i - 1, u - 1). \tag{106}$$

This follows by adding the intervals $[w_i, w_i+u-1]$ in order: the next interval adds $\min(w_{i+1}-w_i, u)$ new points. For a progression with arbitrary integer step, apply this formula separately to each occupied residue class. If there are t classes, the sumset excess is at least $(t-1)(u-1)$.

For positive integers $x_1 < \cdots < x_b$, the largest consecutive gap in their subset-sum support is

$$\max_{1 \le i \le b} \left(x_i - \sum_{j<i} x_j \right). \tag{107}$$

Indeed, adjoining $[0, L-1]_{\mathbb{Z}}$ to the support produces one interval exactly when $x_i \le L + \sum_{j<i} x_j$ for every i. Sufficiency is induction on the two overlapping translated intervals. If a test fails, the interval between the previous total and x_i cannot be filled by any later coefficient. This proves necessity and (107). If the support has exactly one gap of length at least u, a prefix gap attaining that size and its reflection about half the total must be the same gap. Its endpoints are the previous total and x_i; equality with the reflected gap forces $i = b$. Thus x_b exceeds the preceding total, the last two supports are disjoint, and

$$\mu(X) \ge 2(1 + T_{b-1}) - 1 - T_b = T_{b-2}. \tag{108}$$

We now apply these facts at size 91. Write $a = 37 - k \le b = 37 + k$ and

$$d = 83, \quad e = 75, \quad m = 74, \quad \Lambda = 1407 + k^2, \quad \Omega = 516 - k^2, \quad C = -945 - k^2.$$

Here $C = 2m + 4e + 14 - \Lambda$ is the coefficient in the pre-lattice comparisons (85) and (84). The deficient-sign argument gives $|B| = 8$: otherwise $1 + T_{75} = 2851 > 24 \cdot 91 + 77 = 2261$. The old uniform defect estimate first gives $k^2 \le 534$, hence $a \ge 14$ and the threshold-14 theorem applies. The unpaired-value bound is $q^2 \le 2648$; parity gives $q \le 50$, so there are at least 12 opposite pairs and zero is not a diagonal exception. The nonnegative exact budget then gives $k \le 22$, hence $a, b \ge 15$.

Put $U = \mathrm{FS}(P)$, $V = \mathrm{FS}(Q)$, $u = |U|$, $v = |V|$. Reflection of V is a translate of V by subset complementation. If both supports are non-progressions, the sufficient small-side inverse margin is

$$T_a - 2 - \Omega + 2a + b - 31 > 0 \qquad (0 \le k \le 22),$$

whose minimum is 18. This uses $\mu_P \geq a-3$ and $\rho_P+\rho_Q \geq a+b-28$. The larger-side comparison follows from $T_b - 2 > \Omega$. Thus the strict simultaneous inverse theorem gives one core lattice with $h_P, h_Q \leq \kappa$ and strict support-cardinality hole bounds.

Suppose instead that one support is a progression. Call it $U = t[0, u-1]_{\mathbb{Z}}$, reflecting the names of the sides if necessary. If this side has at least 37 coefficients, then $u-1 \geq T_{37} = 703 > \Omega$. Formula (106) forces the other support into one residue class modulo t, with no gap as large as u. On that lattice its hull holes equal κ; the progression has no holes, so $z = 0$.

It remains that the progression side has $15 \leq a \leq 37$, with $37 \leq b \leq 59$. Throughout this range direct quadratic arithmetic gives

$$\Omega < 2T_a, \qquad \Omega - T_a \leq 47, \qquad u \geq T_a + 1 \geq 121 > 2b. \tag{109}$$

Since $\kappa \leq \Omega$, at most two residue classes of V occur. If there is one class, normalize t. Formula (106) allows at most one gap of V as large as u. If such a gap existed, (108) and $\delta_0(Q) \leq b$ would give

$$\rho_Q \geq T_{b-2} + 2 - 2b \geq 558 > 516 \geq \Omega,$$

which is impossible. Hence $h_P = 0$, $h_Q = \kappa$ and $z = 0$.

If two residues occur, exactly one off-lattice coefficient would give two disjoint translated supports and the same contradiction. At least two coefficients therefore have the nonzero residue r. Their sum has residue $2r \in \{0, r\}$, so r has order two. Normalize the half-step: $U = 2[0, u-1]_{\mathbb{Z}}$ and Q has $s \geq 2$ odd coefficients. Let w be the least odd coefficient and let h_0, h_1 be the holes within the even and odd fiber hulls of V, measured with step two. By (106) and $\kappa < 2(u-1)$, no fiber gap is as large as u, and therefore

$$\kappa = u - 1 + h_0 + h_1, \qquad h_P = u - 1, \qquad h_Q = \kappa - u + w, \qquad z = w - 1.$$

For the formula for h_Q, the least odd support element is w. Complement symmetry gives the opposite fiber endpoints; the number of integer positions outside their two hulls is $w-1$, whether the total sum is even or odd.

We need $w \leq u$. If $w \geq u + 1 > 2b$, positivity and distinctness give

$$\sum Q \geq (b-s)(b-s+1) + sw + s(s-1).$$

Since $|V| = \sum Q + 1 - h_Q$, the resulting lower bound is minimized at $s = 2$, and it yields $|V| \geq b^2 - 3b + 2u + 6 - \kappa$. But the exact budget and $\rho_Q \geq \mu_Q + 2 - 2b$ give $\mu_Q + \kappa \leq \Omega + 2b - 2$, hence $|V| \leq T_b + \Omega + 2b - 1 - \kappa$. The lower minus upper margin is $(b^2 - 11b)/2 + 2u + 7 - \Omega \geq 214$, a contradiction. Thus $h_P, h_Q \leq \kappa$ on this common half-step lattice.

In fact $w \leq 13$. If the odd number w were at least 15, the same mass estimate, minimized over real s, would imply

$$\sum Q - w - T_b \geq \min_s \{T_b + 2s^2 + s(15 - 2b - 2) - 15\} = 7b - \frac{289}{8}.$$

On the other hand the identity $\mu_Q + \kappa = \sum Q + u - w - T_b$ and (109) give $\sum Q - w - T_b \leq \Omega - u + 2b - 2 \leq 2b + 44$. These inequalities contradict each other for $b \geq 37$. Hence $z \leq 12$.

In this two-residue case a sharper span estimate forces the outer layers onto the lattice without using the post-lattice constant. The progression side has $R_P \leq 2(a + \mu_P)$, by subtracting the least possible sum T_{a-1} from its normalized full support total. The other side satisfies $R_Q \leq b + \mu_Q + \kappa - T_a - \mu_P + 12$. Substitution in the exact D_1 comparison gives

$$|\operatorname{FS}(D_1)| - 2s \leq 2543 + 2a - 4T_a - 2T_b \leq -1157.$$

Also $\Lambda \geq 1407 > 4 \cdot 83 - 10 + 516 = 838$, so the first-layer comparison forces D onto the same lattice. Only now is $G = 156$ available. It gives $T \geq \kappa + 156 - 3z \geq T_a + 120$, contradicting $\Omega - T_a \leq 47$. Thus the two-residue case is excluded. All remaining progression-support cases have $z = 0$ and are forced integral by the old comparison $C + \Omega = -429 - 2k^2 < 0$ and the same first-layer inequality.

28.2 A defect-sensitive span bound and the remaining forcing step

Lemma 28.1. *For a positive real set Y of size at least 24, $\rho(Y) \ge \delta_0(Y) - 4$. Consequently, for a positive integer set X of size $r \ge 25$ with hull-hole count h,*

$$\max X \le r + \rho(X) + h + 7 - \delta_0(X).$$

Proof. Write $s = |Y|$ and $\eta = \rho(Y)$. If $\eta < s - 14$, the existing flexible inverse bound with $\beta = 2$ gives $\delta_0(Y) \le \eta + 4$. If $s - 14 \le \eta < 2(s - 14)$, use $\beta = 3$ to get $\delta_0(Y) \le \eta/2 + 6 \le \eta + 4$, since $\eta \ge 10$. In the remaining range, $\eta \ge 2(s - 14) \ge s - 4 \ge \delta_0(Y) - 4$. Now remove $R = \max X$, putting $Y = X \setminus \{R\}$. If v old diagonal defects are destroyed on adjoining R, the exact recurrence is

$$R = r + \rho(X) + h(X) + 2 - \rho(Y) - h(Y) - 2v.$$

Since $\delta_0(Y) \ge \delta_0(X) - 1$ and $h(Y), v \ge 0$, the first part gives the stated bound. No primitivity assumption is used. □

Summing the new span bound for the two cores and using $2(\delta_0(P) + \delta_0(Q)) = \mu + 4 - \rho$ gives, before D_1 is known to be integral,

$$|\operatorname{FS}(D_1)| - 2s \le C + 16 + 3\rho - 2\mu + \kappa + 2z + \ell_2 \le C + 3\Omega + 16 - 2\mu. \tag{110}$$

The final step uses only $z \le \kappa$ and $\rho + \kappa + \ell_2 \le \Omega$, with each nonnegative term charged once.

For the remaining non-progression supports the old one-budget comparison is at most $-429 - 2k^2 + 2z$, so it is strict for $z \le 214 + k^2$. Otherwise $z \ge 215 + k^2$. Since $\rho + \mu \ge 2m - 34 = 114$, the old three-budget comparison reduces the possible failures to $0 \le k \le 11$. Both cores then have at least 26 elements. The effective-hole rows below give $\mu(X) \ge r + C_h + 1$; each (C_h, β) satisfies the two strict tests in (80), and $2\beta < 215$ throughout:

r	26	27	28	29	30	31	32	33	34	35	36	37
C_h	48	53	59	64	70	76	83	89	96	103	111	118
β	5	6	6	6	6	7	7	7	7	8	8	8
r	38	39	40	41	42	43	44	45	46	47	48	
C_h	126	134	143	151	160	169	179	188	198	208	219	
β	8	9	9	9	9	10	10	10	10	11	11	

The resulting full list of forcing bounds in (110) is

k	0	1	2	3	4	5	6	7	8	9	10	11
$\mu_P + \mu_Q \ge$	312	313	313	315	316	319	321	325	328	333	337	343
$\lvert\operatorname{FS}(D_1)\rvert - 2s \le$	−5	−11	−23	−47	−77	−119	−167	−227	−293	−371	−455	−551

All are strict. Two disjoint s-point translates would result from an off-lattice element of $D_1 \setminus E$, so D_1 is integral. The first-layer inequality already displayed then forces D to be integral.

Theorem 28.2. *A zero-containing feasible integer set of size 91 has the common-core lattice structure of Theorem 20.1, with $d = 83$, $e = 75$, $m = 74$ and $G = 156$. Both core supports satisfy the strict support-cardinality hole bounds after this reduction.*

Proof. The preceding arguments cover both non-progression supports and every progression-support case. After D is integral, hole propagation and Theorem 19.1 give (81) with $G = 156$. In a remaining $z = 0$ case, $\kappa \le \Omega - 156$. The minimum of $T_a - 2 - \Omega$ is -49, so $T_a + \mu_P - \kappa \ge 109 > 0$; the larger-side comparison is stronger. Thus both hole counts are strictly below the corresponding support cardinalities minus one. Positive-hole cases already had these strict inverse bounds. □

The exact companion `verify_size91_lattice.py` checks the finite inverse parameter tables and forcing margins. The uniform progression-support repair above is an analytic proof, and no use of G precedes D-integrality. This theorem alone excludes no cardinality.

29 Fixed-score losses and an exact exclusion of size 91

Theorem 29.1. *No feasible real set has 91 elements. Consequently*

$$\boxed{N \le 90.}$$

Proof. The preceding upper bound forces zero in a hypothetical size-91 set. Integralization and Theorem 28.2 give the common core lattice. The complete necessary-model classification below has 432560 roots. Its finite certificates give 432758 exact leaf contradictions, using the new fixed-score inequalities proved below. This excludes size 91 and proves the general upper bound. □

29.1 Eight additional losses and exact fixed-score indicators

The original covering has 110 labeled fixed scores, with at most 84 positive ones. Its 26 disjoint deletion groups have coordinatewise nonpositive coefficient sums in the eight nonnegative outer gap variables; each group therefore supplies at least one nonpositive score. These original groups and all 110 coefficient vectors are regenerated exactly by the accompanying fixed-score checker. We retain these labels even when two labels happen to represent the same value: this weakens an upper bound for a union and preserves the capacity accounting.

Suppose $z = 0$. Hole propagation makes $\mathrm{FS}(D) = [-S_q, S_p]_{\mathbb{Z}}$. Assume the largest positive own slot $u + D$, $u = c_p + p_1$, is empty, its center is integral, and a lower own slot $v + D$, $v = c_p + p_i$ with $i > 1$, contributes. Then $u > S_p$. Indeed, $0 < u \le S_p$ would give the contribution $u + 0$. If $u \le 0$, choose $d' \in D$ with $1 \le v + d' \le S_p$. Since $u > v$, one has $1 \le u + d' \le d' \le M_p \le S_p$, again an integral contribution. Thus the four fixed labels

$$C^{+}_{1i} = c_p + p_1 + p_i \qquad (1 \le i \le 4)$$

all exceed S_p. They are positive and hence distinct from the nonpositive deletions. The reflected argument supplies four more losses if the largest negative own slot is empty with a lower contributor. These eight labels can therefore be charged as extra fixed losses. The scalar definitions of c and $\delta = c - jd$ still use the original fixed capacity 84; the new loss is an additional term in its allocation.

For any other fixed label f, introduce $H_f \in [0, 1]$, realized by an actual configuration as the indicator that its score exceeds its own target maximum S_p or S_q. The high-loss auxiliary may equal $\sum H_f$ and be added to every covering-deficit budget. For the seven roots using the eight losses above, only the other 102 labels enter this sum. Integral scores have the exhaustive alternatives

$$f \le S_{\mathrm{side}}, \qquad \text{or} \quad f \ge S_{\mathrm{side}} + 1 \text{ and } H_f \ge 1.$$

All score coefficients and their target sides are fixed by the original covering, so each such split is reconstructed directly.

At the balanced root an additional nonpositive-loss auxiliary is used. For all 110 labels introduce $N_f \in [0, 1]$, realized as the indicator of $f \le 0$. Each of the 26 disjoint groups imposes $\sum_{f \in \mathcal{G}} N_f \ge 1$. Thus

$$L_{\le 0} = \sum_f N_f - 26 \ge 0$$

is a valid realization of the extra nonpositive loss beyond the already charged 26. It may be combined with high losses and the eight separately charged positive labels, since these three sets of labels are disjoint. The binary alternatives $f \le 0$, $N_f \ge 1$, or $f \ge 1$ cover every integral score. The LP may relax these indicators to real variables: the stated actual realizations prove all necessary rows and both branches.

These rows are used only after individual retained-center identities $WC = I_8$ establish outer integrality. The seven applicable zero-hole roots have $a = 34, \dots, 40$ and missing units `pair1,pair2,p0,q0`; all required lower own slots are retained. No statement about an arbitrary empty-largest slot is needed.

29.2 Complete roots and the weak outer-order geometry

At this endpoint $d = 83$, $e = 75$, $m = 74$, $I = (a-37)^2$, $\Omega = 516 - I$ and $G = 156$. Enumerate low-hole sides $15 \le a \le 59$ and empty-unit counts $j = 0, \dots, 4$, retaining exactly

$$B_0 = 360 - I - 83j \ge 0 \quad (Z), \qquad B_0 - 18 \ge 0 \quad (P).$$

The underlying bounds are

$$\delta \le B_0 \quad (Z), \qquad \delta \le B_0 - 46 + 2z \quad (P), \qquad 1 \le z \le 14 \quad (P),$$
$$R_p + R_q + \delta \le 78 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1578 - 83j + 4z.$$

The positive-hole four-empty wrapper starts with the more permissive zero-hole cap, then installs the positive-hole mass penalties and exact z coefficients. Consequently its inherited optional quantile and leading rows remain valid. The exact counts are

j	0	1	2	3	4
Z_j	148	2310	15606	62832	119680
P_j	148	2310	15606	56848	76160

Their sum is 351648.

For high holes, $(a, b) = (37 - k, 37 + k)$ up to reflection and

$$C_{91}(a, z) = 516 - k^2 - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(52, z, 156 - 2z).$$

Since $\rho \ge 46$, $T \ge 52$ gives $c \le 418 - k^2$, so $0 \le k \le 20$ suffices. Enumerate $15 \le z \le 516 - k^2$: there are 7672 integer states, with 646 directly validated effective-hole constants. The largest cover cap is 268, permitting $j \le 3$. The all-active mass upper bound is $1620 + 4z$. The 5614 retained states have counts $(2568, 2014, 944, 88)$ for $j = 0, 1, 2, 3$. Their 110 canonical intervals have counts $(21, 52, 29, 8)$. Both orientations are used except at $a = b = 37$. The resulting high roots number $(164, 7070, 31790, 41888)$, totaling 80912. All roots together number $351648 + 80912 = 432560$.

The leading, nested and core-quantile thresholds are respectively 165, 157 and 79. With $j \le 2$, Lemma 27.4 justifies strict outer gaps. For $j \ge 3$ the ordinary and geometric models keep weak real outer order. The actual residual caps are at most 111 in the low regime and 19 in the high regime; both are below $K \ge 165$. Therefore both leading components contribute and c_p, c_q are integral. Every retained own or cross center is integral, aligning the selected outer endpoints needed for Propositions 27.2 and 27.3. Missing outer coordinates need not be integral. The checker tests the caps at both ends of each affine interval. Among $j \ge 3$ roots, only the seven fixed-score roots restore all strict outer gaps through their separate $WC = I_8$ certificates.

29.3 Exact finite lemma for size 91

The required finite assertion has 432560 canonical roots and 432758 leaf contradictions. The data are in `proofs/exact90`. There are 428966 ordinary roots, 3546 direct geometric roots, 41 geometric tree roots with 84 splits and 125 leaves, and seven fixed-score tree roots with 114 splits and 121 leaves. The 198 splits comprise 74 normal/outside, four core/large-loss, 30 inner-extreme, 58 high-score, and 32 nonpositive-score splits. No integer-coordinate split is selected. The 110

fixed labels, all 26 disjoint groups, exact score rows, and all retained-center identities must be reconstructed.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact90.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

30 A sharper non-progression defect and the size 90 lattice

The next structural reduction uses a sharper non-progression penalty and applies the effective-hole theorem to the cores with their maxima removed. The side of size 14 requires a separate argument.

30.1 A sharp non-progression penalty

Theorem 30.1. *If X is a positive real set of size $r \geq 21$ and $\mathrm{FS}(X)$ is not an arithmetic progression, then $\rho(X) \geq r - 5$. The constant is sharp.*

Proof. First consider a deleted interval $Y = [1, j+q]_{\mathbb{Z}} \setminus H$, where $j \geq 10$, $|H| = q \leq 3$, and the maximum $j+q$ is retained. The case $q = 0$ has full support. For $q \geq 1$, Lev's central-interval theorem applies because $j + q \leq 3j/2 - 2$. It gives

$$[2q+1, \textstyle\sum Y - (2q+1)]_{\mathbb{Z}} \subseteq \mathrm{FS}(Y).$$

Thus its total hole count is twice the number of missing subset sums in $[1, 2q]_{\mathbb{Z}}$, determined entirely by the deleted positions in that range. The density of Y forces gcd one, so a progression support is exactly a full interval.

Every diagonal defect lies in $[1, q]$ or $[j, j+q]$. To see this, an interior element x with $q+1 \leq x \leq j-1$ has more than q disjoint pairs of distinct positions from $[0, j+q]$ centered at x; deleting only q positions cannot destroy all pairs. The low defects depend only on positions through $2q$, and the high defects only on depths through $2q$ below the retained maximum.

For $j \geq 14$, these two bands are disjoint. Preserve the deleted low positions and high depths, and let l, u be their respective counts. For the other $t = q - l - u$ deletions choose the depths $2q+1, \ldots, 2q+t$ in a new interval of 14 retained elements. They fit between the two bands for every $q \leq 3$. The resulting set Y_0 has exactly the same hole count and the same defect count. Writing $h = j + q$ and measuring each deleted depth as $h - d$, one has

$$\mu(Y) = \sum_{d \in H} (h - d) - T_{q-1} - \mathrm{holes}(\mathrm{FS}(Y)).$$

Each low deletion contributes an additional depth $j - 14$ in Y; the original middle-depth sum is at least the chosen one. If the support is not full, at least one low deletion is necessary, so $l \geq 1$. Consequently

$$\rho(Y) - j \geq \rho(Y_0) - 14 + (l-1)(j-14) \geq \rho(Y_0) - 14.$$

This reduces all sizes to the finite cases $j = 10, 11, 12, 13, 14$. The finite checker must enumerate every $q = 1, 2, 3$ and every $H \subseteq [1, j+q-1]$ with $|H| = q$. It forms the subset-sum support by repeated unions with its translates, and tests every diagonal by all possible distinct-pair representations. The complete counts are

j	10	11	12	13	14
All deletion sets	285	363	454	559	679
Non-full supports	130	154	180	208	238
Minimum $\rho - j$ for non-full support	-5	-5	-5	-5	-5

Thus all 2340 cases, including all 910 non-full ones, are covered. This finite exhaustive check and the preceding signature reduction prove $\rho(Y) \geq j - 5$ for every non-progression deleted interval in this range.

For an arbitrary positive real set, use its increasing prefixes X_i and increments y_i from Section 20. The exact recurrence gives $\rho_i \geq \rho_{i-1} + y_i - 2$, and globally $\rho_i \geq -8$. If every $y_i \geq 4$ for $10 \leq i \leq r$, then $\rho_r \geq -8 + 2(r-9) = 2r - 26 \geq r - 5$. Otherwise take the last $j \geq 10$ with $y_j \leq 3$. The small-increment inverse theorem, valid since $3 \leq j - 5$, makes X_j a scaled deleted interval with at most three deletions and its maximum retained. If its support is not a progression, the result just proved and all later $y_i \geq 4$ give $\rho_r \geq r - 5$. If its support is a progression, let $k > j$ be the first non-progression support. A progression and its translate, if they intersect, have their centers on the same progression step and their union is a progression. Thus this first non-progression step has disjoint supports, and

$$y_k = |\operatorname{FS}(X_{k-1})| - k \geq 1 + T_{k-1} - k, \qquad \rho_k \geq T_{k-1} - k - 9 \geq k - 5$$

for $k \geq 11$. Later increments again increase ρ by at least two. Finally $X = \{1, 3, 4, \ldots, r+1\}$ has two diagonal defects and $\mu = r - 3$, so $\rho = r - 5$. □

The companion `verify_small_deleted.py` checks precisely this finite base. It is separate from the earlier length-70 AP-free computation.

30.2 The common lattice and the size 14 boundary

At size 90 put

$$d = 82, \quad e = 74, \quad m = 73, \quad a = 36 - k, \quad b = 37 + k,$$
$$I = k(k+1), \quad \Lambda = 1370 + I, \quad \Omega = 531 - I, \quad C = -914 - I.$$

The coarse unpaired bound gives $a \geq 11$. For $a = 11, 12, 13$, the exact values of Ω are $-119, -69, -21$, each below the small-side global bound -8 plus the other side's nonnegative defect. They are impossible. Hence $a \geq 14$, both defects are nonnegative, and $k \leq 22$.

Set $v_0(r) = r - 5$ for $r \geq 21$ and $v_0(r) = r - 14$ for $14 \leq r < 21$. If both supports are non-progressions, the strict inverse test follows from

$$(a-3) + v_0(a) + v_0(b) > \Omega - T_a + 2 \qquad (0 \leq k \leq 22).$$

The minimum margin is 7; the larger-side test is automatic. Thus the common core lattice has $h_P, h_Q \leq \kappa$, with strict support-cardinality hole bounds.

If a progression side has at least 37 elements, the proof of Section 28 gives one residue and $z = 0$, since $T_{37} = 703 > 531$. For a smaller progression side, the same repair applies with the following endpoint-specific checks. One has $\Omega < 2T_a$ and $\Omega - T_a \leq 75$. A sole off-lattice generator or a unique gap of length at least u would again imply $\rho_Q \geq 558 > 531$, so neither is possible. In the two-residue case, $\kappa \geq u - 1 \geq T_a$ forces $a \geq 18$, whence $u \geq 172 > 2b$ and $37 \leq b \leq 55$. The lower-minus-upper mass margin excluding $w > u$ is at least 301. Also $\Omega - u \leq 74$. If $w \geq 15$, the same quadratic estimate would give $\sum Q - w - T_b \geq 7b - 289/8$, while the budget gives at most $2b + 72$, a contradiction for $b \geq 37$. Hence $w \leq 13$ and $z \leq 12$. The improved progression-side span now gives

$$|\operatorname{FS}(D_1)| - 2s \leq 2513 + 2a - 4T_a - 2T_b \leq -1089.$$

The first-layer inequality is $1370 > 4 \cdot 82 - 10 + 531 = 849$. Only after these establish D-integrality may we use $G = 154$; then $T \geq \kappa + 154 - 3z \geq T_a + 118$ contradicts $\Omega - T_a \leq 75$. All remaining progression-support cases have $z = 0$.

The boundary $a = 14$, $b = 59$ has $I = 506$, $\Omega = 25$. It must be resolved before using Lemma 9.4: its prefix has size 13 and need not have nonnegative ρ. The exact maximum-removal identity gives instead

$$R \le r + \rho_X + h_X + 2 - \rho_{X\setminus\{R\}} \le r + \rho_X + h_X + 10,$$

using only the global lower bound -8 for that prefix. The other side keeps the usual constant 2. Therefore the D_1 comparison gains 16 and is still at most

$$-383 - 2 \cdot 506 + 2 \cdot 25 + 16 = -1329 < 0.$$

The first layer is strict as above, so D is integral. Now $T \ge \lceil 154/3 \rceil = 52 > \Omega = 25$ excludes this boundary entirely. All subsequent uses of the ordinary prefix span have $a \ge 15$.

30.3 Applying the effective-hole bound to the prefixes

For $k \le 21$, the one-budget D_1 comparison is at most $-383 - 2I + 2z$. It is strict for $z \le 191 + I$. Assume otherwise $z \ge 192 + I$. The bound $\kappa + \rho_P + \rho_Q \le \Omega$, together with v_0 above, excludes $k \ge 12$. Thus $0 \le k \le 11$ and both core sizes lie between 25 and 48.

For either core X of size r, remove its maximum R and write $Y = X \setminus \{R\}$. The last two subset-sum translates cannot be disjoint. If they were, (108) would give $\rho_X \ge T_{r-2} + 2 - 2r$. Adding the other core's effective-hole defect bound and $\kappa \ge 192 + I$ exceeds Ω in every case; the minimum surplus for the two choices is shown below. The translates therefore intersect. Their hulls overlap, and every hole of their union is a hole of one of the two constituent hull supports, so $h_X \le 2h_Y$. Hence $h_Y \ge \lceil z/2 \rceil \ge 96$. The exact increment identity also gives

$$h_Y = h_X - R + r + (\mu_X - \mu_Y) \le \kappa + \mu_X \le \Omega - \rho_{\text{other}} + 2r - 2.$$

The latter upper bound is strictly below $T_{r-1} \le |\operatorname{FS}(Y)| - 1$, with the minimum strict margin recorded in the table. Thus the effective-hole theorem applies to both prefixes on the same unit lattice.

For $26 \le r \le 48$, use the effective-hole rows in Section 28, supplemented by $(r, C_h, \beta) = (24, 39, 5)$ and $(25, 43, 5)$. All have $2\beta < 96$. Let $u_r = r + C_h + 1$ and obtain v_r from the already proved flexible inverse bounds. Explicitly one may take the maximum of $r - 14$, $u_r + 2 - 2r$ and, for $2 \le \beta \le r - 5$,

$$\min((\beta - 1)(r - \max(14, \beta + 4)),\ (\beta - 1)q + \min(t, \beta - 1)),$$

where $\max(0, u_r - 4\beta + 2) = (\beta + 1)q + t$, $0 \le t \le \beta$. The same inverse argument extends to the displayed range of β. Start after $L = \max(14, \beta + 4)$. If every later increment exceeds β, then $\rho \ge (\beta - 1)(r - L)$. Otherwise the last prefix with increment at most β has at most 2β diagonal defects by the small-increment inverse theorem. If s elements follow it, then $s \le \lfloor \rho/(\beta - 1) \rfloor$ and the final defect count is at most $2\beta + s$. Thus $\mu - 4\beta + 2 \le \rho + 2\lfloor \rho/(\beta - 1) \rfloor$. Exact integer inversion yields $(\beta - 1)q + \min(t, \beta - 1)$ above. Taking the minimum with the no-small-increment alternative proves the formula for every $2 \le \beta \le r - 5$. The size-90 arithmetic checker repeats all its parameter tests.

The exact maximum-removal identity now strengthens the forcing bound to

$$|\operatorname{FS}(D_1)| - 2s \le C + 3\Omega - \rho - \mu - 2(\rho_{P'} + \rho_{Q'}), \tag{111}$$

where P', Q' have their maxima removed. Put $U = u_a + u_b$, $V = v_a + v_b$, $V' = v_{a-1} + v_{b-1}$. The complete finite checks are

k	U	V	V'	Disjoint surplus	Prefix-hole margin	Forcing upper bound
0	304	190	177	284	127	−169
1	304	190	177	263	103	−177
2	305	191	178	247	82	−197
3	307	193	180	237	65	−229
4	309	196	182	232	51	−270
5	311	198	185	232	40	−320
6	314	201	188	237	32	−380
7	318	205	192	248	28	−452
8	322	210	196	264	27	−533
9	326	214	201	285	29	−623
10	331	219	206	311	34	−723
11	337	225	212	343	43	−835

The surplus and hole-margin columns are the minima over the two cores. The last column is $679 - 4I - U - V - 2V'$, the bound in (111). Every value is strict. It forces D_1 onto the core lattice, and the first-layer inequality forces D.

Theorem 30.2. *A zero-containing feasible integer set of size 90 has the common-core lattice structure of Theorem 20.1, with $d = 82$, $e = 74$, $m = 73$ and $G = 154$. The side of size 14 is impossible, and the two remaining cores satisfy the strict support-cardinality hole bounds.*

Proof. The preceding cases establish D-integrality before using G. Hole propagation and Theorem 19.1 then give the full post-lattice budget. In a remaining $z = 0$ case, $\kappa \le \Omega - 154$ and $\Omega - T_a \le 75$ imply $T_a - \kappa \ge 79 > 0$. This supplies both strict hole bounds; the positive-hole cases already had them. The half-step progression case and the size-14 boundary were excluded above. □

The `check90.py` verifies all displayed finite arithmetic. The theorem is a structural reduction, not by itself a cardinality exclusion.

31 The prefix lattice reduction and an exact exclusion of size 90

Theorem 31.1. *No feasible real set has 90 elements. Consequently*

$$\boxed{N \le 89.}$$

Proof. Theorem 29.1 justifies adjoining zero in a hypothetical size-90 example. Integralization and Theorem 30.2, including its separate size-14 argument, give the common lattice and every required structural hypothesis. The complete necessary-model classification below has 622492 roots, all excluded by 624102 exact leaf contradictions. The preceding descending exclusions therefore yield $N \le 89$. This is an intermediate endpoint conclusion; Section 33 excludes size 89 as well. □

31.1 Complete endpoint classification

Use the structural parameters $d = 82$, $e = 74$, $m = 73$, $I = T_a + T_b - T_{36} - T_{37}$, $\Omega = 531 - I$ and $G = 154$. The conservative low list considers $14 \le a \le 59$ and every $j = 0, \dots, 4$, retaining precisely

$$B_0 = 377 - I - 82j \ge 0 \quad (Z), \qquad B_0 - 17 \ge 0 \quad (P).$$

The size-14 structural boundary is already excluded, and its negative low cap supplies no root. The necessary simultaneous bounds are

$$\delta \le B_0 \quad (Z), \qquad \delta \le B_0 - 45 + 2z \quad (P), \qquad 1 \le z \le 14 \quad (P),$$
$$R_p + R_q + \delta \le 77 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1560 - 82j + 4z.$$

The positive-hole four-empty wrapper uses the same conservative-template argument as at size 91. All modes and contributor choices give

j	0	1	2	3	4
Z_j	152	2380	17340	65824	152320
P_j	152	2380	16184	65824	130560

Their sum is 453116.

For high holes, write $(a, b) = (36 - k, 37 + k)$ up to reflection and use

$$C_{90}(a, z) = 531 - k(k + 1) - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(52, z, 154 - 2z).$$

The deliberately conservative domain is $0 \le k \le 22$, $15 \le z \le 531 - k(k + 1)$, comprising 7843 integer states. All 713 effective-hole constants are rechecked directly. The largest cover cap is 289, so $j \le 3$. The all-active mass bound is $1602+4z$. The 6564 retained unoriented states have counts $(2920, 2223, 1156, 265)$; their 163 intervals have counts $(33, 67, 45, 18)$ for $j = 0, 1, 2, 3$. Both side orientations are distinct. All modes then give high-root counts $(264, 9380, 52020, 107712)$, totaling 169376. Thus there are $453116 + 169376 = 622492$ original roots.

The leading, nested and core-quantile thresholds are 163, 155 and 78. For $j \ge 3$, the low residual cap is at most 131 and the high cap at most 43, both below $K \ge 163$. Hence leading centers are integral, and the retained endpoints suffice for full-D geometry while missing outer coordinates keep weak order. For $j \le 2$, Lemma 27.4 applies, with the redundant 578-mode integer-left-inverse check matched to every ordinary two-empty model.

31.2 Selected geometric and fixed-score proofs

Of the original roots, 616746 have ordinary exact contradictions. Their 4626-vector pool has 103362 positive integer weights, maximum weight 1554 and minimum contradiction magnitude 1. The other 5746 original roots divide into 5663 direct geometric proofs and 83 complete trees. The direct models use only the full-D raw and clipped group inequalities already proved. Every used labeled row is matched against its freshly regenerated complete model.

Every branch root has an explicit retained-center identity $WC = I_8$. Only after checking it are strict outer gaps and fixed-score integer dichotomies used. The 71 geometric trees have 236 splits and 307 leaves: 178 normal or outside alternatives, five core or large-loss alternatives, and 53 inner-extreme alternatives. The 12 fixed-score trees have 1374 splits and 1386 leaves: 715 nonpositive-score, 635 high-score, and 24 inner-extreme alternatives. These are exactly the proved above split types; no integer-coordinate or integer-form split is selected.

The fixed-score loss realization is the one in Section 29. Missing-largest charges are used only for zero-hole roots with the required retained lower contributor. The two positive-hole fixed roots receive no such charge. Their high and nonpositive indicators remain valid, since a score above S_p or S_q or at most zero is outside its respective positive target even when that target has holes. Every family excludes from its high-indicator sum any label already charged separately, so the deficit allocation does not double count. All 110 score formulas and the 26 disjoint deletion groups are regenerated.

The direct proof records contain 123118 positive weights, with largest weight 15258. The 307 geometric leaf vectors have 5410 weights and largest weight 348; the 1386 fixed-score leaf vectors have 42822 weights and largest weight 2082. All satisfy the exact nonnegative-column and negative-right-side contradiction criterion.

31.3 Exact finite lemma for size 90

The required finite assertion has 622492 canonical roots and 624102 leaf contradictions. The data are in `proofs/exact89`. The complete ordinary/direct/tree partition gives

$$616746 + 5663 + 307 + 1386 = 624102.$$

These are the leaf contradictions. There are 83 tree roots and 1610 binary splits. The exact verification reconstructs every root, every branch, all retained-center identities and high-hole intervals. It also verifies the complete 2340-case deleted-interval base and the size-90 structural scalar arithmetic, including the size-14 boundary.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact89.py` regenerates the complete ordered root universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

32 The common core lattice at size 89

The next endpoint requires a further structural reduction. The crude simultaneous inverse criterion can fail even when both core supports are non-progressions. A dense large summand repairs precisely those cases. The proof below uses no finite AP70 table and no effective-hole table to obtain the lattice. Effective-hole bounds enter only the later finite high-hole classification.

Theorem 32.1. *Suppose that a feasible integer set A contains zero and has size 89. Then $|B| = 8$, and the two peeled layers lie on a common core lattice. After normalization, D is integral and*

$$d = 81, \quad e = 73, \quad m = 72, \quad (a, b) = (36 - k, 36 + k), \quad I = k^2,$$

$$\Lambda = 1333 + I, \qquad \Omega = 546 - I, \qquad C = -883 - I.$$

The core supports are individually primitive on that lattice. Their hole counts satisfy

$$h_P, h_Q \le \kappa, \qquad z = h_P + h_Q - \kappa, \qquad 0 \le z \le \min(h_P, h_Q),$$

and the post-lattice conclusions are

$$w \le 4z, \qquad T \ge \kappa + 152 - 3z, \qquad T \ge z, \qquad T \ge 51, \qquad T + \rho \le \Omega. \tag{112}$$

Every surviving model has $a, b \ge 15$. This theorem does not assert that the original eight outer elements all belong to the core lattice.

32.1 Bootstrap and the side of size 13

If $|B| \le 7$, one sign is absent from D and the one-sided core has at least 73 nonzero elements. Its support would satisfy

$$s \ge 1 + T_{73} = 2702 > 24 \cdot 89 + 77 = 2213,$$

contrary to (76). Hence $|B| = 8$. The unpaired-value bound gives $q^2 \le 96 \cdot 73 + 1073 - 73^2 = 2752$. Since q is even, $q \le 52$; thus the core has at least ten opposite pairs and $a, b \ge 10$. Zero is not a diagonal exception. The algebraic identity (77) is valid already here; only the small-side ρ term may be negative. Since $b \ge 36$, the universal lower bound $\rho_P \ge -8$ and $\rho_Q \ge 0$ imply $\Omega \ge -8$. Therefore $k \le 23$ and $a \ge 13$.

At $a = 13, b = 59, k = 23$, one has $\Omega = 17$. If the larger support were not an AP, the non-progression theorem of Section 30 would give $\rho_Q \geq 54$, incompatible with $\rho_P \geq -8$ and the budget. Thus the larger support is an AP. Its size is at least 1771, whereas $\kappa \leq 25$. The residue bound following (106) forces the smaller support into its lattice, and the same gap formula gives $h_P = \kappa, h_Q = 0, z = 0$.

The maximum-deleted small prefix has size 12, so its ρ-value must be bounded below by -8, not by zero. The exact maximum-removal recurrence from Lemma 9.4 gives $+10$ on the small side in place of $+2$. The resulting $+16$ correction to (85) yields

$$|\operatorname{FS}(D_1)| - 2s \leq C + \Omega + \rho - \mu + 16 \leq -337 - 2 \cdot 529 + 16 = -1379 < 0.$$

For the outer layer its available slack is at most $\Omega - \rho \leq 25$; the strict comparison is $\Lambda = 1862 > 4 \cdot 81 - 10 + 25 = 339$. Thus both layers are integral. Only now may one invoke $G = 152$, giving $T \geq 51$, whereas the budget gives $T \leq 25$. This excludes $a = 13$. Henceforth $a, b \geq 14$, $0 \leq k \leq 22$, and all pre-lattice budget terms are nonnegative.

32.2 A progression support and its half step alternative

Put $U = \operatorname{FS}(P)$, $V = \operatorname{FS}(Q)$ and $u = |U|$, $v = |V|$. If an AP side has at least 36 coefficients, its support has at least 667 points. Since $\Omega \leq 546$, the residue and gap arguments above force the other support onto its lattice, with no long gap. Thus the AP side has zero holes, the other has κ holes, and $z = 0$.

Otherwise name the smaller AP side P. Exact quadratic arithmetic for $14 \leq a \leq 36$ gives

$$\Omega < 2T_a, \qquad \Omega - T_a \leq 102. \tag{113}$$

The residue bound permits at most two classes for V modulo the AP step. With one residue, (106) permits at most one gap of length at least u. If it exists, the reflection argument in (107)–(108) gives $\mu_Q \geq T_{b-2}$ and $\rho_Q \geq T_{b-2} + 2 - 2b$. A long gap also requires $\kappa \geq u - 1 \geq T_a$, so $\Omega \geq T_a$. These rows have $16 \leq a \leq 32$ and $40 \leq b \leq 56$, and

$$T_{b-2} + 2 - 2b - \Omega \geq 133 > 0.$$

This contradicts the budget. Therefore this case has $h_P = 0$, $h_Q = \kappa$ and $z = 0$.

With two residues, exactly one off-lattice coefficient would likewise give two disjoint copies of the remaining support and the same contradiction. At least two have a common nonzero residue r, so $2r = 0$ modulo the AP step. Normalize the half-step. Then $U = 2[0, u-1]_{\mathbb{Z}}$ and Q has $t \geq 2$ odd coefficients. If w_0 is the least odd coefficient and η is the sum of the holes in the two parity-fiber hulls, the fiber version of (106) gives

$$\kappa = u - 1 + \eta, \qquad h_P = u - 1, \qquad h_Q = \kappa - u + w_0, \qquad z = w_0 - 1.$$

Here no fiber gap reaches u, since $\kappa < 2(u-1)$; reflection leaves exactly $w_0 - 1$ ordinary lattice positions outside the two fiber hulls.

We first show $w_0 \leq u$. In these rows $u \geq T_a + 1 > 2b$. If $w_0 \geq u+1$, positivity and distinctness imply

$$\sum Q \geq (b-t)(b-t+1) + tw_0 + t(t-1).$$

After subtracting w_0, this quadratic is minimized over integers $t \geq 2$ at $t = 2$. It follows that

$$|V| \geq b^2 - 3b + 2u + 6 - \kappa.$$

But $\mu_Q + \kappa \leq \Omega + 2b - 2$ gives $|V| \leq T_b + \Omega + 2b - 1 - \kappa$. Their difference is at least

$$\frac{b^2 - 11b}{2} + 2u + 7 - \Omega \geq 1065 > 0,$$

a contradiction. Consequently $h_P, h_Q \le \kappa$. In fact $w_0 \le 13$. If $w_0 \ge 15$, minimize the same mass expression over real t to obtain

$$\sum Q - w_0 - T_b \ge 7b - \frac{289}{8}.$$

The identity $\mu_Q + \kappa = \sum Q + u - w_0 - T_b$, the budget and (113) give the opposite bound $\sum Q - w_0 - T_b \le \Omega - u + 2b - 2 \le 2b + 99$. These are inconsistent for $b \ge 40$, so $z \le 12$.

The half-step case needs its sharper AP span estimate before lattice forcing. Since P is even, $R_P \le 2(a + \mu_P)$, while

$$R_Q \le b + \mu_Q + \kappa - T_a - \mu_P + 12.$$

Substitute these into the exact pre-integrality comparison, using $\mu \le \rho + 2m - 4$ and $\rho + \kappa + \ell_2 \le \Omega$ once. This gives

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\le 8m + 2a - 2T_a + 30 - \Lambda + \Omega \\ &= 2483 + 2a - 4T_a - 2T_b \le -1021 < 0. \end{aligned}$$

The outer layer is strict because $\Lambda \ge 1333 > 4 \cdot 81 - 10 + 546 = 860$. Thus D is integral. Now $G = 152$ gives $T \ge \kappa + 152 - 3z \ge T_a + 116 > \Omega$, contradicting (113). All surviving AP cases therefore have $z = 0$; their ordinary lattice forcing is completed below.

32.3 The dense large summand repair

Suppose both supports are non-progressions. Write

$$\nu(r) = \begin{cases} r - 5, & r \ge 21, \\ r - 14, & 14 \le r < 21. \end{cases}$$

The large-side inverse condition is automatic since $\kappa \le \Omega < T_b - 1$. A deliberately conservative sufficient small-side condition for Stanchescu's Corollary 2 is

$$\mu_P + \rho_P + \rho_Q > \Omega - T_a + 2. \tag{114}$$

Using $\mu_P \ge a - 3$ and $\rho_X \ge \nu(|X|)$ shows this condition unless $9 \le k \le 17$. Whenever it holds, the corollary gives the common-lattice hull bounds, including the strict inequalities $h_P < u - 1$ and $h_Q < v - 1$.

When (114) actually fails, subtracting the same lower bounds gives the following exhaustive list. The last column bounds $b - \lfloor \rho_Q/3 \rfloor$ from below.

k	a	b	Ω	Upper bound on ρ_Q	Prefix size
9	27	45	465	43	31
10	26	46	446	53	29
11	25	47	425	60	27
12	24	48	402	64	27
13	23	49	377	65	28
14	22	50	350	63	29
15	21	51	321	58	32
16	20	52	290	59	33
17	19	53	257	48	37

Lemma 32.2. *Let X be a positive real r-set with $45 \le r \le 53$, $\eta = \rho(X) \le 65$ and $r - \lfloor \eta/3 \rfloor \ge 27$. There is a scale $t > 0$ such that X/t is integral and primitive, and*

$$\operatorname{FS}(X/t) = [0, L]_{\mathbb{Z}} \setminus (H \cup (L - H)), \quad H \subseteq [1, 8]_{\mathbb{Z}}, \quad |H| \le 4.$$

In particular, the support has at most eight holes and contains $[9, L - 9]_{\mathbb{Z}}$.

Proof. Let X_i be the increasing prefixes, with nonnegative excess increments y_i. The recurrence is $\rho_i \ge \rho_{i-1} + y_i - 2$; threshold 14 gives $\rho_i \ge 0$ for $i \ge 14$. If $y_i \ge 5$ for all $15 \le i \le r$, then $\eta \ge 3(r-14) > 65$. Let $j \ge 15$ be the last index with $y_j \le 4$. The recurrence gives $j \ge r - \lfloor \eta/3 \rfloor \ge 27$. Set $f = y_j \le 4$. The counted-hole lemma applies because $j \ge 2f + 5$:

$$X_j = t([1, j+f]_{\mathbb{Z}} \setminus F), \quad |F| = f, \quad \mathrm{FS}(X_j/t) = [0, L_j]_{\mathbb{Z}} \setminus (H \cup (L_j - H)),$$

where $H \subseteq [1, 2f]$ and $|H| \le f$. The retained j coefficients have gcd one, since a divisor at least two would allow at most $\lfloor (j+f)/2 \rfloor < j$ positions.

No later coefficient can leave $t\mathbb{Z}$. At the first such index i, the two subset-sum translates are disjoint, and therefore

$$y_i = |\mathrm{FS}(X_{i-1})| - i \ge 1 + T_j - r, \qquad \rho_i \ge T_j - r - 1 \ge 324.$$

All later increments exceed four, so $\rho_r \ge \rho_i$, a contradiction. The full coefficient set is integral and remains primitive on this scale.

For $i > j$, monotonicity gives $\rho_i \le \eta$ and hence $y_i \le 67$. We verify directly that the endpoint holes persist. Suppose there are $m \ge 27$ current coefficients, their support is $[0, L] \setminus (H \cup (L - H))$, and the next integer coefficient is x. Then $L \ge T_m$ and $x > \max X_m \ge m > 8$. If $x \ge L - 16$, at least $\min(x, L+1) - 8 \ge L - 24 \ge T_m - 24 > m + 68$ support points of $x + \mathrm{FS}(X_m)$ lie above L. This exceeds the allowed increment $m + 1 + y_{m+1} \le m + 68$. Thus $x \le L - 17$. The central intervals $[9, L-9]$ and $[x+9, x+L-9]$ overlap or are adjacent, covering $[9, L+x-9]$. Because $x > 8$, the lower holes remain exactly H; reflection supplies the upper holes $L + x - H$. Induction proves the claim. □

Apply Lemma 32.2 to Q and normalize its primitive scale. If U occupied c residue classes modulo this scale, the ordered sumset bound in each class would give $|U + V| \ge u + c(v-1)$, hence $\kappa \ge (c-1)(v-1)$. But $v - 1 \ge T_b > \Omega \ge \kappa$, so $c = 1$ and all of P lies on the same lattice. We already have $h_Q \le 8$.

The interval $J = [9, L_Q - 9] \subseteq V$ has length $\ell = L_Q - 17 = v + h_Q - 18 \ge v - 18$. If g_i are consecutive gaps of U, (106) and $U + J \subseteq U + V$ imply

$$\sum_i \min(g_i - 1, \ell - 1) \le \kappa + 18.$$

Here $v \ge T_{45} + 1 = 1036$, $\kappa \le 465$, and $\kappa + 18 < \ell - 1$. Thus no term is truncated, proving $h_P \le \kappa + 18$. Consequently

$$\operatorname{diam} U \le u + \kappa + 17 \le u + v - 3, \qquad \operatorname{diam} V \le v + 7 \le u + v - 3.$$

The first uses $\kappa + 20 \le v$, and the second $u \ge 191$. Stanchescu's Theorem 4 now applies directly: the stronger bound $u + v - 3$ covers its equal- and unequal-diameter alternatives. This short-diameter theorem has no additional gcd premise. It gives $|U+V| \ge u+v-1+\max(h_P, h_Q)$, hence

$$h_P, h_Q \le \kappa, \qquad z = h_P + h_Q - \kappa \le \min(h_P, h_Q) \le 8.$$

This repairs every failed inverse row without using a post-lattice inequality to establish the lattice.

32.4 Forcing the peeled layers and proving primitivity

For a positive side X of size r, maximum R, and maximum-deleted prefix Y, let v_0 count old diagonal defects destroyed on reinsertion. The exact recurrence is

$$R = r + \rho_X + h_X + 2 - \rho_Y - h_Y - 2v_0. \tag{115}$$

For $r \geq 15$, $\rho_Y \geq 0$. The ordinary one-budget comparison gives

$$|\operatorname{FS}(D_1)| - 2s \leq C + \Omega + \rho - \mu + 2z \leq -337 - 2I + 2z. \tag{116}$$

It is strict if $z \leq 168 + I$, including every surviving AP branch and every dense-summand branch. At $a = 14, b = 58$, the prefix of size 13 requires the same $+16$ correction as before. Since $\Omega = 62$,

$$|\operatorname{FS}(D_1)| - 2s \leq -337 - 2 \cdot 484 + 2 \cdot 62 + 16 = -1165 < 0.$$

It remains that $a \geq 15$ and $z \geq 169 + I$. This cannot be an AP or a repaired dense-summand case. Both supports are non-progressions. The bounds $\kappa \geq z$ and $\rho_X \geq \nu(|X|)$ rule out $k \geq 13$; thus $0 \leq k \leq 12$ and both side sizes are at least 24. For either side of size r, with other side of size r', its final two subset-sum translates cannot be disjoint. Otherwise (108) gives

$$\rho_X \geq T_{r-2} + 2 - 2r, \qquad \rho_{X'} \geq r' - 5,$$

and, for every such k and either side,

$$T_{r-2} + 2 - 2r + (r' - 5) + 169 + I - \Omega \geq 80 > 0,$$

contradicting the budget. The final hulls therefore overlap and $h_X \leq 2h_Y$. Retaining $-h_Y$ in (115) yields $R_X \leq r + \rho_X + h_X/2 + 2$. Substitution in the original D_1 comparison now gives

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\leq C + 2\rho - \mu + z + \ell_2 \\ &\leq C + \Omega + \rho - \mu \leq -337 - 2I < 0, \end{aligned}$$

using $z \leq \kappa$ and $\rho + \kappa + \ell_2 \leq \Omega$ once. Thus D_1 lies on the core lattice in every case. Since $\Lambda \geq 1333 > 4 \cdot 81 - 10 + 546$, the outer peeled layer is strict as well. The standard overlap and hole propagation arguments now give (112) with $G = 152$.

The $a = 14$ row is now excluded entirely. A non-AP large support costs $\rho_Q \geq 53$, so $T + \rho \geq 51 + 53 = 104 > 62$. An AP large support has $z = 0$ and $T \geq 152 > 62$. No subsequent actual size-89 model uses a universal core-span row at side size 14.

Finally, the strict Corollary 2 cases are already primitive. In the surviving AP cases, $z = 0$ and $\kappa \leq \Omega - 152 < T_a$ by (113). In the dense-summand cases $z \leq 8$ gives $\kappa \leq \Omega - 128 < T_a$. In either case also $\kappa < T_b$. A coefficient gcd at least two would force at least $u - 1$ or $v - 1$ holes in its support, contradicting $h_X \leq \kappa < T_{|X|}$. This proves Theorem 32.1. The checks in `check89.py` and `check_dense_repair.py` reproduce every finite margin used above; the universal assertions are proved in the text.

33 Translated core overlap and the size 89 endpoint

Theorem 33.1. *There is no feasible set of 89 real numbers. Consequently,*

$$\boxed{N \leq 88.}$$

Proof. Theorem 31.1 already excludes all larger cardinalities. If a feasible size-89 set omitted zero, adjoining zero would give a feasible size-90 set, a contradiction. Integralize without changing its cardinality and apply Theorem 32.1. The complete necessary-model partition and exact contradictions are described below. Their finite certificate excludes every model, so $N \leq 88$. □

33.1 Complete low and high hole classification

At this endpoint let $I = (a-36)^2$, $b = 72-a$ and $B_0 = 394 - I - 81j$. The low list uses $15 \le a \le 57$ and $j = 0, \dots, 4$, retaining $B_0 \ge 0$ for zero holes and $B_0 - 16 \ge 0$ for $1 \le z \le 14$. Its simultaneous necessary bounds are

$$\delta \le B_0 \quad (Z), \qquad \delta \le B_0 - 44 + 2z \quad (P),$$
$$R_p + R_q + \delta \le 76 + B_0 + 4z, \qquad S_p + S_q + \delta - 2K \le 1542 - 81j + 4z.$$

As at the preceding endpoint, the positive-hole four-empty wrapper inherits optional rows from a more permissive zero-hole template before installing its positive-hole mass penalties. This preserves their gates. The mode counts, including selected contributor choices, are

$$(4, 70, 578, 2992, 10880) \qquad (j = 0, 1, 2, 3, 4).$$

The root counts are

j	0	1	2	3	4
Z_j	156	2450	17918	74800	184960
P_j	156	2450	16762	68816	163200

Their sum is 531668.

For high holes, set $(a, b) = (36-k, 36+k)$ up to reflection. With the previously defined effective-hole excess bounds u_a, u_b and flexible defect bounds ρ_*, the cover cap is

$$C_{89}(a, z) = 546 - k^2 - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(51, z, 152 - 2z).$$

Enumerate $0 \le k \le 21$ and $15 \le z \le 546 - k^2$, comprising 8393 raw integer states. All 656 effective-hole constants used in this classification are directly checked against their strict hypotheses. The maximum cover cap is 311, so $j \le 3$. The all-active mass upper bound is $1584 + 4z$. The 8319 retained unoriented states have counts $(3527, 2684, 1534, 574)$; the 195 intervals have counts $(43, 73, 52, 27)$, for $j = 0, 1, 2, 3$. Use both side orientations unless $a = b = 36$. The resulting high-root counts are $(340, 9940, 57800, 149600)$, totaling 217680. Thus the canonical original list contains exactly

$$531668 + 217680 = 749348 \quad \text{roots.}$$

The leading, nested and core-quantile thresholds are respectively 161, 153 and 77. For $j \ge 3$, ordinary models keep weak real outer order; their residual caps are at most 151 in the low regime and 68 in the high regime, below $K \ge 161$. The leading components therefore contribute, and retained centers align the endpoints actually used in full-D geometry. They do not put missing outer coordinates on the lattice. For $j \le 2$, Lemma 27.4 applies without a leading-contribution premise, and all 578 two-empty center matrices have independently reconstructed integer left inverses. The checker validates all relevant gates on every ordinary model; 2660 fast/full representatives provide a separate parameter check.

33.2 Actual group deficits and the final overlap inequality

The direct selected models use the established full-D group geometry, raw overlap and clipped-hull inequalities. Their auxiliary variables have one simultaneous realization against the actual positive target. We record the small additional inequality needed for the final five roots.

Lemma 33.2. *Let D be integral, $R_p, R_q \in \mathbb{Z}_{\ge 0}$, $E \subseteq D \cap [-R_q, R_p]$, $|D| = d$ and $|E| = e$. Let $M_p = \max D > 0$ and let an outer element $p > M_p$ have $p - M_p \in \mathbb{Z}$. For any center c, put*

$$A_0 = c + 2D, \qquad A_1 = c + p + D, \qquad K = |2D|, \qquad H = 2e - R_p - R_q - p + M_p - 1.$$

For a common target F, let $l_0 = K - |A_0 \cap F|$ and $l_1 = d - |A_1 \cap F|$. Then the two-component capacity deficit satisfies

$$K + d - |(A_0 \cup A_1) \cap F| \geq H + \max(l_0, l_1).$$

The same conclusion holds with any lower bound $\lambda \leq l_1$ in place of l_1.

Proof. The sets A_0, A_1 contain $c + M_p + E$ and $c + p + E$, respectively. Those two e-element sets lie in a translate of an integer interval of cardinality $R_p + R_q + p - M_p + 1$. Hence $|A_0 \cap A_1| \geq H$. This remains valid when $H < 0$. The exact identity

$$K + d - |(A_0 \cup A_1) \cap F| = |A_0 \cap A_1| + |(A_0 \cup A_1) \setminus F|$$

and $|(A_0 \cup A_1) \setminus F| \geq \max(l_0, l_1)$ prove the claim. No contribution by A_0 is required. □

For a larger labeled component family, partition off this pair and add the individual lower losses of all remaining components. Capacity deficits are superadditive under such a partition: the union has cardinality at most the sum of the subunions' cardinalities. The positive and negative families use disjoint covering labels and combine through the established compact full-D budget. Reflection swaps M_p, p with M_q, q.

In the implementation, start from the ordinary baseline

$$-G_s + L_s + \sum_{i \in J_s} \lambda_i \leq 0,$$

where G_s is the actual full-family deficit, L_s is the leading loss, and λ_i is the sum of the ith D component's two endpoint losses. For each retained own component i, Lemma 33.2 gives the two alternative rows

$$-G_s + L_s + \sum_{j \neq i} \lambda_j - R_p - R_q - p_i + M_p \leq 1 - 2e,$$
$$-G_s + \lambda_i + \sum_{j \neq i} \lambda_j - R_p - R_q - p_i + M_p \leq 1 - 2e,$$

and their negative-side reflections. These rows impose alternatives on the same actual deficit; they do not sum overlapping pair groups. They use the raw-overlap identity, so no member loss is added to an already clipped hull bound. At size 89, $1 - 2e = -145$. Three retained own slots on each sign produce exactly twelve added rows. The required relative integrality follows from the selected roots' checked integer left inverses.

Two other selected roots use whole-side deficits. On each sign, group the leading component of capacity K and all seven retained unpaired D components. If G_s^{all} is their actual target deficit and the positive target has cardinality at most S_s, then

$$G_s^{\mathrm{all}} \geq K + 7 \cdot 81 - S_s, \qquad G_s^{\mathrm{all}} \geq L_s + \sum_{i \in J_s} \lambda_i.$$

The two sign groups are disjoint in covering labels, giving $G_p^{\mathrm{all}} + G_q^{\mathrm{all}} \leq \delta$. These use separate whole-side variables, replacing rather than adding the old smaller-family losses. No maximum-member loss is added to the clipped target-cardinality bound. The separately proved fixed-score losses may further strengthen this whole-side covering budget: their fixed labels are disjoint from the grouped components and from all precharged labels.

33.3 Selected roots and exhaustive binary proofs

The ordinary proof excludes 739722 roots. Its 5370 integer Farkas vectors contain 119594 positive weights, with maximum weight 1440 and minimum contradiction magnitude 1. The other 9626 roots partition into 9472 direct geometric proofs and 154 complete binary trees. The direct roots comprise 6462 low and 3010 high cases. Their 9472 proof records use 618 distinct vectors and 194302 positive weights. Every used labeled row is matched to a regenerated model.

All 154 tree roots have checked retained-center identities $WC = I_8$. Strict outer gaps and integer fixed-score splits are used only after that root-specific check. Twelve high roots include selected p/q variants; their selected beta centers are included in the matrices. The two selected both/sole paired-component branches use the global 81/80 quantiles and reserve loss coordinates 52 and 53. Individual paired-member losses are charged only in the applicable sole branch. Every split keeps both children and is a proved catalog disjunction. The complete census is 500 normal, 13 core, 682 inner-extreme, two both-or-sole, 1300 fixed-high and 1515 fixed-nonpositive splits. No arbitrary integer-coordinate or new midpoint split is selected.

The fixed-score rows retain the zero/positive-hole distinction. Four of the thirteen low fixed-score roots have zero holes and the required missing largest slots and retained lower contributors; they receive eight missing-largest losses and 102 remaining high-label candidates. Nine positive-hole roots, and the single high fixed-score root, receive no such saving and use 110 high-label candidates. Any additional nonpositive-label group losses are disjoint from the high-label and separately precharged labels. The two whole-side roots and five translated-overlap roots have zero holes and the same valid eight-loss saving. No criterion sufficient only for contribution is used to choose a contributor.

The complete selected tree partition is

Selected class	Roots	Nodes	Binary splits	Leaves
Low minimal geometry	56	360	152	208
High minimal geometry	77	1579	751	828
Low fixed score	13	4813	2400	2413
High fixed score	1	133	66	67
Low whole side	2	626	312	314
Low translated core overlap	5	667	331	336
Total	154	8178	4012	4166

The whole-side roots are original low indices 2427 and 2433. The final five are indices 2428 through 2432, corresponding to $a = 34, 35, 36, 37, 38$, zero holes, branch A, missing units `pair1`, `pair2`, `p0`, `q0`, contributor choice zero, and no variant. Their added dependency consists exactly of the twelve rows in Lemma 33.2, on top of the already proved minimal geometry and fixed-score models. Exploratory midpoint, dense-mass, expanded-side, restricted-pair and nine-hull techniques are not dependencies of this proof.

33.4 Exact finite lemma for size 89

The required finite assertion has 749348 canonical roots and 753360 leaf contradictions. The data are in `proofs/exact88`. The complete partition is 739722 ordinary roots, 9472 direct geometric roots, and 154 tree roots. The trees have 4012 splits and 4166 leaves, so 749348=739722+9472+154 and 753360=739722+9472+4166. The source-to-model reconstruction includes the dense-summand repair, all high-hole intervals, every translated-core row, all 578 two-empty witnesses, and the 2340-case non-AP finite base.

Every leaf is checked by the integer criterion of Lemma 1.2; every path row must be one of the proved alternatives. The verifier `verify_exact88.py` regenerates the complete ordered root

universe and all required children, rather than accepting stored counts or an optimizer status. These assertions are finite mathematical proof obligations in the supplied data.

34 The common core lattice at size 88

The next structural reduction retains the order of implications used above: obtain a common core lattice, force both peeled layers onto it, and only then use the strengthened post-lattice budget. A primitive large-side effective-hole argument repairs the new failed inverse rows.

Theorem 34.1. *Let A be a feasible integer set of size 88 containing zero. Then $|B| = 8$, and after a common normalization the peeled set D is integral. With $a \le b$, the parameters are*

$$d = 80, \quad e = 72, \quad m = 71, \quad (a, b) = (35 - k, 36 + k), \quad I = k(k + 1),$$

$$\Lambda = 1297 + I, \qquad \Omega = 560 - I, \qquad C = -853 - I.$$

The core supports are individually primitive and satisfy

$$h_P, h_Q \le \kappa, \qquad z = h_P + h_Q - \kappa, \qquad 0 \le z \le \min(h_P, h_Q).$$

Every surviving case has $a, b \ge 15$, and

$$w \le 4z, \qquad T \ge \kappa + 150 - 3z, \qquad T \ge z, \qquad T \ge 50, \qquad T + \rho \le \Omega. \tag{117}$$

The original eight covering elements need not all be integral on this lattice merely by this theorem.

34.1 Bootstrap and the three small side boundaries

The deficient-sign argument gives $|B| = 8$, since otherwise

$$s \ge 1 + T_{72} = 2629 > 24 \cdot 88 + 77 = 2189.$$

The unpaired-value bound is $q^2 \le 2801$. Here q is odd, so $q \le 51$; there are at least ten opposite pairs in E and $a, b \ge 10$. The exact budget (77) applies algebraically; its only possibly negative terms are the two ρ values. Since $b \ge 36$, $\rho_Q \ge 0$ and $\rho_P \ge -8$ imply $k \le 23$ and $a \ge 12$. As before, $\rho \le \mu$.

The boundary calculations are

a	b	k	I	Ω	Lower ρ_P	Upper κ
12	59	23	552	8	-8	16
13	58	22	506	54	-8	62
14	57	21	462	98	0	98

Write K_0 for the last column. In all three rows, $K_0 < T_a - 1$ and $K_0 < T_b - 1$. Stanchescu's strict Corollary 2 therefore gives the common lattice, $h_P, h_Q \le \kappa$, and both strict support-cardinality hole bounds. The maximum-deleted small prefix has size 11, 12 or 13; use $\rho_Y \ge -8$ in (115), adding 16 to the ordinary D_1 comparison. Hence

$$|\operatorname{FS}(D_1)| - 2s \le -293 - 2I + 2K_0 + 16,$$

whose three values are $-1349, -1165, -1005$. The outer-layer comparisons $\Lambda > 4 \cdot 80 - 10 + K_0$ are also strict. Both peeled layers are consequently integral before $G = 150$ is invoked.

Now $T \ge 50$ excludes $a = 12$, since its available budget is at most 16. For $a = 13$ or 14 with a non-AP larger support, respectively,

$$T + \rho \ge 50 - 8 + 53 = 95 > 54, \qquad T + \rho \ge 50 + 52 = 102 > 98.$$

If that larger support is an AP, its strict hole bound forces step one on the common lattice, so $h_Q = 0$ and $z = 0$. Then $T \geq 150$ exceeds the available budgets 62 and 98. All three boundaries are excluded. Thus $0 \leq k \leq 20$, $a, b \geq 15$, and every later maximum-deleted prefix has size at least 14.

34.2 Progression supports with the necessary extra defect term

An AP side with at least 36 coefficients has support size at least 667, whereas $\kappa \leq \Omega \leq 560$. The residue and gap formula (106) forces the other support into the same lattice, with no long gap: $h_{\mathrm{AP}} = 0$, $h_{\mathrm{other}} = \kappa$, $z = 0$. This also handles two AP supports by choosing the larger side. Otherwise P is the smaller AP side and Q is non-AP. For every $0 \leq k \leq 20$,

$$\Omega < 2T_a, \qquad \Omega - T_a \leq 128. \tag{118}$$

At most two residue classes of $V = \mathrm{FS}(Q)$ occur modulo the AP step.

In one residue, at most one gap can have length at least $u = |\mathrm{FS}(P)|$. A unique long gap would give two disjoint final translates by (107)–(108), hence $\rho_Q \geq T_{b-2} + 2 - 2b$. It also costs $\kappa \geq u - 1 \geq T_a$. The rows with $\Omega \geq T_a$ have $15 \leq a \leq 32$, $39 \leq b \leq 56$, and

$$T_{b-2} + 2 - 2b - \Omega \geq 79 > 0.$$

Thus no long gap exists and $z = 0$.

With two residues, one off-lattice coefficient would give the same contradiction. At least two have the nonzero residue; their sum forces that residue to have order two. Normalize the half-step. Exactly as in Section 32, if w_0 is the least odd coefficient, then

$$h_P = u - 1, \qquad h_Q = \kappa - u + w_0, \qquad z = w_0 - 1.$$

The mass inequality for $t \geq 2$ odd coefficients is

$$\sum Q \geq (b - t)(b - t + 1) + tw_0 + t(t - 1).$$

If $w_0 > u$, use $u \geq T_a + 1 > 2b$ to minimize at $t = 2$. The lower-minus-upper support-size margin becomes

$$(b^2 - 11b)/2 + 2u + 7 - \Omega \geq 1003 > 0.$$

Therefore $w_0 \leq u$ and $h_P, h_Q \leq \kappa$. If $w_0 \geq 15$, minimizing over real t gives $\sum Q - w_0 - T_b \geq 7b - 289/8$; the budget and (118) give the upper bound $2b + 125$. These conflict for $b \geq 39$. Hence $w_0 \leq 13$ and $z \leq 12$.

The AP-specific span estimates and the same one-budget substitution give

$$|\mathrm{FS}(D_1)| - 2s \leq 2453 + 2a - 4T_a - 2T_b \leq -955 < 0.$$

Also $\Lambda \geq 1297 > 4 \cdot 80 - 10 + 560 = 870$, forcing D integral. Only now $G = 150$ yields $T \geq T_a + 114$. At this endpoint that inequality alone is insufficient, since $\Omega - T_a$ reaches 128. Retain the fact that Q is non-AP:

$$T + \rho \geq T_a + 114 + (b - 5), \qquad T_a + 114 + (b - 5) - \Omega \geq 28 > 0.$$

This excludes the half-step case. Every surviving AP case has $z = 0$.

34.3 The primitive large summand repair

Assume both supports are non-progressions, and retain the notation $\nu(r)$ from Section 32. The large-side inverse condition follows from $\Omega < T_b - 1$. The sufficient small-side condition is again

$$\mu_P + \rho_P + \rho_Q > \Omega - T_a + 2.$$

The lower bounds $\mu_P \geq a - 3$ and $\rho_X \geq \nu(|X|)$ establish it except possibly at $k = 7, \ldots, 18$. In the strict cases Stanchescu supplies the common lattice and strict support-cardinality hole bounds. Suppose instead that this sufficient condition actually fails. Put $r_p = \nu(a)$, $r_q = b - 5$, and

$$R = \Omega - T_a + 2 - (a-3) - r_p, \qquad U_{\max} = \Omega - r_p - r_q + 3,$$
$$K_{\max} = \Omega - r_p - r_q, \qquad V_{\min} = T_b + b - 2, \qquad U_{\min} = T_a + a - 2.$$

Then $\rho_Q \leq R$, $u \leq U_{\max}$, $\kappa \leq K_{\max}$, $v \geq V_{\min}$, and $u \geq U_{\min}$.

First scale by the coefficient gcd of Q in the initial integer realization. The set Q is now primitive and integral; P has not yet been shown integral on this scale. The effective-hole lemma of Section 20 has the following primitive form: for admissible parameters β, C_h, if a positive integer r-set is primitive and its support has $h > 2\beta$ holes, then $\mu \geq r + C_h + 1$. Indeed, in the proof of Lemma 10.2, the extra hypothesis $h < |\,\mathrm{FS}\,| - 1$ is used to obtain coefficient primitivity. When primitivity is given directly, a positive-hole support cannot be an AP, and the remaining tail, counted-hole, off-scale and endpoint-bridge arguments apply with the same parameter tests. Thus this use on the intrinsic primitive lattice of Q does not assume an unproved upper bound on its holes.

The required explicit witnesses are

k	a	b	R	β	C_h	Forced lower ρ_Q	$H = 2\beta$
7	28	43	52	3	51	53	6
8	27	44	66	3	70	67	6
9	26	45	77	4	82	78	8
10	25	46	85	4	93	86	8
11	24	47	90	4	97	91	8
12	23	48	92	4	100	93	8
13	22	49	91	4	98	92	8
14	21	50	87	4	91	88	8
15	20	51	89	4	92	90	8
16	19	52	79	3	79	80	6
17	18	53	66	2	61	67	4
18	17	54	50	2	38	51	4

Every C_h is checked directly against both strict inequalities in (80), using its explicit K_0, D_0, M_0, J_0 definitions. If $h_Q > 2\beta$, insert $u_0 = b + C_h + 1$ into the old flexible defect inverse: for each $\gamma = 2, \ldots, 10$, write

$$\max(0, u_0 - 4\gamma + 2) = (\gamma + 1)q + s, \qquad 0 \leq s \leq \gamma,$$

and use

$$\rho_Q \geq \min\big((\gamma - 1)(b - 14), (\gamma - 1)q + \min(s, \gamma - 1)\big),$$

together with $b - 14$ and $u_0 + 2 - 2b$. The displayed maximum lower bounds all exceed R. Consequently $h_Q \leq H \leq 8$ on its primitive lattice.

Partition U into c residue classes modulo that lattice. The ordered sumset bound gives $\kappa \geq (c-1)(v-1)$. Since $K_{\max} < V_{\min} - 1$, $c = 1$ and P is integral there too. Let $A_0 = \operatorname{diam} U$, $B_0 = \operatorname{diam} V$, and $h = h_Q \leq H$. The checked inequality

$$V_{\min} > U_{\max} + K_{\max} - 1$$

has minimum margin 99. If $A_0 > B_0$, the two disjoint sets V and $A_0 + V$ would give $|U + V| \geq 2v > u + v - 1 + \kappa$. Hence $A_0 \leq B_0$. Their two endpoint hulls cover $[0, A_0 + B_0]$, so

$$|U + V| \geq |V \cup (A_0 + V)| \geq A_0 + B_0 + 1 - 2h = A_0 + v - h.$$

Thus $h_P \leq \kappa + h_Q \leq \kappa + H$. The additional verified margins

$$V_{\min} - K_{\max} - H - 2 \geq 536 > 0, \qquad U_{\min} - H - 2 \geq 162 > 0$$

put both support diameters at most $u + v - 3$. Stanchescu's short-diameter Theorem 4, which has no extra gcd premise, now yields $h_P, h_Q \leq \kappa$ and $0 \leq z = h_P + h_Q - \kappa \leq H \leq 8$.

34.4 Layer forcing and final primitivity

All remaining sides have size at least 15. The ordinary one-budget comparison gives

$$|\operatorname{FS}(D_1)| - 2s \leq C + \Omega + \rho - \mu + 2z \leq -293 - 2I + 2z.$$

This is strict for $z \leq 146 + I$, including the AP and repaired cases. If $z \geq 147 + I$, both supports are non-AP, and the defect lower bounds with $\kappa \geq z$ exclude $k \geq 13$. For $0 \leq k \leq 12$, both side sizes are at least 23. If either side's final translates were disjoint, its size r and the other side's size r' would satisfy

$$T_{r-2} + 2 - 2r + (r' - 5) + 147 + I - \Omega \geq 30 > 0,$$

contradicting the exact budget. Their hulls therefore overlap. For the maximum-deleted prefix Y this gives $h_X \leq 2h_Y$; (115) then yields $R_X \leq r + \rho_X + h_X/2 + 2$. Substitute once to obtain

$$|\operatorname{FS}(D_1)| - 2s \leq C + 2\rho - \mu + z + \ell_2 \leq C + \Omega + \rho - \mu \leq -293 - 2I < 0.$$

Hence D_1 is integral. The already stated strict outer-layer bound forces D onto that lattice. Hole propagation and the sharp four-extreme theorem now give (117) with $G = 150$.

The strict inverse cases were already primitive. In a surviving AP case, $z = 0$ and $\kappa \leq \Omega - 150 < T_a \leq u - 1$, and similarly for Q. In a repaired case, Q is intrinsically primitive and $\rho_Q \geq b - 5 \geq 38$; with $z \leq 8$, (118) gives

$$\kappa \leq \Omega - 126 - \rho_Q \leq T_a + 128 - 126 - 38 = T_a - 36 < u - 1.$$

A nontrivial coefficient gcd for P would contradict $h_P \leq \kappa$. This proves Theorem 34.1. The standalone `check88.py` verifies every finite margin and effective-hole witness used here; it does not itself exclude size 88.

35 The size 88 endpoint and exhaustive loss covers

Theorem 35.1. *There is no feasible set of 88 real numbers. Consequently,*

$$\boxed{N \leq 87.}$$

Proof. Theorem 33.1 already excludes every larger cardinality. A feasible size-88 set omitting zero would become a forbidden size-89 set after adjoining zero. Therefore a hypothetical size-88 example contains zero; integralize it and apply Theorem 34.1. The complete classification and necessary inequalities below assign it to a root of the canonical finite universe. The required exact contradiction at every root excludes it, giving $N \leq 87$.

□

35.1 Three missing units give alignment without contribution

At size 88 a low three-empty residual cap may reach 170, above $|2D| \geq 159$. Consequently the old leading-contribution shortcut is not enough to obtain every outer coordinate's integrality. An additional finite algebraic theorem resolves exactly this issue.

Recall the twenty center forms

$$\begin{aligned} c_p &= p_1 - p_2 - p_3 - p_4, & \mathtt{p0}, \ldots, \mathtt{p3} &= c_p + p_1, \ldots, c_p + p_4, \\ \mathtt{p4}, \ldots, \mathtt{p7} &= c_p - q_1, \ldots, c_p - q_4, & \mathtt{p8} &= p_2 - p_3 - p_4, \\ \mathtt{p9} &= 2p_2 - p_1 - p_3 - p_4, \end{aligned}$$

and their reflected q forms. The paired indices are 4 and 9 in branch A, and 4 and 5 in branch B. The existing classification has eighteen covering units. Reflect to choose an actual positive contributor of the first nonempty pair; include each legitimate selected contributor of the second pair, with the appropriate beta variant where required. For exactly three missing units, the complete finite enumeration has 2992 ordered modes. For each, reconstruct the matrix C of actually contributing centers in the eight outer coordinates and check a supplied integer matrix W satisfying

$$WC = I_8.$$

The checker enumerates the complete ordered mode list and verifies every product entry with integers; the largest absolute witness coefficient is six. Since Cx is integral, $x = W(Cx)$ is integral. The earlier analytic theorem covers at most two missing units. Thus all eight outer coordinates are integral when $j \leq 3$. This theorem supplies alignment, not leading contribution: the leading loss inequalities still require their own strict cap gates. For $j \geq 4$, missing outer coordinates require a separate root-specific witness before strict outer gaps or fixed-score integer branches are used.

35.2 The canonical low and high families

Put $b = 71 - a$, $I = T_a + T_b - T_{35} - T_{36}$, and $B_0 = 410 - I - 80j$. The low zero-hole families retain $B_0 \geq 0$ for $j = 0, \ldots, 5$. Positive low-hole families use $1 \leq z \leq 14$ and $B_0 - 15 \geq 0$ for $j = 0, \ldots, 4$. The side range is $15 \leq a \leq 56$ and the simultaneous rows include

$$\begin{aligned} \delta &\leq B_0 \quad (Z), & \delta &\leq B_0 - 43 + 2z \quad (P), \\ R_p + R_q + \delta &\leq 75 + B_0 + 4z, & S_p + S_q + \delta - 2K &\leq 1524 - 80j + 4z. \end{aligned}$$

The positive four-empty wrapper again starts from a more permissive zero-hole template before installing its correct positive-hole rows. The complete low counts are

j	0	1	2	3	4	5
Z_j	160	2520	18496	77792	217600	177072
P_j	160	2520	17340	71808	195840	0

They total 781308. The complete mode counts for $j = 0, \ldots, 5$ are $4, 70, 578, 2992, 10880, 29512$.

For high holes, $(a, b) = (35 - k, 36 + k)$ up to reflection, and

$$C_{88}(a, z) = 560 - k(k+1) - \rho_*(a, u_a) - \rho_*(b, u_b) - \max(50, z, 150 - 2z).$$

The domain $0 \leq k \leq 20$, $15 \leq z \leq 560 - k(k+1)$ has 8386 integer states. All 630 effective-hole constants are checked against their strict hypotheses. The maximum cover cap is 333, so $j \leq 4$. The all-active mass upper bound is $1566 + 4z$. The retained state-and-empty-count pairs number $(3882, 2881, 1737, 780, 47)$, totaling 9327. Their 188 intervals have counts $(51, 60, 45, 26, 6)$ for

$j = 0, \dots, 4$. Both reflected orientations are distinct because $m = 71$ is odd. The high-root counts are $(408, 8400, 52020, 155584, 130560)$, totaling 346972. Hence the canonical universe has

$$781308 + 346972 = 1128280 \quad \text{original roots.}$$

Leading, nested and core-quantile gates are respectively $\delta < 159$, $\delta < 151$ and $\delta < 76$, interpreted through the relevant uniform residual cap. Ordinary $j \geq 3$ models retain weak outer order. For $j \geq 4$, low caps are at most 90 and 10 for $j = 4, 5$; the high four-empty cap is at most 13. Thus both leading components contribute, and retained centers supply the relative alignment used in full-D geometry. They need not put missing outer coordinates on the lattice. The finite gate check covers 2736 fast/full parameter representatives; every certified current row is regenerated in the complete arithmetic replay.

35.3 Raw star overlap and a simultaneous denser core

Let A_0 be a leading component and $A_1, \dots, A_t$ distinct labeled retained components in its group. Pointwise incidence counting gives

$$\sum_{i=1}^{t} |A_0 \cap A_i| \leq \sum_{i=0}^{t} |A_i| - \left| \bigcup_{i=0}^{t} A_i \right|. \tag{119}$$

Indeed, a point in the hub and r leaves contributes r on both sides; a point outside the hub contributes zero on the left and a nonnegative amount on the right. For an actual common target F, define $G_F = \sum_{i=0}^{t} |A_i| - |(\bigcup_{i=0}^{t} A_i) \cap F|$. Its group deficit is therefore at least

$$G_F \geq \sum_{i=1}^{t} |A_0 \cap A_i| + |A_j \setminus F| \qquad (0 \leq j \leq t). \tag{120}$$

Outside F, adding one chosen member incidence to the hub-leaf count never exceeds total component multiplicity. Endpoint-hull losses may replace actual target losses as lower bounds. Each choice of j gives an alternative lower bound for the same actual deficit; these alternatives are not added together.

Lemma 33.2 and its cross-component reflection supply

$$H_{\text{own}} = 2e - 1 - R_p - R_q - p_i + M_p, \qquad H_{\text{cross}} = 2e - 1 - R_p - R_q - q_j + M_q.$$

The adopted size-88 full model sums at most one own and one cross raw bound per row, retains one member loss, and adds unmatched component losses. Its cheap scalar budget rows choose any nonempty subset of at most four distinct overlap clusters across the two target signs and drop other nonnegative losses. Negative H values remain valid lower bounds. The relevant alignment follows from the three-empty theorem, or from leading and retained contribution when $j \geq 4$.

Only the eight zero-hole loss-cover parents below need a denser nested core and all retained leaves at once. Write the positive peeled points as $x_1 < x_2 < x_3 < x_4 = M_p$, with $x_0 = R_p$, and similarly $y_0 = R_q < y_1 < \cdots < y_4 = M_q$. Define

$$g_p = \max\bigl(0, \max_{1 \leq j \leq 4} (2j + R_p - x_j)\bigr), \quad g_q = \max\bigl(0, \max_{1 \leq k \leq 4} (2k + R_q - y_k)\bigr).$$

Integral gaps give $0 \leq g_p, g_q \leq 4$. Choose maximizing indices, allowing zero. Adjoining exactly those first j positive and first k negative peeled points gives one simultaneous set $E' \subseteq D$ with

$$2|E'| - \operatorname{diam} E' = 2e - R_p - R_q + g_p + g_q.$$

Using E' in the two translated copies improves every own or cross raw overlap bound to $H_i + g_p + g_q$ simultaneously. Nonnegative overlap variables may be realized by the actual raw intersections. Formula (120) permits their sum plus one leading or leaf loss. No disjointness of the individual intersections is assumed. The separately charged fixed-label losses remain in the overall covering budget, not in these raw star rows.

35.4 Retained-unit consequences and the fourteen-unit loss cover

If a retained unpaired positive component $v + D$ contributes to the actual target inside $[1, S_p]$, then v is integral and

$$v + M_p \geq 1, \qquad v - M_q \leq S_p, \qquad \lambda_{\mathrm{low}}(v) + \lambda_{\mathrm{high}}(v) \leq 79.$$

The two hull-loss sets are disjoint, and at least one of the 80 points remains in the target hull. If a retained cross component contributes, $c_p - q_j + M_p \geq 1$ and $p_i > M_p$, so every retained own center is positive and its low-normal quantile holds. Conversely, a contributing own component gives $c_p + p_i - M_q \leq S_p$; since $q_j > M_q$, every retained cross center is below S_p, giving its high-normal quantile. Reflection supplies the negative side. These statements apply only to actual unpaired contributors, not arbitrarily chosen members of a pair.

The fixed-score argument of Section 29 supplies the same mandatory saving $F = 8$ for the eight zero-hole four-empty parents under consideration. It uses their checked outer lattice and missing-largest slots with retained lower contributors. In the common covering budget, fixed labels are disjoint from component losses; hence the uniform remaining cap C bounds their sum. The usual quantiles may be recomputed from that cap: global constants are 80/79, core constants 76/75 require $C < 76$, and normal quantiles require $C < \min(a, b) + 5$. This does not change any original lattice provenance.

Both paired units are missing in these eight roots, leaving fourteen actual unpaired contributors. Let L_i be each component's two-endpoint hull-loss sum. Then L_i are nonnegative integers and

$$\sum_{i=1}^{14} L_i \leq C < 152.$$

If $C < 76$, the all-small case is exhaustive. Otherwise use all fifteen children: the all-small child $L_i \leq 75$ for every i, and one child $L_i \geq 76$ for each possible selected i. At most one component can be large. The all-small child receives the existing core quantiles for every component. In a selected-large child, every other loss is at most $C - 76 < \min(a, b) + 5$, so those thirteen components receive normal quantiles. No unsupported new normal row is imposed on the selected large component.

The two end parents have $C = 70$ and one child each; the other six have $C = 76, 80$ or 82 and fifteen children each. All 92 children are required. They use the already proved dense-star, fixed-aware and retained-unit rows, with fixed-score binary splits licensed by each root's original $WC = I_8$ certificate. This finite cover is separate from the binary branch count and cannot be replaced by one selected large-loss case.

35.5 Positive holes and the three interval refinement

For positive holes, $h_P, h_Q \geq z \geq 1$. Individual primitivity implies both supports are non-AP: a primitive AP support has step one and no holes. If both side sizes are at least 21, the non-AP theorem gives

$$\rho_P + \rho_Q \geq (a - 5) + (b - 5) = 61.$$

The older low-positive template used 43. The exact budget therefore permits the additional row

$$\delta - 2z \leq \Omega - 150 - 80j - 61,$$

improving its right side by exactly 18. The original labeled row remains unchanged; new quantile and leading rows are appended only under their new valid cap gates. A smaller cap does not retroactively alter the center polynomials used in an existing lattice certificate. This strengthening directly excludes 106 parents and supports 26 three-empty branch proofs. The latter first check

the original-cap $WC = I_8$ witness, then apply the positive-defect row. They use no fixed-score layer.

Fourteen positive four-empty parents remain, with $29 \le a \le 42$, branch A, missing units `pair1`, `pair2`, `p0`, `q0`, and contributor choice zero. Partition their integer holes into exactly three closed intervals

$$[1,2]_{\mathbb{Z}}, \qquad [3,4]_{\mathbb{Z}}, \qquad [5,14]_{\mathbb{Z}}.$$

Every parent requires all three children. In a child with lower endpoint l, use each validated effective-hole witness with $l > 2\beta$ to obtain $\mu_X \ge r + C_h + 1$. At $l = 1$ use only $\mu_X \ge r - 3$; at $l = 3$ use $\beta = 1$; at $l = 5$ use $\beta = 1, 2$. The usual strict hole hypothesis $h_X < |\mathrm{FS}(X)| - 1$ is explicitly available here: strict inverse cases already have it; in the repaired cases $h_Q \le 8$ and the post-lattice estimate gives $h_P \le \kappa < T_a$; AP cases have $z = 0$ and are absent. Primitivity alone is not substituted for that hypothesis in this small-hole application.

The 28 required witnesses are listed below. Here C_1, C_2 correspond to $\beta = 1, 2$, and $u_r(l)$ is the resulting excess lower bound at hole lower endpoint l.

r	C_1	C_2	$u_r(1)$	$u_r(3)$	$u_r(5)$
29	13	28	26	43	58
30	14	30	27	45	61
31	15	31	28	47	63
32	16	32	29	49	65
33	17	33	30	51	67
34	18	35	31	53	70
35	18	37	32	54	73
36	19	39	33	56	76
37	19	41	34	57	79
38	20	41	35	59	80
39	21	42	36	61	82
40	22	44	37	63	85
41	23	46	38	65	88
42	24	48	39	67	91

All strict parameter tests are explicit finite arithmetic obligations. Replacing the old per-side excess $r - 3$ by u_r changes the S_s mass row by $-2(u_r - r + 3)$ and the core mass row by $-(u_r - r + 3)$, preserving their hole coefficients. Recompute endpoint quantiles on the same restricted child with the single 18-unit defect improvement; then restore the old labeled cap and hole rows. The $[3,4]$ and $[5,14]$ children are 28 direct raw-model contradictions. The fourteen $[1,2]$ children use fixed-score trees with their original outer-lattice witnesses. Their positive holes mean the mandatory missing-largest saving is zero and all 110 high-label candidates remain. No stronger non-AP inverse, seven-interval refinement, conditional-span improvement or experimental core-mass table is used.

35.6 The full exact partition and verification standard

The ordinary scanner records an exact current-row certificate whenever it certifies a root. Its terminal ledger contains 1127649 certified roots and 631 explicit gaps, with no unattempted root. A numerical proposal without a successful integer certificate remains a gap. The complete gap list has 242 low and 389 high parents, partitioned as follows:

Supplemental class	Parents	Child models	Nodes	Binary splits	Leaves
Zero-hole three-empty	20	20	68	24	44
Zero-hole four/five-empty	68	68	548	240	308
Zero-hole loss cover	8	92	5112	2510	2602
Direct positive defect	106	106	106	0	106
Positive three-empty	26	26	144	59	85
Positive three-interval cover	14	42	876	417	459
High geometry first group	41	41	223	91	132
High geometry second group	170	170	922	376	546
High geometry terminal group	178	178	1448	635	813
Total	631	743	9447	4352	5095

The three high packages use only the adopted geometry and checked outer-integrality wrapper. Their selected splits comprise 963 normal, 70 core-or-large-loss and 69 inner-quantile alternatives. They use no fixed-score or additional inequality layer. All selected contributor variants and original-cap gates remain linked to their exact canonical descriptors. The last 178 IDs are checked to be exactly the terminal 389 high gaps minus the previous 211.

The low union is checked against the 242 actual ordinary-low gaps, not inferred from its count. Its 354 child models give 3604 leaves. The high union has 389 child models and 1491 leaves. The finite covers add 112 children beyond the original 631 parents; therefore the exact accounting identities are

$$\begin{aligned} 743 &= 631 + 112, \\ 5095 &= 743 + 4352, \\ 1132744 &= 1127649 + 5095 = 1128280 + 4352 + 112. \end{aligned}$$

Counts alone do not prove infeasibility. Every ordinary and supplemental leaf is checked on its regenerated current matrix by positive integer weights w with $w^{\mathsf{T}}A \geq 0$ and $w^{\mathsf{T}}b < 0$ on $x \geq 0$. Every required cover child and both children of every binary split must be present and exactly linked to its original parent.

The portable ordinary certificate carries 10689 positive-integer labeled Farkas vectors and exactly 1128280 unsigned 16-bit little-endian assignment entries, totaling 2256560 bytes. Zero marks a supplemental parent. A nonzero code is $2i + 1 + \varepsilon$, where i is the zero-based vector index and $\varepsilon = 0, 1$ selects the fast or full necessary model. The decoder checks exact stream length, integer encoding, pool range, vector identities and the 631 zero positions. All canonical descriptors are regenerated in order and their complete manifest hash checked. For every nonzero assignment, the verifier rebuilds the selected current model and proves the exact Farkas contradiction. It does not need a stored per-root matrix or the raw discovery journal to prove that contradiction. The fast/full counts are 1086923 and 40726, summing to 1127649.

Supplemental verifiers reconstruct the full canonical manifest, actual center polynomials, integer left inverses, strengthened rows and branch formulas. Pure adapters are checked against frozen mathematical source functions, and numerical optimizers are excluded from replay. The integration verifies exact disjoint equality between the supplemental parent IDs and the 631 compact zero positions, as well as all source and proof identities. The earlier size-89 exclusion remains a separate descending prerequisite.

The finite lemma requires exact equality between the supplemental parent-ID set and all 631 zero positions in the ordinary assignment stream, with disjoint supplemental parents and every required cover child. The full entry point `proofs/exact87/verify_exact87.py` must check structural and classification arithmetic, all nine supplemental families, their exact complement, and every ordinary current-matrix contradiction. A supplements-only check would leave the ordinary finite lemma unproved.

36 The common core lattice at size 87

This is a necessary-structure theorem, separate from the endpoint infeasibility lemma. The last nonprimitive AP branch is discharged by the fully printed analytic combinations in Appendix G.

36.1 Bootstrap and the order of normalization

The deficient-sign case has $|\mathrm{FS}(E)| \geq 1 + T_{71} = 2557 > 24 \cdot 87 + 77 = 2165$, so $d = 79$, $e = 71$ and $m = 70$. The inherited unpaired-element bound is $q^2 \leq 2848$. Its even parity gives $q \leq 52$, hence at least nine opposite pairs in E and at least nine elements on each sign. In particular zero is not a diagonal defect. Reflect so that $a = 35 - k \leq b = 35 + k$. Then

$$\Lambda = 1261 + k^2, \qquad \Omega = 574 - k^2, \qquad C = -823 - k^2.$$

The exact pre-lattice budget is

$$\rho_P + \rho_Q + \kappa + 2\theta + c + \ell_1 + \ell_2 + g = \Omega. \tag{121}$$

All terms other than a possible small-side ρ are nonnegative. Since $b \geq 35$, $\rho_Q \geq 0$; the global bound $\rho_P \geq -8$ gives $k \leq 24$, so $a \geq 11$.

Write $U = \mathrm{FS}(P)$, $V = \mathrm{FS}(Q)$, $u = 1 + T_a + \mu_P$ and $v = 1 + T_b + \mu_Q$. Subset-sum reflection identifies $-V$ with a translate of V, so $\kappa = |U + V| - u - v + 1$. Once a common lattice is proved, put $h_P = \sum P - T_a - \mu_P$, $h_Q = \sum Q - T_b - \mu_Q$, and $z = h_P + h_Q - \kappa$. The inverse conclusion $h_P, h_Q \leq \kappa$ then gives $0 \leq z \leq \min(h_P, h_Q)$.

For a positive side X of size r, with maximum R_X, let $Y = X \setminus \{R_X\}$ and let ν count its old diagonal defects destroyed by adjoining R_X. The exact identity is

$$R_X = r + \rho_X + h_X + 2 - \rho_Y - h_Y - 2\nu. \tag{122}$$

Thus $R_X \leq r + \rho_X + h_X + 2$ for $r \geq 15$. At the smaller boundaries one must retain $\rho_Y \geq -8$, which adds eight to that side's bound and sixteen to the first peeled-layer comparison.

Only after D is integral and the requisite extreme translates intersect may we use the sharp four-extreme theorem with $G = 148$. Writing $\mathcal{T} = \kappa + \ell_1 + \ell_2 + g$, its consequences are

$$\mathcal{T} \geq \kappa + 148 - 3z, \quad \mathcal{T} \geq z, \quad \mathcal{T} \geq 50, \quad c + \rho_P + \rho_Q + \mathcal{T} \leq \Omega. \tag{123}$$

The proof below respects this order; these post-lattice inequalities are not used to obtain the initial common lattice.

36.2 The four small-side boundaries

The relevant $(a, b, \Omega, \kappa \text{ cap})$ are

$$(11, 59, -2, 6),\ (12, 58, 45, 53),\ (13, 57, 90, 98),\ (14, 56, 133, 133).$$

For $a = 11, 12$ the cap is less than both $T_a - 1$ and $T_b - 1$, so the strict simultaneous inverse applies. For $a = 13, 14$, a non-AP larger support has $\rho_Q \geq b - 5$ and reduces the caps to $46, 82$; the same inverse again applies. If instead V is an AP, its length exceeds the raw cap plus one. Residue counting aligns P with its step, and the AP gap formula gives $h_P = \kappa$, $h_Q = z = 0$.

Using the extra sixteen required by the deleted small prefix, the first-layer comparison is bounded by

$$-249 - 2k^2 + 2\,\kappa_{\max} + 16,$$

whose four values are $-1373, -1185, -1005, -849$. The other layer is strict from $\Lambda > 4d - 10 + \kappa_{\max}$. These comparisons prove D-integrality and overlap before (123) is used. At $a = 11$, $\mathcal{T} \geq 50 > 6$

is impossible. At $a = 12, 13$, the non-AP case gives $\mathcal{T} + \rho_P + \rho_Q \geq 50 - 8 + (b - 5) > \Omega$; the AP case has $\mathcal{T} \geq 148$ and is also impossible. For $a = 14$ the AP case is excluded identically. Otherwise $\rho_Q \geq 51$ and $\rho_P \geq 0$ imply $z \geq 33$. The strict inverse hole bound permits the effective-hole witness $(r, \beta, C_h) = (56, 5, 143)$. It gives $\mu_Q \geq 200$, and the old flexible inverse gives $\rho_Q \geq 134 > 133$. Hence all surviving sides have $a, b \geq 15$.

36.3 AP supports and the half-step case

If the larger support is an AP, it has at least $T_{35} + 1 = 631$ points, whereas $\kappa \leq 574$. Residue counting and the exact AP gap formula give a common lattice, $h_{\mathrm{AP}} = 0$, $h_{\mathrm{other}} = \kappa$ and $z = 0$. Now suppose that only the smaller support U is an AP. For $0 \leq k \leq 20$, $\Omega < 2T_a$, so the residue count permits at most two residues of V modulo its step.

In one residue, the AP gap formula is $\kappa = \sum_i (\min(u, g_i) - 1)$. A gap of length at least u would be unique and central by reflection. The largest-gap argument then forces the final coefficient of Q to exceed the sum of its predecessors. The two final subset-sum translates are disjoint, giving $\mu_Q \geq T_{b-2}$ and $\rho_Q \geq T_{b-2} + 2 - 2b$. The same disjoint-copy bound holds if exactly one coefficient is off-lattice. In either case the essential extra cost is $\kappa \geq u - 1 \geq T_a$. Over the relevant range $k = 2, \ldots, 20$,

$$T_a + T_{b-2} + 2 - 2b - \Omega \geq 549 > 0.$$

Thus neither exception occurs. The one-residue case has only short gaps and gives $h_P = 0$, $h_Q = \kappa$, $z = 0$.

If two or more coefficients occupy the nonzero residue, their sum forces that residue to have order two. Normalize $U = 2[0, u - 1]_{\mathbb{Z}}$ and let w be the least odd coefficient of Q. The two parity-fiber gap formulas, and reflection of their endpoints, give exactly

$$h_P = u - 1, \qquad h_Q = \kappa - u + w, \qquad z = w - 1.$$

These formulas alone do not yet give $h_Q \leq \kappa$. For $s \geq 2$ odd coefficients, distinctness yields

$$\sum Q \geq (b - s)(b - s + 1) + sw + s(s - 1).$$

If $w > u$, the quadratic is minimized at $s = 2$ in this range; comparison with the budget upper bound gives the positive margin $(b^2 - 11b)/2 + 2u + 7 - \Omega \geq 943$. Hence $w \leq u$, establishing both exact hole bounds. If $w \geq 15$, minimize the same quadratic over real s to obtain

$$\sum Q - w - T_b \geq 7b - 289/8.$$

But the exact mass identity and budget bound its left side by $2b + \Omega - T_a - 3$; the minimum contradiction margin is $343/8$. Thus $w \leq 13$ and $z \leq 12$.

The AP-specific estimates

$$R_P \leq 2(a + \mu_P), \qquad R_Q \leq b + \mu_Q + \kappa - T_a - \mu_P + 12$$

make the first peeled-layer comparison at most $2423 + 2a - 4T_a - 2T_b \leq -891$. The other layer is strict because $1261 > 4 \cdot 79 - 10 + 574 = 880$. Therefore D is integral before G is used. The post-lattice lower bound $T_a + 148 - 36 + (b - 5) - \Omega$ is positive except for equality at $k = 11, 12$. If $z \leq 11$, it increases by three. At $z = 12$, the effective-hole witness $(\beta, C_h) = (2, 33)$ gives $\rho_Q \geq 42, 43$ for $b = 46, 47$, respectively, exceeding $b - 5$ by one. This excludes the equalities too. Every surviving AP branch therefore has $z = 0$.

36.4 Both supports non-AP and the inverse gap

Put

$$n(r) = \begin{cases} r-5, & r \geq 21, \\ r-14, & 15 \leq r \leq 20. \end{cases}$$

The large-side strict inverse threshold is automatic. The sufficient small-side test

$$\mu_P + \rho_P + \rho_Q > \Omega - T_a + 2$$

implies the strict simultaneous inverse bounds. Substitution of the valid lower floors leaves possible failures only at $k = 6, \ldots, 19$. If the test fails, define

$$\begin{aligned} R_* &= \Omega - T_a + 2 - (a-3) - n(a), & \rho_Q &\leq R_*, \\ U_{\max} &= \Omega - n(a) - n(b) + 3, & u &\leq U_{\max}, \\ K_{\max} &= \Omega - n(a) - n(b), & \kappa &\leq K_{\max}, \\ V_{\min} &= T_b + b - 2, & v &\geq V_{\min}, \qquad u \geq U_{\min} = T_a + a - 2. \end{aligned}$$

For $k = 7, \ldots, 19$, work first on the intrinsic primitive lattice of Q. The already proved primitive extension of the effective-hole lemma, followed by the old flexible inverse, gives $h_Q \leq H$ using the following complete list. The "forced" column is the contradictory old inverse floor if $h_Q > H$.

k	R_*	β	C_h	Forced ρ_Q	H
7	73	4	88	79	8
8	88	5	108	96	10
9	100	6	132	115	12
10	109	6	133	117	12
11	115	6	137	120	12
12	118	6	143	127	12
13	118	6	149	132	12
14	115	5	132	120	10
15	118	5	136	123	10
16	109	5	139	127	10
17	97	4	118	111	8
18	82	3	95	93	6
19	64	2	66	72	4

Each parameter pair satisfies the strict effective-hole tests. For these rows the four margins

$$V_{\min} - 1 - K_{\max}, \quad V_{\min} - U_{\max} - K_{\max} + 1,$$

$$V_{\min} - K_{\max} - H - 2, \qquad U_{\min} - H - 2$$

have minima $477, 11, 468, 144$, respectively. The first aligns P by residue counting. The second forces $\operatorname{diam} U \leq \operatorname{diam} V$: otherwise two disjoint endpoint translates of V would exceed $u+v-1+\kappa$. The two endpoint hulls then give $h_P \leq \kappa + h_Q \leq \kappa + H$. The last two margins put both diameters at most $u + v - 3$. Stanchescu's short-diameter theorem therefore recovers the exact $h_P, h_Q \leq \kappa$, with $z \leq H \leq 12$. No primitivity of P has been assumed here.

At $k = 6$ the diameter-order margin is negative, so a separate bridge is required. Here $(a, b, \Omega, R_*) = (29, 41, 538, 55)$. If every excess increment y_i for $15 \leq i \leq 41$ were at least five, the recurrence would give $\rho_Q \geq 81$. Thus the last $y_j \leq 4$ has $j \geq 41 - \lfloor 55/3 \rfloor = 23$. On its primitive scale, the counted-hole lemma puts that prefix's support holes in $H_0 \cup (L - H_0)$, with $H_0 \subset [1, 8]_{\mathbb{Z}}$ and $|H_0| \leq 4$. A first later off-scale coefficient would force $\rho \geq T_{23} - 41 - 1 = 234 > 55$. Later increments cannot undo this. Every subsequent $y_i \leq 57$; for a current prefix of size $t \geq 23$, a new coefficient $x \geq L - 16$ would add at least $L - 24 \geq T_t - 24 > t + 58$ support points, too many.

Hence $x \le L - 17$. The central intervals join, and $x > 8$ preserves the lower holes; reflection preserves the upper holes. The final V therefore contains $[9, \sum Q - 9]_{\mathbb{Z}}$ and has at most eight endpoint holes. Residue counting aligns P, and this central interval gives $h_P \le \kappa + 18$. The checked inequalities $\kappa + 18 < v - 19$, $\kappa + 20 \le v$ and $u \ge 10$ imply both short-diameter hypotheses, so Stanchescu again gives the exact hole bounds, now with $z \le 8$.

36.5 Forcing both layers and proving individual primitivity

At this point every branch has a common core lattice and exact hole bounds, and both maximum-deleted prefixes have nonnegative ρ. The first-layer comparison is

$$|\operatorname{FS}(D_1)| - 2s \le C + \Omega + \rho - \mu + 2z \le -249 - 2k^2 + 2z.$$

It is strict for $z \le 124 + k^2$, including all AP and dense-repair branches. Otherwise $z \ge Z = 125 + k^2$. The pre-lattice budget excludes $k \ge 14$; this case retains the original strict inverse hole bounds. If the last two translates for a side of size r were disjoint, then $\rho_X \ge T_{r-2} + 2 - 2r$. Add $\kappa \ge Z$ and the other side's non-AP floor. The elementary floor suffices except for a potentially smaller disjoint side at $k = 4, \dots, 8$. For these five cases the strict hole hypotheses justify the effective-hole witnesses

$$(k, \beta, C_h) = (4, 2, 41), (5, 3, 53), (6, 3, 58), (7, 3, 54), (8, 2, 45),$$

which give other-side floors $43, 52, 56, 55, 49$. Both side choices and all $k = 0, \dots, 13$ now have positive contradiction margin, with minimum one at $k = 4$ and disjoint-side size 31. Thus the actual supports intersect, not merely their hulls.

The maximum-deleted support hulls consequently cover the full side hull, so $h_X \le 2h_Y$. Equation (122) improves to $R_X \le r + \rho_X + h_X/2 + 2$. Substitution gives

$$|\operatorname{FS}(D_1)| - 2s \le C + 2\rho - \mu + z + \ell_2 \le C + \Omega + \rho - \mu \le -249 - 2k^2 < 0.$$

Here $z \le \kappa$ and $\rho + \kappa + \ell_2 \le \Omega$ are used once. An off-lattice peeled element would give two disjoint s-point translates, contrary to this strict bound. The outer layer is likewise strict from $1261 > 880$. The same comparisons force the extreme translate intersections needed for propagation. Thus D is integral and $G = 148$ is now available without circularity.

The original strict inverse branches are individually primitive: a nonunit gcd would give at least $|\operatorname{FS}(X)| - 1$ holes. In the dense repairs Q was primitive by construction; (123) and both non-AP floors give

$$\kappa \le \Omega - (148 - 3H) - n(a) - n(b) < T_a \le u - 1.$$

The strict margins are at least eighteen in the table and eighty-one at $k = 6$, so P is primitive too. In an AP branch whose smaller side defines the step, the other side follows from $\kappa \le \Omega < T_b$. When the larger AP side defines the step, a separate exception is needed, as the next paragraph proves.

In the remaining branch the larger AP side defines the unit lattice. All common-lattice, D-integrality, overlap and propagation premises have already been proved, so Corollary G.2 applies. Its proof derives the integer cap $c \le 6$ from any hypothetical nonprimitive smaller side, then contradicts that cap with the four analytic covering combinations of Lemma G.1. No primitive-core premise, finite side-size enumeration, or leading contribution is used. This proves individual primitivity in the last branch.

We have proved the size-87 structural theorem: $a, b \ge 15$, common integral D, both primitive cores, exact core-hole bounds and the required overlap and propagation. It asserts neither general outer-B integrality nor contribution of a leading $2D$ component.

37 The size 87 endpoint and selected actual-set refinements

37.1 Canonical roots and strict contribution gates

Use $I = (a-35)^2$, $\Omega = 574 - I$, $G = 148$, and $\delta = c - 79j$, where j counts missing ordinary cover units. The canonical descriptors retain the previous covering branch, contributor variants and missing-unit choices. Every matrix variable is relaxed to a nonnegative real. An integer disjunction additionally requires a proof that its actual-set quantity is integral; no linear-programming solution is assumed to be realized by a set.

For $z = 0$, put $B_0 = 426 - I - 79j$; then $\delta \le B_0$. For $1 \le z \le 14$, both primitive core supports are non-AP because $h_P, h_Q \ge z > 0$. Thus

$$\rho_P + \rho_Q \ge n(a) + n(b), \qquad \delta - 2z \le B_0 - n(a) - n(b).$$

The floor sum is 60 when both sides have at least 21 elements and 51 otherwise. The uniform cap is $B_0 - n(a) - n(b) + 28$. The scalar budget excludes zero-hole $j \ge 6$ and positive-hole $j \ge 5$. In particular the balanced positive $j = 5$ cap is -1; the smaller-side range only reduces it further.

If u_s is a proved lower bound on the side excess, the main positive mass row is

$$S_s \ge T_r + 6 + u_s + z + M_s + 3R_s. \tag{124}$$

Here S_s is the sum of the positive magnitudes on that side of D, M_s its outermost member, and R_s its core maximum. The actual core mass is $C_s = S_s - M_s - x_{s,1} - x_{s,2} - x_{s,3}$, with the three inner peeled magnitudes written in increasing order. Hence $C_s \ge T_r + u_s + z$, and $x_{s,i} \ge R_s + i$ proves (124). This explains the constant six and the factor three; doubling the core inequality alone would not give the main row. The conservative span row is $R_P + R_Q + \delta - 4z \le 74 + B_0$. No conditional improved span is used.

For $z \ge 15$, use the effective-hole floor $u_s = u_{\mathrm{old}}(r_s, z)$ and $v_s = \max(n(r_s), v_{\mathrm{old}}(r_s, u_s))$. The complete scalar cap is

$$c \le \Omega - v_P - v_Q - \max(50, z, 148 - 2z).$$

All $0 \le k \le 20$ and $15 \le z \le \Omega$ are enumerated. The established strict mass test omits only proved impossible $j = 0$ states; it omits no $j \ge 1$ state. Consecutive integers are merged only when all active formulas and excess floors agree. The required finite reconstruction has 8890 raw (k, z) states, 11374 retained (k, z, j) states and 277 intervals, with maximum $c = 354$. Thus $j = 0, \dots, 4$ is complete. The balanced high orientation is emitted once. All 605 effective-hole constants are checked by strict integer inequalities, without requiring optimality of the search that found them.

The full canonical counts are

Family	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	Total
Zero holes	164	2590	19074	80784	228480	324632	655724
Positive low	164	2590	17918	74800	184960	0	280432
High holes	604	10780	69360	239360	359040	0	679144
All roots							1615300

Both low families together have 936156 roots. Their descriptors, including complete high interval data, are compared individually; agreement of aggregate counts alone is insufficient.

The final residual cap is recomputed after every budget change. Leading endpoint rows require cap < 157; nested leading rows require cap < 149; core quantiles require cap < 75; normal quantiles require cap $< \min(a, b) + 5$. Equality does not license the stronger row. Balanced positive $j = 3$ has cap 157 and zero-hole $j = 3$ can have cap 189, so neither is granted a leading contributor by this test. The retained-center theorem gives alignment instead. For $j \ge 4$ the

maximal caps are $110, 31, 78, 38$ for zero $j = 4$, zero $j = 5$, positive $j = 4$, high $j = 4$, respectively. These permit leading contribution but still do not make every missing outer coordinate integral.

A supplemental geometry wrapper may restore all strict outer gaps only after reconstructing genuine contributing centers and checking an integer matrix identity $WC = I_8$. A center with an integral target point and an integral D point is integral; W then proves the outer-coordinate assertion. The original cap governs whether leading centers may enter C. A later scalar strengthening cannot retroactively add a leading center to this witness.

The exact cardinality identity, common to every hole regime, is

$$\delta + S_P + S_Q - 2K = (18 - j)79 + 84 + w, \qquad 0 \le w \le 4z, \tag{125}$$

where $K = |D + D|$. Its upper and lower inequalities are both retained. The lower one is an identity restriction, not a cover-loss budget: fixed-label or overlap losses must never be injected into it.

37.2 Zero-hole and small-positive-hole finite covers

In the selected zero-hole loss cover, exactly `pair1,pair2,p0,q0` are missing. The full outer integer witness and a retained smaller own contributor on each side justify four extra high fixed-label deletions per side. These eight losses are disjoint from the 26 baseline nonpositive groups. Fourteen retained unpaired ordinary units remain, seven on each sign. Their actual integral hull losses satisfy

$$\sum_u L_u \le \delta - 8 \le C_0 := 102 - I < 150.$$

At most one has loss at least 75. Thus either all have loss at most 74, licensing the 75/74 core quantiles, or one specified unit has loss at least 75 and every other has loss at most $C_0 - 75$, below the normal quantile threshold. Every possible named-large alternative is retained; when $C_0 < 75$ only the all-small alternative is needed. The eleven selected parents require all 165 child models, not eleven selected successful children. The converse row (125) remains unchanged.

For the eleven remaining positive low parents, the full integer domain $1 \le z \le 14$ is partitioned into $[1, 2]$, $[3, 4]$, $[5, 14]$. All three intervals are required for each parent. Here $j = 4$, $a = 30, \dots, 40$ in the selected component, and the original budget is $\delta - 2z \le 50 - I$. No extra eighteen-unit saving is subtracted. The broader applicability range $27 \le a \le 43$ gives $\kappa \le 92 - I < T_{27}$, proving the strict effective-hole premise. The old floor at each interval's lower endpoint raises u_s in the main and core mass rows, with multipliers two and one in their respective inequality encodings. The three residual caps are $54 - I, 58 - I, 78 - I$, with the same strict quantile gates.

Positive-hole fixed saving is zero and all 110 high-label candidates remain. Seventeen interval children are direct leaves. Sixteen use checked outer witnesses and complete fixed-score trees; five of these, only on $[1, 2]$, use the mass-trimming refinement below. A replacement of one interval child never deletes its two siblings. The 26 disjoint nonpositive groups and high positive losses are charged once, and the cardinality converse is protected throughout.

37.3 Upper core mass and one-point trimming

For r distinct positive integers $x_1 < \cdots < x_r = R$, unit gaps give $x_i \le R - r + i$. Hence the elementary necessary inequality is

$$C = \sum_i x_i \le rR - T_{r-1}. \tag{126}$$

It needs integer core points, not primitivity, zero holes or integral initial outer coordinates. In the model it is $S_s - M_s - x_{s,1} - x_{s,2} - x_{s,3} - rR_s \le -T_{r-1}$.

Let U be the second-largest core element and $r \geq 2$. The other $r-2$ core elements have sum at least T_{r-2}, so

$$C \geq R + U + T_{r-2}, \qquad R - U - 2 \geq 2R - C + T_{r-2} - 2.$$

Deleting the maximum changes $2|F| - \operatorname{diam} F - 1$ by exactly $R - U - 2$. On each sign choose independently either the untrimmed core plus a prefix of zero to four peeled points, or the core with its maximum deleted and no peeled point added. The two maximizing options form one actual $F \subset D$ and give

$$2|F| - \operatorname{diam} F - 1 = 2e - R_P - R_Q - 1 + g_P + g_Q,$$

where $g_s \geq 0$ dominates all original bounds $2i + R_s - x_{s,i}$ and also $2R_s - C_s + T_{r_s-2} - 2$. The new gains can exceed four, so exactly the two old gain upper bounds are removed. Their domain remains nonnegative real; no integer gain branch is authorized.

For each retained own or cross leaf, the leading component and the leaf contain the corresponding translates of this same F. If their relative displacement is the proved integer t, the intersection has at least $2|F| - \operatorname{diam} F - |t| - 1$ points. At a point in the raw union, the number of hub-leaf incidences is at most total component multiplicity minus one. Outside the target, one selected member-loss indicator may additionally be charged. Summing pointwise proves that the actual group deficit dominates all star intersection bounds plus any one member loss. The same F simultaneously realizes every star auxiliary. Other fixed and unmatched losses keep their original disjoint allocation. This argument is valid with positive holes and does not assume that the trimmed set is non-AP.

37.4 Dense spans and the stronger non-AP inverse

Lev's theorem gives, for a positive integer r-set of maximum $R \leq 3r/2 - 2$, writing $k = R - r$ and $C = \sum X$,

$$[2k+1, C-(2k+1)]_{\mathbb{Z}} \subseteq \mathrm{FS}(X).$$

A lower hole lies in $[1, 2k]_{\mathbb{Z}}$ and must be a missing generator, so there are at most k; reflection gives at most k upper holes. Thus $h_X \leq 2(R-r)$. In a high interval with lower endpoint z_0, set $q = \lfloor 3r/2 - 2 \rfloor$. If $z_0 > 2(q-r)$, the structural inequality $h_X \geq z \geq z_0$ excludes $R \leq q$, giving $R \geq q+1$. The strict test is essential. This uses the same lattice throughout and adds no gcd hypothesis to Lev's theorem.

The final ten high parents also use the following separately proved non-AP inverse. Let $r \geq 21$, $\mathrm{FS}(X)$ be non-AP and $\mu(X) \geq u$. For $21 \leq L \leq r$, put $M = T_L - r$. If $\mu \leq M$, every prefix of size at least L has non-AP support: otherwise the first subsequent non-AP prefix is a disjoint translate of its AP predecessor and adds excess at least $1 + T_L - r = M + 1$. Fix $3 \leq \beta \leq L - 5$. If there is no later small increment, the non-AP prefix floor and recurrence imply

$$\rho \geq A := r - 5 + (\beta - 2)(r - L).$$

If $i \geq L$ is the last increment of excess at most β, the small-increment inverse gives at most 2β defects there. With $t = r - i$, one has $d_0(X) \leq 2\beta + t$ and $\rho \geq r - 5 + (\beta - 2)t$. Eliminating t using $\mu = \rho - 2 + 2d_0$ yields

$$\beta\rho \geq (\beta - 2)(\mu - 4\beta + 2) + 2(r - 5).$$

Thus this case has the floor

$$B := \max\left(r - 5, \left\lceil \frac{(\beta-2)(u - 4\beta + 2) + 2(r-5)}{\beta} \right\rceil\right).$$

The complementary case $\mu \geq M+1$ uses the old inverse, giving $C := \max(r-5, v_{\mathrm{old}}(r, M+1))$. The minimum of A, B, C is universally valid; maximize it over the finite parameter choices and the old floor. This derivation needs no AP-free enumeration and does not assume that the search is optimal.

The odd-pivot bound $d_0(X) \leq R - r + 2$ gives independently $\rho \geq u + 2r - 2R - 2$. In the selected compositions, each side of one original budget may use its justified constant floor or this odd-pivot floor. Each choice is a separate necessary inequality on the same actual set; adding the budgets to each other would double-count. The eight-parent component uses original-floor odd masks plus a separate strengthened constant-floor budget. The final two also have the derived joint forms. Every original row is retained.

37.5 A full-D small-doubling alternative

The structural theorem gives $D \subset \mathbb{Z}$, $|D| = 79$, $0 \in D$ and primitive $P \subset D$. Thus $\gcd(D-D) = 1$. Applying Stanchescu's Corollary 2 to D, D uses its equal-diameter parameter one and gives

$$K < 3 \cdot 79 - 3 = 234 \quad \Longrightarrow \quad \operatorname{diam} D = M_P + M_Q \leq K - 79.$$

Every actual configuration therefore satisfies the exhaustive pair

$$\{K \leq 233,\ M_P + M_Q - K \leq -79\} \quad \text{or} \quad K \geq 234. \tag{127}$$

The low child has both inequalities; the high child has no inverse width restriction. This argument concerns the ordinary repeated-sum set $D + D$, and needs no general outer-B integrality or leading contribution. In a wrapper those separate gates remain necessary for any inherited stricter outer geometry.

37.6 The final two parents use actual structural scalars

The last two selected parents, canonical IDs 1570114 and 1570147, are balanced high roots. Their refinement has one common actual-set realization. For each core side of size r, assign

$$\mu_s = |\operatorname{FS}(X_s)| - 1 - T_r, \quad h_s = C_s - T_r - \mu_s, \quad d_s = d_0(X_s), \quad \rho_s = \mu_s + 2 - 2d_s.$$

Let $D_0 = \Delta(D)$ be the actual full-D diagonal-defect count and set

$$\kappa = h_P + h_Q - z, \qquad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g, \qquad \theta = d_P + d_Q + 8 - D_0.$$

These variables are nonnegative integers under the structural theorem. Zero is nondefective because E has at least nine opposite pairs. The original $\delta = c - 79j$ and its cardinality identity are preserved. The prior common-F construction simultaneously realizes the star auxiliaries, which remain nonnegative real coordinates.

The scalar rows include the following actual identities and bounds:

$$\begin{gathered}
\mu_s \geq u_s, \quad \rho_s \geq v_s, \quad \rho_s - \mu_s + 2d_s = 2, \quad 1 \leq d_s \leq r, \\
C_s - \mu_s \geq T_r + z, \qquad C_s - 2\mu_s \leq 2T_r - 1, \\
C_s \leq rR_s - T_{r-1}, \qquad d_s \leq R_s - r + 2, \\
\kappa + \mu_P + \mu_Q + z - C_P - C_Q = -T_a - T_b, \\
\delta + \mathcal{T} + \rho_P + \rho_Q + 2\theta \leq 574 - I - 79j, \\
D_0 - d_P - d_Q + \theta = 8, \quad 2 \leq D_0 \leq M_P + M_Q - 75, \quad K \leq 79 \cdot 80/2.
\end{gathered}$$

The bounds on $\mathcal{T}$ are those in (123); in addition $\mathcal{T} \geq \kappa$ follows from its definition and nonnegative slack. The strict hole bound $h_s < |\operatorname{FS}(X_s)| - 1 = T_r + \mu_s$ explains the integer minus one in the

second line. In these two balanced roots its hypothesis follows directly from $h_s \le \kappa \le 574 < T_{35} + \mu_s$. To prove the full-$D$ parity bound, translate its minimum to zero, apply the odd-pivot bound to its 78 positive members, and add the defective minimum. Both parities occur since D has gcd of differences one. The two endpoints are always defective, explaining $D_0 \ge 2$. The budget is the original exact budget with justified nonnegative slack dropped, not a second copy of a fixed floor budget.

For each of the six distinct peeled inner points, choose its actual nondefect indicator $\nu_{s,i} \in \{0,1\}$, $i = 1,2,3$. Adding points cannot create an old core defect. Of the eight newly peeled points, at most $8 - \sum_{s,i} \nu_{s,i}$ are defective, hence

$$\theta \ge \sum_{s,i} \nu_{s,i}.$$

A point with several midpoint witnesses still uses this one indicator. Opposite-sign inner labels refer to distinct positive/negative points, so no point is charged twice.

For the five same-side endpoints $R < x_1 < x_2 < x_3 < M$, write $w = M - R$. Their integer gaps and midpoint witnesses imply

$$w + \sum_{i=1}^{3} \nu_i \ge 7, \qquad w + 2\sum_{i=1}^{3} \nu_i \ge 8, \qquad \nu_i + x_{i+1} - x_{i-1} \ge 3, \tag{128}$$

where $x_0 = R$, $x_4 = M$. For the local inequality the neighboring span is at least two, and equality forces a midpoint witness. For $w \ge 8$ the first two are immediate. For $w = 4,5,6,7$, translating R to zero leaves respectively $\binom{w-1}{3} = 1,4,10,20$ profiles. Directly testing the distinct endpoint pairs gives minimum numbers of witnessed inner points $3,2,1,1$, respectively. These 35 elementary profiles prove both global constants. They use only witnesses among these five points; such witnesses are also valid in D.

37.7 The selected AP-free bounds and width alternatives

The defective elements of any integer set contain no nontrivial three-term AP: its middle element would not be defective. For a positive core, adjoining zero to its defective set remains AP-free. With $r_3(L)$ the maximal AP-free size in an L-point interval,

$$d_s + 1 \le r_3(R_s + 1).$$

Here is the boundary-localization argument used in the next inequalities. Let $S \subset [0,L]_{\mathbb{Z}}$ contain $0, L$ and have n elements. If a diagonal-defective $x \in S$ satisfies $x \le L/2$, none of the x disjoint pairs $\{x-i, x+i\}$, $1 \le i \le x$, has both members in S. Counting the points in $[0,2x]$ and then those above $2x$ gives

$$n \le x + 1 + (L - 2x) = L - x + 1,$$

so $x \le L - n + 1$. Reflection gives $x \ge n - 1$ for a defective point in the other half. All defects therefore lie in

$$[0, L-n+1]_{\mathbb{Z}} \ \cup \ [n-1, L]_{\mathbb{Z}},$$

two intervals of $L - n + 2$ points. The defective set is AP-free, so its cardinality is at most $2r_3(L-n+2)$; overlap of the intervals only weakens this upper bound. Apply this to $\{0\} \cup X_s$, where zero itself is defective, and then to a translate of D. This gives

$$d_s \le 2r_3(R_s - r + 1) - 1, \qquad D_0 \le 2r_3(M_P + M_Q - 77).$$

These facts require neither an odd element nor primitivity.

The finite dependencies are the exhaustive $r_3(0), \dots, r_3(30)$ table and the following required finite upper bounds

$$r_3(31) \le 12,\ r_3(35) \le 13,\ r_3(39) \le 14,\ r_3(40) \le 15,$$

$$r_3(50) \le 16, \qquad r_3(58) \le 19, \qquad r_3(70) \le 20.$$

The complete inclusion/exclusion verifier maintains an increasing AP-free chosen set and forbids every future third term $2x - y$. Branching on the least available point covers all possibilities; available-position and contiguous-block capacity bounds justify every prune. The finite upper bounds are mathematical proof dependencies, not numerical optimizer conclusions.

Restricting an AP-free set to consecutive blocks and summing upper capacities proves every partition bound $g(L)$ used by the model. Checking $7r_3(L) \le 2L + 30$ through 70 and adjoining 70-blocks of capacity 20 proves the same inequality for every positive L. Together with boundary localization this gives

$$7d_s - 2R_s \le 25, \qquad 7d_s - 4R_s \le 57 - 4r,$$
$$7D_0 - 2(M_P + M_Q) \le 32, \qquad 7D_0 - 4(M_P + M_Q) \le 68 - 4 \cdot 79.$$

For an integer core span R_s, the pair $R_s \le t$ or $R_s \ge t + 1$ is exhaustive. In the first child,

$$d_s \le \min\{r, g(t+1) - 1, 2g(t - r + 1) - 1, t - r + 2\}.$$

This uses monotonicity of the true r_3, not an assumed monotonicity of a computed partition bound.

Only $r = 35$, $t = 77, 79$ occur in the final two selected trees. The explicit partitions $78 = 70+8$ and $80 = 70 + 10$, with capacities $20 + 4$ and $20 + 5$, give $d_s \le 23$ and $d_s \le 24$ after subtracting one for zero. The complementary children are $R_s \ge 78$ and $R_s \ge 80$. All six selected AP-cap splits retain both children. The trees also have two full-D Stanchescu splits (127). Their remaining splits are the already proved normal, inner and core-or-large alternatives. No optional conditional-span split, full-D finite-cap split, peeled-D tail cut or arbitrary integer-form split appears in these selected trees. The final-two composition has no fixed-score layer. Its 188 nodes, 93 binary splits and 95 exact leaves satisfy $188 = 2 + 2 \cdot 93$ and $95 = 2 + 93$.

37.8 Exact selected partition and dependency list

The required ordinary classification has 1614434 exact contradictions and 866 supplemental parents: 251 low and 615 high IDs. The following is the complete supplemental partition. The actual canonical ID lists, not the displayed totals, determine the parent-set union.

Selected component	Parents	Children	Splits	Leaves
Low geometry 27	27	27	44	71
Low fixed 77	77	77	228	305
Zero-hole five-missing 42	42	42	707	749
Zero-hole loss cover 11	11	165	2621	2786
Positive three-missing 24	24	24	69	93
Positive fixed 59	59	59	759	818
Positive three-interval 11	11	33	2959	2992
High geometry 9	9	9	10	19
High geometry 28	28	28	86	114
High geometry 85	85	85	143	228
High core mass 3	3	3	90	93
High geometry 107	107	107	274	381

Selected component	Parents	Children	Splits	Leaves
Recovered high geometry 121	121	121	342	463
Recovered high core mass 1	1	1	15	16
Recovered high core mass 2	2	2	51	53
High geometry 86	86	86	312	398
High dense spans 9	9	9	251	260
New high core mass 2	2	2	49	51
High geometry 150	150	150	644	794
Final high core mass 2	2	2	28	30
High inverse and odd-pivot 8	8	8	388	396
High actual scalars and AP caps 2	2	2	93	95
Total	866	1042	10163	11205

The low totals are 427 child models, 7387 splits and 7814 leaves; the high totals are 615 child models, 2776 splits and 3391 leaves. The covers add 176 child models beyond their parent count. Thus

$$11205 = 1042 + 10163, \qquad 1625639 = 1614434 + 11205 = 1615300 + 176 + 10163.$$

The actual canonical parent IDs must form exactly the ordinary zero positions, with pairwise disjointness. All required cover children and both children of every binary split must be included.

The logical dependencies of the new endpoint are precisely:

1. The retained foundational real-to-integer reduction, covering, peeling and budget identities; the descending exclusions proved in order through size 88.
2. The complete size-87 lattice proof above, including the analytic four-combination non-primitive AP argument in Appendix G, the strengthened non-AP defect theorem, old effective-hole and flexible-inverse lemmas, and Stanchescu's common-lattice results.
3. The complete size-87 root classification, 605 effective-hole inputs, original-cap quantile gates, 2992-mode retained-center identities, and full nonnegative-variable realization.
4. The already proved normal, inner and core-loss disjunctions, checked outer integer witnesses, and the fixed-score 110-label/26-group accounting used by the selected low forests.
5. The actual-set zero-hole loss cover and three-interval positive cover; common-F dense stars, one-point mass trimming and retained-unit consequences, only at their selected levels.
6. The upper core-mass inequality, qualified dense-span exclusion from Lev, strengthened non-AP inverse, original and joint odd-pivot budget substitutions, and full-D Stanchescu disjunction, where selected.
7. For the final two parents, the actual scalar and shared-indicator identities, all 35 elementary five-point profiles, the AP-free finite upper bounds and their block partitions, and exactly the selected 77/79 core-span alternatives described above.
8. Exact current-matrix Farkas arithmetic for every ordinary and supplemental leaf, exhaustive covers and binary topology, immutable source linkage, compact assignment equivalence, and final disjoint integration. These are separate obligations, all required.

The optional tail-eight and conditional-span research refinements are not selected dependencies. Neither floating-point infeasibility, source-code similarity alone, nor a partial successful run replaces any item on this list.

38 Composition of the ten exclusions

Theorem 38.1 (Global upper bound)**.** *Every feasible finite real set has at most* 86 *elements.*

Proof. The earlier chapters prove $N \leq 96$. The size-96 actual-set reduction and its finite lemma give $N \leq 95$. Repeating the proved descending step for sizes $95, 94, 93, 92, 91, 90, 89, 88$ gives $N \leq 87$. A hypothetical feasible size-87 set therefore contains zero, since otherwise adjoining zero would give a forbidden size-88 set. Integralization supplies an integral size-87 set with zero. The complete structural theorem maps it to one of the 1615300 canonical necessary models. The ordinary assignment and all 22 supplemental families cover that model and every required alternative. Their integer row combinations yield the contradiction of Lemma 1.2. Thus size 87 is impossible, so $N \leq 86$. None of these steps invokes SIZE86 or any subsequent exclusion of size 86. □

39 Finite verification record

The finite premises used in the preceding proof were freshly recomputed on 5 October 2026. The complete 19-stage endpoint replay exited with status zero, and every added prerequisite and independent check passed. Together with the actual-set reductions and exhaustive alternatives proved above, these calculations discharge the finite premises of Theorem 38.1 through Lemma 1.2.

39.1 Companion files and reproduction

The complete electronic companion is `subset_sums_upper86_complete_verification_20261005.zip`. Extract it completely and run the following command from its top-level directory:

```
python3 -I -S -B verify_complete_upper86.py
```

On Windows the corresponding command is

```
py -3 -I -S -B verify_complete_upper86.py
```

Python 3.10 or later and its standard library suffice. No network access, numerical solver, optimization package, or original research directory is required. A complete run may take several hours. The supplied `README_COMPLETE.txt` gives extraction, storage and launcher details.

The original generators, root descriptors, coefficient pools, compact assignments, trees and manifests are retained under `payload/subset_sum_verification/`. The paths beginning `proofs/` in the mathematical chapters are relative to that directory. The new complete exposition is included as `proof/complete_upper86_proof.tex`, one independently compilable LaTeX file. Its compiled PDF is delivered separately alongside the archive.

The default entry point runs the earlier endpoint stages through size 88, then the complete size-87 controller, then the added finite prerequisites and independent checks. Each invocation creates its own new result directory; it does not reuse endpoint receipts from a prior run. The resulting `result.json`, aggregate log and individual step logs record completion. A failed or interrupted run does not establish the finite lemmas. The optional integrity-only and auxiliary-only modes do not establish the complete chain either.

39.2 The execution actually performed

The validation of this edition was compositional:

1. One fresh invocation of the unchanged 19-stage endpoint replay began at 16:53:13 UTC and finished at 20:21:50 UTC on 5 October 2026. Every stage passed, the process exited zero, and the final source-integrity check passed. Its elapsed time was 12 517.34 seconds, approximately 3 hours 28 minutes 37 seconds.

2. All fourteen added auxiliary, independent and control subprocesses were executed through the new public wrapper and exited zero. This auxiliary run took 303.96 seconds. The public runner, all mathematical checkers and all proof data in the delivered version are byte-identical to those tested. The later changes to packaging and documentation do not alter their mathematical computations.

3. Thirteen wrapper rejection controls and twelve mathematical negative controls passed. Extraction and execution paths containing spaces and non-ASCII characters were tested. The executed environment was Linux/POSIX with Python 3.12.14. The Windows launcher and paths were statically checked; no native-Windows execution is claimed.

A second multi-hour invocation of the new wrapper in its default mode was not performed. Its routing and failure behavior were tested separately, and the completed component executions above cover all required mathematical stages. The exact commands, elapsed times, source-equivalence record and scope are recorded in `REPRODUCTION_REPORT.json`. The companion's `validation/historical_19_stage.log` and `validation/auxiliary_result.json` preserve the corresponding execution records. They document the calculations; the calculations, not the existence of a record, supply the finite premises.

39.3 Coverage of the final endpoint

The size-87 replay regenerated and covered all 1 615 300 canonical roots. Exactly 1 614 434 received ordinary contradictions; the remaining 866 parents are the disjoint union of the 22 specified supplemental families. These have 1 042 child models, 10 163 exhaustive binary splits and 11 205 contradictory leaves. Thus the final endpoint has

$$1\,614\,434 + 11\,205 = 1\,625\,639$$

exact leaf contradictions. The ordinary replay used 65 consecutive ranges whose union is exactly $[0, 1615300)$; the final composition checked the complete range union and every supplemental identity. It was a complete fresh replay, rather than a sampled root test.

The additional checks also recomputed the finite structural and strict-gate prerequisites, the additional AP-free bounds and block partitions, fixed-label ownership, canonical root ordering, exact supplemental-parent equality, selected split rows and independent stored-row contradictions. The 605 effective-hole witnesses, all 2 992 three-missing-unit integer-center identities, and all 35 five-point profiles were checked in their stated domains. The earlier AP-free and non-AP finite bases are included in the 19-stage replay. The four analytic combinations in Appendix G were independently expanded; the former 32-case exception calculation is retained only as an additional regression check.

At each finite leaf, its current necessary rows and the supplied nonnegative integer multipliers satisfy the exact criterion in Lemma 1.1. Exact child and parent identities, both children of every selected split, and every leaf are checked; census totals alone are not the test. The universal reductions, cited mathematical inputs and applicability of the rows have their written proofs in this supplement. No proof-assistant formalization is claimed.

A The fixed-slot deletion certificate

Let

$$p_1 > p_2 > p_3 > p_4 > 0, \qquad q_1 > q_2 > q_3 > q_4 > 0,$$

and define the eight positive gaps

$$r_1 = p_1 - p_2,\ r_2 = p_2 - p_3,\ r_3 = p_3 - p_4,\ r_4 = p_4,$$
$$s_1 = q_1 - q_2,\ s_2 = q_2 - q_3,\ s_3 = q_3 - q_4,\ s_4 = q_4.$$

Put

$$c_p = p_1 - p_2 - p_3 - p_4, \quad \alpha_p = p_2 - p_3 - p_4, \quad \beta_p = 2p_2 - p_1 - p_3 - p_4,$$
$$c_q = q_1 - q_2 - q_3 - q_4, \quad \alpha_q = q_2 - q_3 - q_4, \quad \beta_q = 2q_2 - q_1 - q_3 - q_4.$$

The two ordered eight-tuples are

$$B^+ = (p_1, p_2, p_3, p_4, -q_1, -q_2, -q_3, -q_4),$$

$$B^- = (q_1, q_2, q_3, q_4, -p_1, -p_2, -p_3, -p_4).$$

For $\varepsilon \in \{+, -\}$, set

$$C_{ij}^{\varepsilon} = c_{\varepsilon} + B_i^{\varepsilon} + B_j^{\varepsilon}, \quad U_i^{\varepsilon} = \alpha_{\varepsilon} + B_i^{\varepsilon}, \quad V_i^{\varepsilon} = \beta_{\varepsilon} + B_i^{\varepsilon},$$

where $1 \le i \le j \le 8$, and U, V use $2 \le i \le 8$. For a decreasing quadruple $x = (x_1, x_2, x_3, x_4)$, define

$$H_1(x) = x_3 - x_4, \qquad H_2(x) = 2x_3 - x_2 - x_4,$$
$$H_3(x) = x_2 + x_3 - x_1 - x_4, \qquad H_4(x) = 2x_3 - x_1 - x_4,$$
$$H_5(x) = 3x_3 - x_1 - x_2 - x_4.$$

Write $H_i^+ = H_i(p)$ and $H_i^- = H_i(q)$. The common slot universe has size

$$2\left(\binom{9}{2} + 7 + 7 + 5\right) = 110.$$

The following table gives 15 templates. Every starred row is also used after interchanging $+$ and $-$. The four unstarred rows and the two orientations of the eleven starred rows give 26 groups.

	sum of scores	$(r_1, r_2, r_3, r_4; s_1, s_2, s_3, s_4)$
1	$C_{25}^+ + C_{25}^-$	$(0, 0, -1, -2; 0, 0, -1, -2)$
2*	$C_{48}^+ + H_3^+$	$(0, 0, 0, -1; 0, 0, 0, -1)$
3	$C_{58}^+ + C_{58}^-$	$(0, -1, -2, -4; 0, -1, -2, -4)$
4*	$C_{77}^+ + H_4^+$	$(0, -1, 0, -2; 0, 0, -2, -2)$
5	$U_6^+ + U_6^-$	$(0, 0, -1, -2; 0, 0, -1, -2)$
6	$V_6^+ + V_6^-$	$(-1, 0, -1, -2; -1, 0, -1, -2)$
7*	$C_{15}^+ + H_1^+ + C_{55}^-$	$(0, -1, -1, -3; 0, -1, -2, -3)$
8*	$C_{26}^+ + C_{57}^- + V_4^-$	$(0, 0, -2, -3; 0, 0, -2, -3)$
9*	$C_{35}^+ + V_3^+ + C_{46}^-$	$(0, 0, 0, -2; 0, -1, -2, -2)$
10*	$C_{36}^+ + U_4^+ + U_5^-$	$(0, 0, -1, -2; 0, 0, -1, -2)$
11*	$C_{44}^+ + U_8^+ + H_5^+$	$(0, -1, 0, -1; 0, 0, 0, -1)$
12*	$C_{45}^+ + V_8^+ + C_{67}^-$	$(0, 0, -3, -4; 0, -1, -2, -4)$
13*	$C_{47}^+ + V_7^+ + H_2^+$	$(0, 0, 0, -2; 0, 0, -2, -2)$
14*	$C_{56}^+ + V_2^+ + C_{66}^-$	$(0, 0, -2, -4; 0, -2, -3, -4)$
15*	$C_{68}^+ + U_3^+ + V_5^-$	$(0, 0, -1, -3; -1, 0, -1, -3)$

Every gap is strictly positive, while each coefficient vector is coordinatewise nonpositive and has a negative coordinate. Hence all slots in one group cannot have positive score. The 26

groups are mutually disjoint in the common universe, as is checked directly by their labels. The accompanying exact-arithmetic script also reconstructs every coefficient vector and checks this disjointness.

For the two translate pairings, first take

$$u_1 = c_p - q_1, \qquad v_1 = -c_q + p_1.$$

Then

$$u_1 - v_1 = -(p_2 + p_3 + p_4 + q_2 + q_3 + q_4) < 0.$$

If $\beta_p + \beta_q \le 0$, take $u_2 = \beta_p$ and $v_2 = -\beta_q$. Otherwise take

$$u_2 = c_p - q_2, \qquad v_2 = -c_q + p_2.$$

The identity

$$u_2 - v_2 = -(\beta_p + \beta_q) - 2(p_3 + p_4 + q_3 + q_4) < 0$$

proves the needed order in the second branch. The two pairs use distinct table slots, so (4) can be applied twice.

B Complete certificates for one empty covering unit

This appendix is part of the proof of Lemma 13.4. It gives the models, a complete case table, and every multiplier; no external certificate database or numerical solver is needed to specify the proof.

B.1 Centers, modes, and constraints

Index the ten positive-side and ten reflected negative-side centers from zero, solely for this appendix:

$$v^p = (c_p + p_1, c_p + p_2, c_p + p_3, c_p + p_4, c_p - q_1, c_p - q_2, c_p - q_3, c_p - q_4, \alpha_p, \beta_p),$$
$$v^q = (c_q + q_1, c_q + q_2, c_q + q_3, c_q + q_4, c_q - p_1, c_q - p_2, c_q - p_3, c_q - p_4, \alpha_q, \beta_q).$$

The first pair consists of indices 4 on the two sides. The second consists of indices $j = 9$ in branch A and $j = 5$ in branch B. The other sixteen slots are unpaired. A mode specifies the one missing unit and, when applicable, the contributing side of the second pair. Labels pi, qi refer to center v_i^p, v_i^q; labels $1p, 2p, 2q$ refer to v_4^p, v_j^p, v_j^q when used as paired units. Such labels identify distinct covering slots even when their numerical center values happen to agree.

Keep every unpaired center except the missing one. Keep $1p$ unless the first pair is missing. Keep the selected member $2p$ or $2q$ unless the second pair is missing. In branch A with missing $p5$ or $q5$, keep neither member of the second pair: this is the deliberate relaxation allowing it to split. These rules define the active list $\mathcal{C}$ for every mode.

All variables

$$x = (p_1, \dots, p_4, q_1, \dots, q_4, M_p, M_q, R_p, R_q, S_p, S_q, \delta, t, (l_v, u_v)_{v \in \mathcal{C}})$$

are nonnegative real numbers. The following row names denote the indicated inequalities in the orientation "left side $\le$ right side."

Row name	Constraint
D_i^p, D_i^q $(1 \le i \le 3)$	$p_{i+1} - p_i \le -1$, $q_{i+1} - q_i \le -1$
D_4^p, D_4^q	$M_p - p_4 \le -1$, $M_q - q_4 \le -1$
E_p, E_q	$R_p - M_p \le -4$, $R_q - M_q \le -4$
G_p, G_q	$-R_p + t \le -39$, $-R_q - t \le -47$
H_p	$2M_p + 6R_p - 2S_p + 87t \le -1556$
H_q	$2M_q + 6R_q - 2S_q - 87t \le -2252$
Z, J	$\delta \le 21$, $t \le 8$
B (branch A)	$\beta_p + \beta_q \le 0$
B (branch B)	$-\beta_p - \beta_q \le -1$
L_v (positive side)	$-v - l_v - t \le -52$
U_v (positive side)	$v - u_v - S_p + t \le -43$
L_v (negative side)	$-v - l_v + t \le -44$
U_v (negative side)	$v - u_v - S_q - t \le -51$
Σ	$\sum_{v \in \mathcal{C}} (l_v + u_v) - \delta \le 0$

In the multiplier tables, for example, L_{p5} means the lower row for v_5^p, and U_{2q} means the upper row for the selected second negative paired center. The objective is

$$f(x) = S_p + S_q + \delta - 2(M_p + M_q) - 2(R_p + R_q), \qquad F = 95 + f(x).$$

B.2 How to read and check a certificate

Write the rows as $Ax \le b$. Every displayed multiplier is a nonnegative integer. A certificate with objective multiplier $D > 0$ has row-weight vector w satisfying

$$Df + wA \ge 0 \quad \text{coefficientwise.}$$

Since $x \ge 0$, it proves $f(x) \ge -wb/D$, hence $F \ge 95 - wb/D$. A contradiction certificate, marked $D = 0$, satisfies

$$wA \ge 0 \quad \text{coefficientwise,} \qquad wb = -K < 0,$$

and proves $0 \le wAx \le -K$. Each certificate uses only rows present in every mode to which it is assigned. The complete case table below contains 33 branch-A modes and 35 branch-B modes. A dash means no such mode is required; for a missing second pair or a deliberately relaxed pair, the single certificate is placed in the p column without asserting that the pair contributes on that side.

Missing unit	A: second p	A: second q	B: second p	B: second q
First pair	1	2	10	11
Second pair	3	–	12	–
$p0$	4	5	13	14
$p1$	4	6	12	14
$p2$	4	6	12	14
$p3$	4	6	12	14
$p5$	3	–	–	–
$p6$	4	6	12	14
$p7$	4	6	12	14
$p8$	4	7	12	14
$p9$	–	–	15	16
$q0$	4	6	12	14
$q1$	4	6	12	14
$q2$	4	6	12	14
$q3$	4	6	12	14
$q5$	8	–	–	–
$q6$	4	6	17	14
$q7$	4	6	18	19
$q8$	9	6	12	14
$q9$	–	–	20	21

B.3 All twenty-one multiplier combinations

The last column lists the nonzero weights as a formal sum of row names. For $D > 0$ the conclusion is the stated lower bound for F; for $D = 0$ it is the impossible inequality $0 \le -K$.

ID	D	Conclusion	Integer row combination
1	13	$F \ge \frac{25847}{13}$	$10G_p + 10G_q + 6H_p + 6H_q + U_{p0} + 7D_3^p + 14D_4^p + 5L_{p5} + 3L_{2p} + U_{q0} + 7D_3^q + 14D_4^q + 2L_{q5} + 6L_{q8} + 13\Sigma$
2	13	$F \ge \frac{25847}{13}$	$10G_p + 10G_q + 6H_p + 6H_q + U_{p0} + 7D_3^p + 14D_4^p + 2L_{p5} + 6L_{p8} + 3L_{2q} + U_{q0} + 7D_3^q + 14D_4^q + 5L_{q5} + 13\Sigma$
3	14	$F \ge 1982$	$2E_q + 8G_p + 16G_q + 6H_p + 7H_q + 2U_{p0} + 8D_3^p + 16D_4^p + 6L_{p8} + 4L_{1p} + 8D_3^q + 16D_4^q + 4L_{q5} + 4L_{q8} + 14\Sigma + 95J$
4	0	$0 \le -171$	$Z + L_{1p} + L_{2p} + L_{q5} + L_{q8} + \Sigma$
5	11	$F \ge \frac{21575}{11}$	$8G_p + 8G_q + 5H_p + 5H_q + U_{p1} + 6D_3^p + 12D_4^p + 6L_{p8} + L_{1p} + 2L_{2q} + U_{q0} + 6D_3^q + 12D_4^q + 5L_{q5} + 11\Sigma$
6	2	$F \ge 2424$	$20E_p + 22E_q + 16G_p + 24G_q + H_q + 8Z + 2U_{p0} + 12D_3^p + 24D_4^p + 10L_{p8} + 4L_{1p} + 4L_{2q} + 12D_3^q + 24D_4^q + 8L_{q5} + 10\Sigma + 95J$
7	4	$F \ge 2011$	$8E_p + 20E_q + 6G_p + 24G_q + H_p + 2H_q + 2U_{p0} + 6D_2^p + 12D_3^p + 14D_4^p + 4L_{1p} + 4L_{2q} + 12D_3^q + 24D_4^q + 4L_{q5} + 4L_{q6} + 4\Sigma + 95J$
8	10	$F \ge 1940$	$6E_q + 4G_p + 16G_q + 4H_p + 5H_q + 2U_{p0} + 8D_3^p + 12D_4^p + 2L_{p8} + 4L_{1p} + 8D_3^q + 16D_4^q + 4L_{q6} + 4L_{q8} + 10\Sigma + 95J$
9	0	$0 \le -171$	$E_p + G_p + Z + D_4^p + L_{1p} + L_{2p} + L_{q5} + \Sigma$
10	2	$F \ge 2178$	$14E_p + 22E_q + 10G_p + 24G_q + H_q + 4Z + 2U_{p0} + 12D_3^p + 18D_4^p + 2L_{p9} + 6L_{2p} + 12D_3^q + 24D_4^q + 6L_{q6} + 6L_{q9} + 6\Sigma + 95J$
11	2	$F \ge 2178$	$22E_p + 14E_q + 24G_p + 10G_q + H_p + 4Z + 12D_3^p + 24D_4^p + 6L_{p6} + 6L_{p9} + 6L_{2q} + 2U_{q0} + 12D_3^q + 18D_4^q + 2L_{q9} + 6\Sigma$
12	2	$F \ge 2757$	$20E_p + 34E_q + 16G_p + 36G_q + H_q + 4Z + 2U_{p0} + 12D_3^p + 24D_4^p + 2L_{p9} + 6L_{1p} + 18D_3^q + 36D_4^q + 6L_{q6} + 6L_{q7} + 6L_{q9} + 6\Sigma + 95J$
13	2	$F \ge 2101$	$12E_p + 22E_q + 8G_p + 24G_q + H_q + 2Z + 2U_{p1} + 8D_3^p + 16D_4^p + 2L_{p9} + 4L_{1p} + 12D_3^q + 24D_4^q + 4L_{q6} + 4L_{q7} + 4L_{q9} + 4\Sigma + 95J$
14	0	$0 \le -215$	$Z + L_{p9} + L_{1p} + L_{2q} + L_{q7} + L_{q9} + \Sigma$

ID	D	Conclusion	Integer row combination
15	2	$F \geq 2662$	$20E_p + 34E_q + 16G_p + 36G_q + H_q + 2B + 4Z + 2U_{p0} + 12D_3^p + 24D_4^p + 6L_{1p} + 18D_3^q + 36D_4^q + 6L_{q6} + 6L_{q7} + 4L_{q9} + 6\Sigma + 95J$
16	0	$0 \leq -120$	$B + Z + L_{1p} + L_{2q} + L_{q7} + \Sigma$
17	2	$F \geq 2691$	$20E_p + 34E_q + 16G_p + 36G_q + H_q + 10Z + 2U_{p0} + 6D_3^p + 24D_4^p + 2L_{p9} + 6L_{1p} + 18D_3^q + 36D_4^q + 12L_{q7} + 6L_{q9} + 12\Sigma + 95J$
18	2	$F \geq 2697$	$20E_p + 34E_q + 16G_p + 36G_q + H_q + 10Z + 2U_{p0} + 18D_3^p + 24D_4^p + 2L_{p9} + 6L_{1p} + 18D_3^q + 36D_4^q + 12L_{q6} + 6L_{q9} + 12\Sigma + 95J$
19	0	$0 \leq -215$	$Z + L_{p9} + L_{1p} + L_{2q} + L_{q6} + L_{q9} + \Sigma$
20	2	$F \geq 2276$	$16E_p + 26E_q + 12G_p + 28G_q + H_q + 2B + 4Z + 2U_{p0} + 12D_3^p + 20D_4^p + 6L_{1p} + 14D_3^q + 28D_4^q + 6L_{q6} + 2L_{q7} + 4L_{q8} + 6\Sigma + 95J$
21	0	$0 \leq -171$	$Z + L_{p9} + L_{1p} + L_{2q} + L_{q8} + \Sigma$

For example, certificate 8, the weakest lower bound, has $D = 10$ and $wb = -18450$. Its coefficients on the sixteen original variables in $10f + wA$ are all zero, and its remaining coefficients on the loss variables are nonnegative. It therefore gives $F \geq 95 + 18450/10 = 1940 > 1836$. Certificate 4 has $wb = -171$; its non-loss coefficients in wA are $2p_3 + 2p_4 + 2q_3 + 2q_4$, all other terms being nonnegative loss terms. These examples illustrate both certificate types. Expanding every row in the same manner verifies all entries. The companion verifier reconstructs each model, checks that the mode list is exhaustive, and performs every coefficient comparison using integers and rational numbers.

C Complete certificate specification for the upper bound 100

This appendix specifies all 2512 linear systems and all 42 integer combinations used to exclude their feasible-set interpretations. It is independent of floating-point optimization. The companion data file records one successful combination per system; that assignment is a convenience, not an additional premise. Trying the 42 printed combinations in order provides a complete finite verification rule.

C.1 Four regimes and all their modes

Use exactly the centers, branches, paired units, missing-unit choices, and active lists $\mathcal{C}$ defined in Appendix B. There are 68 basic modes. In particular, the two possibly split branch-A second pairs are still deliberately omitted from $\mathcal{C}$. Put $a = a_0 + t$, $b = b_0 - t$, and let f be the same linear form as in that appendix. Now $F = d + f$, where $d = n - 8$ depends on the regime.

Regime	n	a_0	b_0	$t_{\max}$	$\delta_{\max}$	s_0	A_p	A_q	Upper F
I	102	36	49	13	45	85	1308	2413	1874
Z	101	34	50	16	69	85	1138	2498	1800
P	101	36	48	12	41	87	1376	2420	1856
E	101	37	47	10	69	85	1393	2243	1800

The letters designate size 102, zero holes with no retained exception, positive holes, and zero holes with a retained exception, respectively. Regimes I, Z and P use each of the 68 basic modes once. Regime E uses every basic mode and every choice of one retained center and one of its two endpoints. The two split-pair modes have 16 retained centers; the other 66 have 17. Thus regime E has

$$2(2 \cdot 16 + 66 \cdot 17) = 2308$$

systems, and the total is $3 \cdot 68 + 2308 = 2512$.

All original variables and all losses are nonnegative. Keep the order rows D_i^p, D_i^q, span rows E_p, E_q, branch row B, and total-loss row Σ from Appendix B. The rows that depend on the regime are replaced as follows:

Row	Constraint
G_p, G_q	$-R_p + t \le -a_0$, $-R_q - t \le -b_0$
H_p	$2M_p + 6R_p - 2S_p + s_0 t \le -A_p$
H_q	$2M_q + 6R_q - 2S_q - s_0 t \le -A_q$
Z, J	$\delta \le \delta_{\max}$, $t \le t_{\max}$
L_v (positive)	$-v - l_v - t \le -(b_0 + 5)$
U_v (positive)	$v - u_v - S_p + t \le -(a_0 + 4)$
L_v (negative)	$-v - l_v + t \le -(a_0 + 5)$
U_v (negative)	$v - u_v - S_q - t \le -(b_0 + 4)$

The mass rows use $2T_r \ge 85r - 1764$, the tangent at 42. In regime P one also adds $\varepsilon_p \ge a - 2$ and $\varepsilon_q \ge b - 2$; this accounts for its slope 87 and its larger constants.

In regime E, replace the *single* quantile row at the designated exceptional endpoint, and add a row X for its unavoidable loss:

Exceptional endpoint	Replacement quantile row	Additional row X
Positive, lower	$-v - l_v - M_p \le -d$	$-l_v - t \le -(b_0 + 5)$
Positive, upper	$v - u_v - S_p - M_q \le -(d - 1)$	$-u_v + t \le -(a_0 + 5)$
Negative, lower	$-v - l_v - M_q \le -d$	$-l_v + t \le -(a_0 + 5)$
Negative, upper	$v - u_v - S_q - M_p \le -(d - 1)$	$-u_v - t \le -(b_0 + 5)$

The designated quantile row keeps its original name L_v or U_v. Its opposite endpoint row remains unchanged. No inequality asserting the actual location $v < 0$ or $v > S$ is imposed: omitting it makes the model weaker, so a contradiction still excludes the original configuration.

C.2 Exact verification rule

For each of the systems just specified, write its rows as $Ax \le b$, where $x \ge 0$, and regard f as a coefficient row. A candidate in the table below specifies an objective multiplier $D \ge 0$ and a nonnegative integer combination w of named rows. Discard that candidate if it calls for a row absent from the current mode. Otherwise form, using integers,

$$q = Df + wA, \qquad h = wb.$$

It certifies the current mode if $q \ge 0$ coefficientwise and either

$$D = 0,\ h < 0, \qquad \text{or} \qquad D > 0,\ d - h/D > F_{\text{upper}}.$$

The proof is the same elementary row-combination argument as in Appendix B. Importantly, a candidate need not pass in every mode; the assertion is that at least one passes in each mode.

The following pseudocode specifies the full check without relying on a stored assignment of certificates:

```
for regime in (I, Z, P, E):
  for basic_mode in the 68 generated modes:
    for endpoint in this regime's endpoint choices:
      construct A, b, f from the printed rows
      passed = false
      for (D, w) in the 42 printed candidates:
        if all named rows exist and the exact test passes:
          passed = true; break
      assert passed
```

For I, Z and P there is a single choice "no exception." For E the choices are the $2|\mathcal{C}|$ retained endpoints. The delivered verifier reconstructs the same exhaustive list and checks one recorded successful integer combination for each system. Its checked outcomes are:

Regime	Systems	Contradiction combinations	Lower-bound combinations	Minimum certified F
I	68	8	60	$9378/5 > 1874$
Z	68	8	60	$9103/5 > 1800$
P	68	29	39	$1910 > 1856$
E	2308	351	1957	$10957/6 > 1800$

These are counts of certificate types, not claims that a lower-bound system is feasible. All 2512 interpretations as feasible sets are excluded.

C.3 The forty-two integer combinations

Row labels, including L_{1p}, L_{2p}, L_{2q} and their upper counterparts, have the same slot meanings as in Appendix B. The symbol X denotes the extra exception-loss row and is absent outside regime E. All omitted row weights are zero.

ID	D	Nonnegative integer row combination
1	10	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q + 6E_q + 4G_p + 16G_q + 4H_p + 5H_q + 93J + 2U_{p0} + 2L_{p8} + 4L_{q6} + 4L_{q8} + 4L_{1p} + 10\Sigma$
2	0	$D_4^q + Z + B + L_{q2} + L_{1p} + L_{2q} + \Sigma$
3	0	$D_4^p + E_p + G_p + Z + L_{q5} + L_{1p} + L_{2p} + \Sigma$
4	6	$12D_3^p + 18D_4^p + 10E_p + 12D_3^q + 24D_4^q + 12E_q + 10G_p + 18G_q + 2H_p + 3H_q + 93J + 2U_{p0} + 2L_{p9} + 6L_{q6} + 6L_{q9} + 6L_{2p} + 6\Sigma$
5	12	$10D_2^p + 12D_3^p + 14D_4^p + 12D_3^q + 24D_4^q + 12E_q + 6G_p + 24G_q + 5H_p + 6H_q + 95J + 2U_{p0} + 8L_{q5} + 4L_{1p} + 4L_{2q} + 12\Sigma$
6	6	$12D_3^p + 24D_4^p + 12E_p + 12D_3^q + 18D_4^q + 10E_q + 18G_p + 10G_q + 3H_p + 2H_q + 6L_{p6} + 6L_{p9} + 2U_{q0} + 2L_{q9} + 6L_{2q} + 6\Sigma$
7	13	$7D_3^p + 14D_4^p + 7D_3^q + 14D_4^q + 10G_p + 10G_q + 6H_p + 6H_q + U_{p0} + 5L_{p5} + U_{q0} + 2L_{q5} + 6L_{q8} + 3L_{2p} + 13\Sigma$
8	13	$7D_3^p + 14D_4^p + 7D_3^q + 14D_4^q + 10G_p + 10G_q + 6H_p + 6H_q + U_{p0} + 2L_{p5} + 6L_{p8} + U_{q0} + 5L_{q5} + 3L_{2q} + 13\Sigma$
9	11	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q + 8G_p + 8G_q + 5H_p + 5H_q + U_{p1} + 6L_{p8} + U_{q0} + 5L_{q5} + L_{1p} + 2L_{2q} + 11\Sigma$
10	2	$6D_3^p + 24D_4^p + 20E_p + 18D_3^q + 36D_4^q + 22E_q + 16G_p + 24G_q + H_q + 10Z + 93J + 2U_{p0} + 2L_{p9} + 12L_{q7} + 6L_{q9} + 6L_{1p} + 12\Sigma$
11	14	$12D_3^p + 16D_4^p + 12D_3^q + 24D_4^q + 10E_q + 8G_p + 24G_q + 6H_p + 7H_q + 93J + 2U_{p1} + 2L_{p9} + 8L_{q6} + 4L_{q9} + 4L_{1p} + 14\Sigma$
12	10	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q + 2E_q + 4G_p + 12G_q + 4H_p + 5H_q + 87J + 2U_{p0} + 2L_{p8} + 4L_{q6} + 4L_{q8} + 4L_{1p} + 10\Sigma + 6X$
13	0	$D_4^p + Z + L_{q5} + L_{1p} + L_{2p} + \Sigma$
14	6	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q + 10E_q + 16G_q + 2H_p + 3H_q + 95J + 2U_{p0} + 2L_{p8} + 4L_{q6} + 4L_{q8} + 4L_{1p} + 6\Sigma + 2X$
15	1432	$1238D_3^p + 2476D_4^p + 862E_p + 1238D_3^q + 2476D_4^q + 1748G_p + 922G_q + 625H_p + 631H_q + 182U_{p0} + 364L_{p5} + 1056L_{p8} + 170U_{q0} + 874L_{q5} + 534L_{2q} + 1432\Sigma + 558X$
16	0	$2D_4^p + 2E_p + D_4^q + 2G_p + Z + L_{q5} + L_{1p} + L_{2p} + \Sigma$
17	10	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q + 6E_q + 4G_p + 16G_q + 4H_p + 5H_q + 95J + 2U_{p0} + 2L_{p8} + 4L_{q6} + 4L_{q8} + 4L_{1p} + 10\Sigma$
18	8	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q + 8E_q + 2G_p + 16G_q + 3H_p + 4H_q + 99J + 2U_{p0} + 2L_{p8} + 4L_{q6} + 4L_{q8} + 4L_{1p} + 8\Sigma + 6X$
19	657	$657D_3^p + 1314D_4^p + 562E_p + 657D_3^q + 1314D_4^q + 934G_p + 378G_q + 281H_p + 282H_q + 95U_{p0} + 471L_{p5} + 93U_{q0} + 186L_{q5} + 564L_{q8} + 281L_{2p} + 657\Sigma + 93X$
20	1058	$744D_3^p + 1190D_4^p + 744D_3^q + 1488D_4^q + 662G_p + 710G_q + 463H_p + 471H_q + 132U_{p1} + 446L_{p8} + 116U_{q0} + 314L_{q5} + 298L_{q6} + 132L_{1p} + 248L_{2q} + 1058\Sigma + 744X$
21	2	$12D_3^p + 18D_4^p + 12E_p + 12D_3^q + 24D_4^q + 22E_q + 8G_p + 24G_q + H_q + 4Z + 97J + 2U_{p0} + 2L_{p9} + 6L_{q6} + 6L_{q9} + 6L_{2p} + 6\Sigma + 4X$
22	6	$12D_3^p + 24D_4^p + 18E_p + 12D_3^q + 18D_4^q + 4E_q + 24G_p + 4G_q + 3H_p + 2H_q + 6L_{p6} + 6L_{p9} + 2U_{q0} + 2L_{q9} + 6L_{2q} + 6\Sigma$
23	12	$10D_2^p + 12D_3^p + 14D_4^p + 12D_3^q + 24D_4^q + 8E_q + 6G_p + 20G_q + 5H_p + 6H_q + 85J + 2U_{p0} + 8L_{q5} + 4L_{1p} + 4L_{2q} + 12\Sigma + 8X$
24	8	$12D_3^p + 16D_4^p + 6E_p + 12D_3^q + 24D_4^q + 12E_q + 8G_p + 20G_q + 3H_p + 4H_q + 89J + 2U_{p1} + 2L_{p9} + 8L_{q6} + 4L_{q9} + 4L_{1p} + 8\Sigma + 4X$
25	1432	$1238D_3^p + 2476D_4^p + 1238D_3^q + 2476D_4^q + 862E_q + 922G_p + 1748G_q + 631H_p + 625H_q + 170U_{p0} + 874L_{p5} + 182U_{q0} + 364L_{q5} + 1056L_{q8} + 534L_{2p} + 1432\Sigma + 558X$
26	2252	$1310D_3^p + 2620D_4^p + 1310D_3^q + 2620D_4^q + 162E_q + 1766G_p + 1796G_q + 1045H_p + 1023H_q + 162U_{p0} + 898L_{p5} + 206U_{q0} + 412L_{q5} + 1104L_{q8} + 574L_{2p} + 2252\Sigma + 2046X$
27	1868	$1302D_3^p + 2604D_4^p + 1302D_3^q + 2604D_4^q + 538E_q + 1358G_p + 1812G_q + 849H_p + 835H_q + 170U_{p0} + 906L_{p5} + 198U_{q0} + 396L_{q5} + 1104L_{q8} + 566L_{2p} + 1868\Sigma + 1302X$
28	111	$69D_3^p + 138D_4^p + 69D_3^q + 138D_4^q + 16E_q + 84G_p + 94G_q + 51H_p + 50H_q + 9U_{p0} + 18L_{p5} + 60L_{p8} + 11U_{q0} + 51L_{q5} + 29L_{2q} + 111\Sigma + 93X$

ID	D	Nonnegative integer row combination
29	657	$657D_3^p + 1314D_4^p + 657D_3^q + 1314D_4^q + 562E_q + 378G_p + 934G_q + 282H_p + 281H_q + 93U_{p0} + 186L_{p5} + 564L_{p8} + 95U_{q0} + 471L_{q5} + 281L_{2q} + 657\Sigma + 93X$
30	2260	$1322D_3^p + 2644D_4^p + 1322D_3^q + 2644D_4^q + 170E_q + 1750G_p + 1788G_q + 1045H_p + 1023H_q + 170U_{p0} + 340L_{p5} + 1152L_{p8} + 214U_{q0} + 982L_{q5} + 554L_{2q} + 2260\Sigma + 2046X$
31	5	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q + 6E_q + 2G_p + 8G_q + 2H_p + 2H_q + Z + U_{p1} + 6L_{p8} + U_{q0} + 5L_{q5} + L_{1p} + 2L_{2q} + 6\Sigma$
32	1880	$1128D_3^p + 2256D_4^p + 1128D_3^q + 2256D_4^q + 170E_q + 1370G_p + 1432G_q + 855H_p + 837H_q + 170U_{p1} + 1128L_{p8} + 206U_{q0} + 958L_{q5} + 170L_{1p} + 376L_{2q} + 1880\Sigma + 1674X$
33	204	$120D_3^p + 240D_4^p + 120D_3^q + 240D_4^q + 14E_q + 150G_p + 152G_q + 93H_p + 91H_q + 18U_{p1} + 120L_{p8} + 22U_{q0} + 102L_{q5} + 18L_{1p} + 40L_{2q} + 204\Sigma + 186X$
34	8	$10D_2^p + 12D_3^p + 14D_4^p + 12D_3^q + 24D_4^q + 16E_q + 2G_p + 24G_q + 3H_p + 4H_q + 97J + 2U_{p0} + 8L_{q5} + 4L_{1p} + 4L_{2q} + 8\Sigma + 4X$
35	6	$12D_3^p + 18D_4^p + 4E_p + 12D_3^q + 24D_4^q + 18E_q + 4G_p + 24G_q + 2H_p + 3H_q + 93J + 2U_{p0} + 2L_{p9} + 6L_{q6} + 6L_{q9} + 6L_{2p} + 6\Sigma$
36	4	$8D_3^p + 16D_4^p + 10E_p + 12D_3^q + 24D_4^q + 20E_q + 8G_p + 24G_q + H_p + 2H_q + 95J + 2U_{p1} + 2L_{p9} + 4L_{q6} + 4L_{q7} + 4L_{q9} + 4L_{1p} + 4\Sigma$
37	2	$4D_3^p + 16D_4^p + 12E_p + 12D_3^q + 24D_4^q + 22E_q + 8G_p + 24G_q + H_q + 6Z + 85J + 2U_{p1} + 2L_{p9} + 8L_{q7} + 4L_{q9} + 4L_{1p} + 8\Sigma + 8X$
38	4	$8D_3^p + 16D_4^p + 6E_p + 12D_3^q + 24D_4^q + 20E_q + 4G_p + 24G_q + H_p + 2H_q + 93J + 2U_{p1} + 2L_{p9} + 4L_{q6} + 4L_{q7} + 4L_{q9} + 4L_{1p} + 4\Sigma$
39	2	$6D_3^p + 24D_4^p + 18E_p + 18D_3^q + 36D_4^q + 34E_q + 14G_p + 36G_q + H_q + 10Z + 103J + 2U_{p0} + 2L_{p9} + 12L_{q7} + 6L_{q9} + 6L_{1p} + 12\Sigma + 10X$
40	2	$6D_3^p + 24D_4^p + 14E_p + 18D_3^q + 36D_4^q + 34E_q + 10G_p + 36G_q + H_q + 10Z + 99J + 2U_{p0} + 2L_{p9} + 12L_{q7} + 6L_{q9} + 6L_{1p} + 12\Sigma + 6X$
41	4	$2D_2^p + 8D_3^p + 14D_4^p + 8E_p + 12D_3^q + 24D_4^q + 20E_q + 6G_p + 24G_q + H_p + 2H_q + 89J + 2U_{p0} + 4L_{q6} + 4L_{q7} + 4L_{1p} + 4L_{2q} + 4\Sigma + 4X$
42	2	$8D_3^p + 16D_4^p + 10E_p + 12D_3^q + 24D_4^q + 22E_q + 6G_p + 24G_q + H_q + 2Z + 95J + 2U_{p1} + 2L_{p9} + 4L_{q6} + 4L_{q7} + 4L_{q9} + 4L_{1p} + 4\Sigma + 2X$

All multipliers are integers; the largest displayed one is 2644. The finite certificate check uses no approximation, tolerance, or optimization status flag. The exact source matrices and the checked assignment are also supplied as readable Python and JSON in the verification archive.

D Complete certificate specification for the upper bound 99

This appendix gives the complete rule generating the 59206 systems and all 52 integer combinations that certify them. The JSON file assigns a successful combination to each system. That assignment is optional: trying the printed combinations in order is another complete verification procedure. The source model contains no optimization dependency.

D.1 The 64 effective-hole rows

For each row of the table use $(r, C_7, 7)$ and $(r, C_9, 9)$ in Lemma 10.2. For either $\beta \in \{7, 9\}$ define

$$K = \left\lfloor \frac{3 + \sqrt{9 + 8(C+1)}}{2} \right\rfloor, \qquad D = C + \max(3, K),$$

$$M = \max\left(2\beta, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(D + 1 + 6\beta)}}{2} \right\rfloor\right), \qquad J = \max(K + 1, 2\beta + 5, M).$$

Every entry below satisfies $(r - J + 1)\beta > D$ and $T_{J-1} - 1 > D$. Thus $2\beta < h < |\,\mathrm{FS}\,| - 1$ implies $\mu \geq r + C + 1$.

r	C_7	C_9	r	C_7	C_9
26	44	26	42	135	160
27	51	34	43	142	168
28	57	42	44	147	176
29	63	51	45	149	177
30	70	59	46	155	186
31	76	67	47	162	194
32	83	75	48	167	198
33	89	84	49	168	203
34	96	92	50	175	211
35	102	101	51	182	220
36	109	109	52	187	221
37	110	117	53	188	228
38	116	126	54	195	237
39	122	134	55	202	245
40	128	143	56	208	246
41	129	151	57	209	254

D.2 Centers, units, modes, and variables

Use the twenty labeled centers and the two branch definitions in Appendix B. Explicitly, with $c_p = p_1 - p_2 - p_3 - p_4$, $\alpha_p = p_2 - p_3 - p_4$ and $\beta_p = 2p_2 - p_1 - p_3 - p_4$, the positive list is

$$(c_p + p_1, c_p + p_2, c_p + p_3, c_p + p_4, c_p - q_1, c_p - q_2, c_p - q_3, c_p - q_4, \alpha_p, \beta_p).$$

The negative list is obtained by interchanging p, q. Their labels are $p_0, \ldots, p_9$ and $q_0, \ldots, q_9$; these labels are distinct from the variables $p_1, \ldots, p_4$ and $q_1, \ldots, q_4$. The first paired unit uses index 4. The second uses index 9 in branch A ($\beta_p + \beta_q \leq 0$) and index 5 in branch B ($\beta_p + \beta_q > 0$). There are sixteen other, unpaired units.

Choose one missing unit in Z, P, ZE, PE and ZZ, or any unordered pair of missing units in EE. Retain every other unpaired center. If the first pair is present, retain its positive member $\pi_p = c_p - q_1$. If the second pair is present, retain either its positive member σ_p or its negative member σ_q, with both choices considered. There is one exception: in branch A, when p_5 or q_5 is missing, drop the second pair entirely from the retained list and do not duplicate the mode.

Likewise there is only one choice when the second pair is itself missing. Thus the numbers of modes are 68 with one missing unit and 547 with two. With one missing unit, 66 modes retain 17 centers and two retain 16; choosing an exceptional center and an endpoint gives 2308 possibilities for each fixed a.

For each regime take every integer a in the interval in Section 15, put $b = 83 - a$, $k = 41 - \min(a, b)$ and $I = k(k+1)$. In ZE and PE additionally choose each retained center in turn and each of its lower and upper endpoints. There are no exceptional-location choices in ZZ or EE.

All variables $p_i, q_i, M_p, M_q, R_p, R_q, S_p, S_q, \delta, l_v, u_v$ are nonnegative real variables. The implementation also includes a harmless variable $t \geq 0$ and a row $J : t \leq 0$; all other coefficients of t are zero. Unused loss variables can be set to zero. Write

$$f = S_p + S_q + \delta - 2(M_p + M_q) - 2(R_p + R_q).$$

Then $F = 92 + f$ except in EE, where $F = 184 + f$ and $\delta = 0$.

D.3 All coefficient rows

In the first five regimes the order rows are

$$D_i^p : p_{i+1} - p_i \leq -1 \ (1 \leq i \leq 3), \quad D_4^p : M_p - p_4 \leq -1,$$

with reflected D_i^q. In EE replace their right sides by zero. In all regimes retain

$$E_p : R_p - M_p \leq -4, \quad E_q : R_q - M_q \leq -4, \quad G_p : -R_p \leq -a, \quad G_q : -R_q \leq -b.$$

The exact mass rows are

$$H_p : 2M_p + 6R_p - 2S_p \leq -2T_a - 12 - 2\epsilon_p, \quad H_q : 2M_q + 6R_q - 2S_q \leq -2T_b - 12 - 2\epsilon_q,$$

where $(\epsilon_p, \epsilon_q) = (a-2, b-2)$ in P and PE and $(0,0)$ otherwise. The budget row Z is $\delta \leq 92 - I$ in Z, ZE and ZZ, $\delta \leq 65 - I$ in P and PE, and $\delta \leq 0$ in EE. The branch row B is $\beta_p + \beta_q \leq 0$ in A, $-\beta_p - \beta_q \leq -1$ in B outside EE, and $-\beta_p - \beta_q \leq 0$ in branch B of EE. The total-loss row is

$$\Sigma : \sum_{v \text{ retained}} (l_v + u_v) - \delta \leq 0.$$

The ordinary quantile rows for positive centers are

$$L_v : -v - l_v \leq -b - 5, \qquad U_v : v - u_v - S_p \leq -a - 4,$$

with a, b and p, q interchanged for negative centers. In ZZ increase both right sides by 1 for every retained center, and add $K_p : R_p \leq a + 1$, $K_q : R_q \leq b + 1$.

In ZE and PE replace only the designated exceptional endpoint row, and add a row X as follows:

Endpoint	Replacement row	Row X
Positive, lower	$-v - l_v - M_p \leq -92$	$-l_v \leq -b - 5$
Positive, upper	$v - u_v - S_p - M_q \leq -91$	$-u_v \leq -a - 5$
Negative, lower	$-v - l_v - M_q \leq -92$	$-l_v \leq -a - 5$
Negative, upper	$v - u_v - S_q - M_p \leq -91$	$-u_v \leq -b - 5$

The replaced row keeps its name L_v or U_v; the opposite row is unchanged. The actual out-of-hull location inequality is omitted, which is a relaxation. In EE replace all quantile rows by

$$L_v : -v - l_v + M_q \leq -1, \qquad U_v : v - u_v - S_p + M_p \leq 0$$

for positive centers, and their reflected versions for negative centers. Since $\delta = 0$, the total-loss row makes all retained losses zero. These are exactly the full-translate constraints proved in the text.

D.4 The integer verification rule and multiplier pool

For a generated system $Ax \le b$, $x \ge 0$, a table entry gives $D \ge 0$ and a nonnegative row combination w. A term naming a row absent from the current mode makes that entry unavailable. Otherwise compute

$$q = Df + wA, \qquad h = wb.$$

If q is coordinatewise nonnegative, then for $D > 0$ we have $F \ge d_0 - h/D$, where $d_0 = 92$ or 184. It is a certificate when this is strictly greater than 1782, or 1838 in P and PE. For $D = 0$, the condition $h < 0$ is an inconsistency certificate. Every entry below has integer weights; there are 774 nonzero weights in all. The verifier generates all modes independently of the assignment file, checks the recorded case sequence, and evaluates these identities with arbitrary-precision integers and exact fractions.

ID	D	Nonzero row weights (omitted weights are zero)
1	10	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+6E_q + 4G_p + 16G_q + 4H_p$ $+5H_q + 2U_{p_0} + 2L_{p_8} + 4L_{q_6}$ $+4L_{q_8} + 4L_{\pi_p} + 10\Sigma$
2	0	$Z + L_{q_5} + L_{\pi_p} + L_{\sigma_p}$ $+\Sigma$
3	2	$4D_3^p + 16D_4^p + 12E_p + 12D_3^q$ $+24D_4^q + 22E_q + 8G_p + 24G_q$ $+H_q + 6Z + 2U_{p_1} + 2L_{p_9}$ $+8L_{q_7} + 4L_{q_9} + 4L_{\pi_p} + 8\Sigma$ $+8X$
4	6	$6D_3^p + 12D_4^p + 6E_p + 6D_3^q$ $+12D_4^q + 4E_q + 12G_p + 4G_q$ $+3H_p + 2H_q + 6L_{p_8} + 2U_{q_0}$ $+6L_{q_5} + 2L_{\sigma_q} + 6\Sigma$
5	0	$Z + \Sigma + X$
6	6	$12D_3^p + 24D_4^p + 12E_p + 12D_3^q$ $+18D_4^q + 10E_q + 18G_p + 10G_q$ $+3H_p + 2H_q + 6L_{p_6} + 6L_{p_9}$ $+2U_{q_0} + 2L_{q_9} + 6L_{\sigma_q} + 6\Sigma$
7	12	$10D_2^p + 12D_3^p + 14D_4^p + 12D_3^q$ $+24D_4^q + 12E_q + 6G_p + 24G_q$ $+5H_p + 6H_q + 2U_{p_0} + 8L_{q_5}$ $+4L_{\pi_p} + 4L_{\sigma_q} + 12\Sigma$
8	2	$12D_3^p + 18D_4^p + 14E_p + 12D_3^q$ $+24D_4^q + 16E_q + 10G_p + 18G_q$ $+H_q + 4Z + 2U_{p_0} + 2L_{p_9}$ $+6L_{q_6} + 6L_{q_9} + 6L_{\sigma_p} + 6\Sigma$
9	6	$6D_3^p + 12D_4^p + 4E_p + 6D_3^q$ $+12D_4^q + 6E_q + 4G_p + 12G_q$ $+2H_p + 3H_q + 2U_{p_0} + 6L_{p_5}$ $+6L_{q_8} + 2L_{\sigma_p} + 6\Sigma$
10	10	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+2E_q + 4G_p + 12G_q + 4H_p$ $+5H_q + 2U_{p_0} + 2L_{p_8} + 4L_{q_6}$ $+4L_{q_8} + 4L_{\pi_p} + 10\Sigma + 6X$
11	6	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+10E_q + 16G_q + 2H_p + 3H_q$ $+2U_{p_0} + 2L_{p_8} + 4L_{q_6} + 4L_{q_8}$ $+4L_{\pi_p} + 6\Sigma + 2X$
12	12	$8D_3^p + 16D_4^p + 2E_p + 8D_3^q$ $+16D_4^q + 8G_p + 12G_q + 5H_p$ $+6H_q + 2U_{p_0} + 4L_{p_5} + 6L_{p_8}$ $+4L_{q_5} + 4L_{\sigma_q} + 12\Sigma + 8X$

ID	D	Nonzero row weights (continued)
13	0	$D_4^p + Z + L_{q_5} + L_{\pi_p}$ $+L_{\sigma_p} + \Sigma$
14	2	$6D_3^p + 24D_4^p + 20E_p + 18D_3^q$ $+36D_4^q + 22E_q + 16G_p + 24G_q$ $+H_q + 10Z + 2U_{p_0} + 2L_{p_9}$ $+12L_{q_7} + 6L_{q_9} + 6L_{\pi_p} + 12\Sigma$
15	8	$6D_3^p + 12D_4^p + 2E_p + 6D_3^q$ $+12D_4^q + 4G_p + 8G_q + 3H_p$ $+4H_q + 2U_{p_1} + 6L_{p_8} + 4L_{q_5}$ $+2L_{\pi_p} + 2L_{\sigma_q} + 8\Sigma + 4X$
16	2	$8D_3^p + 16D_4^p + 14E_p + 8D_3^q$ $+16D_4^q + 6E_q + 16G_p + 2G_q$ $+H_p + 4Z + 4L_{p_5} + 2U_{q_0}$ $+4L_{q_5} + 6L_{q_8} + 4L_{\sigma_p} + 6\Sigma$
17	2	$8D_3^p + 16D_4^p + 12E_p + 12D_3^q$ $+24D_4^q + 18E_q + 8G_p + 20G_q$ $+H_q + 2Z + 2U_{p_1} + 2L_{p_9}$ $+4L_{q_6} + 4L_{q_7} + 4L_{q_9} + 4L_{\pi_p}$ $+4\Sigma$
18	2	$12D_3^p + 18D_4^p + 8E_p + 12D_3^q$ $+24D_4^q + 22E_q + 4G_p + 24G_q$ $+H_q + 4Z + 2U_{p_0} + 2L_{p_9}$ $+6L_{q_6} + 6L_{q_9} + 6L_{\sigma_p} + 6\Sigma$
19	0	$D_4^q + Z + L_{q_5} + L_{\pi_p}$ $+L_{\sigma_p} + \Sigma$
20	2	$6D_2^p + 12D_3^p + 14D_4^p + 10E_p$ $+12D_3^q + 24D_4^q + 18E_q + 6G_p$ $+20G_q + H_q + 2Z + 2U_{p_0}$ $+4L_{q_5} + 4L_{q_6} + 4L_{\pi_p} + 4L_{\sigma_q}$ $+4\Sigma$
21	10	$4D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$ $+2E_q + 4G_p + 12G_q + 4H_p$ $+5H_q + 2U_{p_1} + 2L_{p_8} + 2L_{q_6}$ $+2L_{q_8} + 2L_{\pi_p} + 2\Sigma$
22	2	$8D_3^p + 16D_4^p + 10E_p + 8D_3^q$ $+16D_4^q + 12E_q + 12G_p + 8G_q$ $+H_p + 4Z + 4L_{p_5} + 2U_{q_0}$ $+4L_{q_5} + 6L_{q_8} + 4L_{\sigma_p} + 6\Sigma$ $+2X$
23	2	$12D_3^p + 24D_4^p + 22E_p + 12D_3^q$ $+18D_4^q + 8E_q + 24G_p + 4G_q$ $+H_p + 4Z + 6L_{p_6} + 6L_{p_9}$ $+2U_{q_0} + 2L_{q_9} + 6L_{\sigma_q} + 6\Sigma$
24	6	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q$ $+4E_q + 6G_p + 4G_q + 3H_p$ $+2H_q + 6L_{p_8} + 2U_{q_0} + 6L_{q_5}$ $+2L_{\sigma_q} + 6\Sigma$
25	2	$8D_3^p + 16D_4^p + 8E_p + 12D_3^q$ $+24D_4^q + 22E_q + 4G_p + 24G_q$ $+H_q + 2Z + 2U_{p_1} + 2L_{p_9}$ $+4L_{q_6} + 4L_{q_7} + 4L_{q_9} + 4L_{\pi_p}$ $+4\Sigma$
26	12	$4D_3^p + 8D_4^p + 2E_p + 4D_3^q$ $+8D_4^q + 14G_p + 6G_q + 6H_p$ $+5H_q + 4L_{p_8} + 2U_{q_0} + 4L_{q_5}$ $+2L_{q_8} + 4\Sigma$
27	20	$6D_3^p + 14D_4^p + 8D_3^q + 16D_4^q$ $+2E_q + 14G_p + 22G_q + 9H_p$ $+10H_q + 2B + 2U_{p_0} + 2L_{q_7}$ $+4L_{q_8} + 6L_{\sigma_p} + 6\Sigma$

ID	D	Nonzero row weights (continued)
28	6	$6D_3^p + 12D_4^p + 4E_p + 6D_3^q$ $+12D_4^q + 4G_p + 6G_q + 2H_p$ $+3H_q + 2U_{p_0} + 6L_{p_5} + 6L_{q_8}$ $+2L_{\sigma_p} + 6\Sigma$
29	10	$4D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+6E_q + 4G_p + 16G_q + 4H_p$ $+5H_q + 2U_{p_0} + 2L_{p_8} + 4L_{q_7}$ $+4L_{q_8} + 4L_{\pi_p} + 10\Sigma + 10X$
30	16	$4D_3^p + 12D_4^p + 4D_2^q + 8D_3^q$ $+12D_4^q + 2E_q + 10G_p + 18G_q$ $+7H_p + 8H_q + 2U_{p_0} + 2L_{p_8}$ $+4L_{q_7} + 4L_{\pi_p} + 4\Sigma$
31	2	$6D_2^p + 12D_3^p + 14D_4^p + 6E_p$ $+12D_3^q + 24D_4^q + 22E_q + 2G_p$ $+24G_q + H_q + 2Z + 2U_{p_0}$ $+4L_{q_5} + 4L_{q_6} + 4L_{\pi_p} + 4L_{\sigma_q}$ $+4\Sigma$
32	2	$6D_3^p + 18D_4^p + 14E_p + 12D_3^q$ $+24D_4^q + 22E_q + 10G_p + 24G_q$ $+H_q + 4Z + 2U_{p_0} + 2L_{p_9}$ $+6L_{q_7} + 6L_{q_9} + 6L_{\sigma_p} + 6\Sigma$ $+6X$
33	2	$8D_3^p + 16D_4^p + 6E_p + 8D_3^q$ $+16D_4^q + 14E_q + 2G_p + 16G_q$ $+H_q + 4Z + 2U_{p_0} + 4L_{p_5}$ $+6L_{p_8} + 4L_{q_5} + 4L_{\sigma_q} + 6\Sigma$
34	9	$12D_3^p + 24D_4^p + 12D_3^q + 24D_4^q$ $+14E_q + 20G_q + 3H_p + 4H_q$ $+3Z + 3U_{p_1} + 12L_{p_8} + U_{q_0}$ $+9L_{q_5} + 3L_{\pi_p} + 4L_{\sigma_q} + 12\Sigma$
35	18	$2D_2^p + 8D_3^p + 10D_4^p + 8D_3^q$ $+16D_4^q + 2E_q + 12G_p + 20G_q$ $+8H_p + 9H_q + 2U_{p_0} + 4L_{q_6}$ $+4L_{q_8} + 4L_{\pi_p} + 4\Sigma$
36	20	$4D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+2E_q + 14G_p + 22G_q + 9H_p$ $+10H_q + 2U_{p_1} + 2L_{p_9} + 4L_{q_7}$ $+4L_{q_9} + 4L_{\sigma_p} + 4\Sigma$
37	8	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+8E_q + 2G_p + 16G_q + 3H_p$ $+4H_q + 2U_{p_0} + 2L_{p_8} + 4L_{q_6}$ $+4L_{q_8} + 4L_{\pi_p} + 8\Sigma + 6X$
38	20	$8D_3^p + 16D_4^p + 2E_p + 6D_3^q$ $+14D_4^q + 22G_p + 14G_q + 10H_p$ $+9H_q + 2B + 2L_{p_7} + 4L_{p_8}$ $+2U_{q_0} + 6L_{\sigma_q} + 6\Sigma$
39	27	$31D_3^p + 62D_4^p + 31D_3^q + 59D_4^q$ $+31E_q + 55G_q + 9H_p + 13H_q$ $+9U_{p_0} + 26L_{p_5} + 3L_{p_6} + U_{q_0}$ $+2L_{q_5} + 27L_{q_8} + 11L_{\sigma_p} + 27\Sigma$ $+X$
40	12	$4D_3^p + 8D_4^p + 4D_3^q + 8D_4^q$ $+2E_q + 6G_p + 14G_q + 5H_p$ $+6H_q + 2U_{p_0} + 4L_{p_5} + 2L_{p_8}$ $+4L_{q_8} + 4\Sigma$
41	2	$12D_3^p + 18D_4^p + 14E_p + 12D_3^q$ $+24D_4^q + 22E_q + 10G_p + 24G_q$ $+H_q + 4Z + 2U_{p_0} + 2L_{p_9}$ $+6L_{q_6} + 6L_{q_9} + 6L_{\sigma_p} + 6\Sigma$

ID	D	Nonzero row weights (continued)
42	2	$8D_3^p + 16D_4^p + 12E_p + 12D_3^q$ $+24D_4^q + 22E_q + 8G_p + 24G_q$ $+H_q + 2Z + 2U_{p_1} + 2L_{p_9}$ $+4L_{q_6} + 4L_{q_7} + 4L_{q_9} + 4L_{\pi_p}$ $+4\Sigma$
43	18	$6D_2^p + 8D_3^p + 10D_4^p + 8D_3^q$ $+16D_4^q + 2E_q + 12G_p + 20G_q$ $+8H_p + 9H_q + 2U_{p_0} + 4L_{p_5}$ $+4L_{q_5} + 4L_{\sigma_q} + 4\Sigma$
44	18	$8D_3^p + 16D_4^p + 2E_p + 2D_2^q$ $+4D_3^q + 10D_4^q + 20G_p + 12G_q$ $+9H_p + 8H_q + 4L_{p_7} + 2U_{q_0}$ $+4L_{q_5} + 4L_{\sigma_p} + 4\Sigma$
45	13	$D_2^p + D_3^p + 6D_4^p + 6D_3^q$ $+12D_4^q + 10G_p + 10G_q + 6H_p$ $+6H_q + U_{p_1} + U_{q_0} + 5L_{q_7}$ $+L_{\pi_p} + 2L_{\sigma_q} + 5\Sigma$
46	13	$3D_3^p + 9D_4^p + 6D_3^q + 9D_4^q$ $+10G_p + 10G_q + 6H_p + 6H_q$ $+U_{p_0} + 3L_{p_6} + L_{p_9} + U_{q_0}$ $+3L_{q_7} + L_{q_9} + 3\Sigma$
47	20	$8D_3^p + 16D_4^p + 2E_p + 4D_3^q$ $+12D_4^q + 22G_p + 14G_q + 10H_p$ $+9H_q + 4L_{p_7} + 4L_{p_9} + 2U_{q_1}$ $+2L_{q_9} + 4L_{\sigma_q} + 4\Sigma$
48	14	$2D_3^p + 8D_4^p + 6D_3^q + 12D_4^q$ $+2E_q + 8G_p + 16G_q + 6H_p$ $+7H_q + 2U_{p_8} + 2L_{p_9} + 4L_{q_7}$ $+2L_{q_9} + 2L_{\pi_p} + 4\Sigma$
49	2	$6D_3^p + 12D_4^p + 28E_p + 18D_3^q$ $+24D_4^q + 30E_q + 24G_p + 32G_q$ $+H_q + 10Z + 2U_{p_0} + 2L_{p_9}$ $+12L_{q_3} + 6L_{q_9} + 6L_{\pi_p} + 12\Sigma$
50	6	$6D_3^p + 12D_4^p + 4E_p + 6D_3^q$ $+12D_4^q + 6E_q + 4G_p + 12G_q$ $+2H_p + 3H_q + 2U_{p_1} + 6L_{p_8}$ $+4L_{q_5} + 2L_{\pi_p} + 2L_{\sigma_q} + 6\Sigma$
51	8	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q$ $+4E_q + 2G_p + 12G_q + 3H_p$ $+4H_q + 2U_{p_1} + 6L_{p_8} + 4L_{q_5}$ $+2L_{\pi_p} + 2L_{\sigma_q} + 8\Sigma + 6X$
52	2	$6D_2^p + 12D_3^p + 14D_4^p + 10E_p$ $+12D_3^q + 24D_4^q + 20E_q + 6G_p$ $+22G_q + H_q + 2Z + 2U_{p_0}$ $+4L_{q_5} + 4L_{q_6} + 4L_{\pi_p} + 4L_{\sigma_q}$ $+4\Sigma + 2X$

E Complete certificate specification for the upper bound 98

This appendix specifies all 82257 systems and prints the 77 integer row combinations certifying them. There are 1568 nonzero weights. The optional JSON assignment selects a successful combination for each system; trying every printed combination is an equivalent finite procedure.

E.1 Effective-hole parameters

Use the definitions of K, D, M, J from Appendix D with each of the following $(r, C_7, 7)$ and $(r, C_9, 9)$ rows. All 66 rows satisfy $(r - J + 1)\beta > D$ and $T_{J-1} - 1 > D$, the strict hypotheses of Lemma 10.2.

r	C_7	C_9	r	C_7	C_9
25	38	18	42	135	160
26	44	26	43	142	168
27	51	34	44	147	176
28	57	42	45	149	177
29	63	51	46	155	186
30	70	59	47	162	194
31	76	67	48	167	198
32	83	75	49	168	203
33	89	84	50	175	211
34	96	92	51	182	220
35	102	101	52	187	221
36	109	109	53	188	228
37	110	117	54	195	237
38	116	126	55	202	245
39	122	134	56	208	246
40	128	143	57	209	254
41	129	151			

E.2 All cases and variables

The labeled centers, eighteen units, branch definitions, reflection, retained-center ordering, and 68 one-empty or 547 two-empty modes are exactly those defined in Appendix D. Here $b = 82 - a$ and $k = |41 - a|$. Use each integer a in the six regime ranges of Section 16. In ZE and PE, choose each retained center and each of its two endpoints in turn as the exceptional location. There are 2308 such mode/location choices for each fixed a. There is no location enumeration in ZZ: its core-corrected rows are imposed on every retained center.

There are precisely two refinements to this enumeration:

1. In ZE at $a = 47$, branch A, missing unit q_5, and exceptional lower endpoint of π_p (the retained first-pair positive member), replace that case by two cases. The first imposes $M_q - R_q \le 4$ and uses upper bound $F \le 1758$; the second imposes $M_q - R_q \ge 5$ and uses $F \le 1764$. The ordinary inequality $M_q - R_q \ge 4$ is always present. Integrality makes this disjunction exhaustive.
2. In EE, branch A with missing units p_0, q_5, replace each of its nine size choices by two cases, adding respectively $\beta_p + M_p \ge 1$ or $\beta_q + M_q \ge 1$. This is the exhaustive nonemptiness disjunction proved in the text.

Thus the case counts are $21 \cdot 68 = 1428$, $19 \cdot 68 = 1292$, $17 \cdot 2308 + 1 = 39237$, $15 \cdot 2308 = 34620$, $11 \cdot 68 = 748$, and $9 \cdot 547 + 9 = 4932$.

All variables are nonnegative. The base variables are $p_i, q_i, M_p, M_q, R_p, R_q, S_p, S_q, \delta$ and the endpoint losses l_v, u_v. A harmless variable t is forced to zero by $J : t \le 0$; unused loss coordinates

have zero coefficients and may be set to zero. For every regime define

$$f = S_p + S_q + \delta - 2M_p - 2M_q - 2R_p - 2R_q.$$

Then $F = 91 + f$ in the five one-empty regimes and $F = 182 + f$ in EE.

EE also has the nonnegative variables $x_1, x_2, x_3, y_1, y_2, y_3, K$ and g^p_{ji}, g^q_{ji} for $0 \le j \le 3$, $j < i \le 4$. Set $x_0 = R_p$, $x_4 = M_p$, $y_0 = R_q$, $y_4 = M_q$ and $W_j = x_j + y_j$. The model uses 56 coordinates before these additions and 83 after (the surplus base coordinates are unused losses).

E.3 All rows, with the labels used in the certificate table

The order and core-size rows are

$$D^p_i : p_{i+1} - p_i \le -1 \ (1 \le i \le 3), \quad D^p_4 : M_p - p_4 \le -1,$$
$$E_p : R_p - M_p \le -4, \qquad G_p : -R_p \le -a,$$

and the reflected q rows. In EE only, replace the right sides of all D^p_i, D^q_i by zero; all other displayed rows retain their constants. The branch row B is $\beta_p + \beta_q \le 0$ in A and $-\beta_p - \beta_q \le -1$ in B; in EE branch B its right side is zero. The mass rows in every regime are

$$H_p : 2M_p + 6R_p - 2S_p \le -2T_a - 12 - 2\epsilon_p,$$
$$\widehat{H}_p : M_p + 4R_p - S_p \le -T_{a-1} - 6,$$

with reflected q rows, where $(\epsilon_p, \epsilon_q) = (a-2, b-2)$ in P and PE and $(0,0)$ otherwise. The budget row Z is $\delta \le 115 - k^2$ in Z, ZE, ZZ, $\delta \le 89 - k^2$ in P, PE, and $\delta \le 24 - k^2$ in EE. The loss row is

$$\Sigma : \sum_{v \text{ retained}} (l_v + u_v) - \delta \le 0.$$

The ordinary quantile rows for a positive center are

$$L_v : -v - l_v \le -b - 5, \qquad U_v : v - u_v - S_p \le -a - 4,$$

with reflected versions for negative centers. In ZE and PE replace only the selected exceptional row and add X as follows:

Endpoint	Replacement row	Row X
Positive, lower	$-v - l_v - M_p \le -91$	$-l_v \le -b - 5$
Positive, upper	$v - u_v - S_p - M_q \le -90$	$-u_v \le -a - 5$
Negative, lower	$-v - l_v - M_q \le -91$	$-l_v \le -a - 5$
Negative, upper	$v - u_v - S_q - M_p \le -90$	$-u_v \le -b - 5$

The actual out-of-hull location is not imposed. In ZZ replace every positive quantile row by

$$L_v : -v - l_v - R_p \le -87, \qquad U_v : v - u_v - S_p - R_q \le -86,$$

with the reflected negative versions, and add

$$X_2 : -\delta \le -2(46 - k), \qquad R : R_p + R_q + \delta \le 199 - k^2.$$

In EE all quantile rows remain ordinary, and add $R : R_p + R_q + \delta \le 106 - k^2$. The special ZE split uses respectively

$$Y_4 : M_q - R_q \le 4, \qquad Y_5 : R_q - M_q \le -5.$$

The following additional rows occur only in EE:

$$I^p_i : x_{i-1} - x_i \le -1 \quad (1 \le i \le 4), \qquad \widetilde{H}_p : R_p + \sum_{i=1}^{4} x_i - S_p \le -T_{a-1},$$

and the reflected negative rows. For $0 \le j \le 3$ and $j < i \le 4$,

$$A^p_{ji} : g^p_{ji} - x_i + x_{i-1} \le 0, \qquad B^p_{ji} : g^p_{ji} - W_j \le 1,$$

with reflected q rows. Put $t_j = 2(4-j)$. The envelope and cardinality rows are

$$C_j : K - 2W_j - \sum_{i=j+1}^{4} (g^p_{ji} + g^q_{ji}) \le 1 + \binom{t_j+1}{2}, \qquad 0 \le j \le 3,$$
$$M : S_p + S_q + \delta - 2K \le 1540.$$

Finally the two special EE branches add respectively

$$P_p : p_1 - 2p_2 + p_3 + p_4 - M_p \le -1, \qquad P_q : q_1 - 2q_2 + q_3 + q_4 - M_q \le -1.$$

These are all rows of the delivered generator; none requires a numerical feasibility assumption.

E.4 Exact verification rule and all multipliers

For $Ax \le b$, $x \ge 0$, each entry gives an integer $D \ge 0$ and nonnegative integer row weights w. A row absent from the current case makes that entry unavailable. Otherwise form

$$q = Df + wA, \qquad h = wb.$$

If every coordinate of q is nonnegative, then $D > 0$ gives $F \ge d_0 - h/D$, where $d_0 = 91$ or 182. It certifies impossibility if that lower bound is strictly greater than 1764, or 1820 in P and PE, or 1758 in the special gap-four ZE case. For $D = 0$, the strict inequality $h < 0$ certifies inconsistency. Every assigned combination passes these tests over the integers. The smallest strict lower-bound margin across all cases is 2/3. There is no use of floating-point tolerance in this verification.

ID	D	Nonzero row weights (omitted weights are zero)
1	10	$8D^p_3 + 12D^p_4 + 8D^q_3 + 16D^q_4$ $+6E_q + 4G_p + 16G_q + 4H_p$ $+5H_q + 2U_{p0} + 2L_{p8} + 4L_{q6}$ $+4L_{q8} + 4L_{\pi_p} + 10\Sigma$
2	0	$Z + L_{q5} + L_{\pi_p} + L_{\sigma_p}$ $+\Sigma$
3	1	$3D^p_3 + 12D^p_4 + 10E_p + 9D^q_3$ $+18D^q_4 + 11E_q + 8G_p + 13G_q$ $+5Z + U_{p0} + L_{p9} + 6L_{q7}$ $+3L_{q9} + 3L_{\pi_p} + 6\Sigma + \widehat{H}_q$
4	6	$5D^p_2 + 6D^p_3 + 7D^p_4 + 6D^q_3$ $+12D^q_4 + 6E_q + 8G_p + 18G_q$ $+U_{p0} + 4L_{q5} + 2L_{\pi_p} + 2L_{\sigma_q}$ $+6\Sigma + 5\widehat{H}_p + 6\widehat{H}_q$
5	3	$6D^p_3 + 12D^p_4 + 9E_p + 6D^q_3$ $+9D^q_4 + 2E_q + 15G_p + 4G_q$ $+3L_{p6} + 3L_{p9} + U_{q0} + L_{q9}$ $+3L_{\sigma_q} + 3\Sigma + 3\widehat{H}_p + 2\widehat{H}_q$
6	5	$3D^p_3 + 6D^p_4 + E_p + 3D^q_3$ $+6D^q_4 + 11G_p + 6G_q + 3L_{p8}$ $+U_{q0} + 3L_{q5} + L_{\sigma_q} + 5\Sigma$ $+5\widehat{H}_p + 4\widehat{H}_q$
7	1	$6D^p_3 + 9D^p_4 + 7E_p + 6D^q_3$ $+12D^q_4 + 11E_q + 5G_p + 13G_q$ $+2Z + U_{p0} + L_{p9} + 3L_{q6}$ $+3L_{q9} + 3L_{\sigma_p} + 3\Sigma + \widehat{H}_q$

ID	D	Nonzero row weights (continued)
8	6	$4D_3^p + 8D_4^p + 4D_3^q + 8D_4^q$ $+E_q + 12G_p + 9G_q + 2L_{p_5}$ $+U_{q_0} + 2L_{q_5} + 3L_{q_8} + 2L_{\sigma_p}$ $+6\Sigma + 4X + 6\widehat{H}_p + 5\widehat{H}_q$
9	1	$4D_3^p + 8D_4^p + 4E_p + 6D_3^q$ $+12D_4^q + 11E_q + 2G_p + 13G_q$ $+Z + U_{p_1} + L_{p_9} + 2L_{q_6}$ $+2L_{q_7} + 2L_{q_9} + 2L_{\pi_p} + 2\Sigma$ $+\widehat{H}_q$
10	5	$4D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$ $+E_q + 6G_p + 11G_q + U_{p_0}$ $+L_{p_8} + 2L_{q_6} + 2L_{q_8} + 2L_{\pi_p}$ $+5\Sigma + 3X + 4\widehat{H}_p + 5\widehat{H}_q$
11	3	$4D_3^p + 8D_4^p + 4E_p + 4D_3^q$ $+8D_4^q + 3E_q + 6G_p + 9G_q$ $+U_{p_0} + 2L_{p_5} + 3L_{p_8} + 2L_{q_5}$ $+2L_{\sigma_q} + 3\Sigma + X + 2\widehat{H}_p$ $+3\widehat{H}_q$
12	5	$4D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$ $+3E_q + 3G_p + 8G_q + U_{p_0}$ $+L_{p_8} + 2L_{q_6} + 2L_{q_8} + 2L_{\pi_p}$ $+2\Sigma + 4\widehat{H}_p + 5\widehat{H}_q + 3X_2$
13	6	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$ $+10E_q + 16G_q + 2H_p + 3H_q$ $+2U_{p_0} + 2L_{p_8} + 4L_{q_6} + 4L_{q_8}$ $+4L_{\pi_p} + 6\Sigma + 2X$
14	18	$12D_3^p + 22D_4^p + 12D_3^q + 24D_4^q$ $+6G_p + 18G_q + 7H_p + 9H_q$ $+4U_{p_1} + 10L_{p_8} + 6L_{q_5} + 2L_{q_6}$ $+4L_{\pi_p} + 4L_{\sigma_q} + 18\Sigma + 12X$
15	3	$4D_3^p + 8D_4^p + 5E_p + 4D_3^q$ $+8D_4^q + E_q + 11G_p + 3G_q$ $+2L_{p_5} + U_{q_0} + 2L_{q_5} + 3L_{q_8}$ $+2L_{\sigma_p} + 3\Sigma + 3\widehat{H}_p + 2\widehat{H}_q$
16	0	$Z + \Sigma + X$
17	1	$6D_3^p + 8D_4^p + 6E_p + 6D_3^q$ $+12D_4^q + 11E_q + 4G_p + 13G_q$ $+3Z + U_{p_1} + L_{p_9} + 4L_{q_6}$ $+2L_{q_9} + 2L_{\pi_p} + 4\Sigma + 4X$ $+\widehat{H}_q$
18	8	$6D_3^p + 12D_4^p + E_p + 6D_3^q$ $+9D_4^q + 17G_p + 12G_q + 3L_{p_6}$ $+3L_{p_9} + U_{q_0} + L_{q_9} + 3L_{\sigma_q}$ $+8\Sigma + 5X + 8\widehat{H}_p + 7\widehat{H}_q$
19	0	$D_4^p + Z + L_{q_5} + L_{\pi_p}$ $+L_{\sigma_p} + \Sigma$
20	0	$D_4^q + Z + L_{q_5} + L_{\pi_p}$ $+L_{\sigma_p} + \Sigma$
21	1	$6D_3^p + 9D_4^p + 7E_p + 6D_3^q$ $+12D_4^q + 8E_q + 5G_p + 10G_q$ $+2Z + U_{p_0} + L_{p_9} + 3L_{q_6}$ $+3L_{q_9} + 3L_{\sigma_p} + 3\Sigma + \widehat{H}_q$
22	1	$4D_3^p + 8D_4^p + 6E_p + 6D_3^q$ $+12D_4^q + 9E_q + 4G_p + 11G_q$ $+Z + U_{p_1} + L_{p_9} + 2L_{q_6}$ $+2L_{q_7} + 2L_{q_9} + 2L_{\pi_p} + 2\Sigma$ $+\widehat{H}_q$

ID	D	Nonzero row weights (continued)
23	1	$6D_3^p + 10D_4^p + 5E_p + 7D_3^q$ $+14D_4^q + 13E_q + 3G_p + 15G_q$ $+2Z + B + U_{p_0} + 3L_{q_6}$ $+L_{q_7} + 2L_{q_8} + 3L_{\pi_p} + 3\Sigma$ $+\widehat{H}_q$
24	1	$3D_2^p + 6D_3^p + 7D_4^p + 5E_p$ $+6D_3^q + 12D_4^q + 9E_q + 3G_p$ $+11G_q + Z + U_{p_0} + 2L_{q_5}$ $+2L_{q_6} + 2L_{\pi_p} + 2L_{\sigma_q} + 2\Sigma$ $+\widehat{H}_q$
25	0	$2D_4^p + 2E_p + D_4^q + E_q$ $+Z + L_{q_5} + L_{\pi_p} + L_{\sigma_p}$ $+\Sigma$
26	1	$3D_3^p + 12D_4^p + 10E_p + 9D_3^q$ $+18D_4^q + 17E_q + 4G_p + 9G_q$ $+5Z + U_{p_0} + L_{p_9} + 6L_{q_7}$ $+3L_{q_9} + 3L_{\pi_p} + 6\Sigma + \widehat{H}_q$
27	0	$8D_3^p + 24D_4^p + 8D_2^q + 16D_3^q$ $+24D_4^q + 24G_p + 38G_q + 3H_q$ $+4U_{p_0} + 4L_{p_8} + 8L_{q_7} + 8L_{\pi_p}$ $+8\Sigma + 8I_1^p + 16\widetilde{H}_p + 6I_1^q$ $+2I_3^q + 4I_4^q + 14\widetilde{H}_q + 8A_{12}^p$ $+8A_{13}^p + 8A_{14}^p + 8A_{12}^q + 8A_{13}^q$ $+8A_{14}^q + 8C_1 + 16A_{23}^p + 16A_{24}^p$ $+16A_{23}^q + 16A_{24}^q + 16C_2 + 16A_{34}^p$ $+16A_{34}^q + 16C_3 + 20M$
28	3	$4D_3^p + 8D_4^p + E_p + 4D_3^q$ $+8D_4^q + 5E_q + 3G_p + 11G_q$ $+U_{p_0} + 2L_{p_5} + 3L_{p_8} + 2L_{q_5}$ $+2L_{\sigma_q} + 3\Sigma + 2\widehat{H}_p + 3\widehat{H}_q$
29	1	$6D_3^p + 9D_4^p + 4E_p + 6D_3^q$ $+12D_4^q + 11E_q + 2G_p + 13G_q$ $+2Z + U_{p_0} + L_{p_9} + 3L_{q_6}$ $+3L_{q_9} + 3L_{\sigma_p} + 3\Sigma + \widehat{H}_q$
30	5	$4D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$ $+2E_q + 6G_p + 12G_q + U_{p_0}$ $+L_{p_8} + 2L_{q_6} + 2L_{q_8} + 2L_{\pi_p}$ $+5\Sigma + 4X + 4\widehat{H}_p + 5\widehat{H}_q$
31	7	$4D_3^p + 8D_4^p + 4D_3^q + 8D_4^q$ $+E_q + 5G_p + 10G_q + U_{p_0}$ $+2L_{p_5} + 3L_{p_8} + 2L_{q_5} + 2L_{\sigma_q}$ $+3\Sigma + 6\widehat{H}_p + 7\widehat{H}_q + 4X_2$
32	6	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q$ $+4E_q + 6G_p + 4G_q + 3H_p$ $+2H_q + 6L_{p_8} + 2U_{q_0} + 6L_{q_5}$ $+2L_{\sigma_q} + 6\Sigma$
33	0	$6D_3^p + 12D_4^p + 3D_3^q + 6D_4^q$ $+8G_p + 12G_q + 2U_{p_0} + 3L_{p_6}$ $+3L_{p_7} + 2L_{p_9} + 3\Sigma + 2I_3^p$ $+8\widetilde{H}_p + 2I_1^q + 10\widetilde{H}_q + 4A_{12}^p$ $+4A_{13}^p + 4A_{14}^p + 4A_{12}^q + 4A_{13}^q$ $+4B_{14}^q + 4C_1 + 10A_{23}^p + 10A_{24}^p$ $+10A_{23}^q + 10A_{24}^q + 10C_2 + 6A_{34}^p$ $+6A_{34}^q + 6C_3 + 10M$
34	4	$5D_2^p + 6D_3^p + 7D_4^p + 6D_3^q$ $+12D_4^q + 8E_q + 4G_p + 16G_q$ $+U_{p_0} + 4L_{q_5} + 2L_{\pi_p} + 2L_{\sigma_q}$ $+4\Sigma + 2X + 3\widehat{H}_p + 4\widehat{H}_q$

ID	D	Nonzero row weights (continued)
35	5	$2D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$
		$+3E_q + 6G_p + 13G_q + U_{p_0}$
		$+L_{p_8} + 2L_{q_7} + 2L_{q_8} + 2L_{\pi_p}$
		$+5\Sigma + 5X + 4\widehat{H}_p + 5\widehat{H}_q$
36	1	$3D_3^p + 9D_4^p + 7E_p + 6D_3^q$
		$+12D_4^q + 11E_q + 5G_p + 13G_q$
		$+2Z + U_{p_0} + L_{p_9} + 3L_{q_7}$
		$+3L_{q_9} + 3L_{\sigma_p} + 3\Sigma + 3X$
		$+\widehat{H}_q$
37	3	$3D_3^p + 6D_4^p + 2E_p + 3D_3^q$
		$+6D_4^q + 4G_p + 6G_q + U_{p_0}$
		$+3L_{p_5} + 3L_{q_8} + L_{\sigma_p} + 3\Sigma$
		$+2\widehat{H}_p + 3\widehat{H}_q$
38	8	$8D_3^p + 12D_4^p + 8D_3^q + 16D_4^q$
		$+8E_q + 2G_p + 16G_q + 3H_p$
		$+4H_q + 2U_{p_0} + 2L_{p_8} + 4L_{q_6}$
		$+4L_{q_8} + 4L_{\pi_p} + 8\Sigma + 6X$
39	3	$3D_3^p + 6D_4^p + 3E_p + 6D_3^q$
		$+12D_4^q + 7E_q + 9G_p + 5G_q$
		$+2U_{q_0} + 3L_{q_6} + 3L_{q_7} + 2L_{q_9}$
		$+3\Sigma + 3\widehat{H}_p + \widehat{H}_q$
40	8	$6D_3^p + 12D_4^p + 4E_p + 6D_3^q$
		$+9D_4^q + 13G_p + 8G_q + 3L_{p_6}$
		$+3L_{p_9} + U_{q_0} + L_{q_9} + 3L_{\sigma_q}$
		$+3\Sigma + 8\widehat{H}_p + 7\widehat{H}_q + 5X_2$
41	10	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q$
		$+2E_q + 4G_p + 12G_q + 4H_p$
		$+5H_q + 2U_{p_1} + 6L_{p_8} + 4L_{q_5}$
		$+2L_{\pi_p} + 2L_{\sigma_q} + 10\Sigma$
42	0	$4D_3^p + 8D_4^p + 3D_3^q + 7D_4^q$
		$+12G_p + 8G_q + H_p + B$
		$+L_{p_7} + 2L_{p_8} + U_{q_0} + 3L_{\sigma_q}$
		$+3\Sigma + I_3^p + 6\widetilde{H}_p + I_1^q$
		$+7\widetilde{H}_q + 4A_{12}^p + 4A_{13}^p + 4A_{14}^p$
		$+4A_{12}^q + 4A_{13}^q + 2A_{14}^q + 2B_{14}^q$
		$+4C_1 + 7A_{23}^p + 7A_{24}^p + 7A_{23}^q$
		$+7A_{24}^q + 7C_2 + 5A_{34}^p + 5A_{34}^q$
		$+5C_3 + 8M$
43	0	$12D_3^p + 24D_4^p + 12D_3^q + 24D_4^q$
		$+34G_p + 12G_q + 4H_p + 12L_{p_8}$
		$+6U_{q_0} + 12L_{q_5} + 6L_{q_8} + 12\Sigma$
		$+2I_2^p + 4I_3^p + 16\widetilde{H}_p + 18\widetilde{H}_q$
		$+3A_{01}^p + 3A_{02}^p + 3A_{03}^p + 3A_{04}^p$
		$+3A_{01}^q + 3A_{02}^q + 3A_{03}^q + 3B_{04}^q$
		$+3C_0 + 15A_{12}^p + 15A_{13}^p + 15A_{14}^p$
		$+15A_{12}^q + 15A_{13}^q + 12A_{14}^q + 3B_{14}^q$
		$+15C_1 + 18A_{23}^p + 18A_{24}^p + 18A_{23}^q$
		$+18A_{24}^q + 18C_2 + 12A_{34}^p + 12A_{34}^q$
		$+12C_3 + 24M$
44	0	$D_2^p + 2D_3^p + 5D_4^p + 4D_3^q$
		$+8D_4^q + 2E_q + 4G_p + 10G_q$
		$+H_q + U_{p_0} + 2L_{q_7} + 2L_{q_8}$
		$+2L_{\pi_p} + 2\Sigma + I_1^p + 3\widetilde{H}_p$
		$+I_3^q + 2I_4^q + 2\widetilde{H}_q + 2A_{12}^p$
		$+2A_{13}^p + 2A_{14}^p + 2A_{12}^q + 2A_{13}^q$
		$+2A_{14}^q + 2C_1 + 3A_{23}^p + 3A_{24}^p$
		$+3A_{23}^q + 3A_{24}^q + 3C_2 + 3A_{34}^p$
		$+3A_{34}^q + 3C_3 + 4M$

ID	D	Nonzero row weights (continued)
45	9	$12D_3^p + 24D_4^p + 12D_3^q + 24D_4^q$
		$+14E_q + 20G_q + 3H_p + 4H_q$
		$+3Z + 3U_{p_1} + 12L_{p_8} + U_{q_0}$
		$+9L_{q_5} + 3L_{\pi_p} + 4L_{\sigma_q} + 12\Sigma$
46	5	$3D_3^p + 6D_4^p + 3D_3^q + 6D_4^q$
		$+E_q + 2G_p + 7G_q + U_{p_0}$
		$+3L_{p_5} + 3L_{q_8} + L_{\sigma_p} + 3\Sigma$
		$+4\widehat{H}_p + 5\widehat{H}_q + 2X_2$
47	5	$3D_3^p + 6D_4^p + 3D_3^q + 6D_4^q$
		$+E_q + 2G_p + 7G_q + U_{p_1}$
		$+3L_{p_8} + 2L_{q_5} + L_{\pi_p} + L_{\sigma_q}$
		$+3\Sigma + 4\widehat{H}_p + 5\widehat{H}_q + 2X_2$
48	1	$6D_3^p + 9D_4^p + 7E_p + 6D_3^q$
		$+12D_4^q + 11E_q + G_p + 6G_q$
		$+2Z + U_{p_0} + L_{p_9} + 3L_{q_6}$
		$+3L_{q_9} + 3L_{\sigma_p} + 3\Sigma + \widehat{H}_q$
49	1	$3D_2^p + 6D_3^p + 7D_4^p + 5E_p$
		$+6D_3^q + 12D_4^q + 11E_q + G_p$
		$+6G_q + Z + U_{p_0} + 2L_{q_5}$
		$+2L_{q_6} + 2L_{\pi_p} + 2L_{\sigma_q} + 2\Sigma$
		$+\widehat{H}_q$
50	1	$4D_3^p + 8D_4^p + 6E_p + 6D_3^q$
		$+12D_4^q + 11E_q + G_p + 6G_q$
		$+Z + U_{p_1} + L_{p_9} + 2L_{q_6}$
		$+2L_{q_7} + 2L_{q_9} + 2L_{\pi_p} + 2\Sigma$
		$+\widehat{H}_q$
51	6	$5D_2^p + 6D_3^p + 7D_4^p + 6D_3^q$
		$+12D_4^q + 4E_q + 8G_p + 16G_q$
		$+U_{p_0} + 4L_{q_5} + 2L_{\pi_p} + 2L_{\sigma_q}$
		$+6\Sigma + 4X + 5\widehat{H}_p + 6\widehat{H}_q$
52	0	$24D_3^p + 72D_4^p + 48D_3^q + 96D_4^q$
		$+70G_p + 124G_q + H_p + 16H_q$
		$+12U_{p_1} + 12L_{p_9} + 24L_{q_7} + 24L_{q_9}$
		$+24L_{\sigma_p} + 24\Sigma + 18I_1^p + 46\widetilde{H}_p$
		$+18I_3^q + 36I_4^q + 28\widetilde{H}_q + 28A_{12}^p$
		$+28A_{13}^p + 28A_{14}^p + 28A_{12}^q + 28A_{13}^q$
		$+28A_{14}^q + 28C_1 + 46A_{23}^p + 46A_{24}^p$
		$+46A_{23}^q + 46A_{24}^q + 46C_2 + 46A_{34}^p$
		$+46A_{34}^q + 46C_3 + 60M$
53	0	$18D_3^p + 42D_4^p + 30D_3^q + 60D_4^q$
		$+46G_p + 76G_q + H_p + 10H_q$
		$+6B + 6U_{p_1} + 6L_{q_6} + 12L_{q_7}$
		$+6L_{q_8} + 12L_{\pi_p} + 12\Sigma + 12I_1^p$
		$+28\widetilde{H}_p + 12I_3^q + 24I_4^q + 16\widetilde{H}_q$
		$+16A_{12}^p + 16A_{13}^p + 16A_{14}^p + 16A_{12}^q$
		$+16A_{13}^q + 16A_{14}^q + 16C_1 + 28A_{23}^p$
		$+28A_{24}^p + 28A_{23}^q + 28A_{24}^q + 28C_2$
		$+28A_{34}^p + 28A_{34}^q + 28C_3 + 36M$
54	2	$2D_3^p + 2D_4^p + 2D_3^q + 4D_4^q$
		$+E_q + 4G_p + G_q + U_{q_0}$
		$+2L_{q_6} + L_{q_8} + 2\Sigma + 2X$
		$+2\widehat{H}_p + \widehat{H}_q$
55	8	$6D_3^p + 12D_4^p + 6D_3^q + 12D_4^q$
		$+4E_q + 2G_p + 12G_q + 3H_p$
		$+4H_q + 2U_{p_1} + 6L_{p_8} + 4L_{q_5}$
		$+2L_{\pi_p} + 2L_{\sigma_q} + 8\Sigma + 6X$

ID	D	Nonzero row weights (continued)
56	0	$6D_3^p + 12D_4^p + 3D_3^q + 6D_4^q$
		$+8G_p + 12G_q + 2U_{p_0} + 3L_{p_6}$
		$+3L_{p_7} + 2L_{\sigma_p} + 3\Sigma + 2I_3^p$
		$+8\widetilde{H}_p + 2I_1^q + 10\widetilde{H}_q + 4A_{12}^p$
		$+4A_{13}^p + 4A_{14}^p + 4A_{12}^q + 4A_{13}^q$
		$+4B_{14}^q + 4C_1 + 10A_{23}^p + 10A_{24}^p$
		$+10A_{23}^q + 10A_{24}^q + 10C_2 + 6A_{34}^p$
		$+6A_{34}^q + 6C_3 + 10M$
57	0	$D_2^p + 2D_3^p + 5D_4^p + 4D_3^q$
		$+8D_4^q + 2E_q + 4G_p + 10G_q$
		$+H_q + U_{p_0} + 2L_{p_5} + 2L_{q_7}$
		$+2L_{\sigma_q} + 2\Sigma + I_1^p + 3\widetilde{H}_p$
		$+I_3^q + 2I_4^q + 2\widetilde{H}_q + 2A_{12}^p$
		$+2A_{13}^p + 2A_{14}^p + 2A_{12}^q + 2A_{13}^q$
		$+2A_{14}^q + 2C_1 + 3A_{23}^p + 3A_{24}^p$
		$+3A_{23}^q + 3A_{24}^q + 3C_2 + 3A_{34}^p$
		$+3A_{34}^q + 3C_3 + 4M$
58	0	$2D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$
		$+2E_q + 4G_p + 10G_q + H_q$
		$+U_{p_1} + 2L_{q_7} + 2L_{q_8} + 2L_{\pi_p}$
		$+2\Sigma + I_1^p + 3\widetilde{H}_p + I_3^q$
		$+2I_4^q + 2\widetilde{H}_q + 2A_{12}^p + 2A_{13}^p$
		$+2A_{14}^p + 2A_{12}^q + 2A_{13}^q + 2A_{14}^q$
		$+2C_1 + 3A_{23}^p + 3A_{24}^p + 3A_{23}^q$
		$+3A_{24}^q + 3C_2 + 3A_{34}^p + 3A_{34}^q$
		$+3C_3 + 4M + P_p$
59	0	$24D_3^p + 32D_4^p + 24D_3^q + 48D_4^q$
		$+38G_p + 60G_q + H_p + 8H_q$
		$+4U_{p_1} + 4L_{p_9} + 16L_{q_6} + 8L_{q_9}$
		$+8L_{\pi_p} + 16\Sigma + 10I_1^p + 22\widetilde{H}_p$
		$+10I_3^q + 20I_4^q + 12\widetilde{H}_q + 12A_{12}^p$
		$+12A_{13}^p + 12A_{14}^p + 12A_{12}^q + 12A_{13}^q$
		$+12A_{14}^q + 12C_1 + 22A_{23}^p + 22A_{24}^p$
		$+22A_{23}^q + 22A_{24}^q + 22C_2 + 22A_{34}^p$
		$+22A_{34}^q + 22C_3 + 28M$
60	0	$4D_3^p + 8D_4^p + 6D_3^q + 12D_4^q$
		$+6E_q + 4G_p + 16G_q + H_q$
		$+2U_{p_8} + 2L_{p_9} + 2L_{q_6} + 2L_{q_7}$
		$+2L_{q_9} + 2L_{\pi_p} + 2\Sigma + 4\widetilde{H}_p$
		$+4\widetilde{H}_q + 4A_{12}^p + 4A_{13}^p + 4A_{14}^p$
		$+4A_{12}^q + 4A_{13}^q + 4A_{14}^q + 4C_1$
		$+4A_{23}^p + 4A_{24}^p + 4A_{23}^q + 4A_{24}^q$
		$+4C_2 + 4A_{34}^p + 4A_{34}^q + 4C_3$
		$+6M$
61	3	$3D_3^p + 6D_4^p + 2E_p + 3D_3^q$
		$+6D_4^q + 3E_q + 4G_p + 9G_q$
		$+U_{p_0} + 3L_{p_5} + 3L_{q_8} + L_{\sigma_p}$
		$+3\Sigma + 2\widehat{H}_p + 3\widehat{H}_q$
62	0	$6D_3^p + 18D_4^p + 12D_3^q + 24D_4^q$
		$+22G_p + 32G_q + H_p + 4H_q$
		$+2B + 2U_{p_0} + 6L_{q_7} + 4L_{q_9}$
		$+6L_{\sigma_p} + 6\Sigma + 4I_1^p + 12\widetilde{H}_p$
		$+4I_3^q + 8I_4^q + 8\widetilde{H}_q + 8A_{12}^p$
		$+8A_{13}^p + 8A_{14}^p + 8A_{12}^q + 8A_{13}^q$
		$+8A_{14}^q + 8C_1 + 12A_{23}^p + 12A_{24}^p$
		$+12A_{23}^q + 12A_{24}^q + 12C_2 + 12A_{34}^p$
		$+12A_{34}^q + 12C_3 + 16M$

ID	D	Nonzero row weights (continued)
63	0	$12D_3^p + 24D_4^p + 12D_3^q + 24D_4^q$
		$+12G_p + 34G_q + 4H_q + 6U_{p_0}$
		$+12L_{p_5} + 6L_{p_8} + 12L_{q_8} + 12\Sigma$
		$+18\widetilde{H}_p + 2I_2^q + 4I_3^q + 16\widetilde{H}_q$
		$+3A_{01}^p + 3A_{02}^p + 3A_{03}^p + 3B_{04}^p$
		$+3A_{01}^q + 3A_{02}^q + 3A_{03}^q + 3A_{04}^q$
		$+3C_0 + 15A_{12}^p + 15A_{13}^p + 12A_{14}^p$
		$+3B_{14}^p + 15A_{12}^q + 15A_{13}^q + 15A_{14}^q$
		$+15C_1 + 18A_{23}^p + 18A_{24}^p + 18A_{23}^q$
		$+18A_{24}^q + 18C_2 + 12A_{34}^p + 12A_{34}^q$
		$+12C_3 + 24M$
64	0	$6D_3^p + 18D_4^p + 18D_3^q + 36D_4^q$
		$+34G_q + 7H_q + 6U_{p_3} + 6L_{q_6}$
		$+6L_{q_7} + 6L_{\pi_p} + 6L_{\sigma_q} + 6\Sigma$
		$+30\widetilde{H}_p + 8I_2^q + 16I_3^q + 22\widetilde{H}_q$
		$+15A_{01}^p + 15A_{02}^p + 15A_{03}^p + 15B_{04}^p$
		$+15A_{01}^q + 15A_{02}^q + 15A_{03}^q + 15A_{04}^q$
		$+15C_0 + 21A_{12}^p + 21A_{13}^p + 12A_{14}^p$
		$+9B_{14}^p + 21A_{12}^q + 21A_{13}^q + 21A_{14}^q$
		$+21C_1 + 30A_{23}^p + 30A_{24}^p + 30A_{23}^q$
		$+30A_{24}^q + 30C_2 + 6A_{34}^p + 6A_{34}^q$
		$+6C_3 + 36M$
65	1	$6D_3^p + 9D_4^p + 6E_p + 6D_3^q$
		$+12D_4^q + 11E_q + 4G_p + 13G_q$
		$+2Z + U_{p_0} + L_{p_9} + 3L_{q_6}$
		$+3L_{q_9} + 3L_{\sigma_p} + 3\Sigma + 2X$
		$+\widehat{H}_q$
66	0	$12D_3^p + 24D_4^p + 18D_3^q + 36D_4^q$
		$+12G_p + 38G_q + 5H_q + 6U_{p_1}$
		$+6L_{p_8} + 6L_{q_6} + 6L_{q_7} + 6L_{\pi_p}$
		$+6\Sigma + 24\widetilde{H}_p + 4I_2^q + 8I_3^q$
		$+20\widetilde{H}_q + 6A_{01}^p + 6A_{02}^p + 6A_{03}^p$
		$+6B_{04}^p + 6A_{01}^q + 6A_{02}^q + 6A_{03}^q$
		$+6A_{04}^q + 6C_0 + 18A_{12}^p + 18A_{13}^p$
		$+12A_{14}^p + 6B_{14}^p + 18A_{12}^q + 18A_{13}^q$
		$+18A_{14}^q + 18C_1 + 24A_{23}^p + 24A_{24}^p$
		$+24A_{23}^q + 24A_{24}^q + 24C_2 + 12A_{34}^p$
		$+12A_{34}^q + 12C_3 + 30M + 6P_q$
67	1	$4D_3^p + 8D_4^p + 6E_p + 6D_3^q$
		$+12D_4^q + 11E_q + 4G_p + 13G_q$
		$+Z + U_{p_1} + L_{p_9} + 2L_{q_6}$
		$+2L_{q_7} + 2L_{q_9} + 2L_{\pi_p} + 2\Sigma$
		$+\widehat{H}_q$
68	1	$D_2^p + 4D_3^p + 7D_4^p + 5E_p$
		$+6D_3^q + 12D_4^q + 11E_q + 3G_p$
		$+13G_q + Z + U_{p_0} + 2L_{q_6}$
		$+2L_{q_7} + 2L_{\pi_p} + 2L_{\sigma_q} + 2\Sigma$
		$+2X + \widehat{H}_q$
69	3	$4D_3^p + 8D_4^p + 5E_p + 4D_3^q$
		$+8D_4^q + 4E_q + 11G_p + 6G_q$
		$+2L_{p_5} + U_{q_0} + 2L_{q_5} + 3L_{q_8}$
		$+2L_{\sigma_p} + 3\Sigma + 3\widehat{H}_p + 2\widehat{H}_q$
70	3	$4D_3^p + 8D_4^p + E_p + 4D_3^q$
		$+8D_4^q + 5E_q + 3G_p + 11G_q$
		$+U_{p_0} + 3L_{p_8} + 2L_{q_5} + 2L_{q_8}$
		$+2L_{\pi_p} + 3\Sigma + 2\widehat{H}_p + 3\widehat{H}_q$

ID	D	Nonzero row weights (continued)
71	1	$4D_3^p + 8D_4^p + 5E_p + 6D_3^q$ $+12D_4^q + 11E_q + 3G_p + 13G_q$ $+Z + U_{p_1} + L_{p_9} + 2L_{q_6}$ $+2L_{q_7} + 2L_{q_9} + 2L_{\pi_p} + 2\Sigma$ $+X + \widehat{H}_q$
72	2	$D_2^p + 3D_3^p + 5D_4^p + 3D_3^q$ $+6D_4^q + 4E_q + 8G_q + U_{p_1}$ $+2L_{p_8} + 2L_{q_5} + L_{\pi_p} + L_{\sigma_q}$ $+2\Sigma + \widehat{H}_p + 2\widehat{H}_q$
73	3	$4D_3^p + 8D_4^p + 4E_p + 4D_3^q$ $+8D_4^q + 5E_q + 6G_p + 11G_q$ $+U_{p_0} + 2L_{p_5} + 3L_{p_8} + 2L_{q_5}$ $+2L_{\sigma_q} + 3\Sigma + 2\widehat{H}_p + 3\widehat{H}_q$
74	3	$4D_3^p + 6D_4^p + 4D_3^q + 8D_4^q$ $+2G_p + 11G_q + U_{p_0} + L_{p_8}$ $+2L_{q_6} + 2L_{q_8} + 2L_{\pi_p} + 3\Sigma$ $+X + 2\widehat{H}_p + 3\widehat{H}_q + 5Y_5$
75	0	$12D_3^p + 24D_4^p + 12D_3^q + 24D_4^q$ $+12G_p + 34G_q + 4H_q + 6U_{p_1}$ $+12L_{p_8} + 6L_{q_5} + 6L_{q_8} + 6L_{\pi_p}$ $+12\Sigma + 18\widetilde{H}_p + 2I_2^q + 4I_3^q$ $+16\widetilde{H}_q + 3A_{01}^p + 3A_{02}^p + 3A_{03}^p$ $+3B_{04}^p + 3A_{01}^q + 3A_{02}^q + 3A_{03}^q$ $+3A_{04}^q + 3C_0 + 15A_{12}^p + 15A_{13}^p$ $+12A_{14}^p + 3B_{14}^p + 15A_{12}^q + 15A_{13}^q$ $+15A_{14}^q + 15C_1 + 18A_{23}^p + 18A_{24}^p$ $+18A_{23}^q + 18A_{24}^q + 18C_2 + 12A_{34}^p$ $+12A_{34}^q + 12C_3 + 24M$
76	0	$2D_3^p + 10D_4^p + 8D_3^q + 16D_4^q$ $+14G_p + 10G_q + U_{p_0} + 2L_{p_5}$ $+L_{p_8} + 3U_{q_0} + 6L_{q_7} + 5L_{q_8}$ $+6\Sigma + 2I_1^p + 12\widetilde{H}_p + 2I_3^q$ $+10\widetilde{H}_q + 6A_{12}^p + 6A_{13}^p + 2A_{14}^p$ $+4B_{14}^p + 6A_{12}^q + 6A_{13}^q + 6A_{14}^q$ $+6C_1 + 12A_{23}^p + 12A_{24}^p + 12A_{23}^q$ $+12A_{24}^q + 12C_2 + 8A_{34}^p + 8A_{34}^q$ $+8C_3 + 13M$
77	0	$8D_3^p + 16D_4^p + 3D_3^q + 10D_4^q$ $+10G_p + 14G_q + 3U_{p_0} + L_{p_6}$ $+5L_{p_7} + 5L_{p_8} + U_{q_0} + 2L_{q_5}$ $+L_{q_8} + 5\Sigma + 2I_3^p + 10\widetilde{H}_p$ $+2I_1^q + 12\widetilde{H}_q + 6A_{12}^p + 6A_{13}^p$ $+6A_{14}^p + 6A_{12}^q + 6A_{13}^q + 2A_{14}^q$ $+4B_{14}^q + 6C_1 + 12A_{23}^p + 12A_{24}^p$ $+12A_{23}^q + 12A_{24}^q + 12C_2 + 8A_{34}^p$ $+8A_{34}^q + 8C_3 + 13M$

F Certificate specification for the upper bound 96

This appendix specifies the additional finite proof in Section 17. The names `upper97` and `upper96` refer to the resulting upper bound, so they exclude sets of sizes 98 and 97 respectively. The two JSON files contain the complete integer multiplier pools, ordered root assignments, and all binary trees. They are required companions to the new finite proof; unlike the earlier multiplier tables, the new pools are not duplicated in print. The LaTeX source itself compiles without the JSON files.

F.1 The 103 effective-hole rows

The following table gives $C_\beta(r)$ in Lemma 10.2, with its parameter $b = \beta$. A dash means that the row is not used. Each entry asserts

$$h > 2\beta, \quad h < |\,\mathrm{FS}(X)| - 1 \quad \Longrightarrow \quad \mu(X) \geq r + C_\beta(r) + 1.$$

For independent verification, compute

$$K = \left\lfloor \frac{3 + \sqrt{9 + 8(C+1)}}{2} \right\rfloor, \qquad D = C + \max(3, K),$$
$$M = \max\left(2\beta, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(D + 1 + 6\beta)}}{2} \right\rfloor\right), \qquad J = \max(K + 1, 2\beta + 5, M).$$

Every listed entry satisfies $(r - J + 1)\beta > D$ and $T_{J-1} - 1 > D$. The verifier uses integer square roots. At size 98 only rows $r \geq 24$ are needed (101 entries); size 97 uses all 103.

r	$C_7(r)$	$C_9(r)$	$C_{10}(r)$
23	25	3	–
24	32	11	–
25	38	18	4
26	44	26	12
27	51	34	21
28	57	42	30
29	63	51	39
30	70	59	48
31	76	67	57
32	83	75	66
33	89	84	75
34	96	92	85
35	102	101	94
36	109	109	103
37	110	117	113
38	116	126	122
39	122	134	132
40	128	143	141
41	129	151	150
42	135	160	160
43	142	168	169
44	147	176	179
45	149	177	188
46	155	186	198
47	162	194	207
48	167	198	215
49	168	203	217
50	175	211	227
51	182	220	236
52	187	221	239
53	188	228	246
54	195	237	255
55	202	245	264

r	$C_7(r)$	$C_9(r)$	$C_{10}(r)$
56	208	246	265
57	209	254	274

F.2 Root variables, center labels, and modes

All variables are nonnegative. The 83 coordinates consist of

$$(p_1, \dots, p_4, q_1, \dots, q_4, M_p, M_q, R_p, R_q, S_p, S_q, \delta, t),$$

40 reserved endpoint-loss coordinates, six inner elements $x_1, x_2, x_3, y_1, y_2, y_3$, the variable K, and twenty envelope variables g_{jis}, where $0 \le j \le 3$, $j < i \le 4$, and $s = p, q$. The coordinate t equals the common hole parameter z in positive-hole low models and all size-97 high models. It is fixed to zero in zero-hole models and in the coarse size-98 H1 family, whose bounds were stated separately. Unused loss coordinates are harmless nonnegative variables.

With the earlier definitions $c_p = p_1 - p_2 - p_3 - p_4$, $\alpha_p = p_2 - p_3 - p_4$, and $\beta_p = 2p_2 - p_1 - p_3 - p_4$, the ordered positive center labels are

label	p_0	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8	p_9
center	$c_p + p_1$	$c_p + p_2$	$c_p + p_3$	$c_p + p_4$	$c_p - q_1$	$c_p - q_2$	$c_p - q_3$	$c_p - q_4$	α_p	β_p

Here a center label p_i is a string used in the programs; in this table it must not be confused with the outer element p_i in the formulas. Negative center labels $q_0, \dots, q_9$ are obtained by swapping p, q. The first pair has index 4 on both sides. The second pair has index 9 in branch A and index 5 in branch B. Remove those indices from each side to obtain the sixteen unpaired units. Enumerate every j-element set of missing units among these sixteen and the two pairs.

If pair 1 is present, its retained member is positive after reflection. If pair 2 is present and cannot split, enumerate its positive and negative member choices, denoted by 0 and 1. If it is missing, keep the dummy choice 0. In branch A, when a missing unit is p_5 or q_5, drop pair 2 from the ordinary retained-center list; if pair 2 is nevertheless present, enumerate the two choices $s = p, q$ of an actually contributing member and add (63). This gives exactly $4,70,578$ modes for $j = 0, 1, 2$.

The code first lists the surviving unpaired centers, then pair 1, then the chosen pair-2 member if retained. Assign the loss coordinates to that list in order. Labels `pair1p`, `pair2p`, and `pair2q` remove ambiguity. The separate member-choice variable for a split pair is called `variant`.

F.3 A complete description of the root rows

The outer orders are $p_i \ge p_{i+1} + 1$, $p_4 \ge M_p + 1$ and their reflections for $j \le 1$; replace these gaps by zero for $j = 2$. Always impose $M_p \ge R_p + 4$, $M_q \ge R_q + 4$, $R_p \ge a$, $R_q \ge b$. Branch A has $\beta_p + \beta_q \le 0$; branch B has $\beta_p + \beta_q \ge 1$ for $j \le 1$ and its relaxation ≥ 0 for $j = 2$. Add the total-loss bound $\sum(l_v + u_v) \le \delta$ and the regime's quantile, mass, loss-budget, span-budget, and hole-parameter rows from Section 17. The original mass rows are multiplied by two in the code; this is just a positive rescaling. The quantities ϵ_p, ϵ_q in the mass formulas are replaced by the affine lower bounds specified for that regime; they are not additional coordinates.

For $x_0 = R_p$, $x_4 = M_p$, $y_0 = R_q$, $y_4 = M_q$, impose $x_i - x_{i-1} \ge 1$, $y_i - y_{i-1} \ge 1$ for $1 \le i \le 4$ and both mass rows (65) on both sides. Write $W_j = x_j + y_j$, $t_j = 2(4 - j)$. For each $j = 0, 1, 2, 3$,

the envelope rows are

$$0 \le g_{jip} \le x_i - x_{i-1}, \qquad g_{jip} \le W_j + 1,$$
$$0 \le g_{jiq} \le y_i - y_{i-1}, \qquad g_{jiq} \le W_j + 1, \qquad j < i \le 4,$$
$$K \le 2W_j + 1 + \frac{t_j(t_j+1)}{2} + \sum_{i=j+1}^{4} (g_{jip} + g_{jiq}).$$

These rows hold for an actual set by choosing each g equal to the indicated minimum of a gap and $W_j + 1$. No equality with that minimum is needed in the relaxed model. Add (66). The only additional root row in an L family is $l_v + u_v \ge d - 4$ for its designated center.

The high-state selection is itself explicit: loop over all $0 \le k \le 16$, $15 \le z \le 395 - k(k+1)$ at size 98, and all $0 \le k \le 17$, $15 \le z \le 414 - k^2$ at size 97. Compute u, v, C, M as in the proof, with exact integer and rational arithmetic. The size-97 loop has 5415 states, including those immediately rejected by $C < 0$. No sampled subrange or numerical tolerance is used.

F.4 The complete list of roots that require a tree

Every branched root is in pairing branch A. In the following table, "choice" is the ordinary second-pair choice, "member" is the actually contributing split-pair member, and a dash means absent. The large center is p_8 in every L row. Each split has both children.

Regime	a	Missing units	Member	Choice	Splits	Depth
ZL	38	p_0	–	1	4	4
ZL	39	p_0	–	1	39	8
PL	37	p_0	–	1	1	1
PL	38	p_0	–	1	1	1
PL	39	p_0	–	1	1	1
PL	40	p_0	–	1	1	1
Z2	38	p_0, p_8	–	1	4	3
Z2	39	p_0, p_8	–	1	64	10
Z2	40	p_0, p_8	–	1	3	3
Z2	40	p_0, q_5	q	0	1	1
Z2	41	p_0, q_5	q	0	1	1
Z2	42	p_0, q_5	q	0	1	1
P2	37	p_0, p_8	–	1	1	1
P2	38	p_0, p_8	–	1	2	2
P2	39	p_0, p_8	–	1	1	1
P2	40	p_0, p_8	–	1	2	2
P2	39	p_0, q_5	q	0	1	1
P2	40	p_0, q_5	q	0	1	1
P2	41	p_0, q_5	q	0	1	1

The 130 internal nodes consist of 13 ordinary/outside splits, 111 inner-extreme/loss splits, and six both/sole-pair splits. The maximum depth is ten. The two additional sole-pair loss variables use reserved coordinates 52 and 53 (zero-based); all ordinary retained losses in these two-empty modes have indices at most 47.

For complete reproducibility, `branches()` enumerates, in retained endpoint-row order, first the ordinary/outside split and then the four inner splits with $i = 1, 2, 3, 4$. It finally adds the both/sole split exactly when $j = 2$ and a split-pair member was selected. A tree node stores the zero-based index in that list. Its left and right children contain the rows in the same order as equations (69)–(72): ordinary versus outside; loss versus inner extreme; both versus sole. The JSON is therefore an explicit finite tree specification, not a record of a solver-dependent search procedure.

F.5 Exact contradiction check and data integrity

Each certificate stores a list of distinct row labels with strictly positive integer weights. All root and branch rows have integer coefficients. At a leaf, add its path rows to its root matrix A and right side b, then compute

$$c_i = \sum_r w_r A_{ri} \quad (1 \le i \le 83), \qquad v = \sum_r w_r b_r.$$

The verifier requires every $c_i \ge 0$ and $v < 0$, and checks the stored positive value $-v$. This suffices because every coordinate is nonnegative. There is no objective-bound denominator in these two new pools: their historical field `D` is always zero.

The checker independently generates the full ordered root sequence, compares every regime, side size, high-hole parameter, missing-unit mode, large center, and member choice with the assignment, and recursively checks both children of every tree. Repeated split indices on a path, duplicate row labels, absent children, and noninteger weights are rejected. The complete verification checks 45928 size-98 roots and 81376 size-97 leaves. The least size-97 contradiction magnitude is 1. All arithmetic uses Python's arbitrary-precision integers or exact fractions.

The proof-model and certificate hashes below identify the exact companion companion data used in the proof. The archive's `SHA256SUMS.txt` also covers the complete source and historical verification dependencies.

`upper97_model.py`

`445525a759e423c2705aa00b1dc5f0ebf5eba5963c6320cc17cf6d87c6f4079a`

`upper97_exact.json`

`b42b641a9154d8607a8b2570d5686628482b6ff0df4fbfb8b10c85aadddcefcd`

`verify_upper97.py`

`cdec064f37a0a65aad991692aa90ffee62f7b28cb86f2eb8e0acad02fb2ebbea`

`upper96_model.py`

`455d71883018823e8eceb4cff557fb69bf108c3130003539f885d3e3c44a573c`

`upper96_exact.json`

`f30e9448567cb33fd08aa9ea108004019122b7f185e8642670d22912607ed411`

`verify_upper96.py`

`2bfd1f4a05557dc1f33371a98ce65311cd6aca74b15d4ea78fe57f5a7bdc2704`

G An analytic certificate for the size-87 AP primitivity exception

The following argument proves individual primitivity in the remaining AP branch of the size-87 common-lattice proof. It uses four explicitly displayed row combinations, valid uniformly in the side sizes. No enumeration or primitive-core hypothesis enters the covering argument.

Lemma G.1 (Small covering slack is impossible at zero holes)**.** *In the size-87 covering setup, suppose D is integral on the common core lattice, $a+b=70$, $a,b\ge 15$, and the extreme-translate propagation conclusion $w\le 4z$ holds with $z=0$. Then the covering slack c cannot satisfy $c\le 6$.*

Proof. Here $|D|=79$. Every empty ordinary covering unit costs 79 in c, so $c\le 6$ forces all eighteen ordinary units to contribute. Write $\delta=c$, and use the outer coordinates

$$p_1>p_2>p_3>p_4>M_p,\qquad q_1>q_2>q_3>q_4>M_q,$$

where M_p, M_q are the positive and negative endpoint magnitudes of D. Put

$$c_p=p_1-p_2-p_3-p_4,\qquad c_q=q_1-q_2-q_3-q_4.$$

The four own centers $b_i=c_p+p_i$ on the positive side are all contributing. Since the actual target and D are integral, each b_i is integral. The identity

$$c_p=-b_1+b_2+b_3+b_4$$

then makes c_p and every p_i integral. Reflection proves the same for c_q, q_i. No leading component is assumed to contribute.

Let R_p, R_q be the core maxima, S_p, S_q the sums of the positive and negative magnitudes in D, and $K=|D+D|$. Distinct integral points give

$$M_p\ge R_p+4,\quad M_q\ge R_q+4,\quad R_p\ge a,\quad R_q\ge b,$$

$$S_p\ge T_a+M_p+3R_p+6,\qquad S_q\ge T_b+M_q+3R_q+6.$$

The last inequalities count the core mass and the three inner peeled points individually; they do not require a primitive core. The integer hull and the exact zero-hole cardinality identity give

$$K\le 2M_p+2M_q+1,\qquad S_p+S_q+\delta-2K=1506.$$

Indeed propagation gives $w=0$, and $18\cdot 79+84=1506$.

The centers and the four actual alternatives. Define

$$\alpha_p=p_2-p_3-p_4,\quad \beta_p=2p_2-p_1-p_3-p_4,$$

with their reflected q versions. For $i=0,\dots,9$, the centers on the positive side are

$$(\gamma_{p,i})_{i=0}^{9}=(c_p+p_1,c_p+p_2,c_p+p_3,c_p+p_4,c_p-q_1,c_p-q_2,c_p-q_3,c_p-q_4,\alpha_p,\beta_p).$$

Reflection interchanges p,q to define $\gamma_{q,i}$. Branch A means $\beta_p+\beta_q\le 0$ and has second-pair index $j=9$; branch B means $\beta_p+\beta_q>0$ and has $j=5$. The sixteen unpaired centers are exactly

$$\gamma_{s,i},\qquad s\in\{p,q\},\quad i\notin\{4,j\}.$$

All these unpaired components contribute. Both paired units in fact have exactly one contributing member. Put $V=p_3+p_4+q_3+q_4$. The first paired centers have sum $-(p_2+p_3+p_4+q_2+q_3+q_4)<$

$-(M_p + M_q)$. If both met their respective open half-lines, their sum would exceed $-(M_p + M_q)$, a contradiction. In branch A, the unpaired centers $c_p - q_2, c_q - p_2$ both contribute, so

$$c_p - q_2 + c_q - p_2 > -(M_p + M_q).$$

It follows that

$$\beta_p + \beta_q = -(c_p - q_2 + c_q - p_2) - 2V < M_p + M_q - 2V < -(M_p + M_q).$$

In branch B, the second paired centers have sum $-(\beta_p + \beta_q) - 2V < -(M_p + M_q)$ directly. Thus at most one member of each pair meets its open half-line, and the nonempty-unit condition makes that member unique. Reflection lets us choose the actual contributing member of the first pair to be $\gamma_{p,4}$. Denote this selected member by 1. Choose an actual contributing member of the second paired unit, denoted by 2. Choice 0 selects $\gamma_{p,j}$ and choice 1 selects $\gamma_{q,j}$. Thus A0, A1, B0, B1 cover every actual configuration. Reflection can interchange a, b, so no order between these two sizes is imposed. The selection is made from actual contribution, not from a necessary inequality satisfied by an arbitrary relaxation point.

For each of these eighteen selected components v, take its actual lower and upper hull losses $l_v, u_v \geq 0$. For a center v on side s, with the opposite side denoted by t, its actual nonempty integral translate satisfies the global endpoint-loss inequalities

$$v + l_v + M_s \geq 79, \qquad S_s - v + u_v + M_t \geq 78.$$

These are the global quantile rows proved in Appendix F; they require a nonempty integral translate, not an in-hull center or a primitive core. Because each pair has a unique contributing member, its other member contributes nothing. Consequently the covering count uses precisely these eighteen actual components. Their hull losses are conservative lower bounds for the corresponding actual-target deficits, so their sum, together with any chosen conservative leading losses, fits the same single covering budget.

Write $L_p, L_q \geq 0$ for the two leading losses and put

$$\mathcal{L} = L_p + L_q + \sum_v (l_v + u_v).$$

The covering inequality gives $\mathcal{L} \leq \delta$. We need no other inequality for L_p, L_q.

All elementary rows. The names below denote inequalities whose left and right sides are added with the displayed positive weights. Let $r_p = a$, $r_q = b$. For $s = p, q$, let s_i mean p_i or q_i accordingly:

$$\begin{aligned}
\mathsf{C} &: \delta \leq E, \qquad E = 6,\\
\mathsf{O}_s &: M_s - s_4 \leq -1,\\
\mathsf{D}_s &: s_4 - s_3 \leq -1,\\
\mathsf{H}_s &: R_s - M_s \leq -4,\\
\mathsf{J}_s &: -R_s \leq -r_s,\\
\mathsf{F}_s &: 2M_s + 6R_s - 2S_s \leq -2T_{r_s} - 12,\\
\mathsf{Q} &: S_p + S_q + \delta - 2K \leq 1506,\\
\mathsf{K} &: K - 2M_p - 2M_q \leq 1,\\
\mathsf{B} &: \mathcal{L} - \delta \leq 0,\\
\mathsf{L}_v &: -v - l_v - M_s \leq -79,\\
\mathsf{U}_v &: v - u_v - S_s - M_t \leq -78.
\end{aligned}$$

The independent coordinates $p_i, q_i, M_s, R_s, S_s, \delta, K, l_v, u_v, L_s$ are nonnegative. Centers v are linear forms and need not be nonnegative. Subscripts p, i and q, i on loss rows refer to $\gamma_{p,i}, \gamma_{q,i}$; subscripts 1 and 2 refer to the selected pair members. Every row used below is among the retained centers in its stated mode.

Mode A0. Add the following rows:

$$\mathsf{C} + 2\mathsf{O}_p + \mathsf{L}_1 + \mathsf{L}_2 + 2\mathsf{O}_q + \mathsf{L}_{q,5} + \mathsf{L}_{q,8} + \mathsf{B}.$$

Exact cancellation gives

$$2p_3 + 2q_3 + \mathcal{L} - l_1 - l_2 - l_{q,5} - l_{q,8} \le E - 320.$$

Each subtracted loss is a distinct term of $\mathcal{L}$, so the left side is nonnegative; the right side is at most -314.

Mode A1. Add

$$\begin{aligned} &56\mathsf{C} + 6\mathsf{H}_p + 24\mathsf{J}_p + 12\mathsf{J}_q + 3\mathsf{F}_p + 2\mathsf{F}_q + 14\mathsf{Q} + 28\mathsf{K} \\ &\quad + 8\mathsf{U}_{p,0} + 74\mathsf{D}_p + 152\mathsf{O}_p + 70\mathsf{L}_{p,8} + 16\mathsf{L}_1 + 26\mathsf{L}_2 \\ &\quad + 10\mathsf{U}_{q,0} + 70\mathsf{D}_q + 148\mathsf{O}_q + 62\mathsf{L}_{q,5} + 70\mathsf{B}. \end{aligned}$$

The result is

$$\begin{aligned} 4p_3 + 8q_3 + 70\mathcal{L} - 8u_{p,0} - 70l_{p,8} - 16l_1 - 26l_2 - 10u_{q,0} - 62l_{q,5} \\ \le 56E + 5434 - 24a - 12b - 6T_a - 4T_b. \end{aligned}$$

The subtracted loss variables are distinct, each with coefficient at most 70. Hence every coefficient on the left is nonnegative. Since $b = 70 - a$ and $E \le 6$, the right side is at most

$$-5a^2 + 267a - 5010 = -5\left(a - \frac{267}{10}\right)^2 - \frac{28911}{20} < 0.$$

Mode B0. Add

$$\begin{aligned} &11\mathsf{C} + 12\mathsf{H}_p + 12\mathsf{H}_q + 12\mathsf{J}_p + 18\mathsf{J}_q + \mathsf{F}_q + 4\mathsf{Q} + 8\mathsf{K} \\ &\quad + 4\mathsf{U}_{p,0} + 19\mathsf{D}_p + 46\mathsf{O}_p + 4\mathsf{L}_{p,9} + 12\mathsf{L}_1 + 2\mathsf{U}_{q,0} \\ &\quad + 6\mathsf{L}_{q,2} + 32\mathsf{D}_q + 74\mathsf{O}_q + 9\mathsf{L}_{q,6} + 15\mathsf{L}_{q,7} + 14\mathsf{L}_{q,9} + 15\mathsf{B}. \end{aligned}$$

This yields

$$\begin{aligned} 2p_3 + 4q_3 + 15\mathcal{L} - 4u_{p,0} - 4l_{p,9} - 12l_1 - 2u_{q,0} \\ - 6l_{q,2} - 9l_{q,6} - 15l_{q,7} - 14l_{q,9} \\ \le 11E + 545 - 12a - 18b - 2T_b. \end{aligned}$$

All subtracted losses are distinct and have coefficients at most 15. The right side, using $a = 70 - b$ and $E \le 6$, is at most

$$-229 - b^2 - 7b < 0.$$

Mode B1. Add

$$\mathsf{C} + 2\mathsf{O}_p + \mathsf{L}_{p,9} + \mathsf{L}_1 + \mathsf{L}_2 + \mathsf{L}_{q,2} + 3\mathsf{O}_q + \mathsf{L}_{q,9} + \mathsf{B}.$$

The sum is

$$2p_3 + 2q_3 + \mathcal{L} - l_{p,9} - l_1 - l_2 - l_{q,2} - l_{q,9} \le E - 400 \le -394.$$

Again all left-side coefficients are nonnegative. The four modes exhaust the actual covering alternatives, proving the lemma. □

Corollary G.2 (The larger-AP branch is individually primitive)**.** *In the larger-side AP branch of the size-87 common-lattice proof, the smaller core side is primitive on the common lattice.*

Proof. All common-lattice, D-integrality and translate-overlap conclusions have already been proved before this step; thus $G = 148$ is available without circularity. Let $r = \min(a, b)$, $k = 35 - r$, and $\Omega = 574 - k^2$. The larger AP support defines the unit lattice, so $h_{\text{large}} = z = 0$ and $h_{\text{small}} = \kappa$. If the smaller side had gcd at least 2, its subset-sum support, of size $1 + T_r + \mu_{\text{small}}$, would have at least $T_r + \mu_{\text{small}}$ holes. Therefore $\kappa \ge T_r$. The post-lattice budget, $\rho_p, \rho_q \ge 0$, and $T \ge \kappa + 148$ give

$$c \le \Omega - 148 - T_r =: E_*(k) = \frac{-408 + 71k - 3k^2}{2} = \frac{145}{24} - \frac{3}{2}\left(k - \frac{71}{6}\right)^2.$$

The integer $E_*(k)$ is at most 6. If it is negative, $c \ge 0$ is already impossible. Otherwise $c \le 6$ contradicts Lemma G.1. This proves primitivity without enumerating side sizes, proving the smaller support AP, or singling out its possible gcd values. □

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